

SEACAR SAV Analysis

Last compiled on 14 May, 2026

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Important Notes

The purpose of this script is to provide a report summary of SAV analysis. The script used for analysis is SEACAR_SAV_BB_script_website.R.

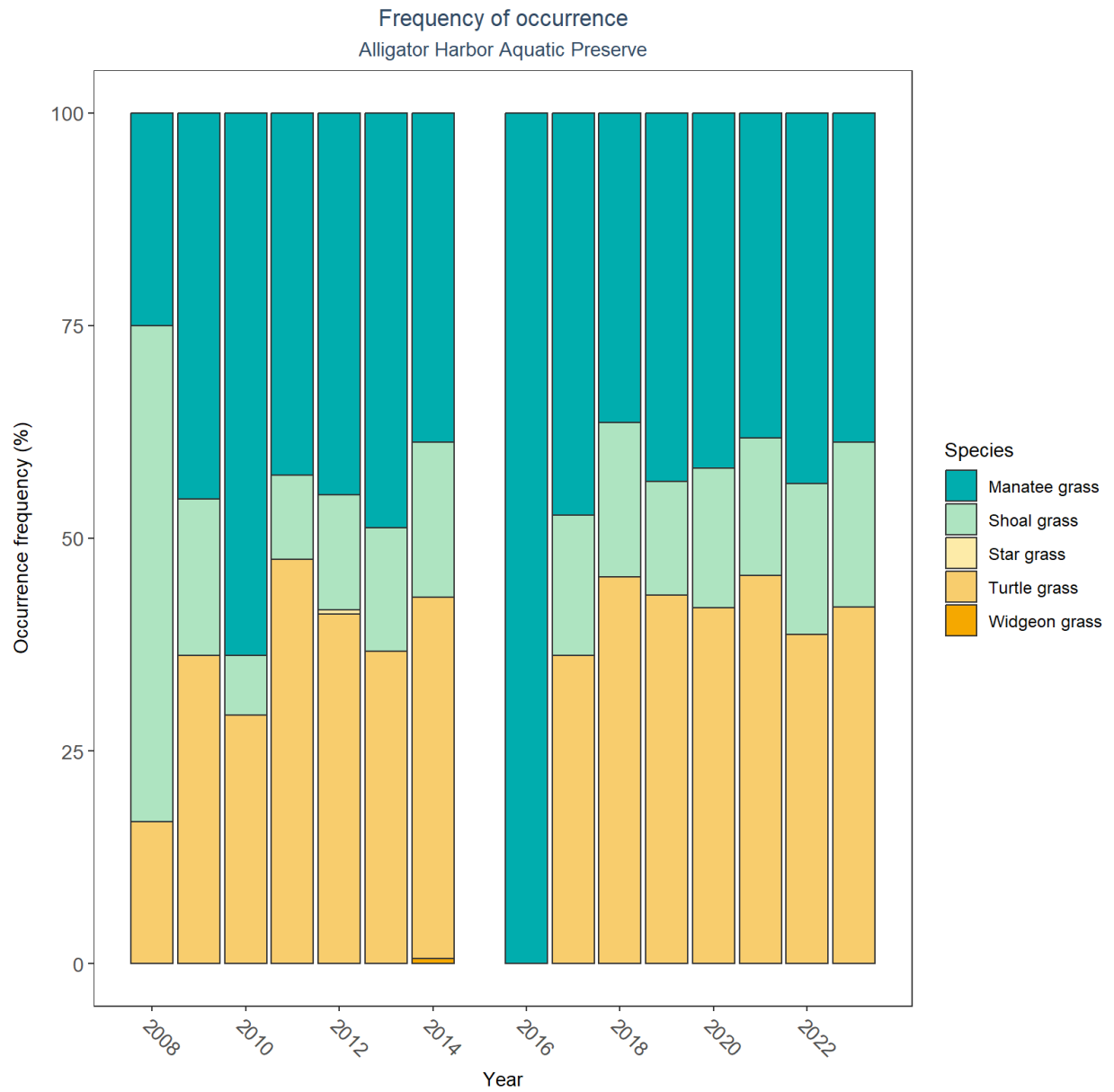
All scripts and outputs can be found on the SEACAR GitHub repository:

https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses

This script is based off of code originally written by Stephen Durham with comments by Marcus W. Beck. Modified by Tyler Hill.

The file being used for the analysis is: **All_SAV_Parameters-2026-Mar-06.txt**

Alligator Harbor Aquatic Preserve



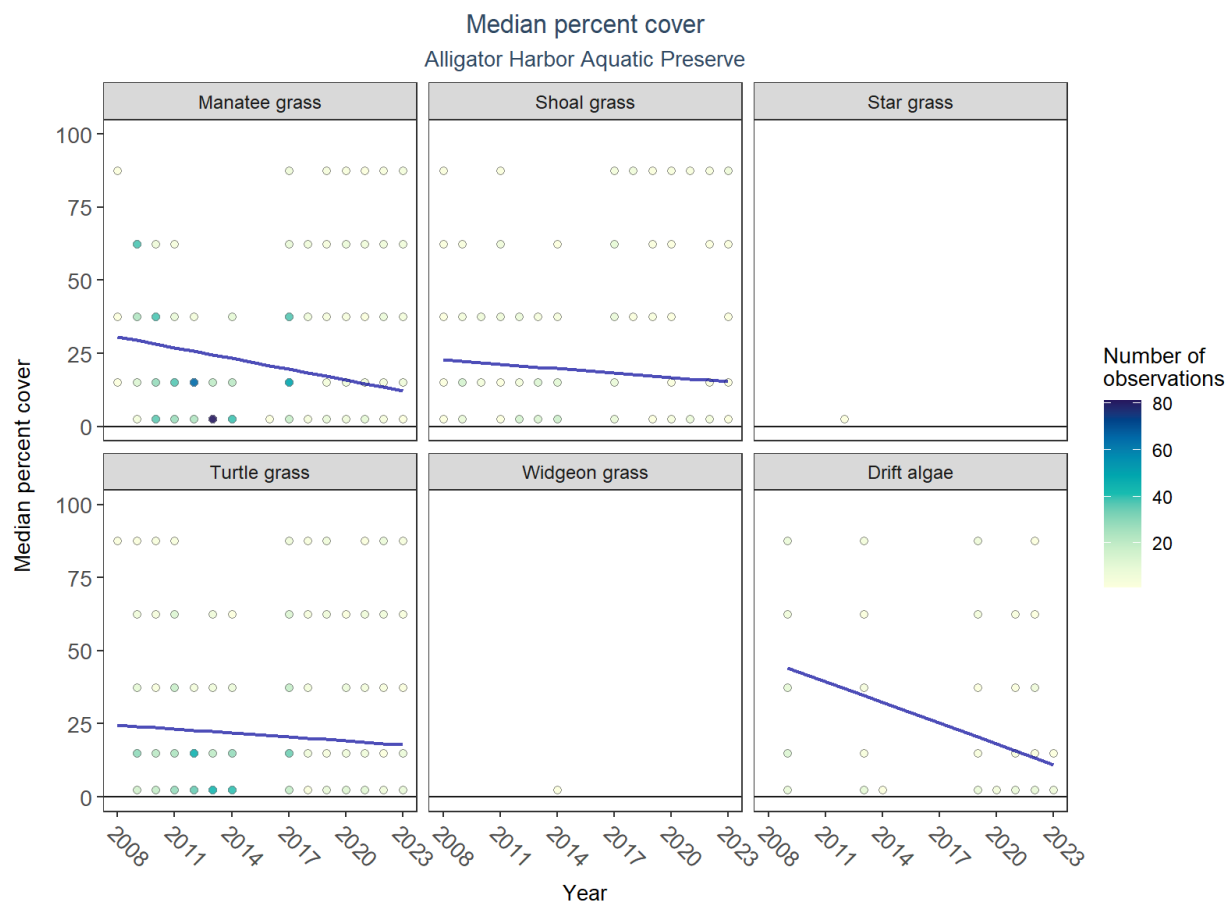
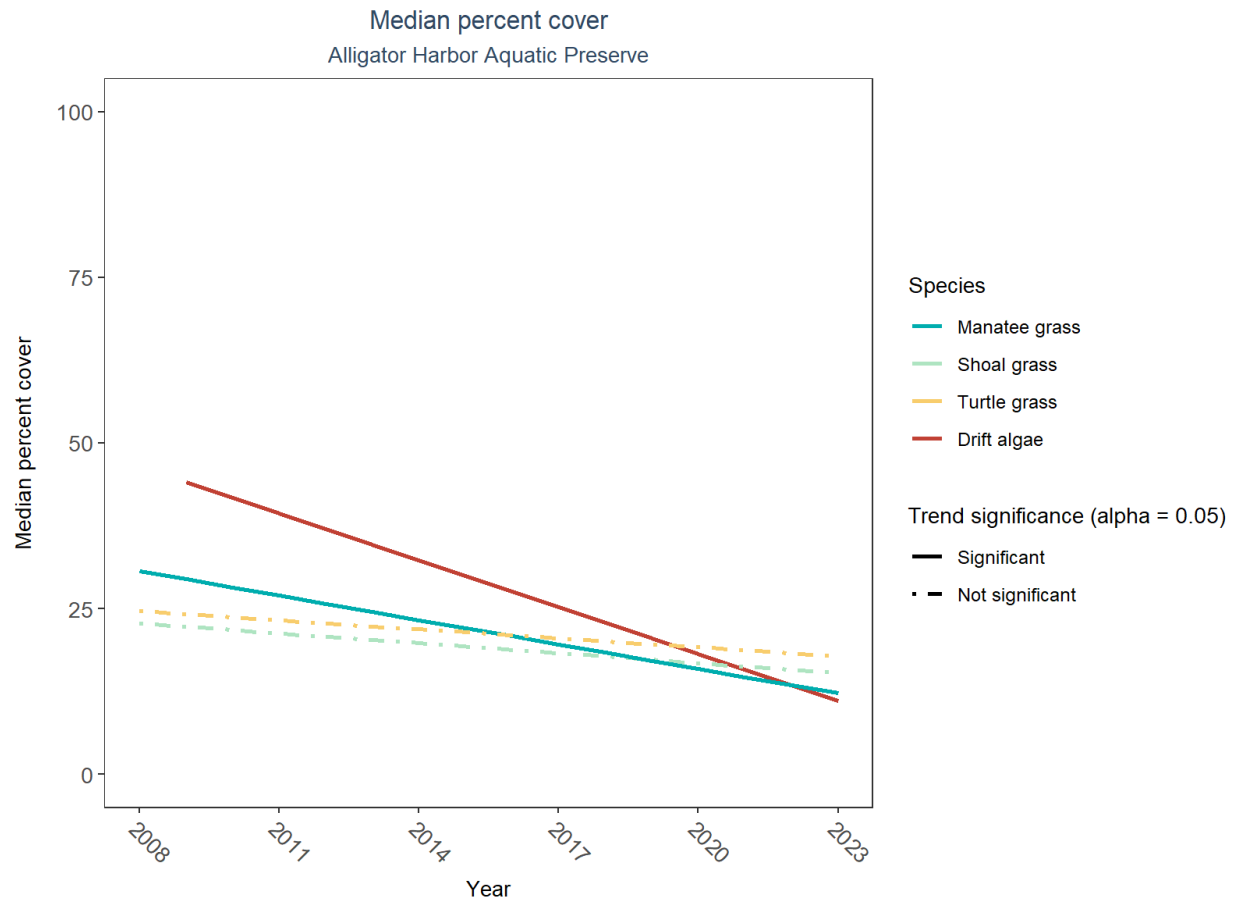


Table 1: SAV LME Results for Alligator Harbor Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2009 - 2023	79.40	-2.36	0.03
Shoal grass	No detectable trend	2008 - 2023	29.79	-0.50	0.47
Star grass	Insufficient data	-	-	-	-
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	Decreasing trend	2008 - 2023	47.86	-1.23	0.03
Turtle grass	No detectable trend	2008 - 2023	30.97	-0.45	0.38

Annual decreases in percent cover were observed for manatee grass (-1.23%) and drift algae (-2.36%). No detectable change in percent cover was observed for shoal grass and turtle grass. Trends in percent cover could not be evaluated for star grass and widgeon grass due to insufficient data.



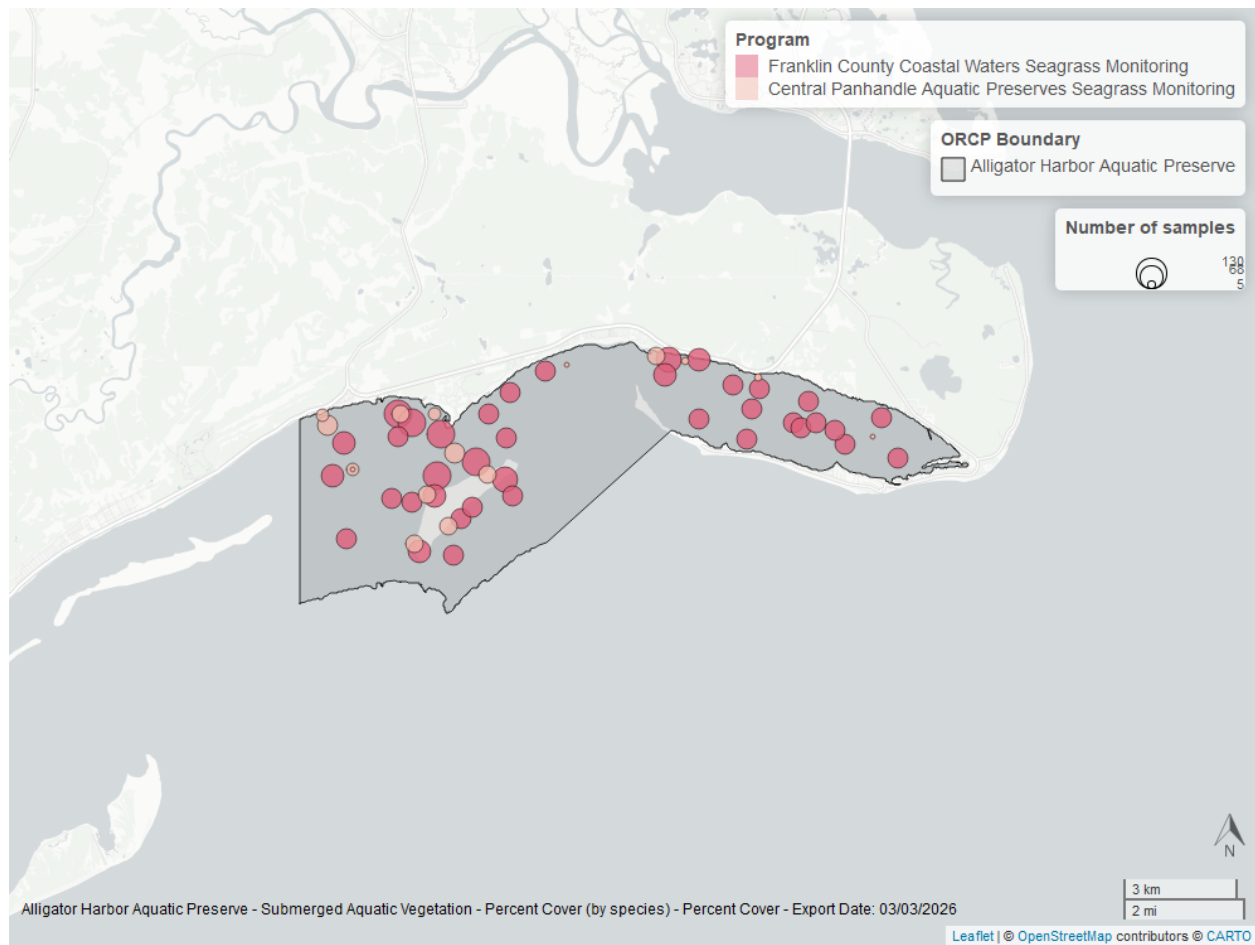
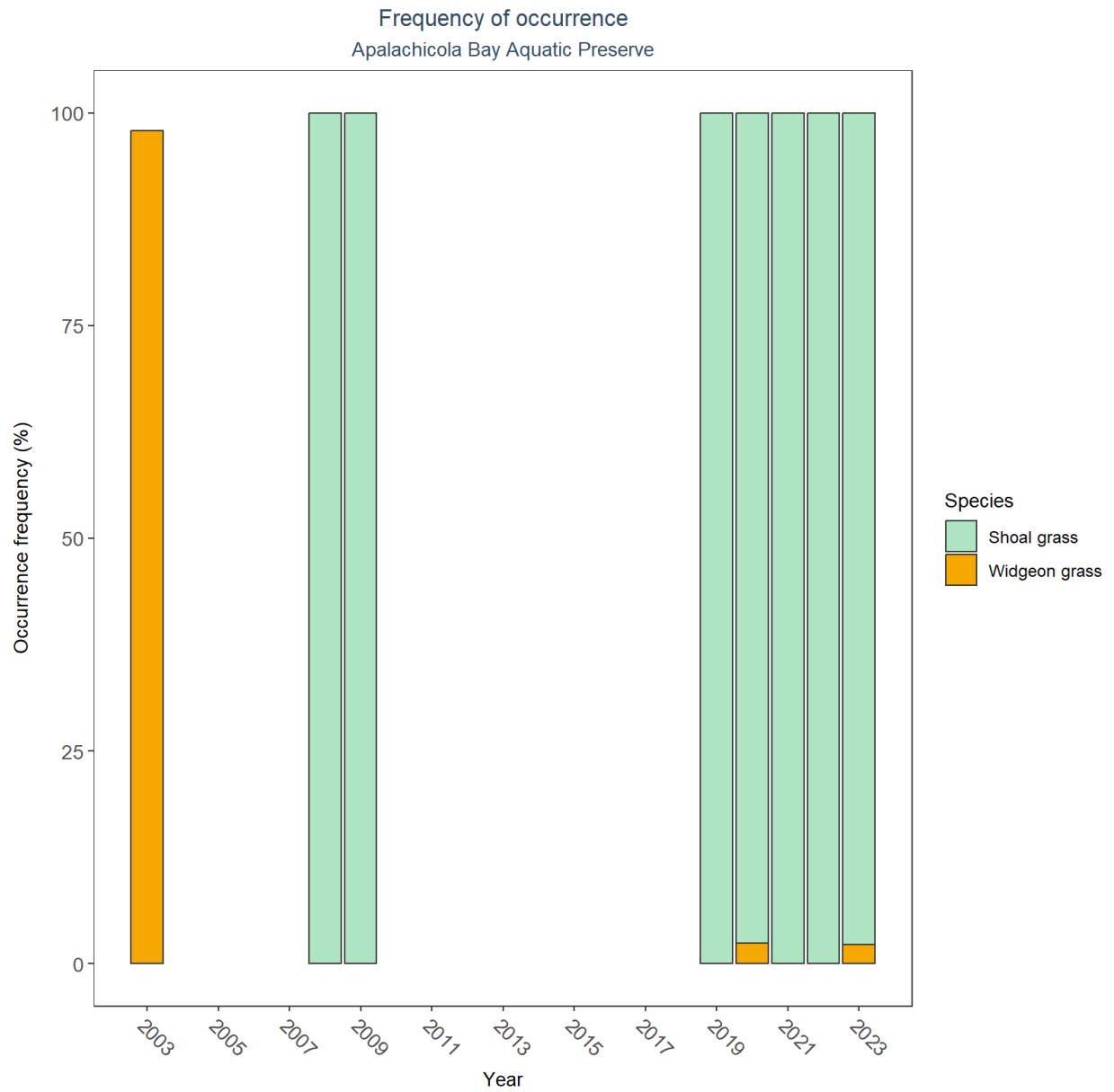


Figure 1: Map showing SAV sampling sites within the boundaries of *Alligator Harbor Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Apalachicola Bay Aquatic Preserve



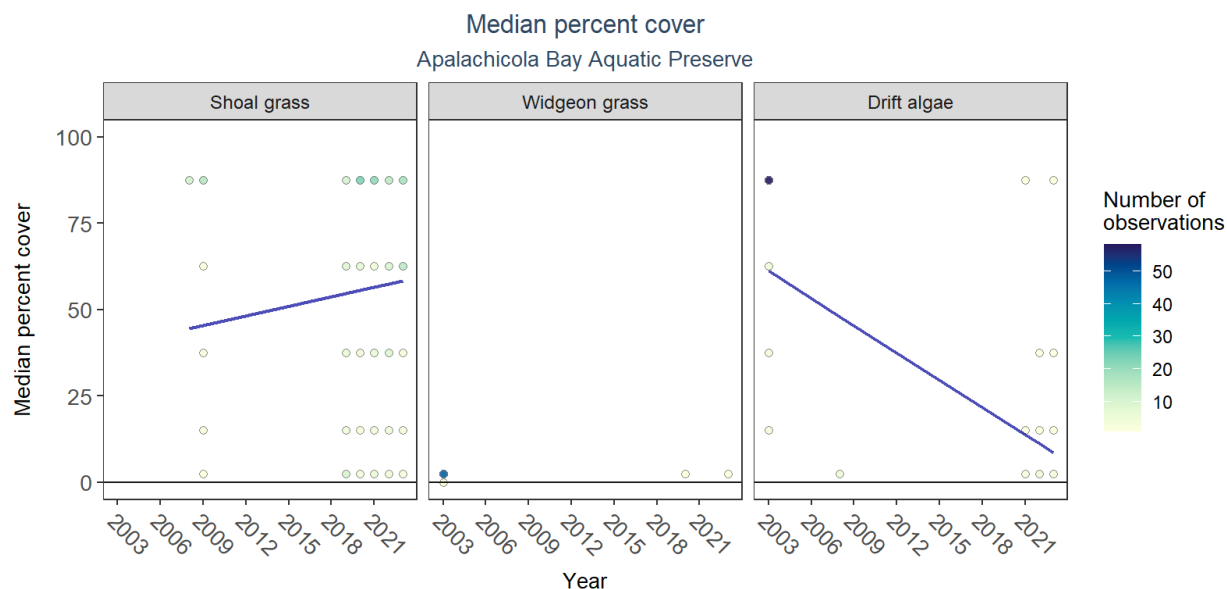
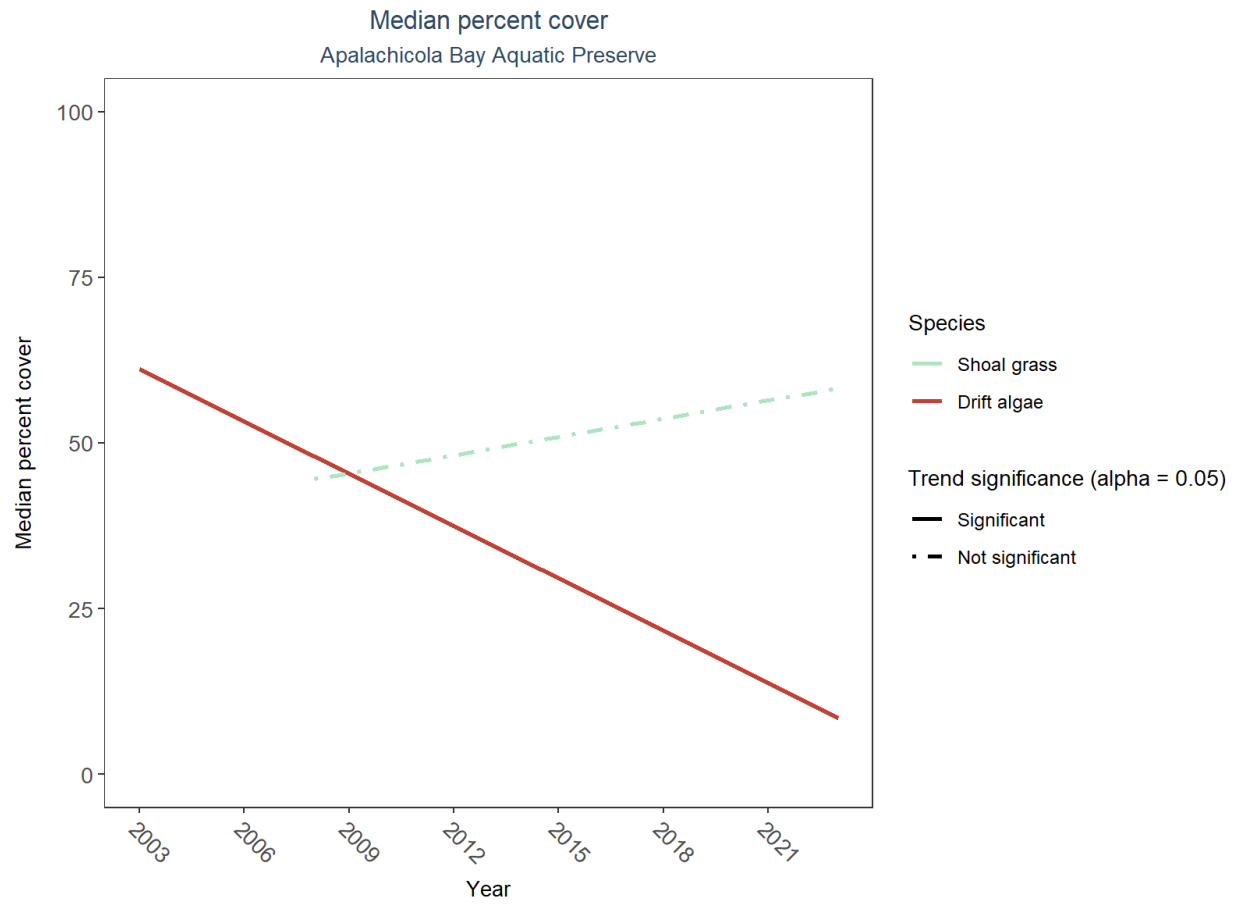


Table 2: SAV LME Results for Apalachicola Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2003 - 2023	84.82	-2.63	0.00
Shoal grass	No detectable trend	2008 - 2023	31.69	0.92	0.49
Widgeon grass	Insufficient data	-	-	-	-

An annual decrease in percent cover was observed for drift algae (-2.63%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for widgeon grass due to insufficient data.



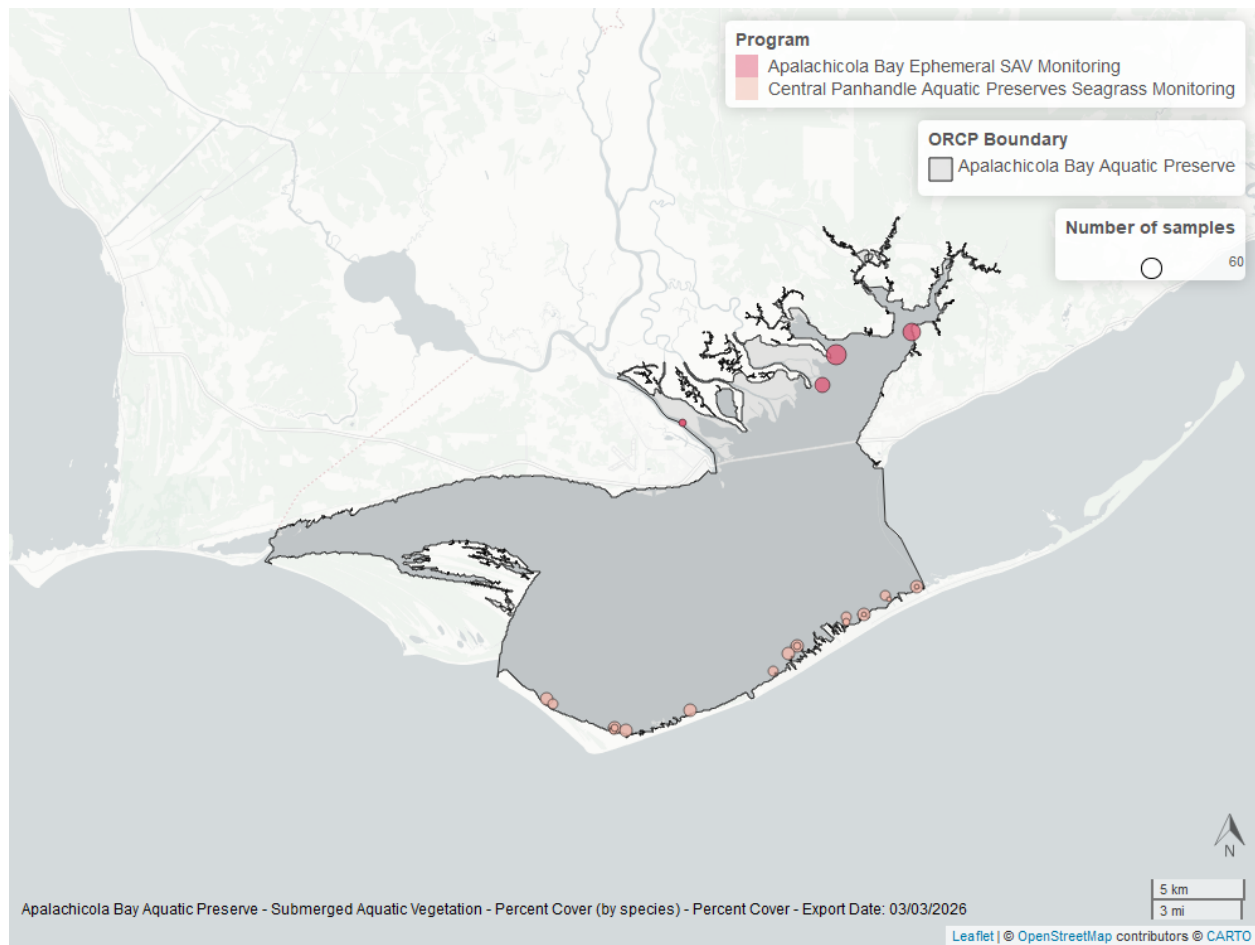
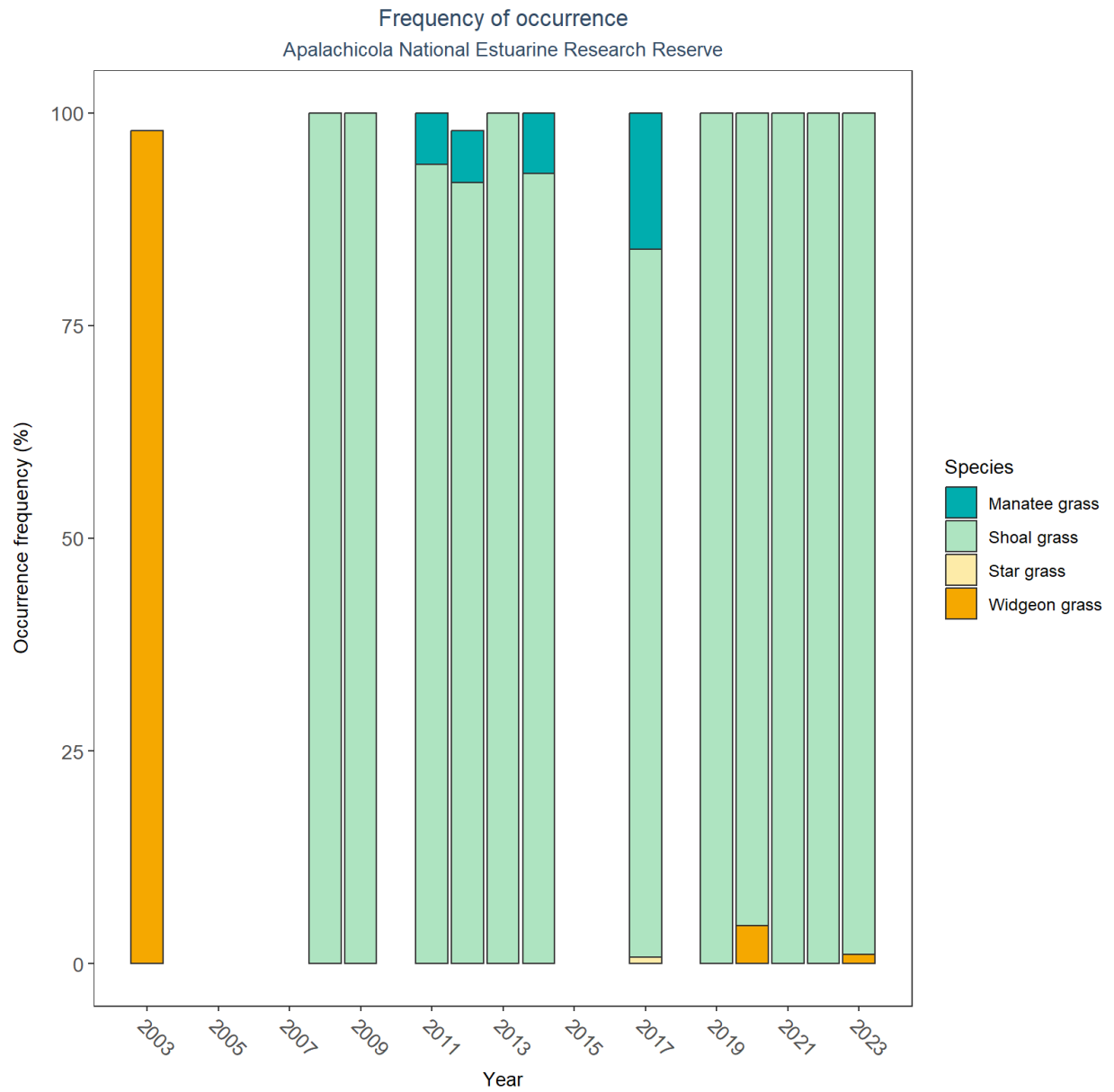


Figure 2: Map showing SAV sampling sites within the boundaries of *Apalachicola Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Apalachicola National Estuarine Research Reserve



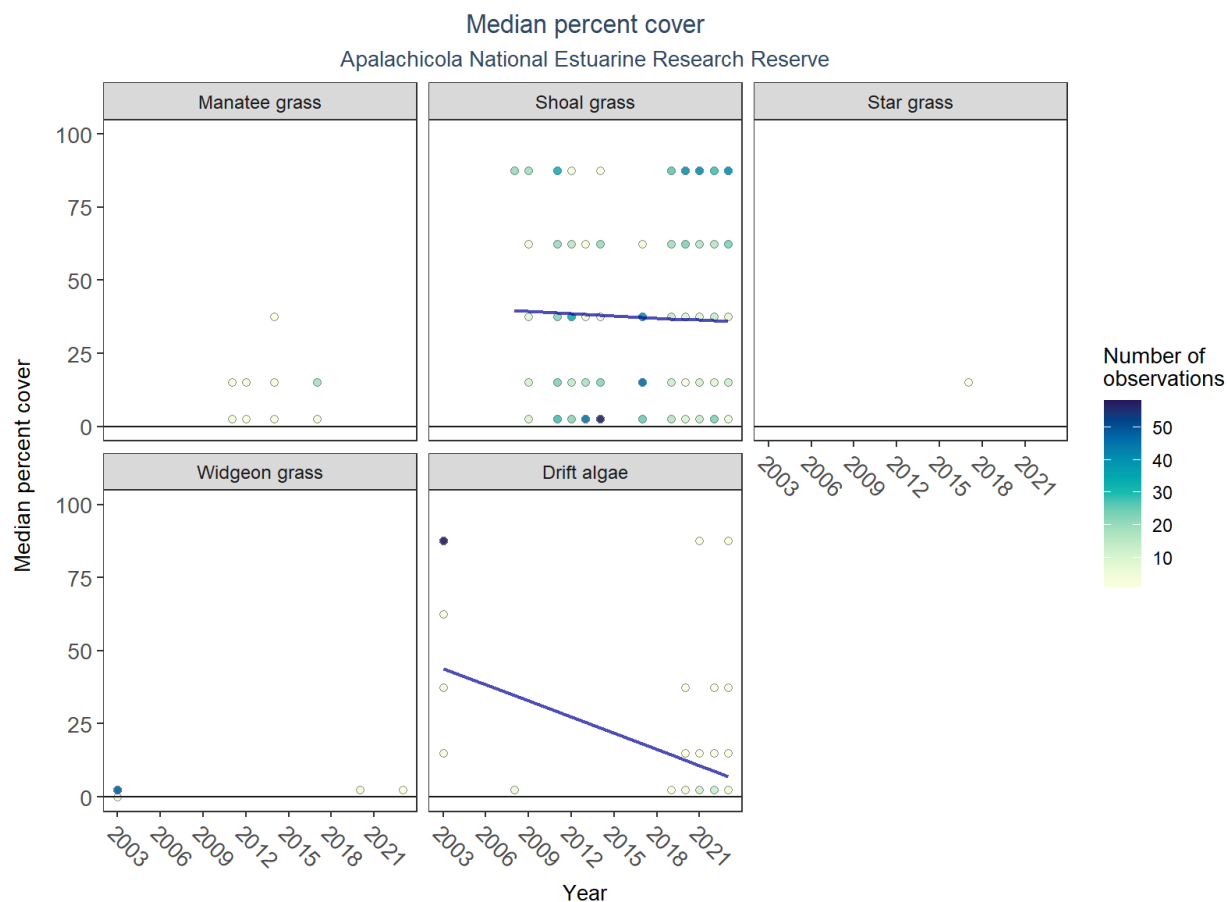
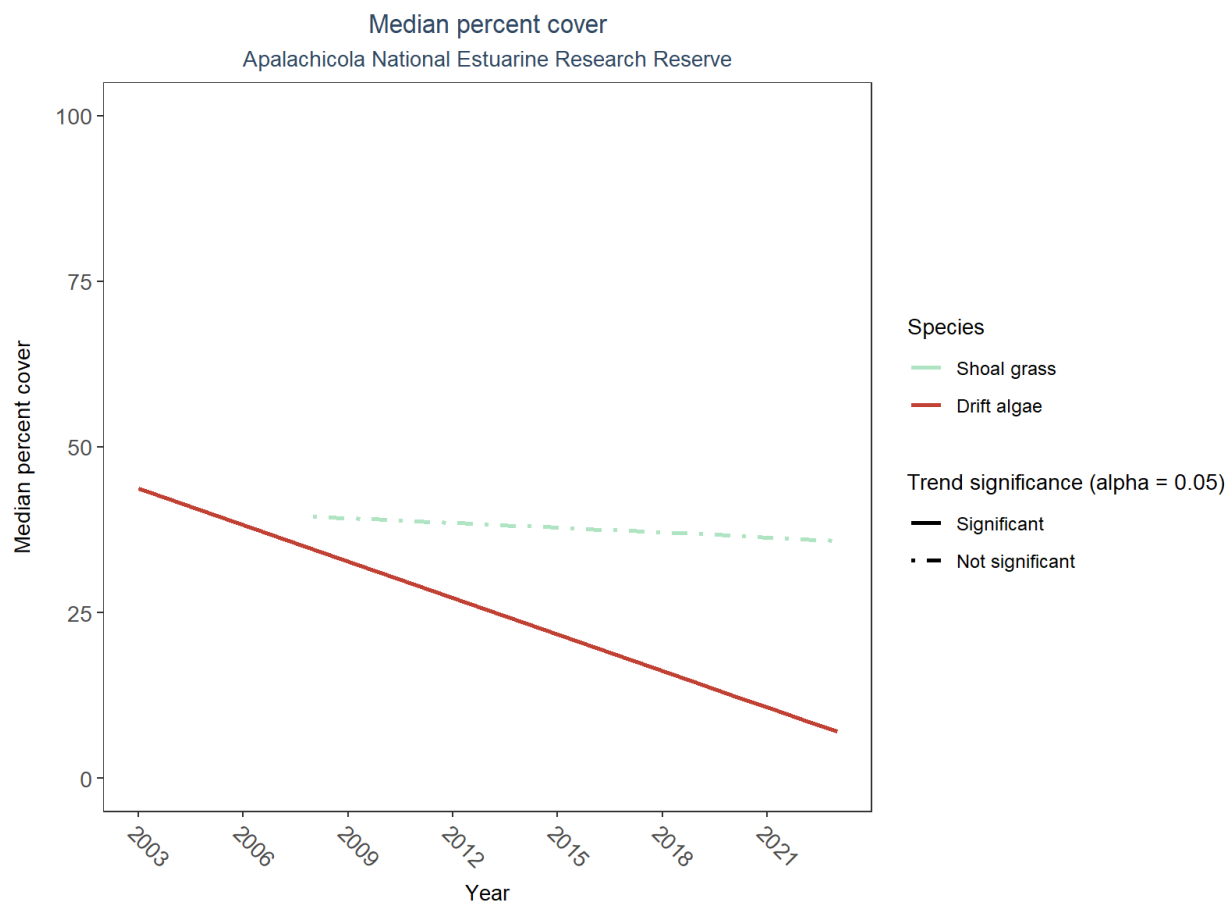


Table 3: SAV LME Results for Apalachicola National Estuarine Research Reserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2003 - 2023	60.31	-1.84	0.03
Shoal grass	No detectable trend	2008 - 2023	42.98	-0.24	0.73
Star grass	Insufficient data	-	-	-	-
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	Insufficient data	-	-	-	-

An annual decrease in percent cover was observed for drift algae (-1.84%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for manatee grass, star grass, and widgeon grass due to insufficient data.



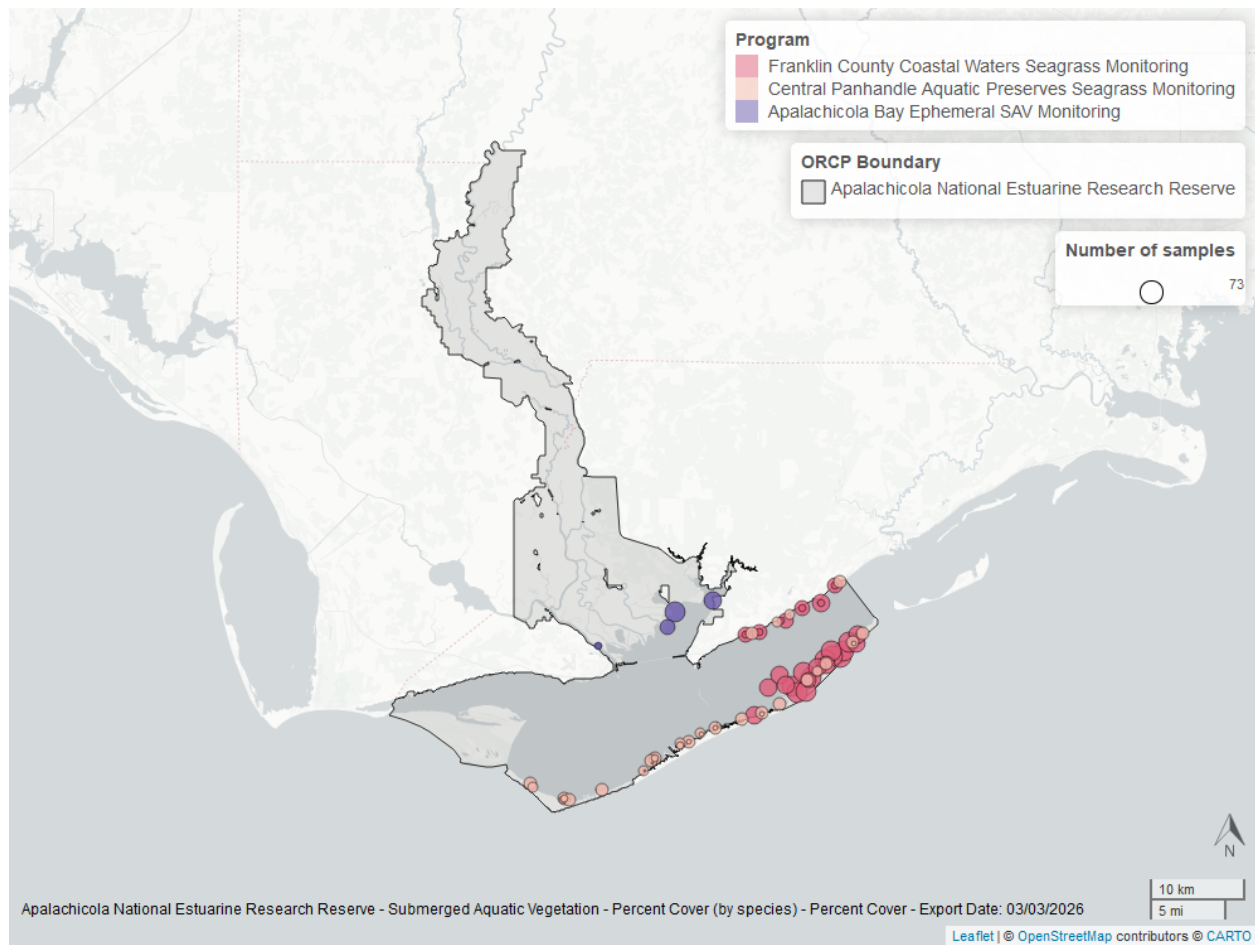
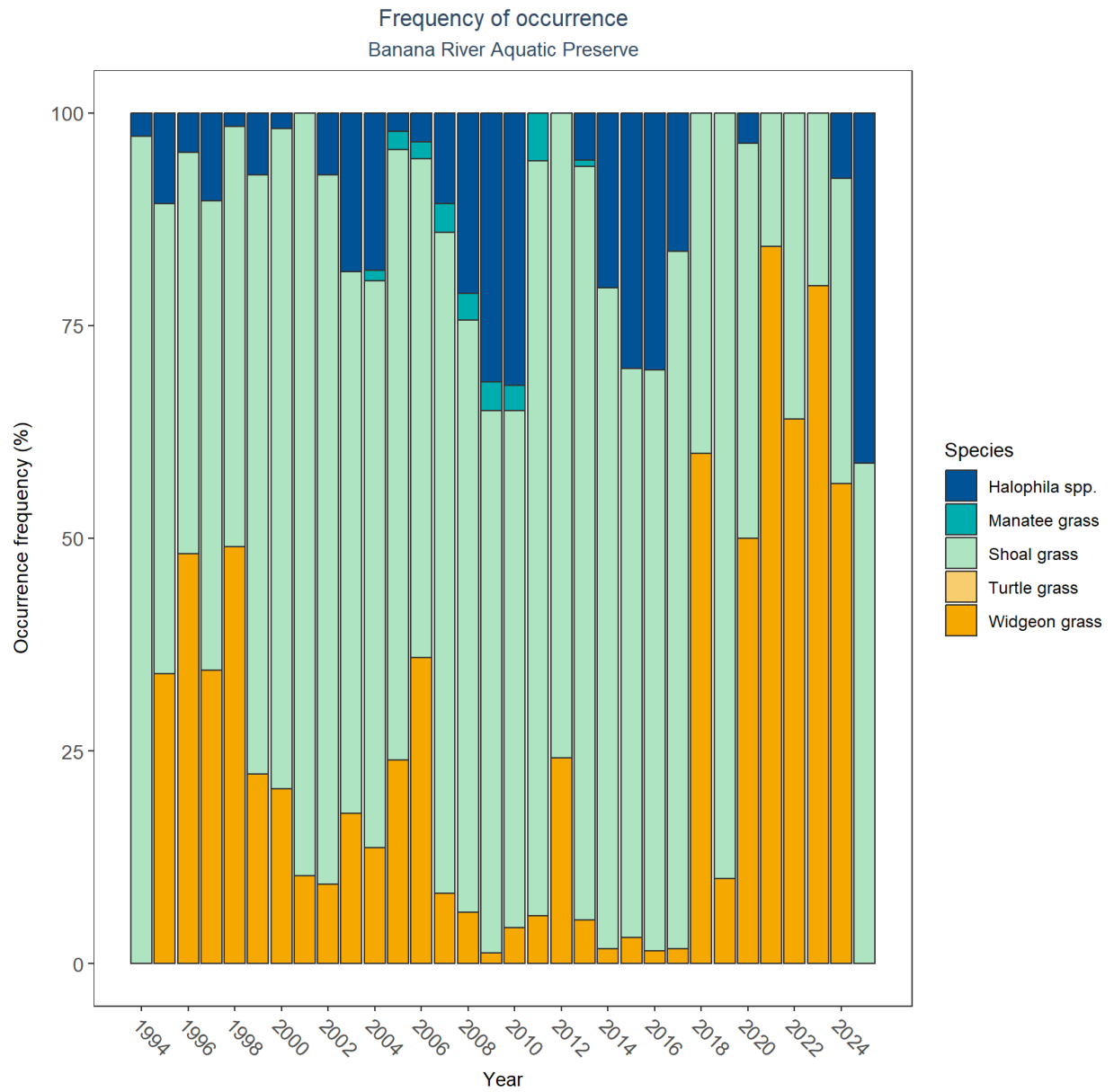


Figure 3: Map showing SAV sampling sites within the boundaries of *Apalachicola National Estuarine Research Reserve*. The point size reflects the number of samples at a given sampling site.

Banana River Aquatic Preserve



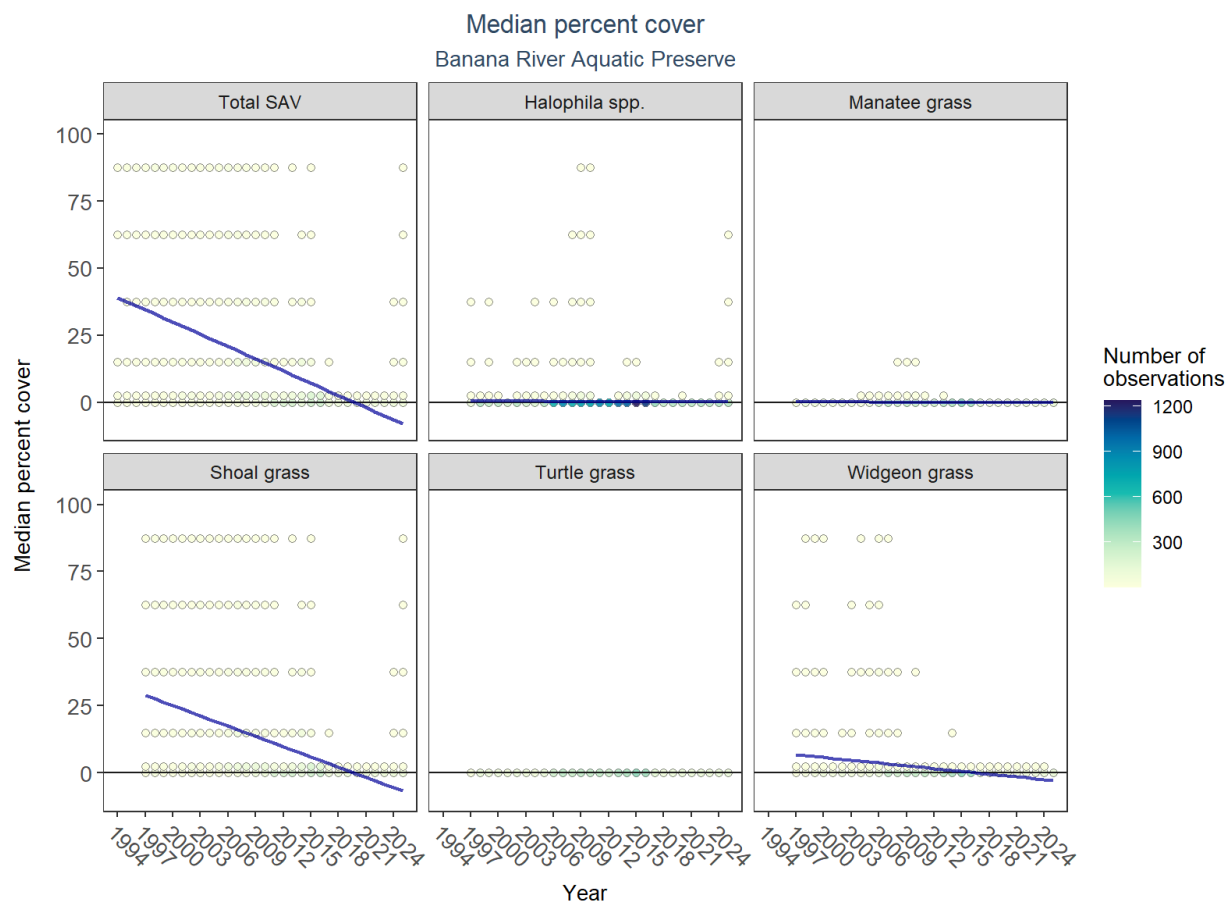
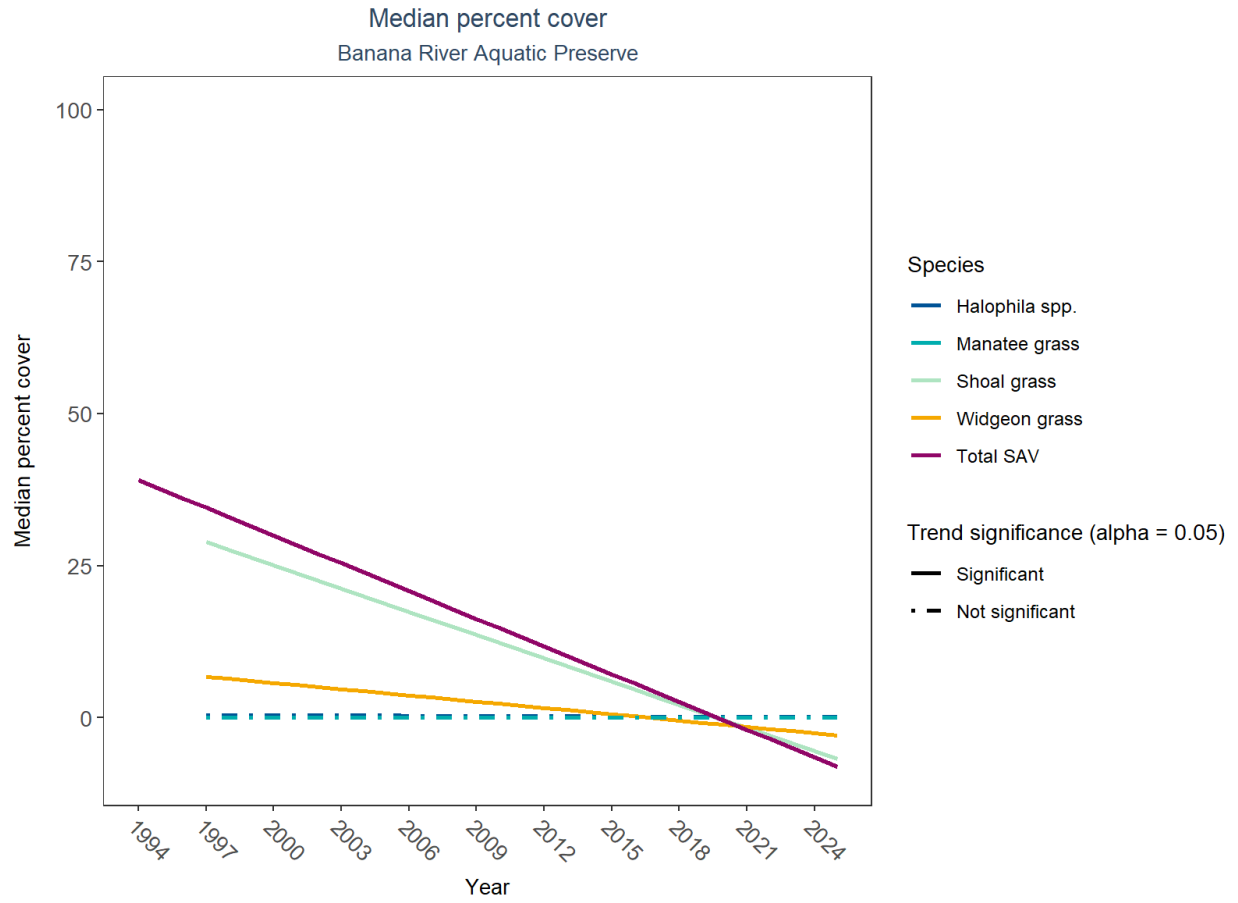


Table 4: SAV LME Results for Banana River Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Decreasing trend	1997 - 2025	32.71	-1.27	0.00
Halophila spp.	No detectable trend	1997 - 2025	0.50	-0.01	0.14
Widgeon grass	Decreasing trend	1997 - 2025	7.80	-0.34	0.00
Manatee grass	No detectable trend	1997 - 2025	0.09	0.00	0.07
Turtle grass	Model did not fit the available data	-	-	-	-
Total SAV	Decreasing trend	1994 - 2025	39.07	-1.52	0.00
Total seagrass	Insufficient data	-	-	-	-

Annual decreases in percent cover were observed for total SAV (-1.52%), shoal grass (-1.27%), and widgeon grass (-0.34%). No detectable change in percent cover was observed for *Halophila* spp. and manatee grass. Trends in percent cover could not be evaluated for total seagrass and drift algae due to insufficient data, and the model could not be fitted for turtle grass.



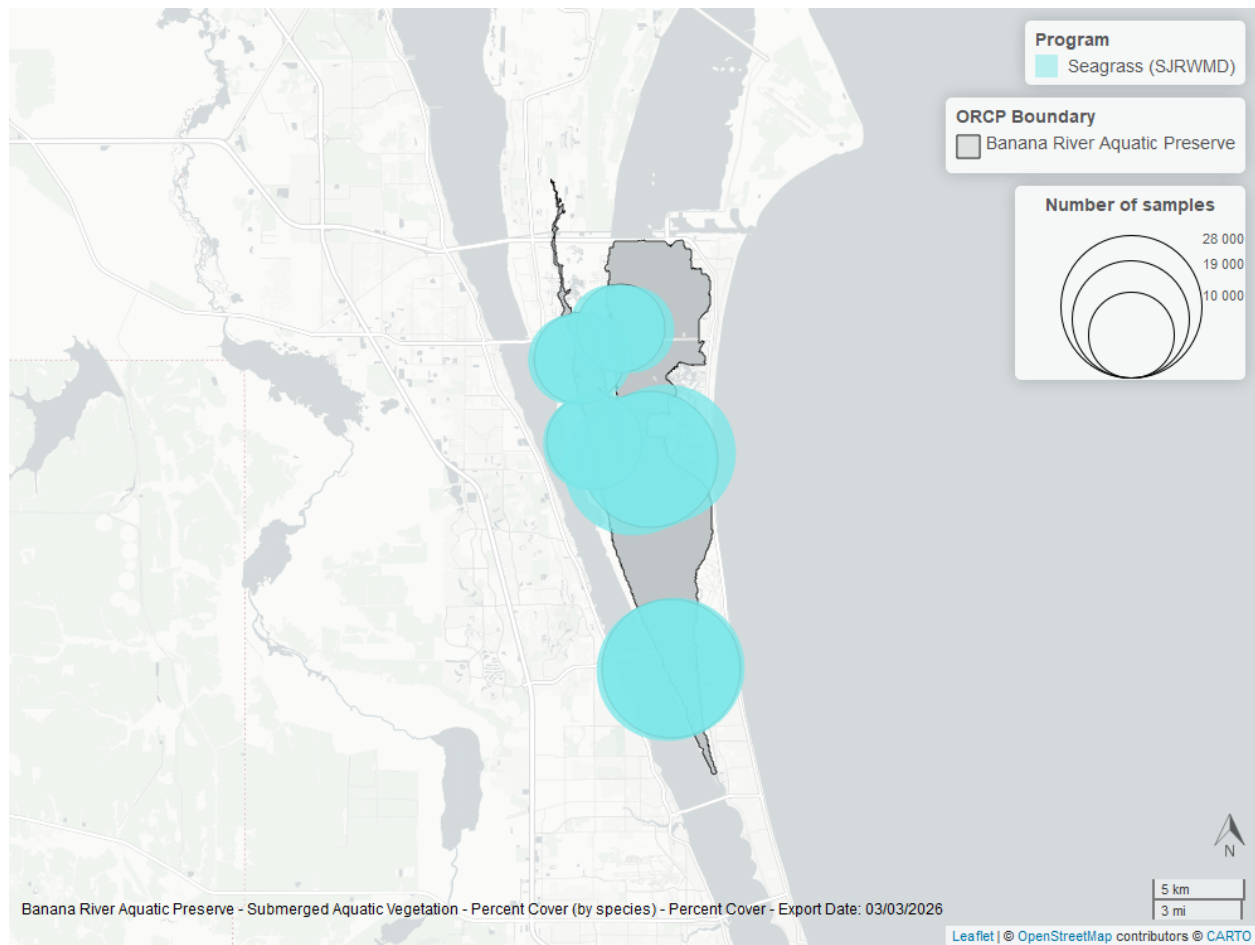
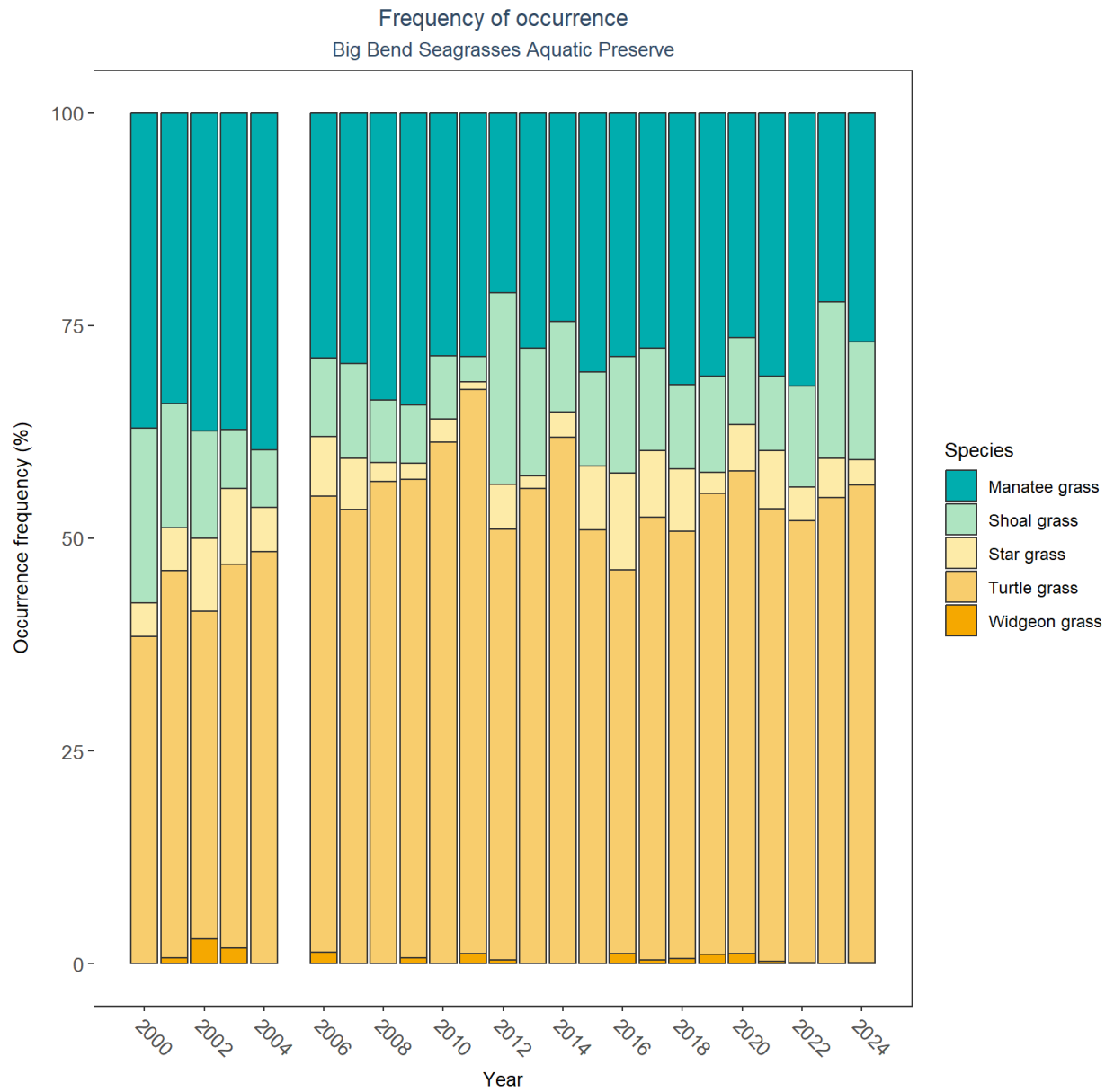


Figure 4: Map showing SAV sampling sites within the boundaries of *Banana River Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Big Bend Seagrasses Aquatic Preserve



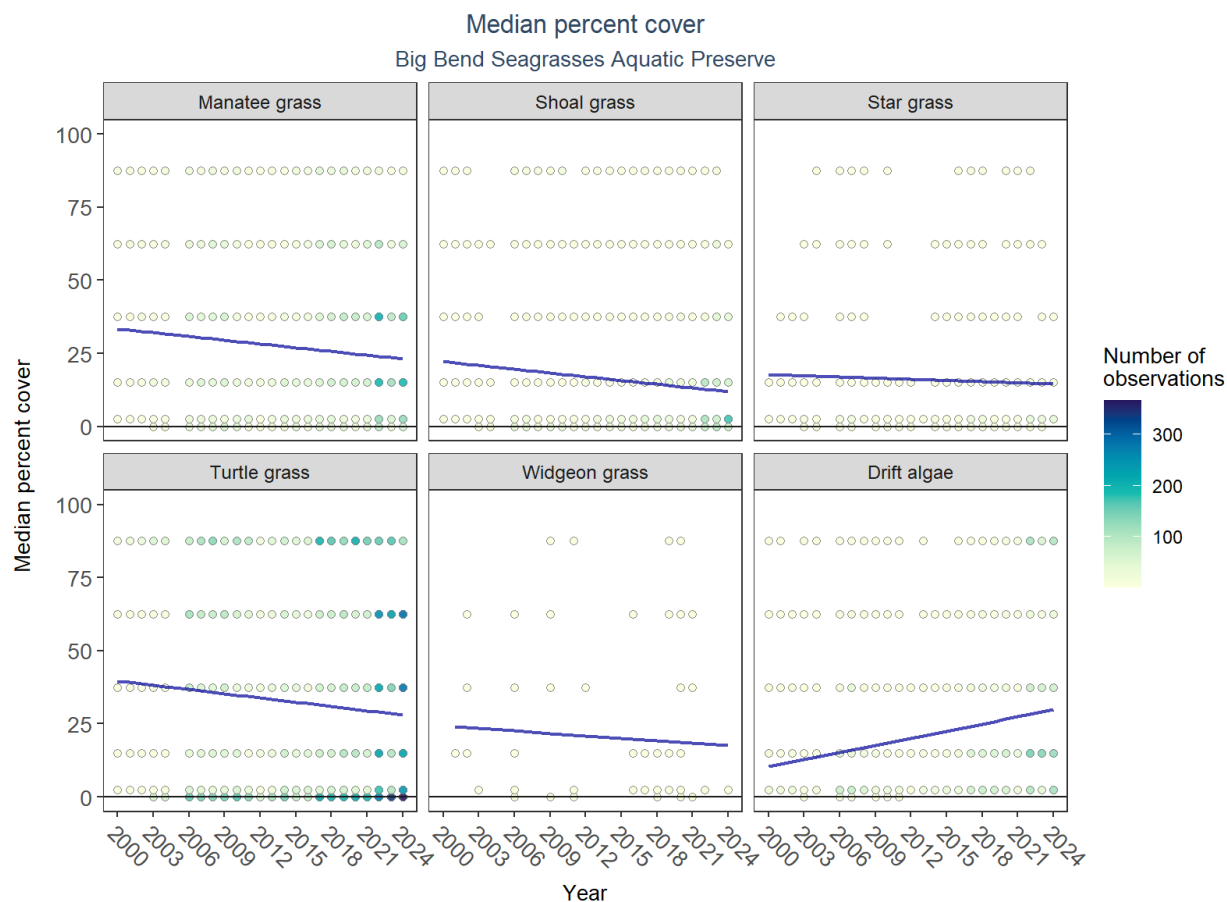
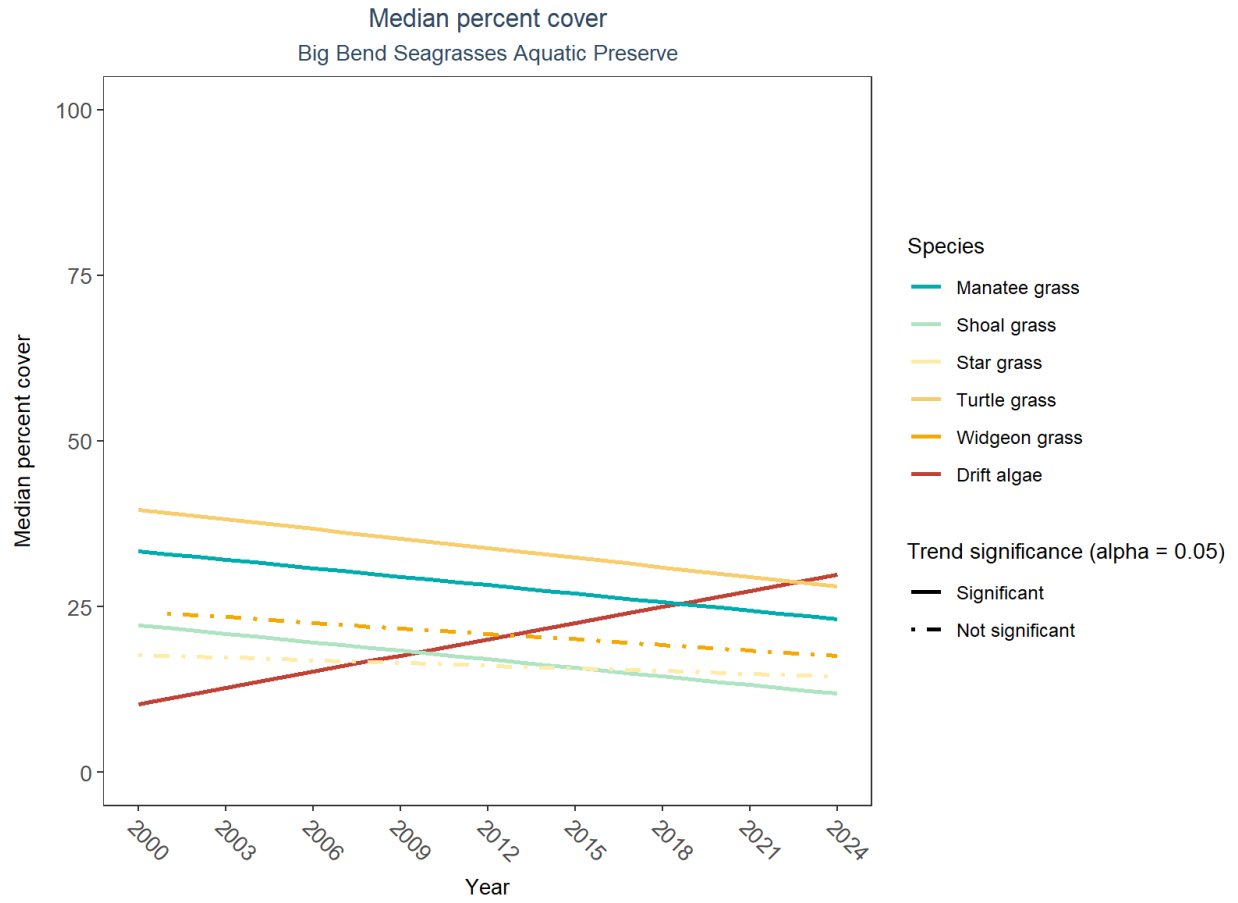


Table 5: SAV LME Results for Big Bend Seagrasses Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Increasing trend	2000 - 2024	5.45	0.81	0.00
Shoal grass	Decreasing trend	2000 - 2024	24.83	-0.43	0.00
Star grass	No detectable trend	2000 - 2024	18.55	-0.13	0.31
Widgeon grass	No detectable trend	2001 - 2024	25.94	-0.28	0.60
Manatee grass	Decreasing trend	2000 - 2024	35.95	-0.43	0.00
Turtle grass	Decreasing trend	2000 - 2024	42.52	-0.48	0.00

An annual increase in percent cover was observed for drift algae (0.81%). Annual decreases in percent cover were observed for manatee grass (-0.43%), shoal grass (-0.43%), and turtle grass (-0.48%). No detectable change in percent cover was observed for star grass and widgeon grass.



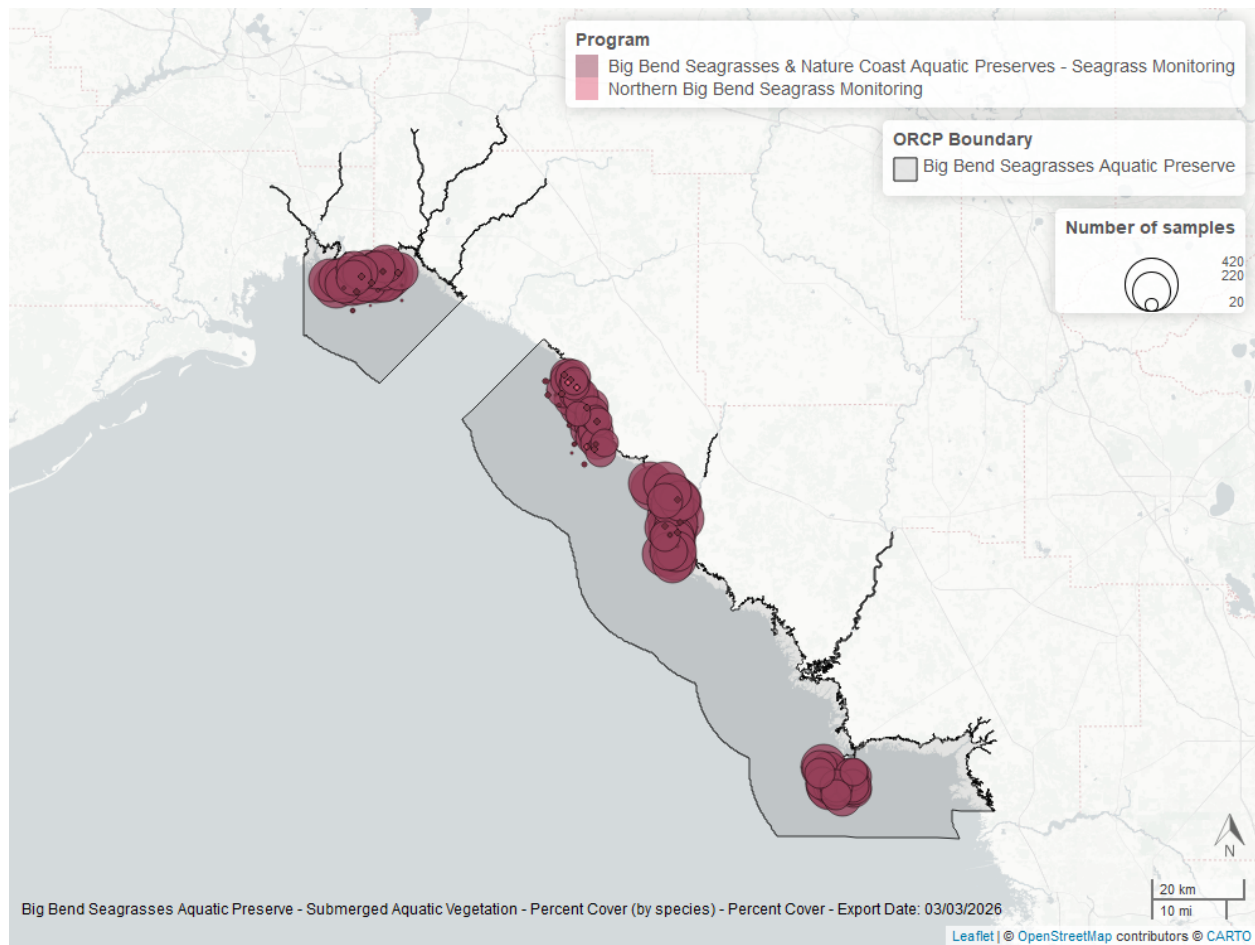
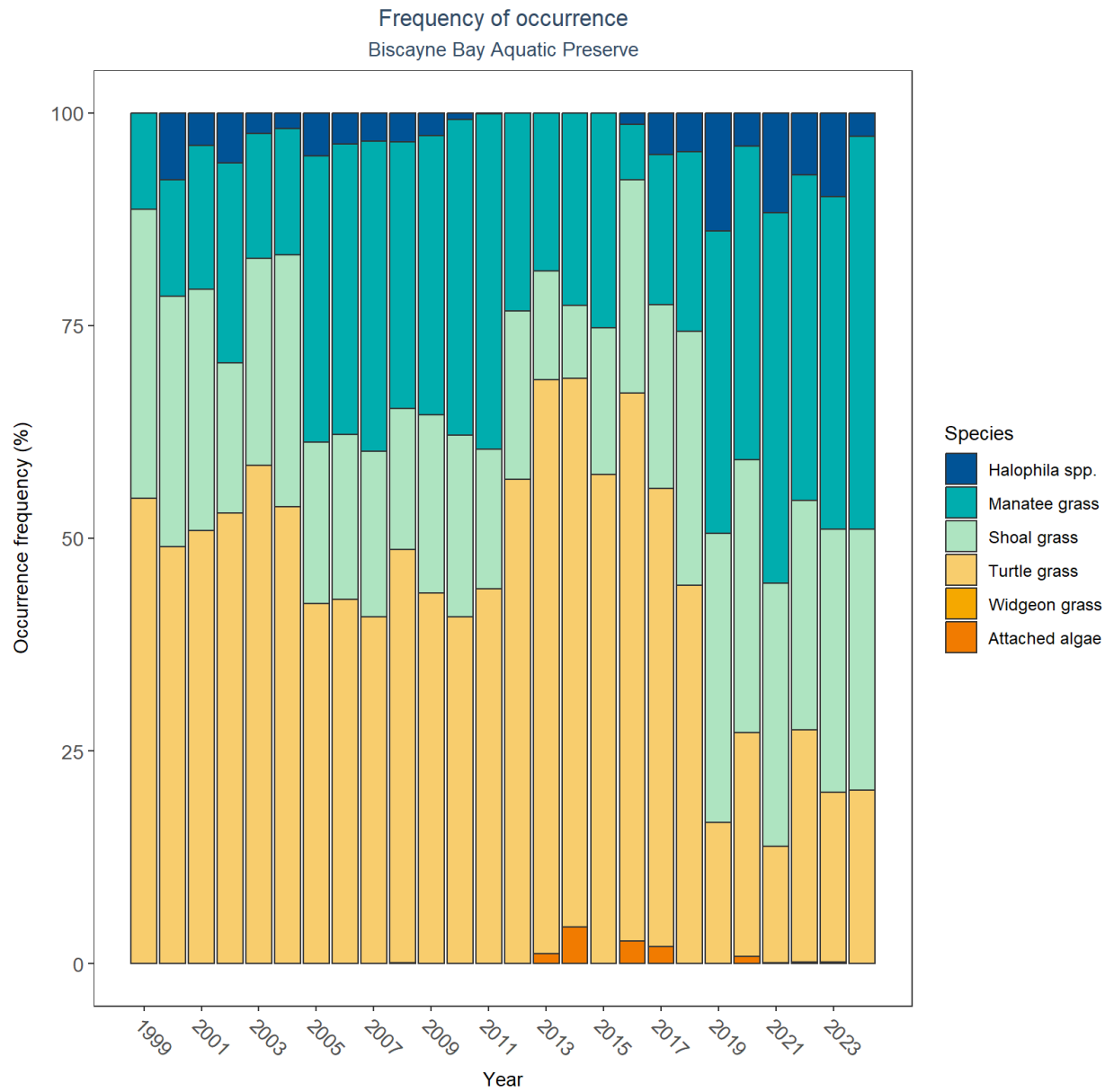


Figure 5: Map showing SAV sampling sites within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Biscayne Bay Aquatic Preserve



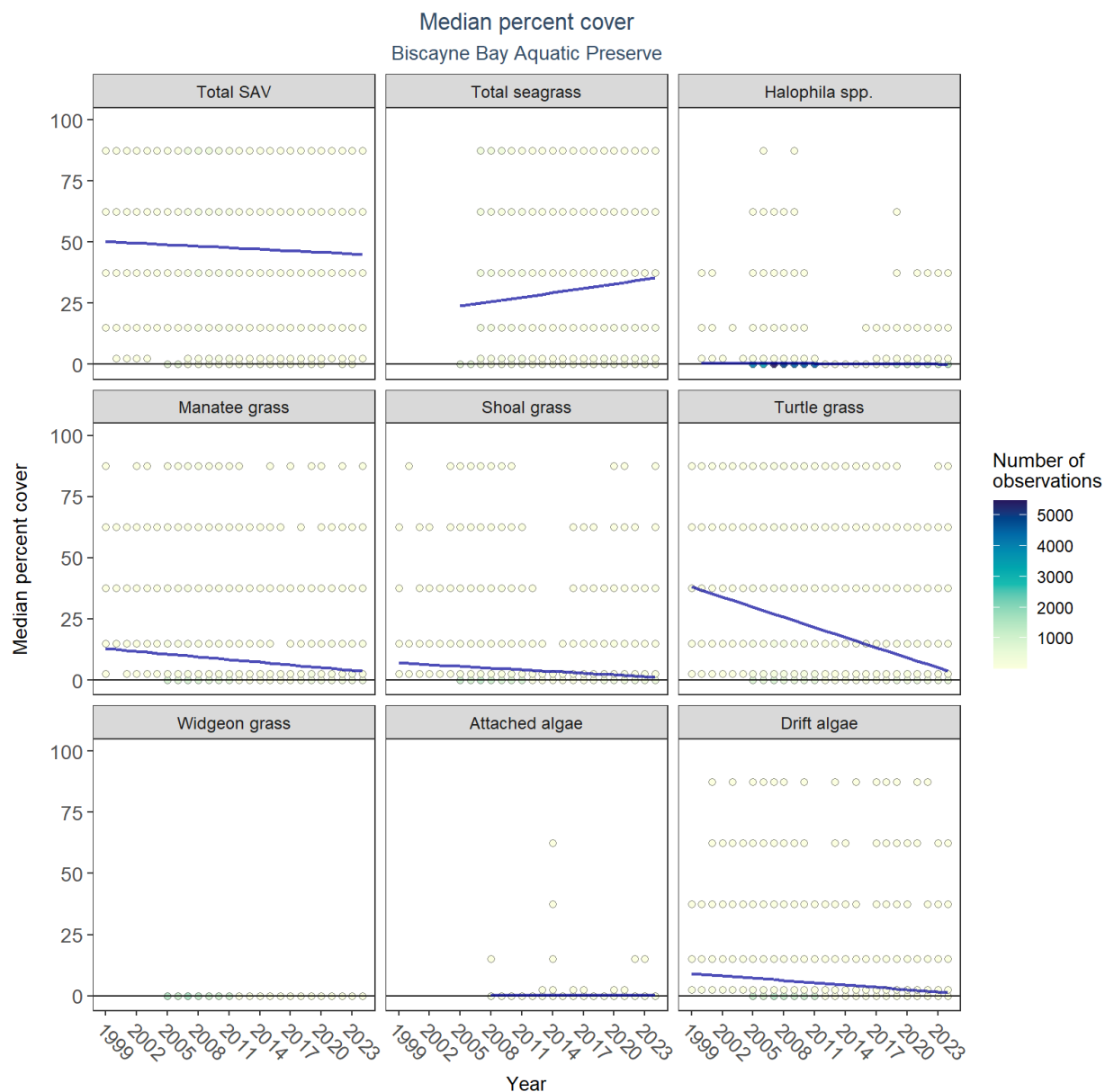
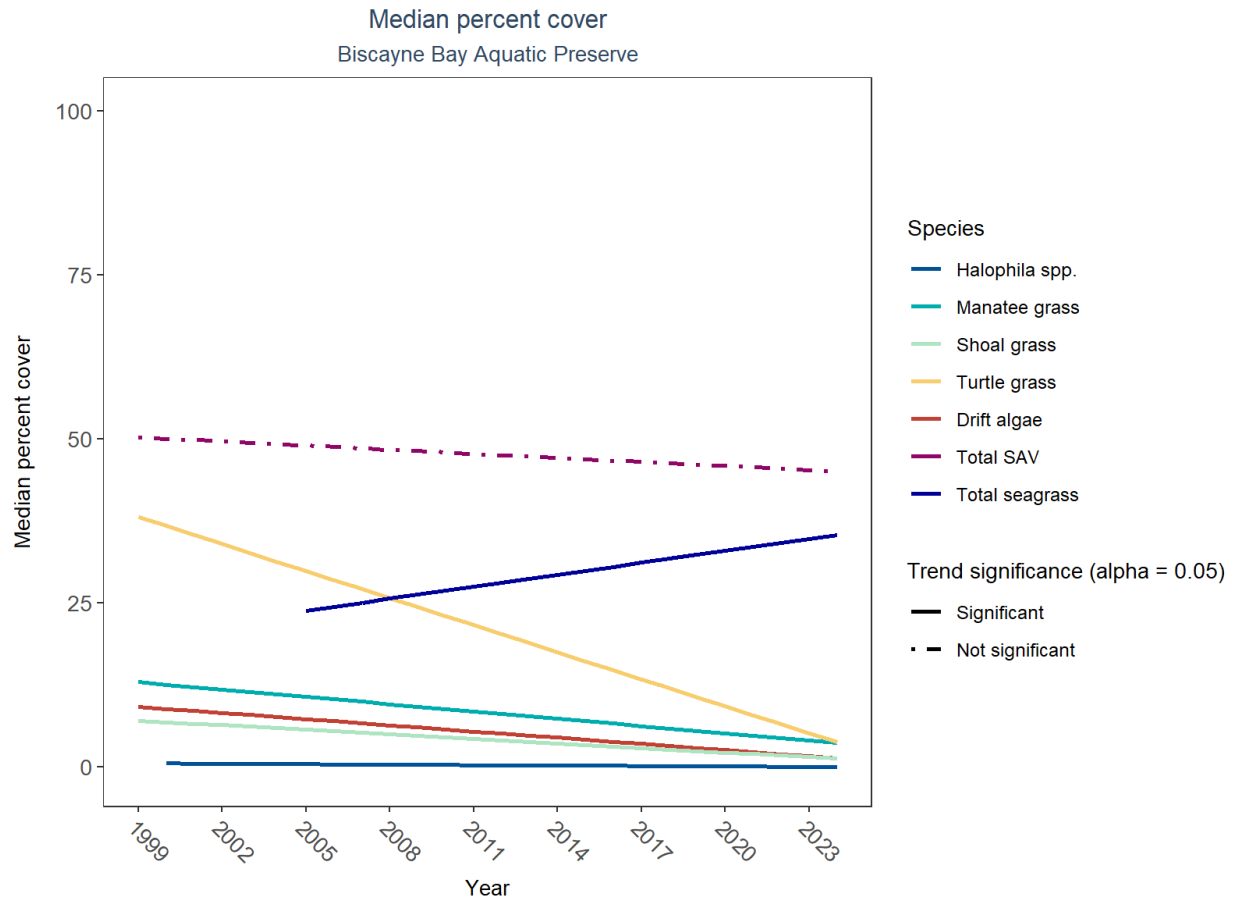


Table 6: SAV LME Results for Biscayne Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2008 - 2024	-0.03	0.01	0.10
Drift algae	Decreasing trend	1999 - 2024	10.73	-0.31	0.00
Shoal grass	Decreasing trend	1999 - 2024	8.21	-0.23	0.00
Halophila spp.	Decreasing trend	2000 - 2024	0.67	-0.02	0.01
Widgeon grass	Model did not fit the available data	-	-	-	-
Manatee grass	Decreasing trend	1999 - 2024	14.75	-0.37	0.00
Turtle grass	Decreasing trend	1999 - 2024	44.97	-1.37	0.00
Total SAV	No detectable trend	1999 - 2024	51.24	-0.21	0.26
Total seagrass	Increasing trend	2005 - 2024	17.14	0.61	0.00

An annual increase in percent cover was observed for total seagrass (0.61%). Annual decreases in percent cover were observed for *Halophila* spp. (-0.02%), manatee grass (-0.37%), shoal grass (-0.23%), turtle grass (-1.37%), and drift algae (-0.31%). No detectable change in percent cover was observed for total SAV and attached algae. The model could not be fitted for widgeon grass.



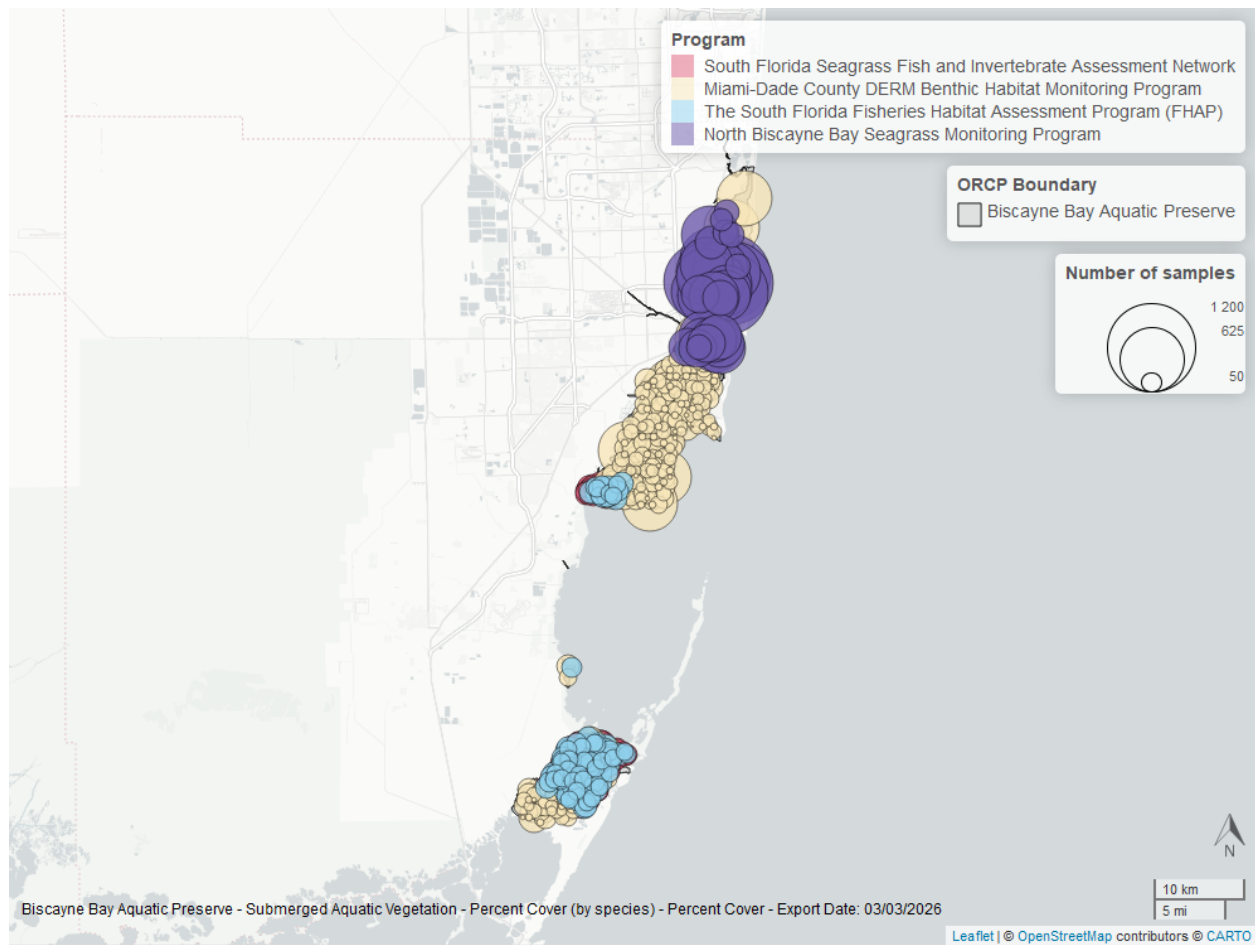
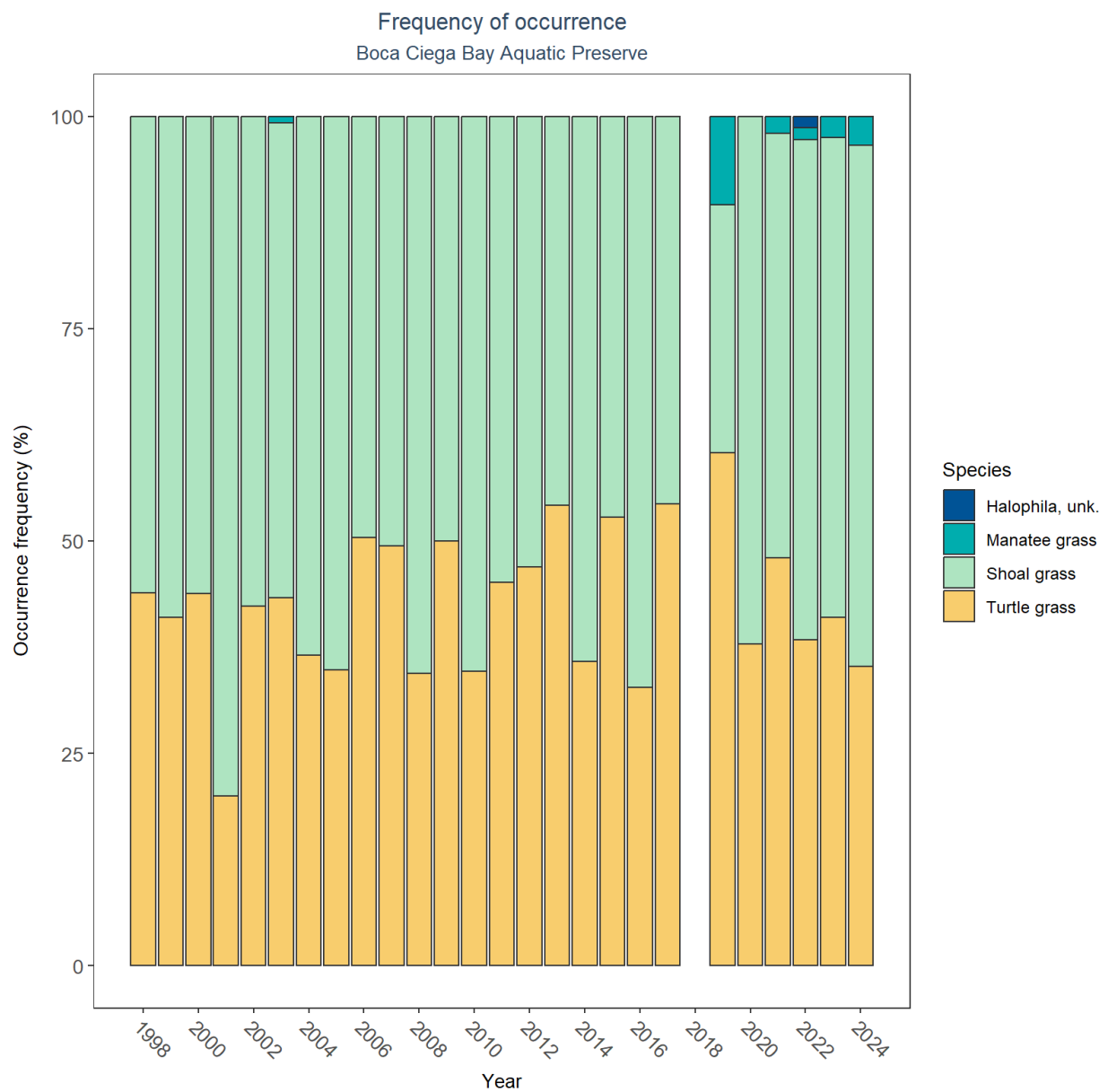


Figure 6: Map showing SAV sampling sites within the boundaries of *Biscayne Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Boca Ciega Bay Aquatic Preserve



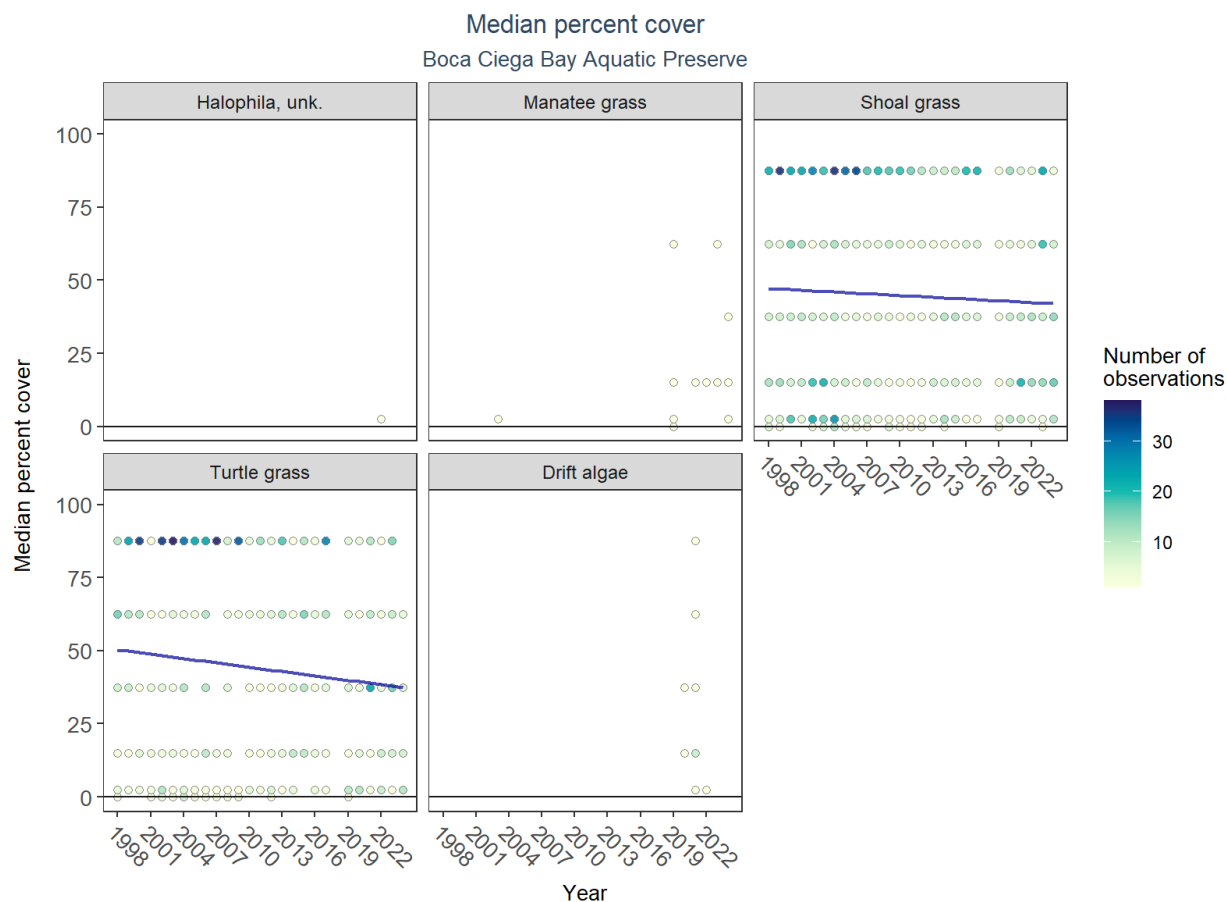
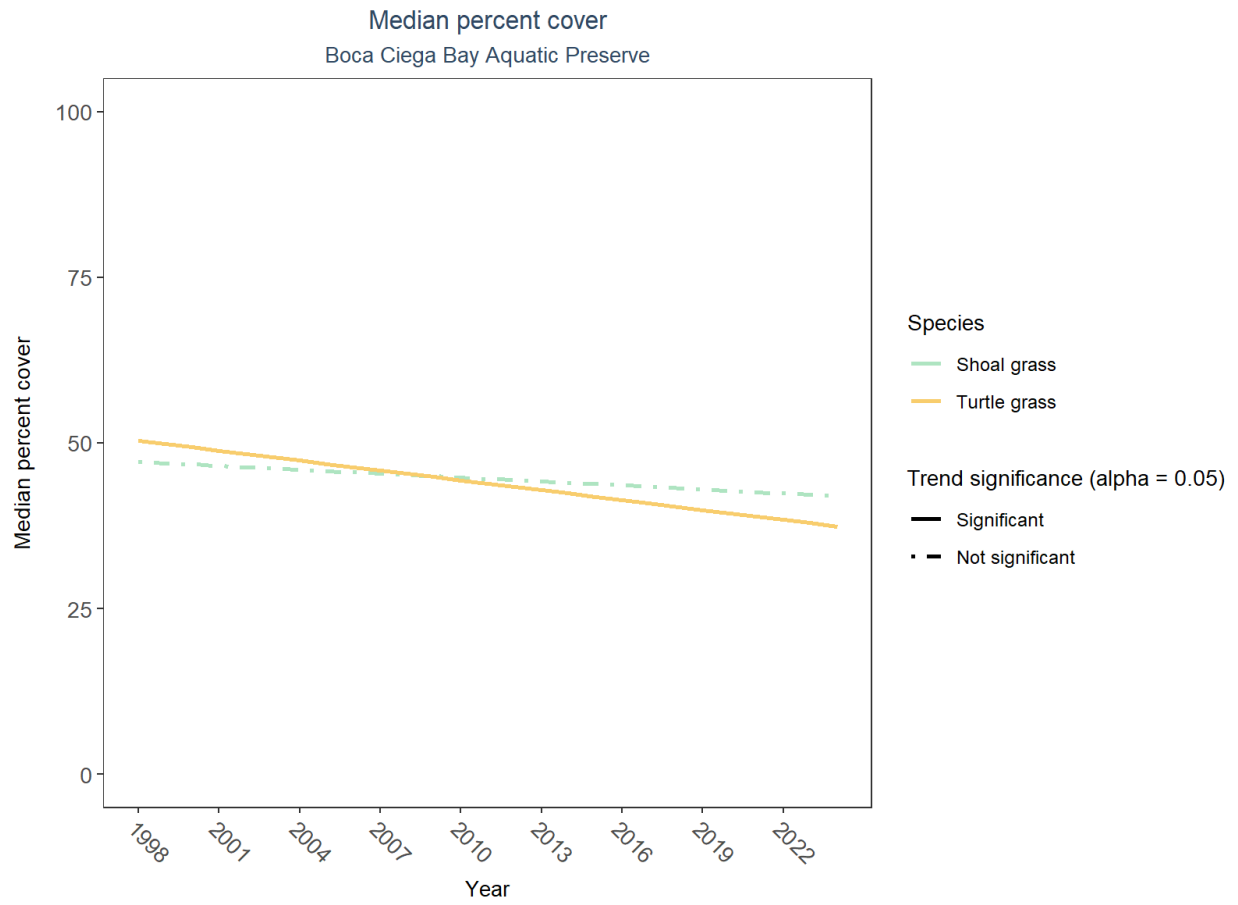


Table 7: SAV LME Results for Boca Ciega Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	No detectable trend	1998 - 2024	47.98	-0.2	0.38
Manatee grass	Model did not fit the available data	-	-	-	-
Turtle grass	Decreasing trend	1998 - 2024	52.33	-0.5	0.05
Halophila, unk.	Insufficient data	-	-	-	-

An annual decrease in percent cover was observed for turtle grass (-0.50%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for unknown *Halophila* and drift algae due to insufficient data, and the model could not be fitted for manatee grass.



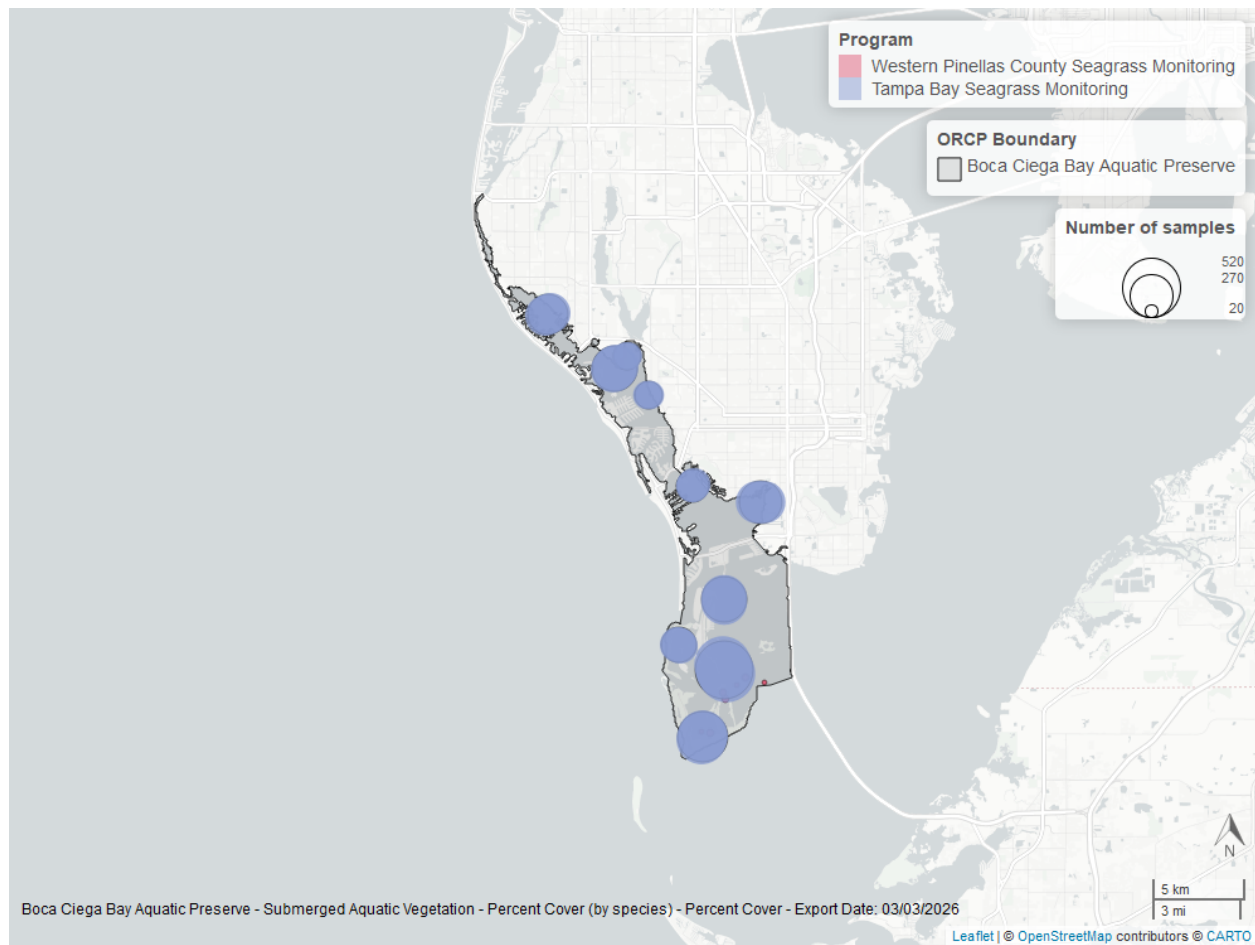
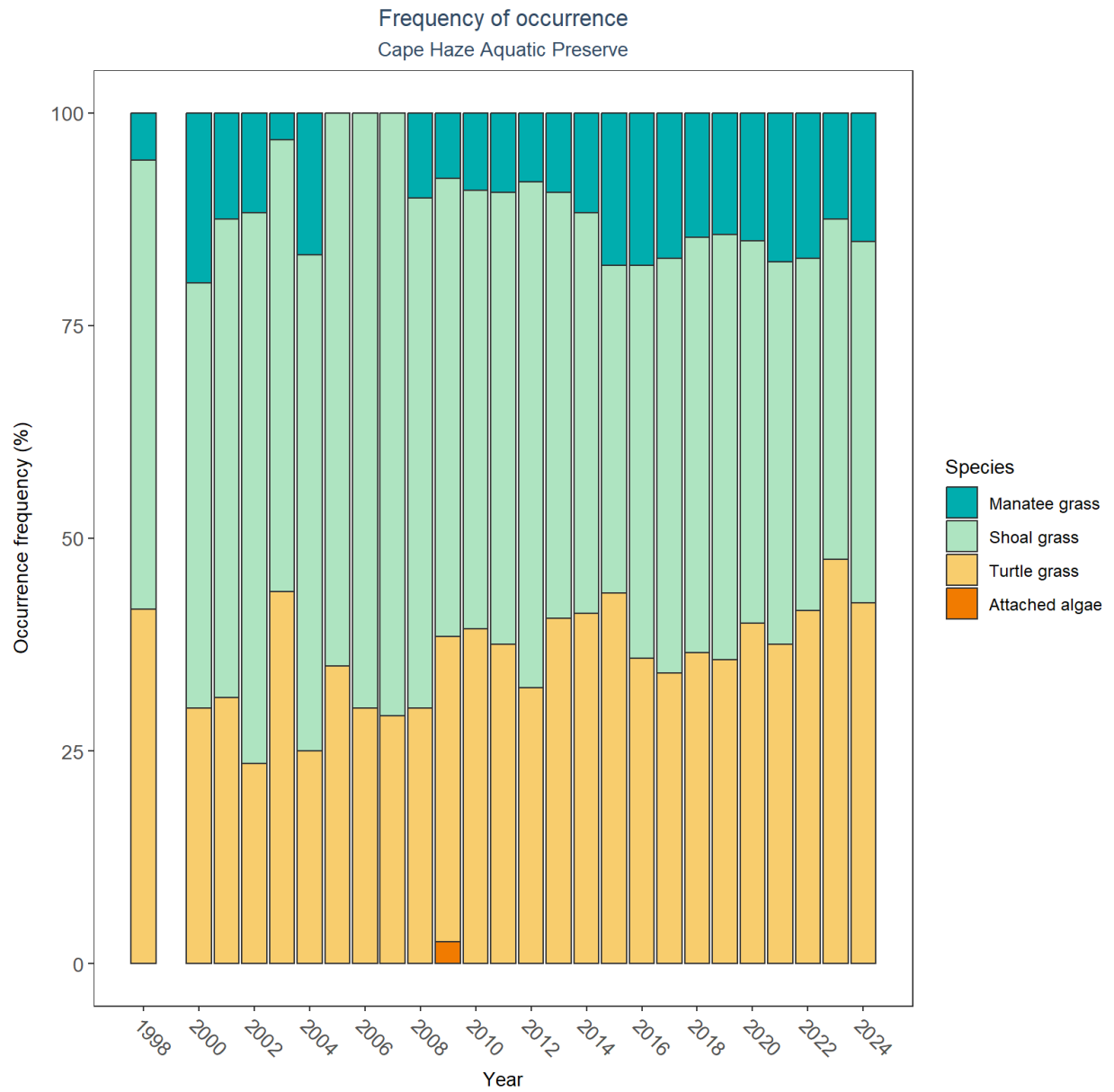


Figure 7: Map showing SAV sampling sites within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Cape Haze Aquatic Preserve



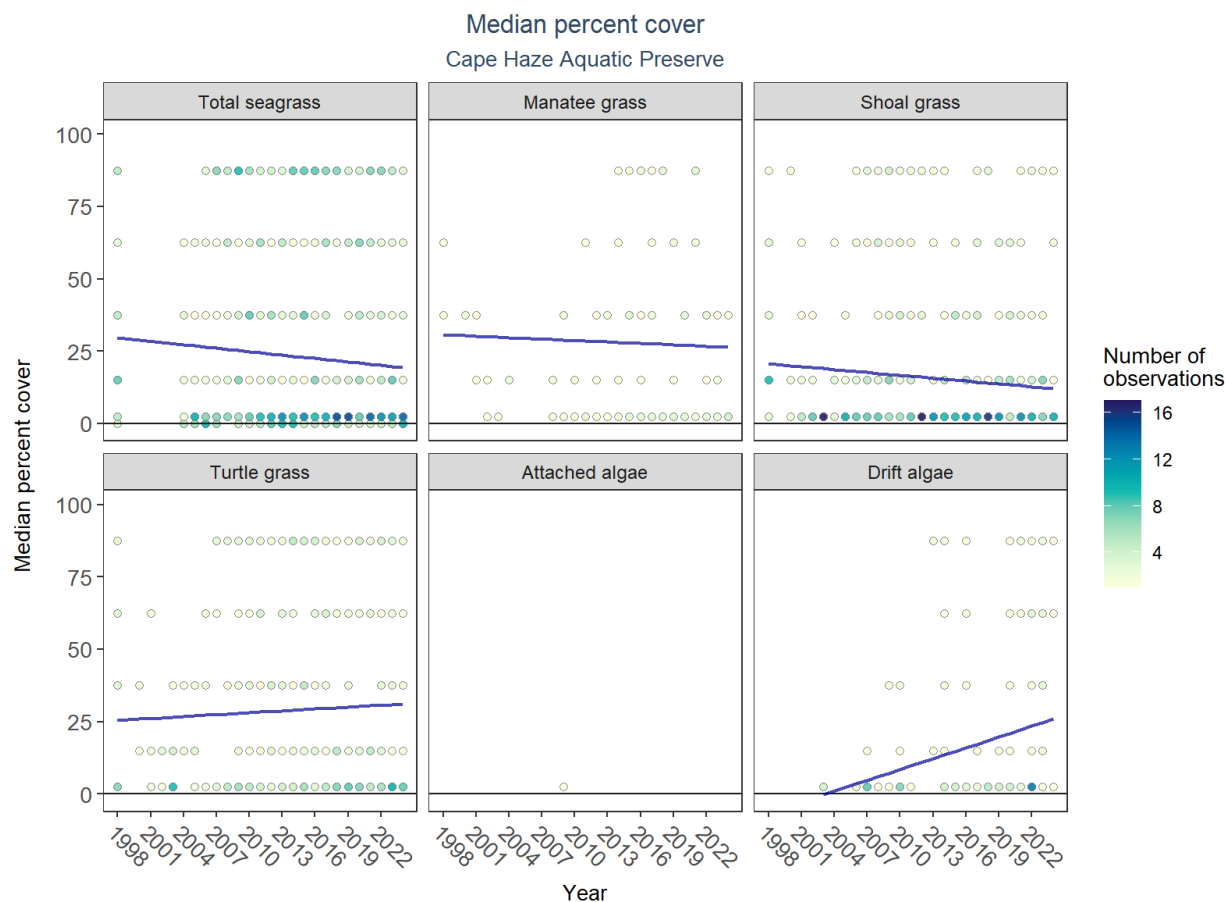
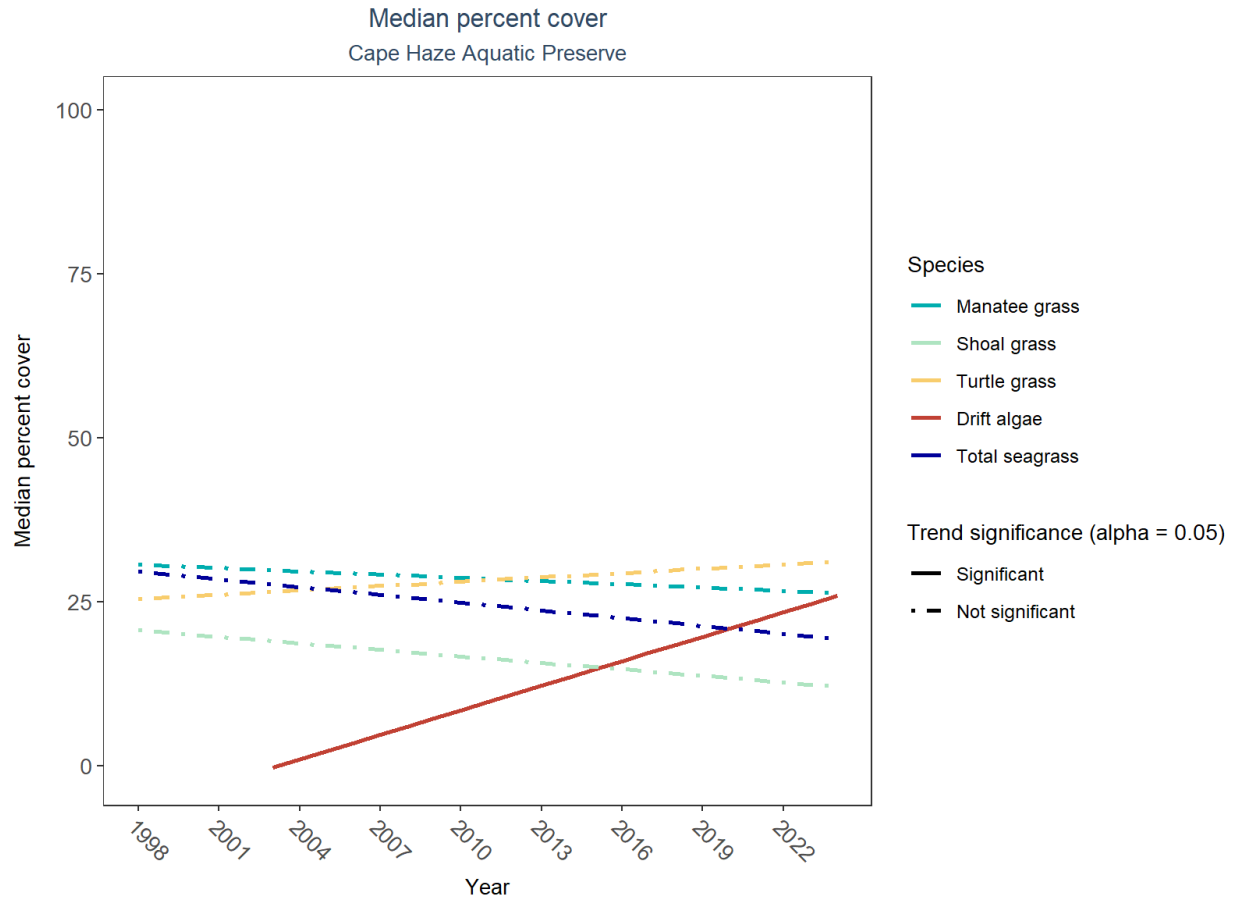


Table 8: SAV LME Results for Cape Haze Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	Insufficient data	-	-	-	-
Drift algae	Increasing trend	2003 - 2024	-11.48	1.25	0.02
Shoal grass	No detectable trend	1998 - 2024	21.98	-0.33	0.35
Manatee grass	No detectable trend	1998 - 2024	31.32	-0.16	0.83
Turtle grass	No detectable trend	1998 - 2024	24.58	0.22	0.48
Total seagrass	No detectable trend	1998 - 2024	31.20	-0.39	0.32

An annual increase in percent cover was observed for drift algae (1.25%). Total seagrass, manatee grass, shoal grass, and turtle grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for attached algae due to insufficient data.



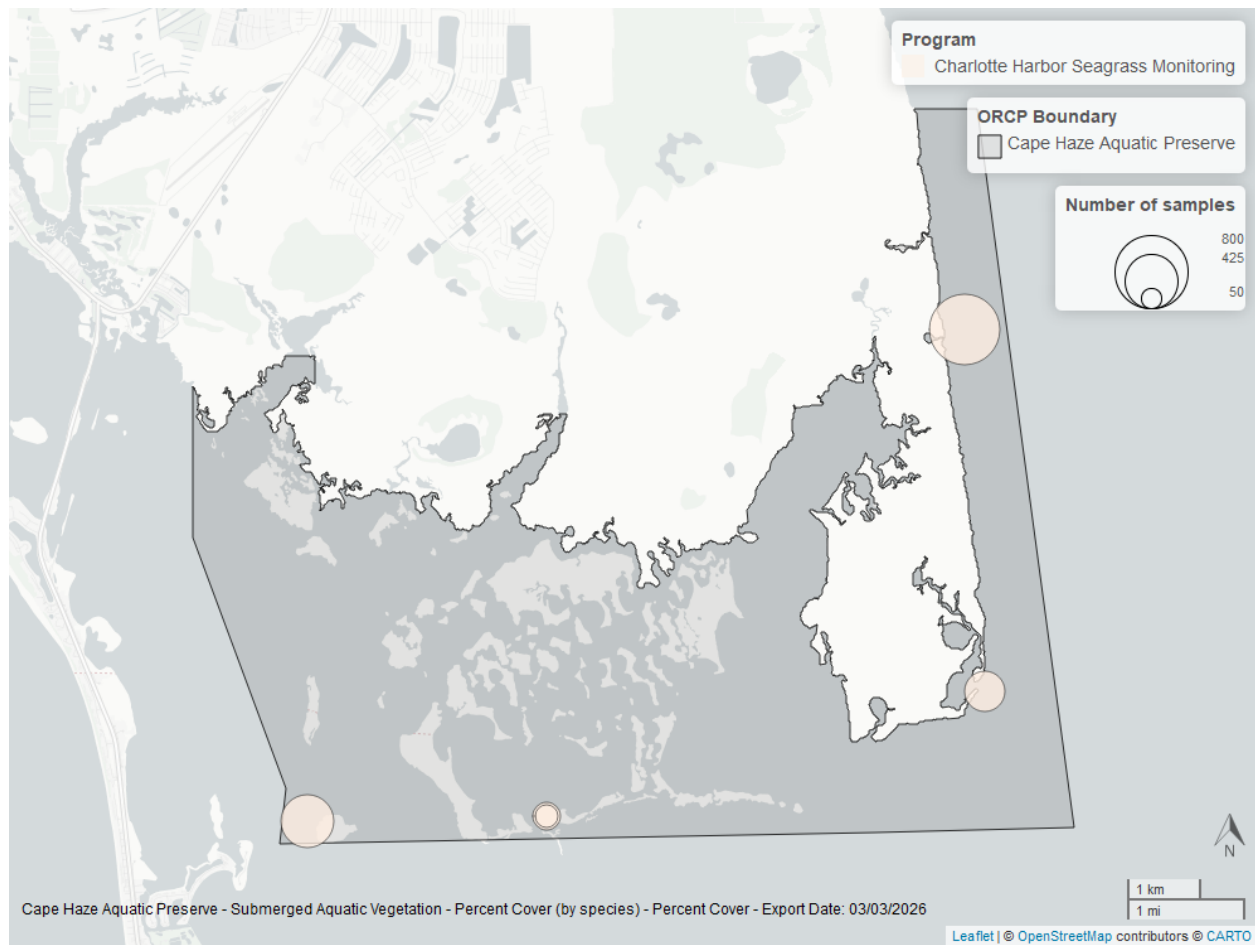
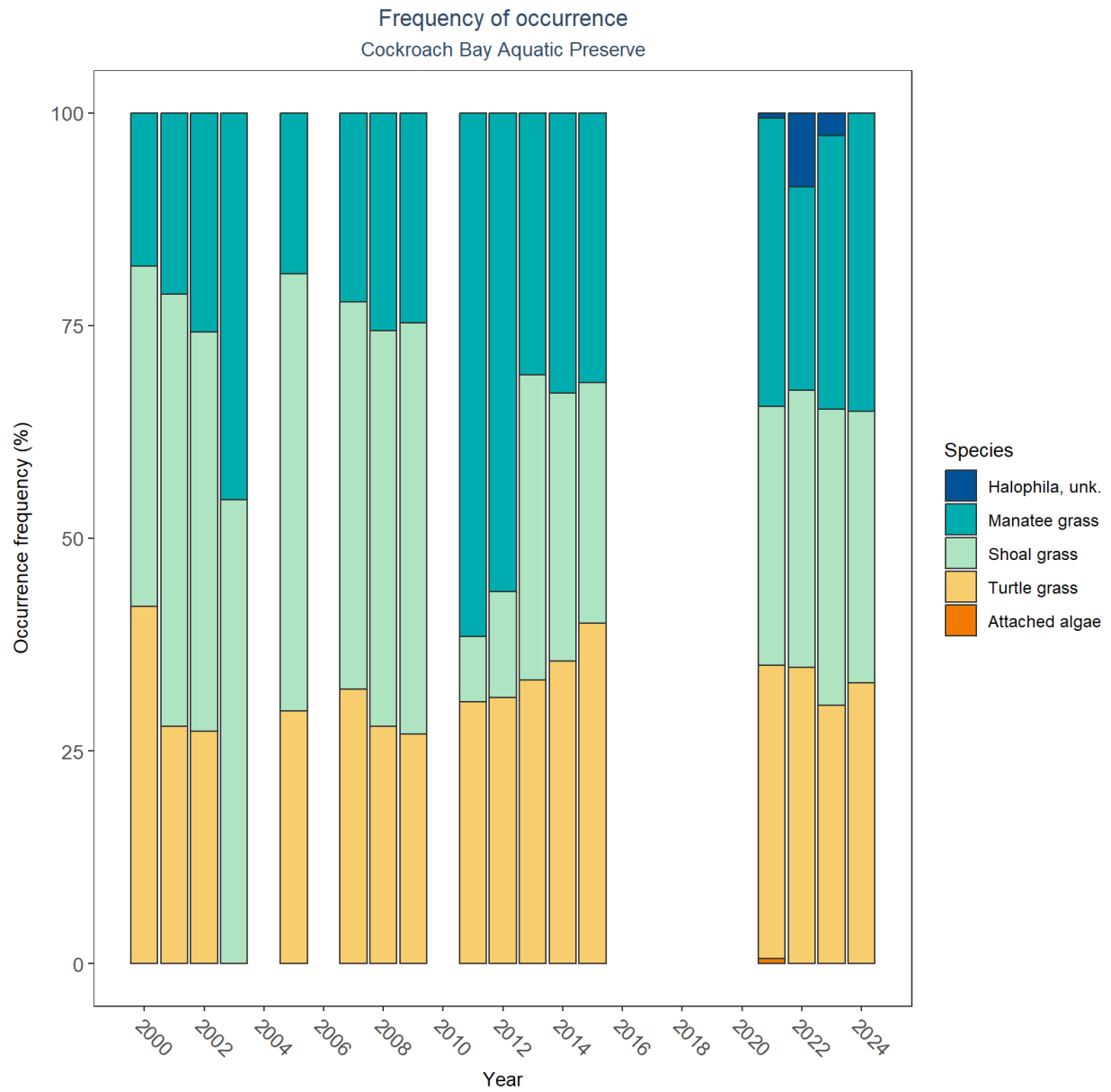


Figure 8: Map showing SAV sampling sites within the boundaries of *Cape Haze Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Cockroach Bay Aquatic Preserve



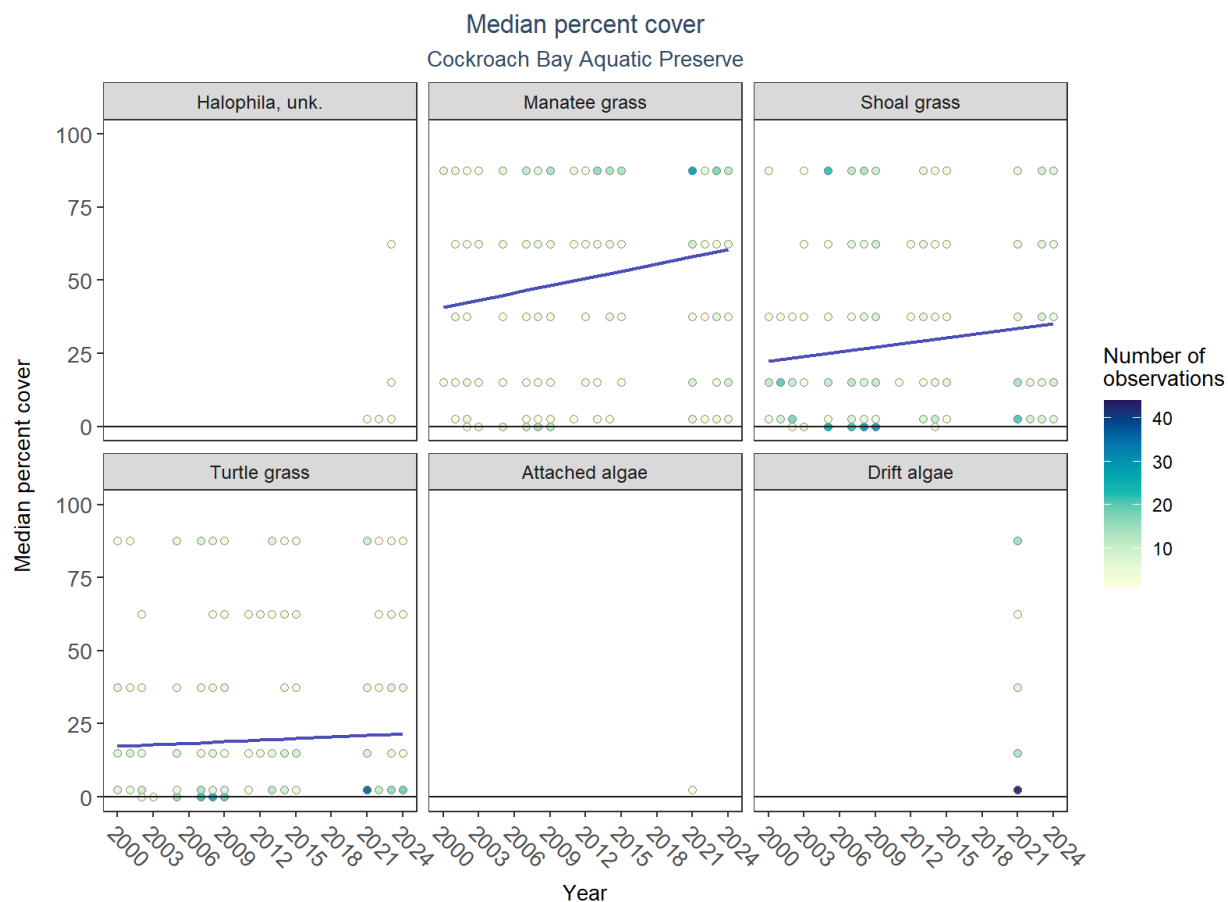
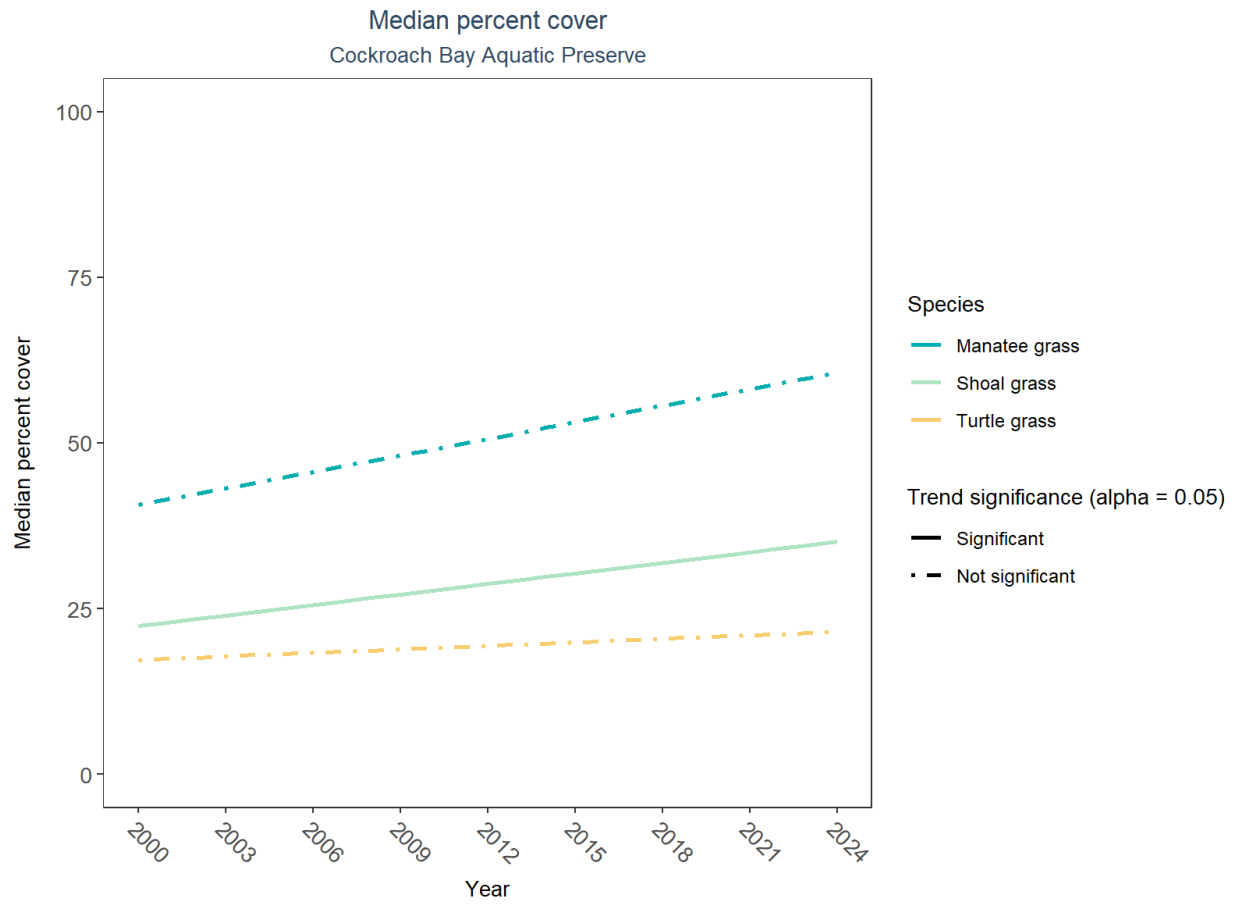


Table 9: SAV LME Results for Cockroach Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	Insufficient data	-	-	-	-
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Increasing trend	2000 - 2024	19.22	0.53	0.02
Manatee grass	No detectable trend	2000 - 2024	35.78	0.83	0.08
Turtle grass	No detectable trend	2000 - 2024	16.24	0.18	0.42
Halophila, unk.	Insufficient data	-	-	-	-

An annual increase in percent cover was observed for shoal grass (0.53%). No detectable change in percent cover was observed for manatee grass and turtle grass. Trends in percent cover could not be evaluated for unknown *Halophila*, attached algae, and drift algae due to insufficient data.



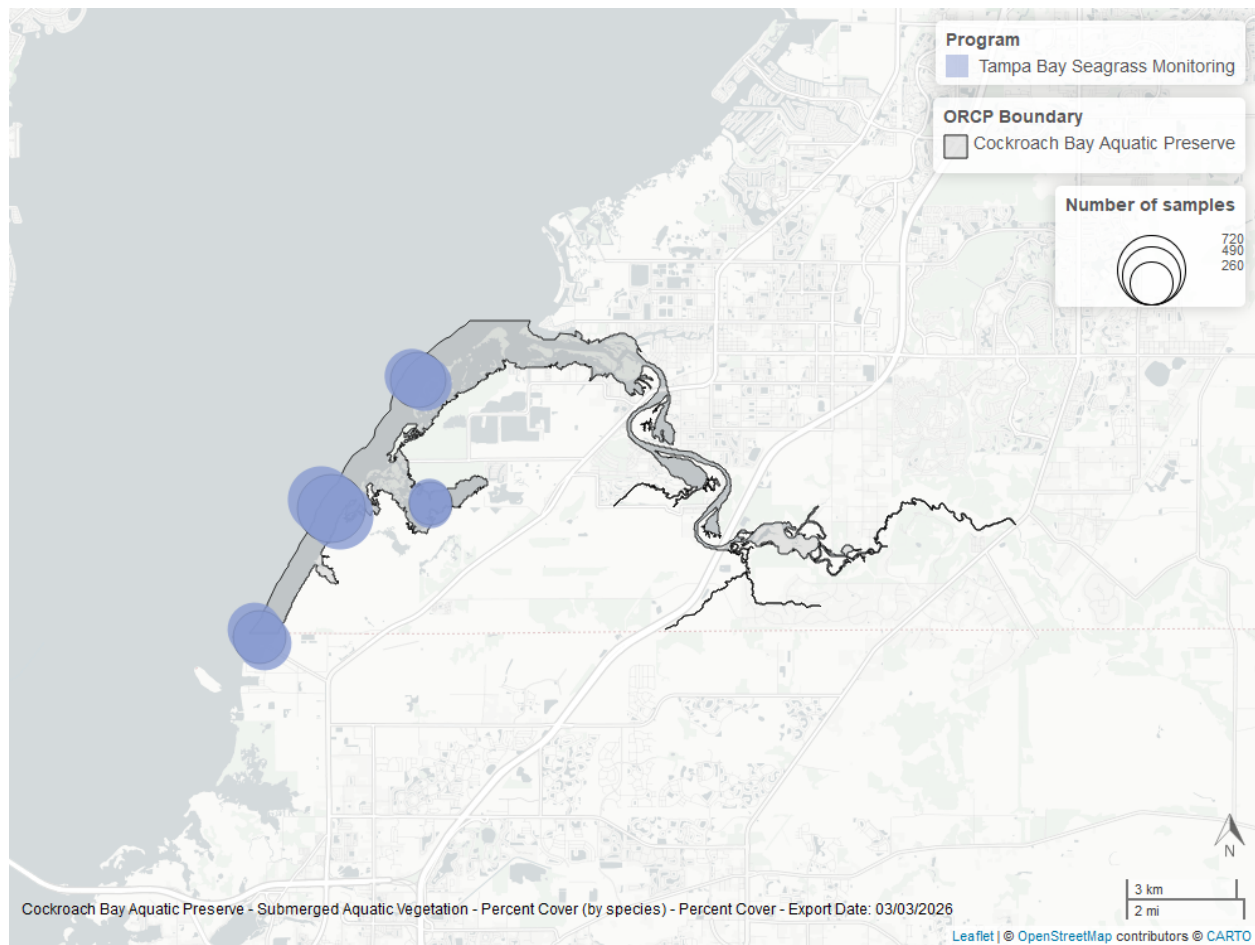
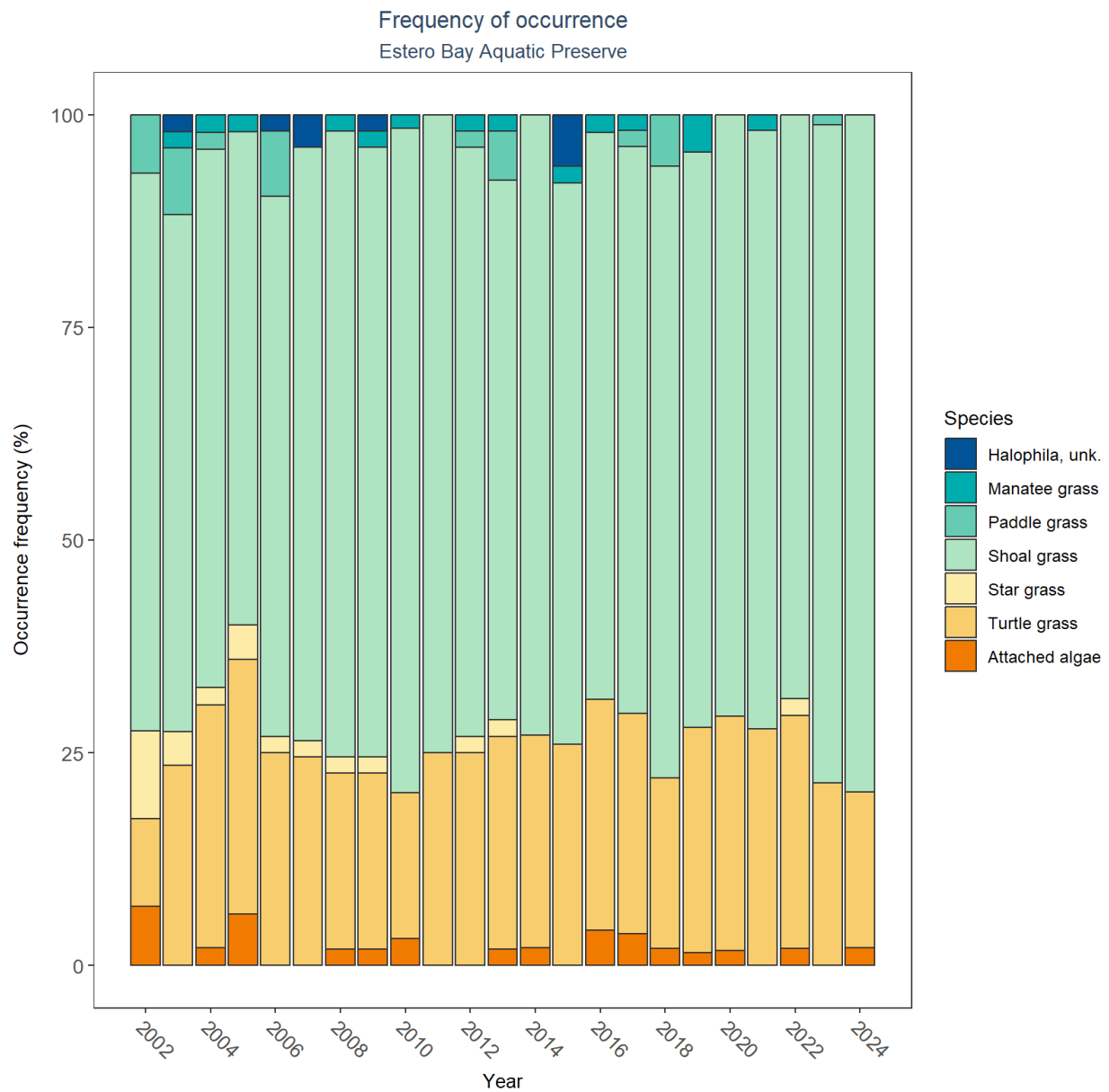


Figure 9: Map showing SAV sampling sites within the boundaries of *Cockroach Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Estero Bay Aquatic Preserve



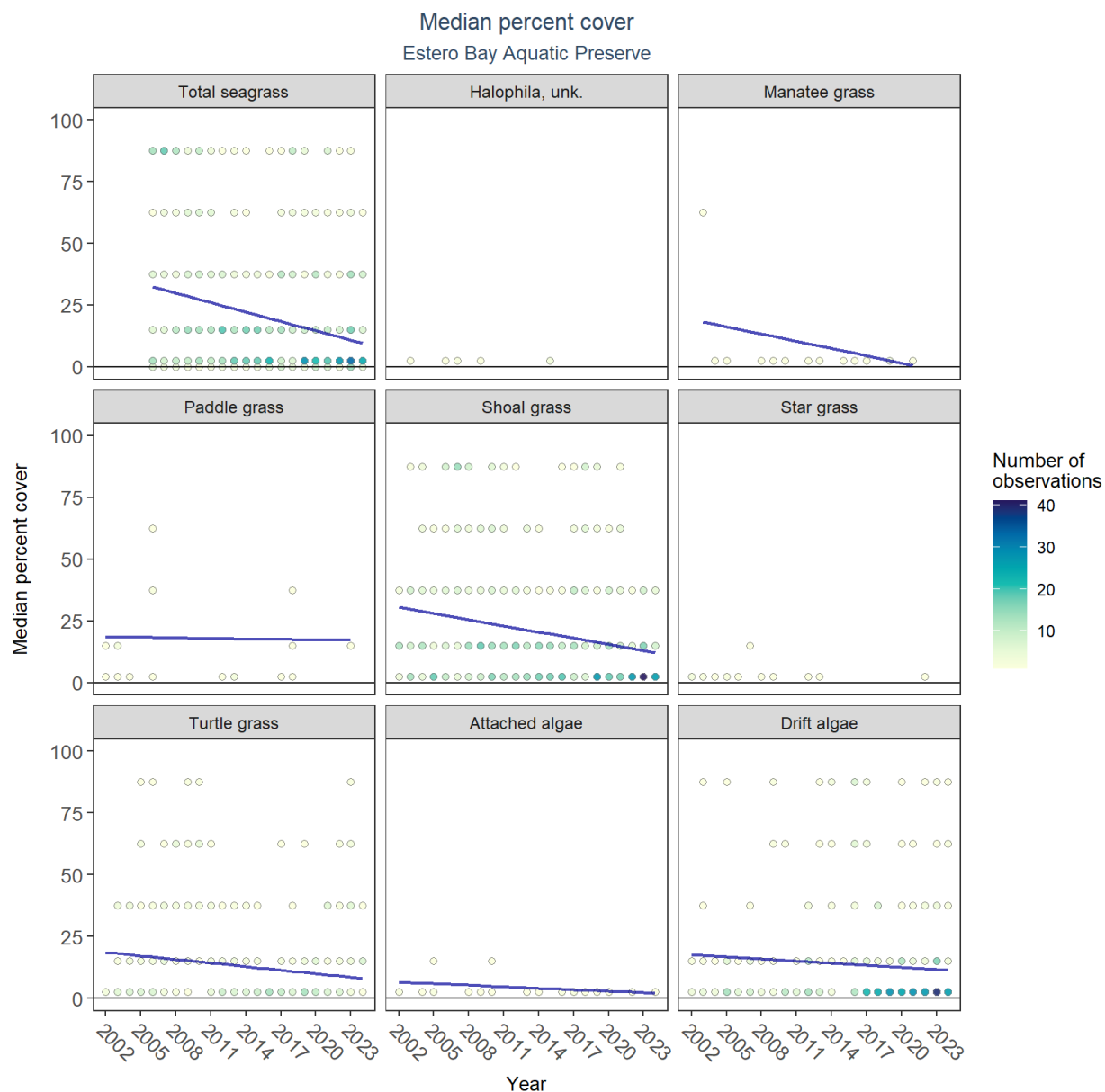
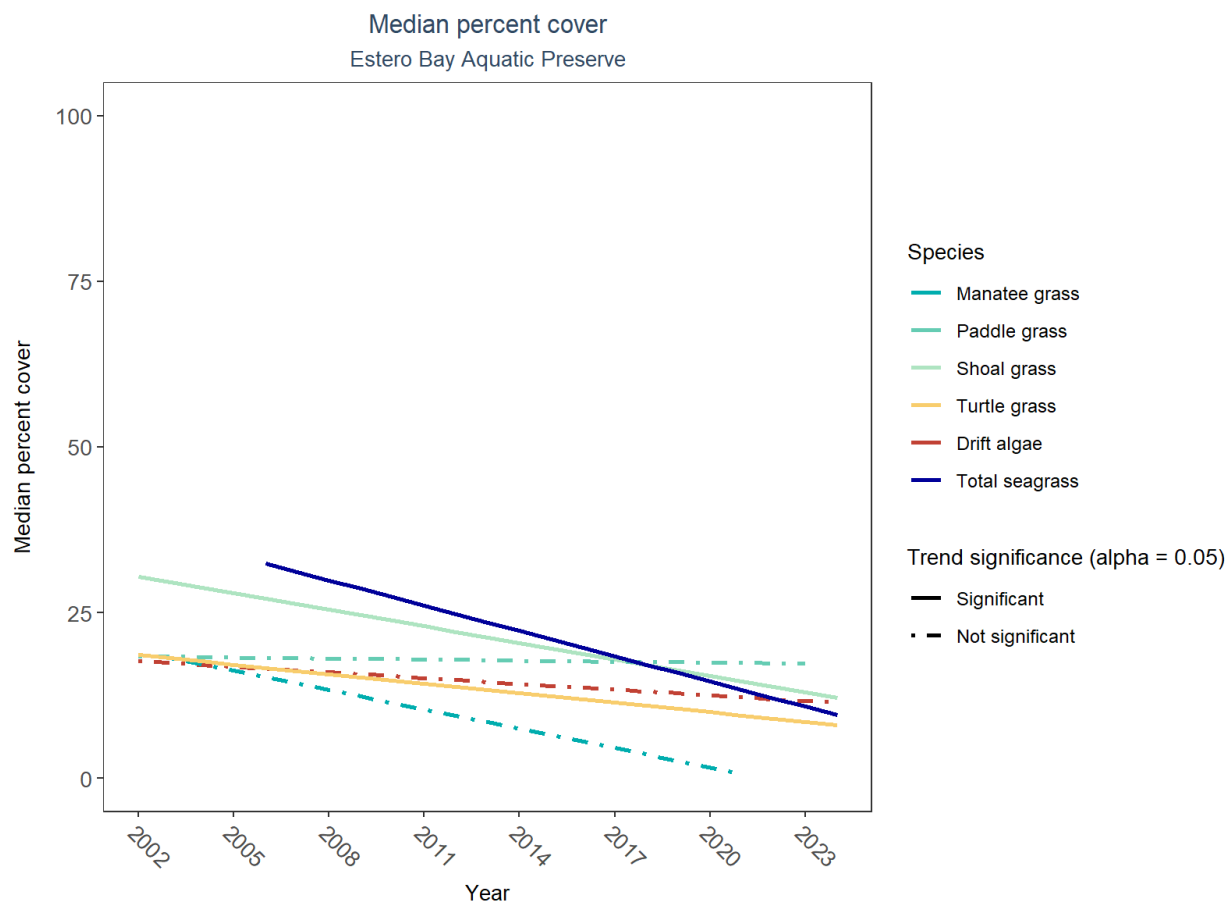


Table 10: SAV LME Results for Estero Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2002 - 2024	8.13	-0.20	0.39
Drift algae	No detectable trend	2002 - 2024	20.04	-0.29	0.38
Shoal grass	Decreasing trend	2002 - 2024	37.07	-0.83	0.00
Paddle grass	No detectable trend	2002 - 2023	18.90	-0.06	0.92
Star grass	Model did not fit the available data	-	-	-	-
Manatee grass	No detectable trend	2003 - 2021	26.94	-0.97	0.35
Turtle grass	Decreasing trend	2002 - 2024	22.45	-0.48	0.01
Total seagrass	Decreasing trend	2006 - 2024	47.66	-1.27	0.02
Halophila, unk.	Model did not fit the available data	-	-	-	-

Annual decreases in percent cover were observed for total seagrass (-1.27%), shoal grass (-0.83%), and turtle grass (-0.48%). Manatee grass, paddle grass, attached algae, and drift algae showed no detectable change in percent cover. A model could not be fitted for unknown *Halophila* and star grass.



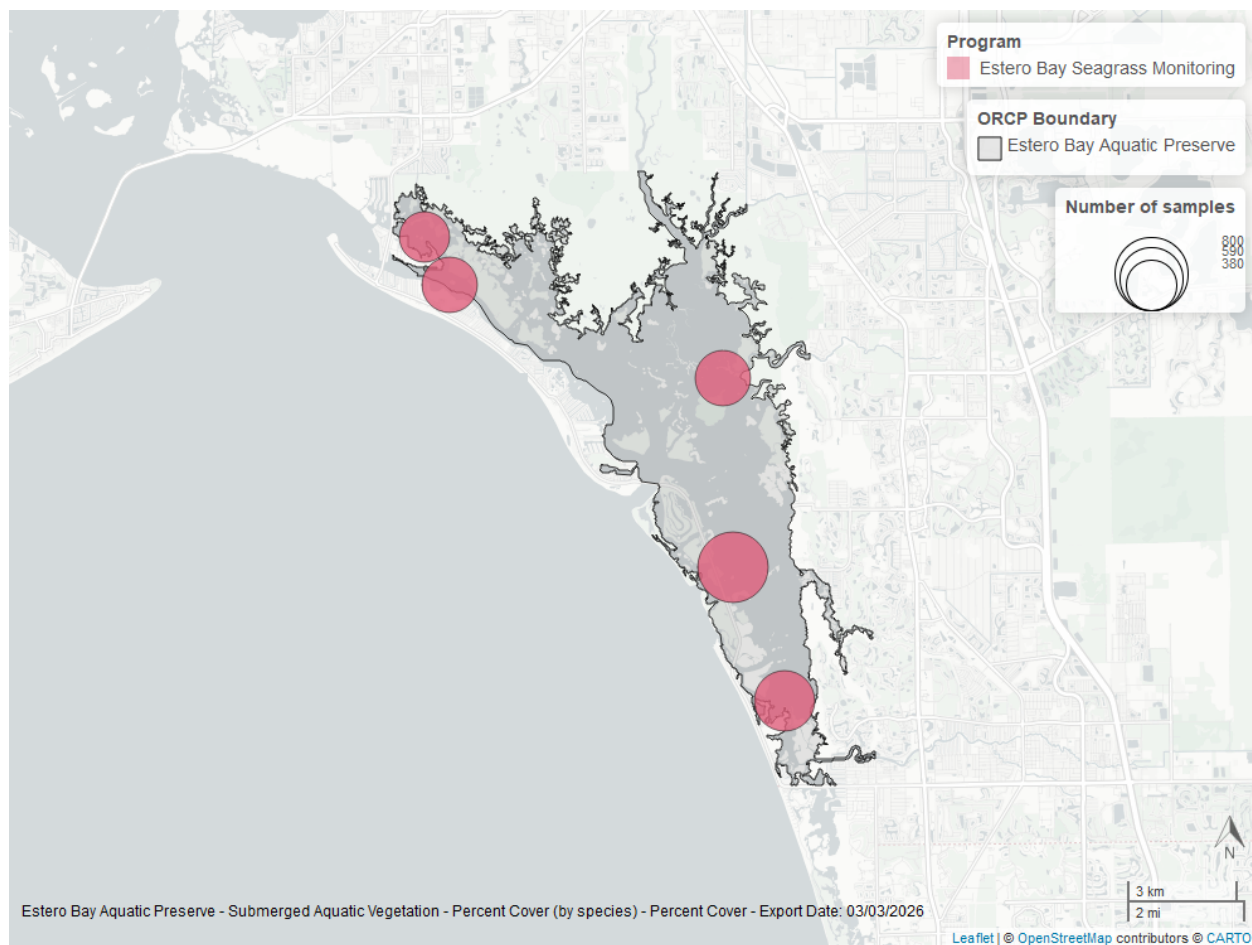
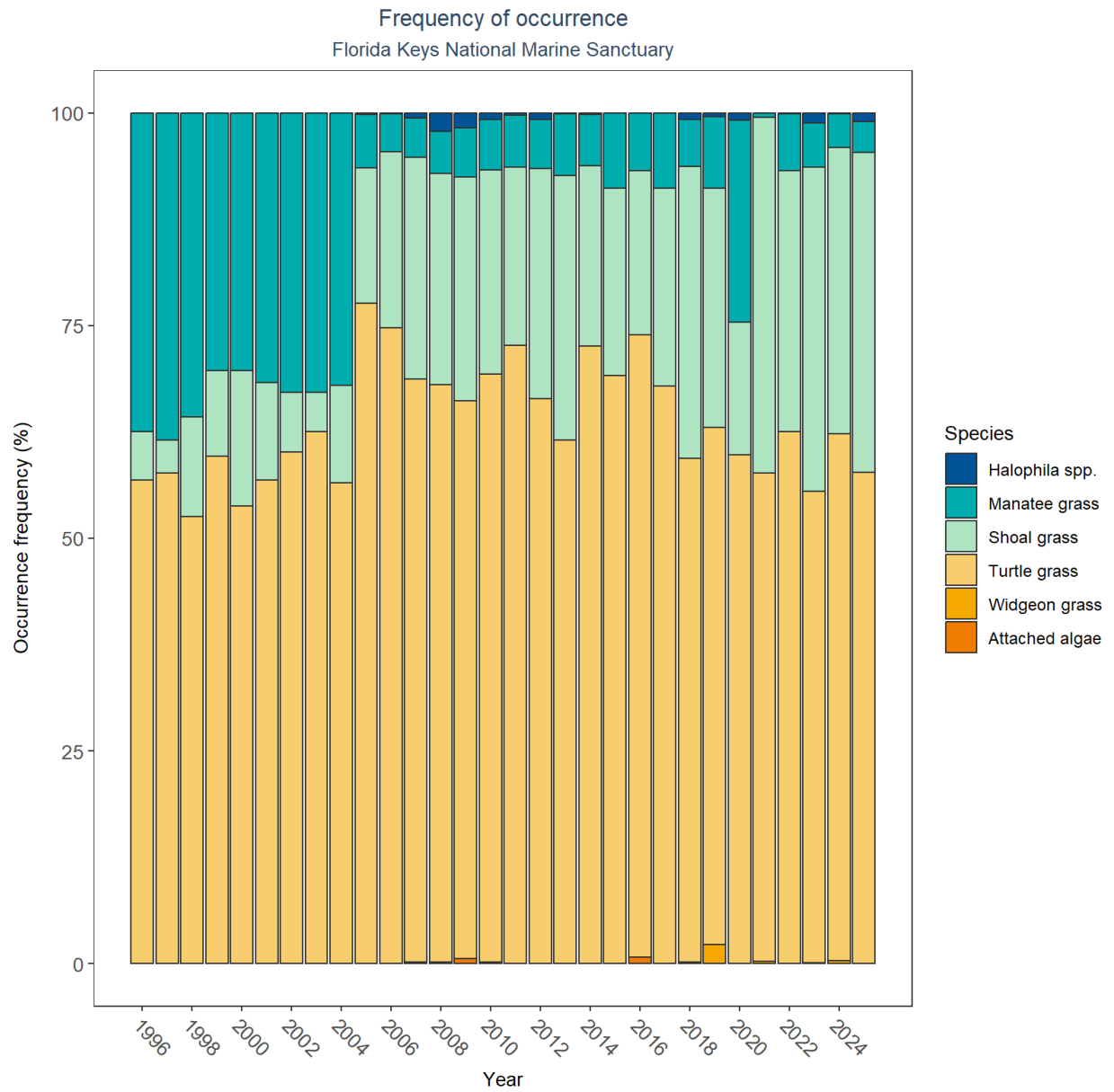


Figure 10: Map showing SAV sampling sites within the boundaries of *Estero Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Florida Keys National Marine Sanctuary



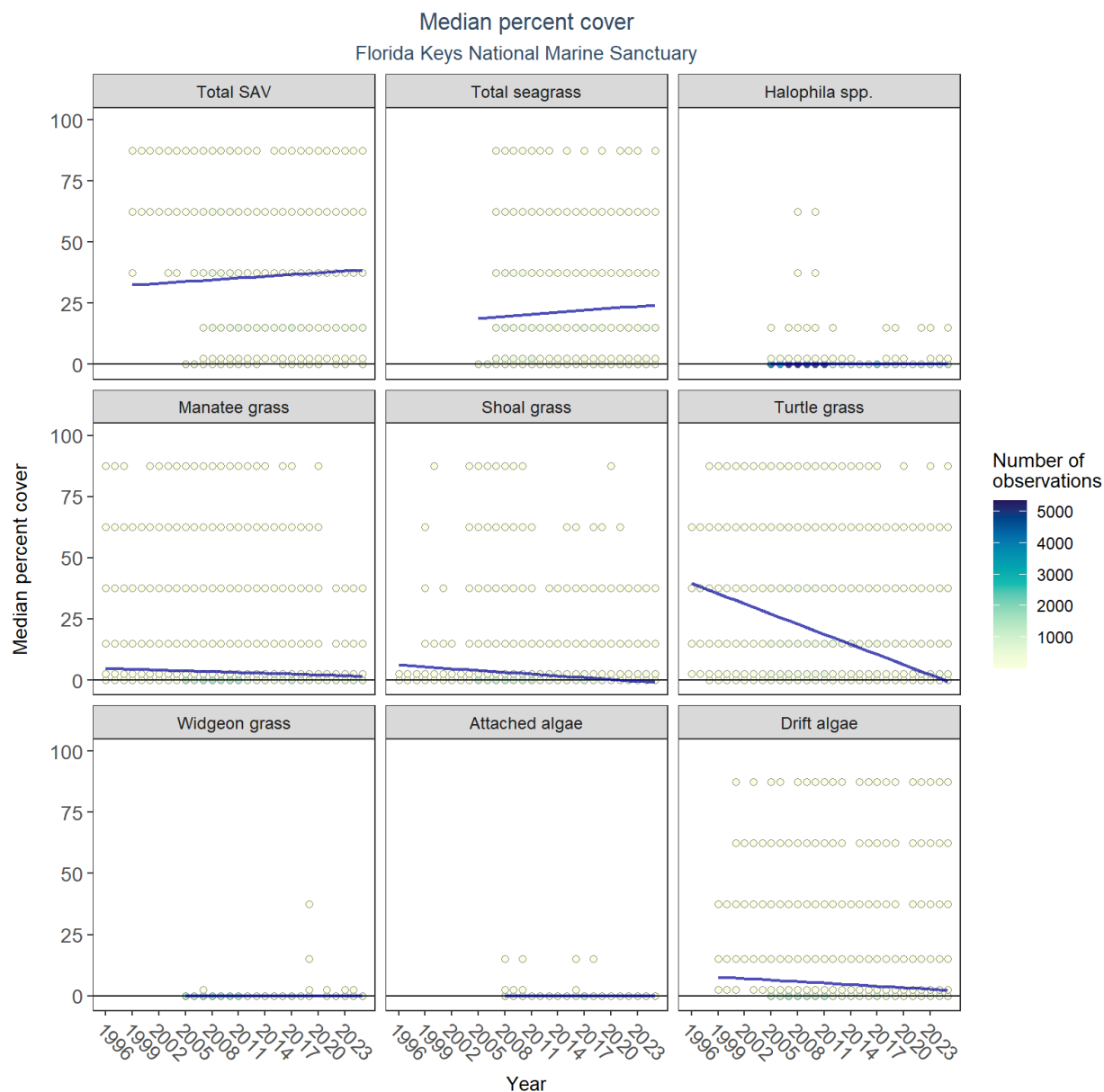
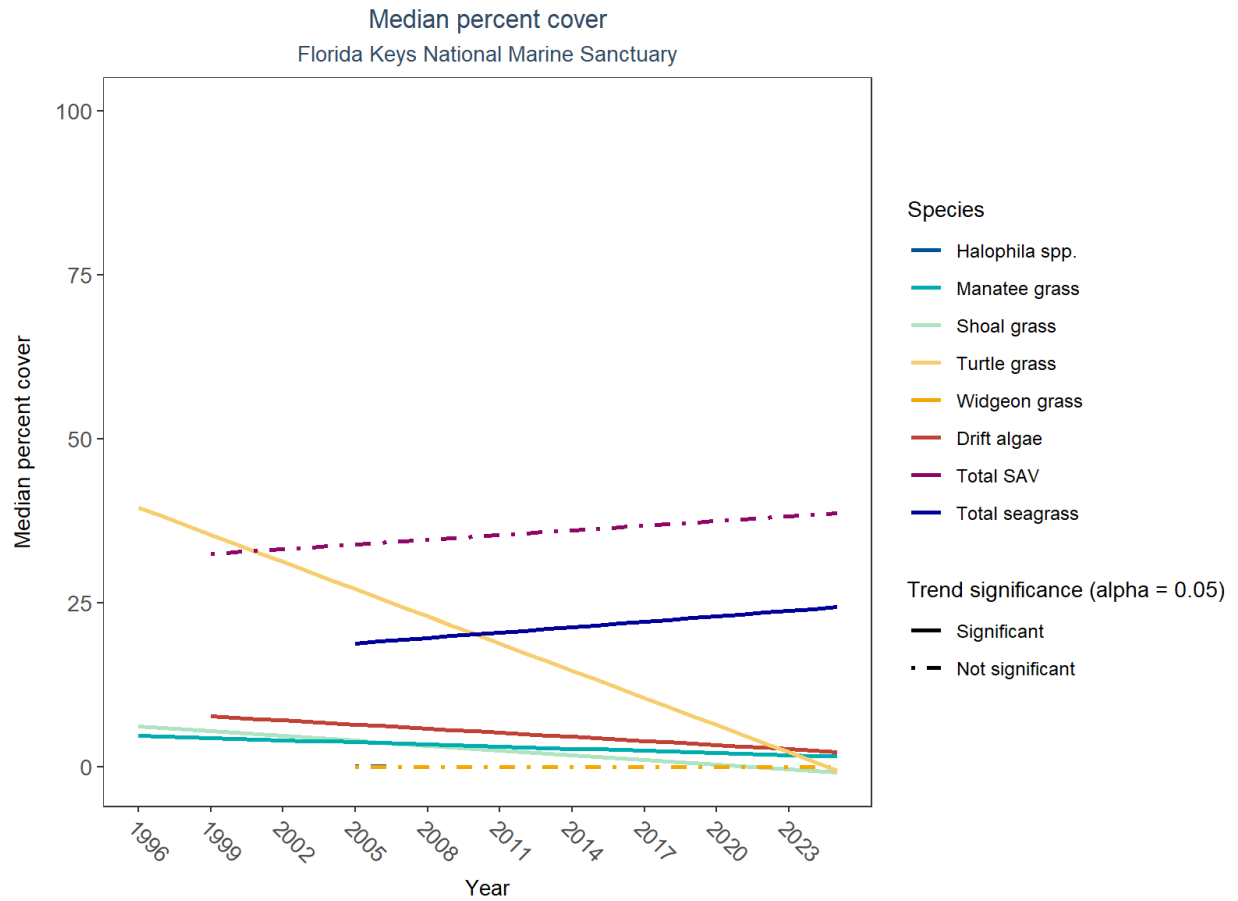


Table 11: SAV LME Results for Florida Keys National Marine Sanctuary

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2008 - 2025	0.07	0.00	0.20
Drift algae	Decreasing trend	1999 - 2025	8.74	-0.21	0.00
Shoal grass	Decreasing trend	1996 - 2025	6.66	-0.24	0.00
Halophila spp.	No detectable trend	2005 - 2025	0.07	0.00	0.41
Widgeon grass	No detectable trend	2005 - 2025	-0.03	0.00	0.13
Manatee grass	Decreasing trend	1996 - 2025	4.94	-0.11	0.02
Turtle grass	Decreasing trend	1996 - 2025	42.29	-1.38	0.00
Total SAV	No detectable trend	1999 - 2025	31.28	0.24	0.12
Total seagrass	Increasing trend	2005 - 2025	15.77	0.28	0.00

An annual increase in percent cover was observed for total seagrass (0.28%). Annual decreases in percent cover were observed for manatee grass (-0.11%), shoal grass (-0.24%), turtle grass (-1.38%), and drift algae (-0.21%). Total SAV, *Halophila* spp., widgeon grass, and attached algae showed no detectable change in percent cover.



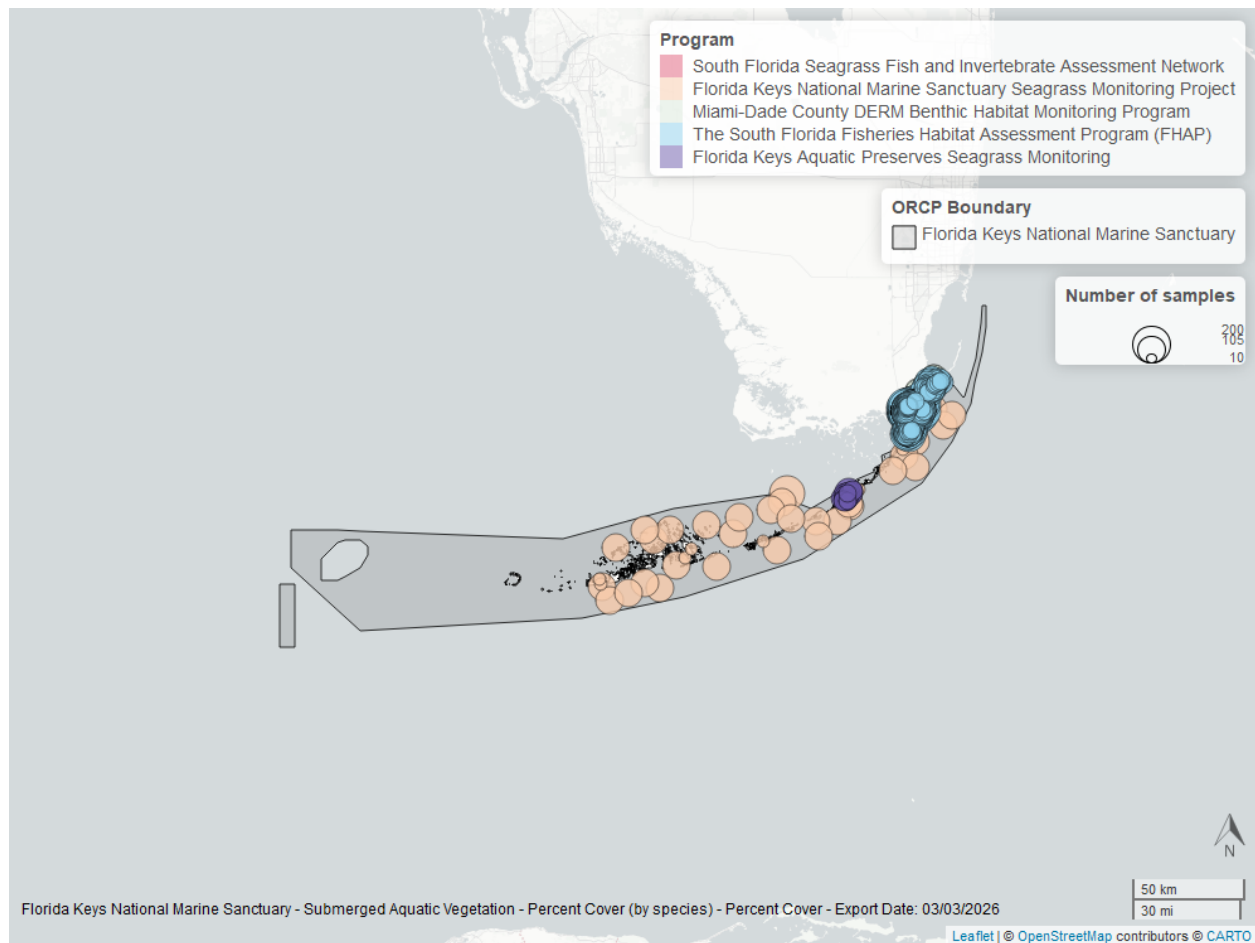
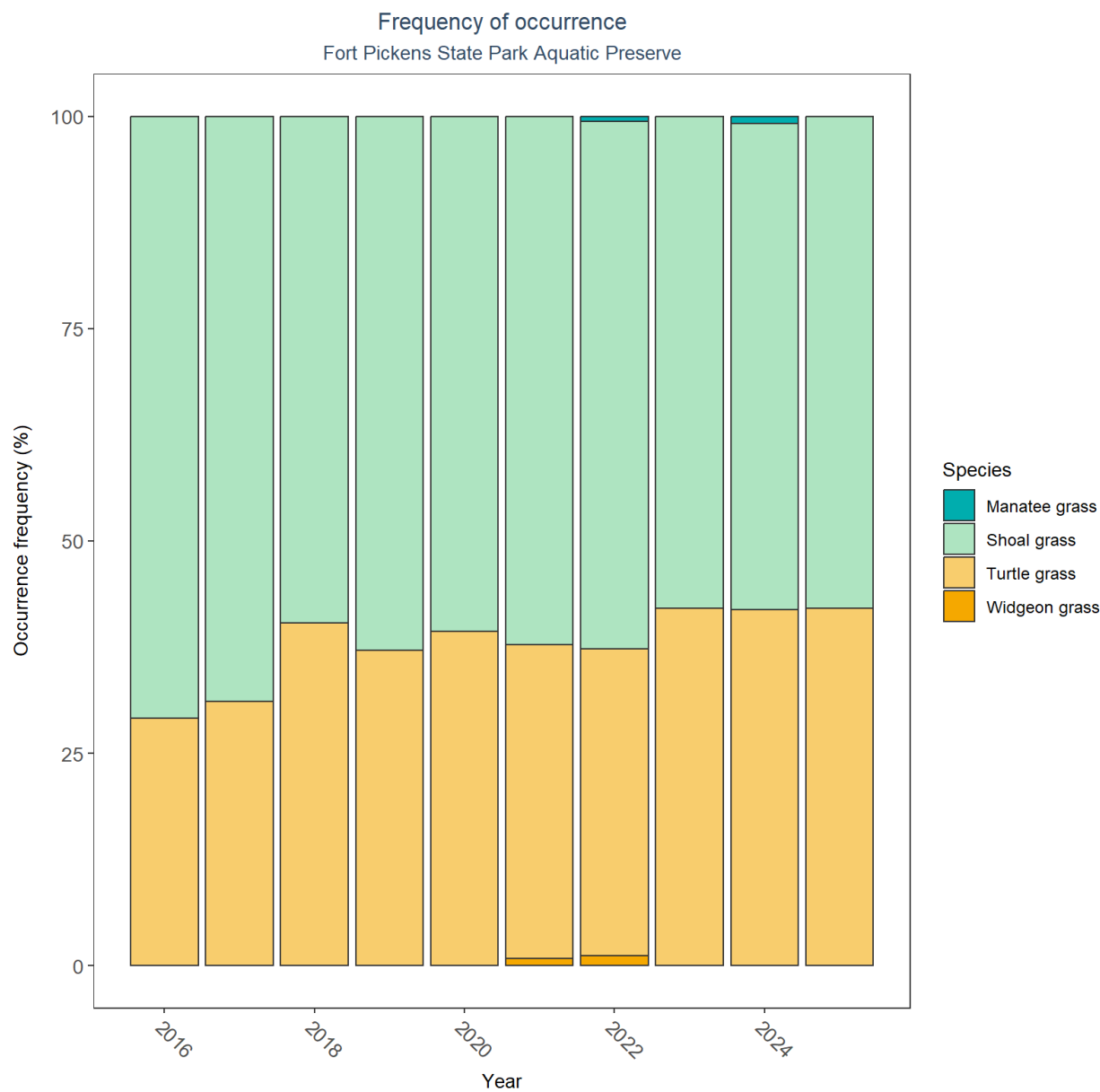


Figure 11: Map showing SAV sampling sites within the boundaries of *Florida Keys National Marine Sanctuary*. The point size reflects the number of samples at a given sampling site.

Fort Pickens State Park Aquatic Preserve



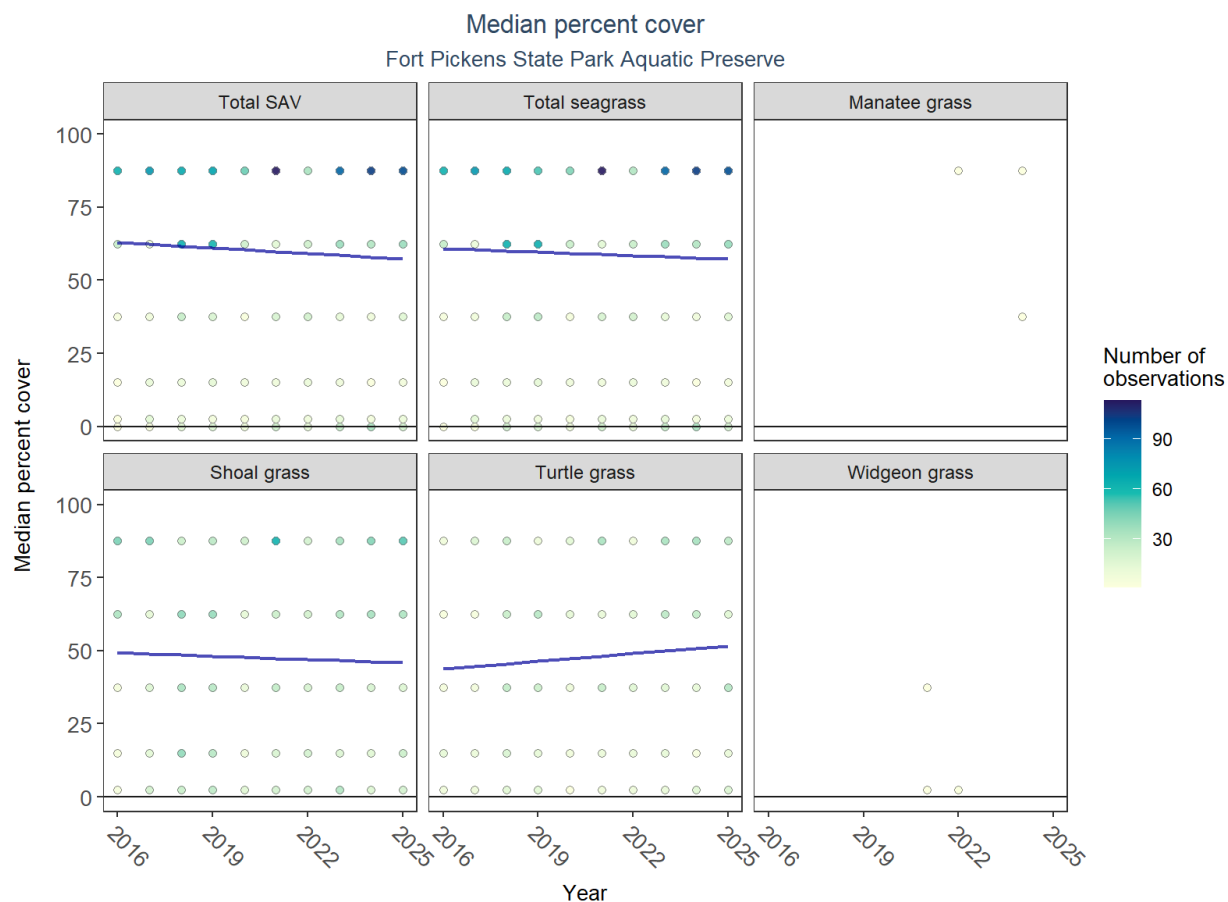
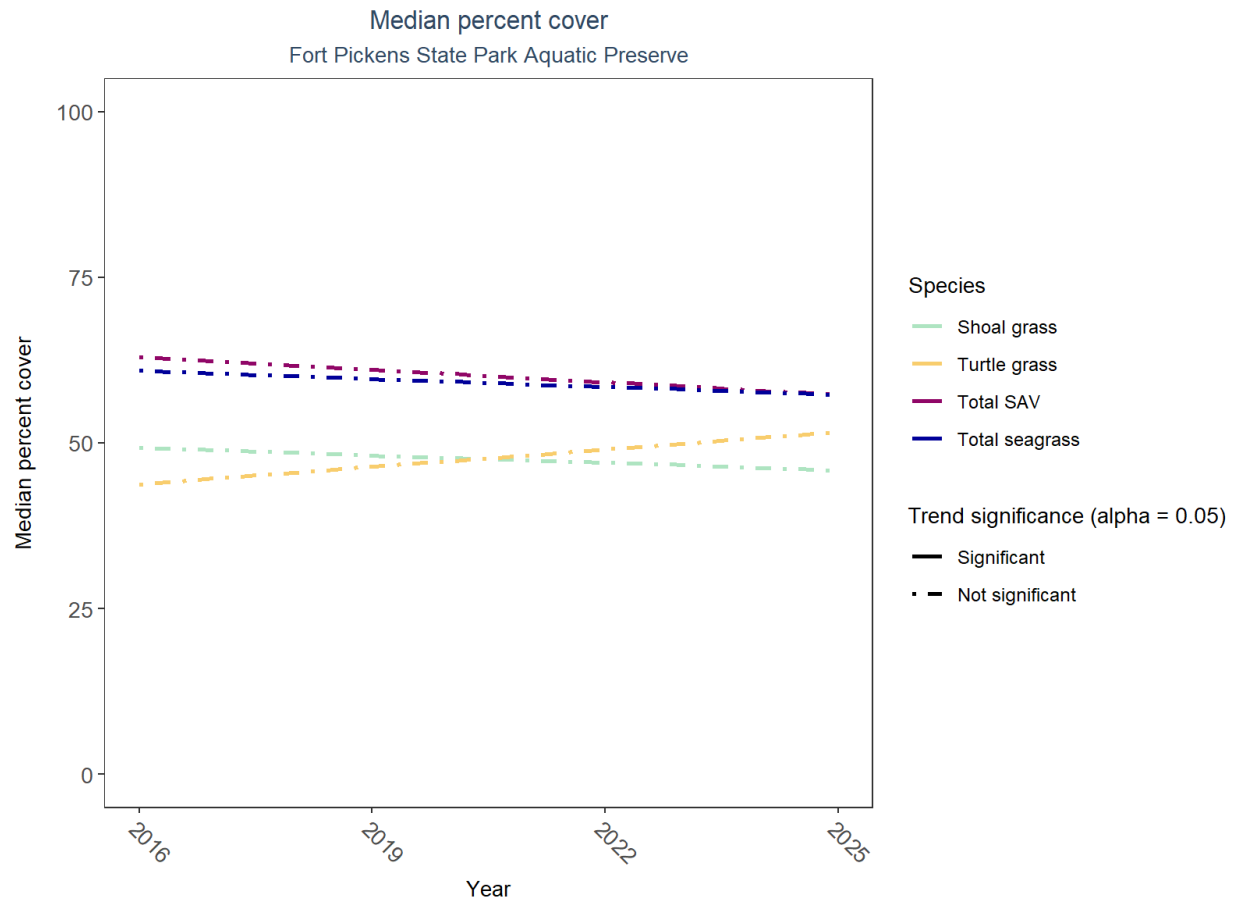


Table 12: SAV LME Results for Fort Pickens State Park Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	2016 - 2025	57.58	-0.38	0.30
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	Insufficient data	-	-	-	-
Turtle grass	No detectable trend	2016 - 2025	24.56	0.87	0.14
Total SAV	No detectable trend	2016 - 2025	76.79	-0.63	0.22
Total seagrass	No detectable trend	2016 - 2025	69.86	-0.41	0.45

Total SAV, total seagrass, shoal grass, and turtle grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for manatee grass and widgeon grass due to insufficient data.



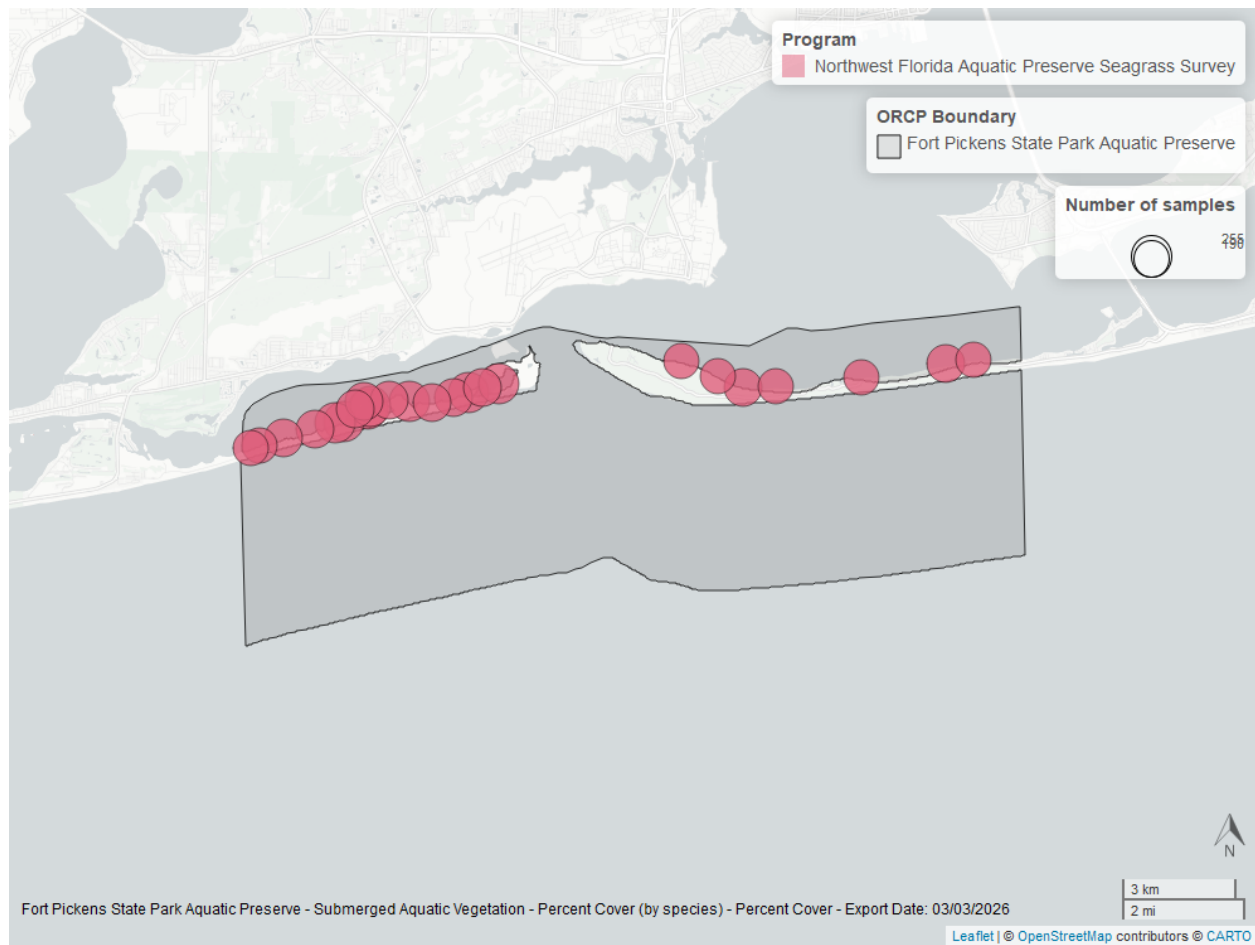
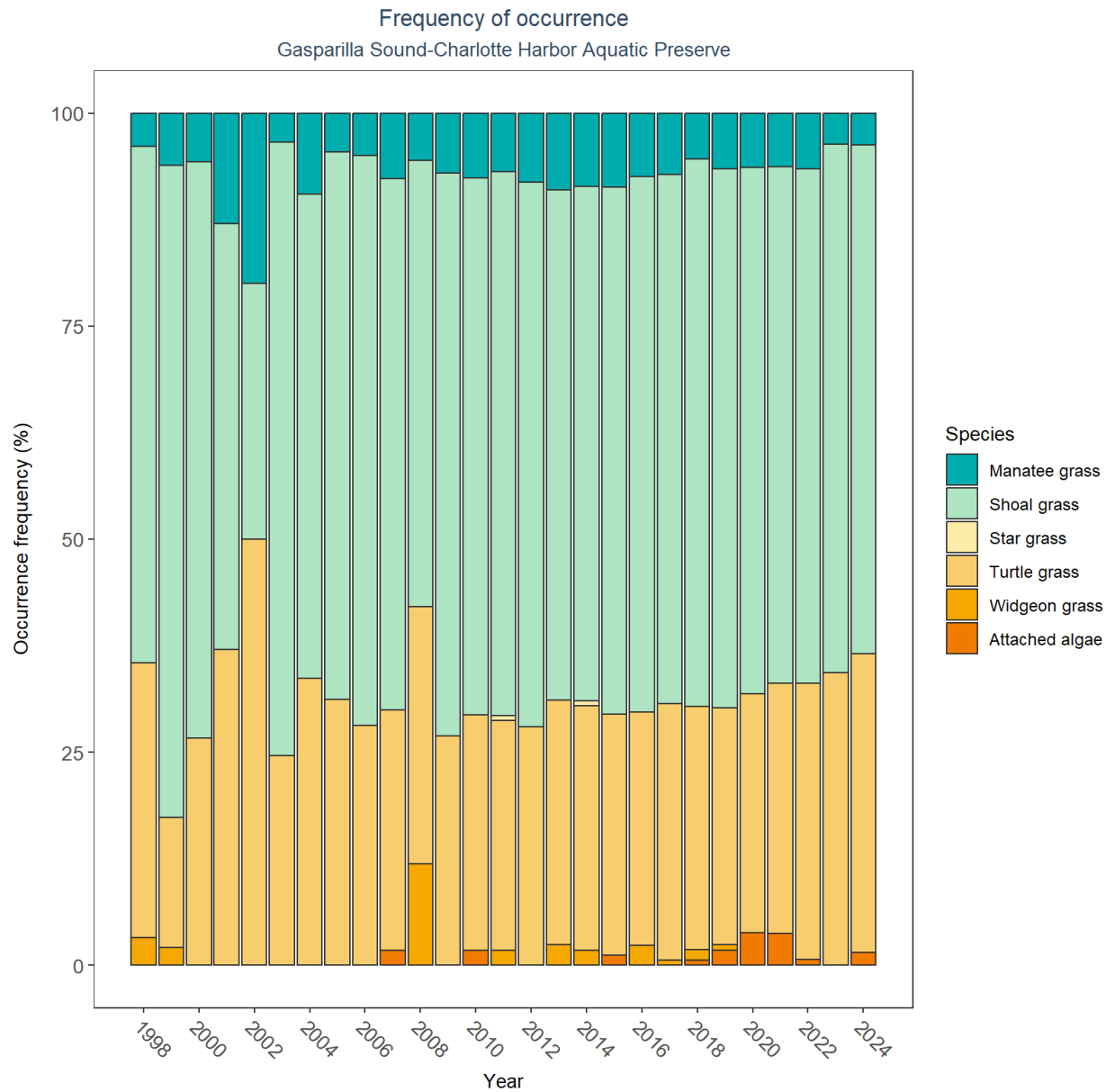


Figure 12: Map showing SAV sampling sites within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Gasparilla Sound-Charlotte Harbor Aquatic Preserve



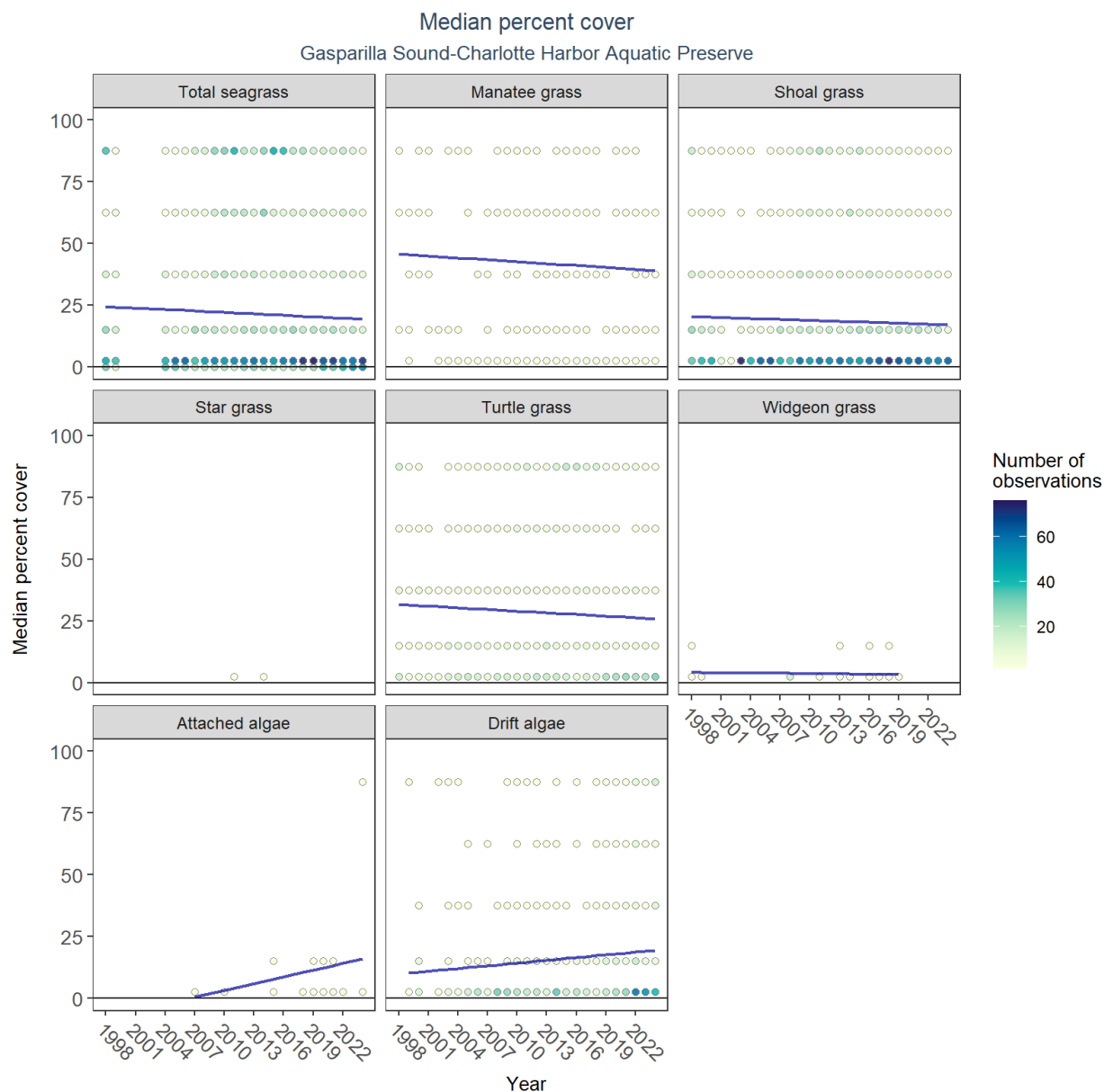
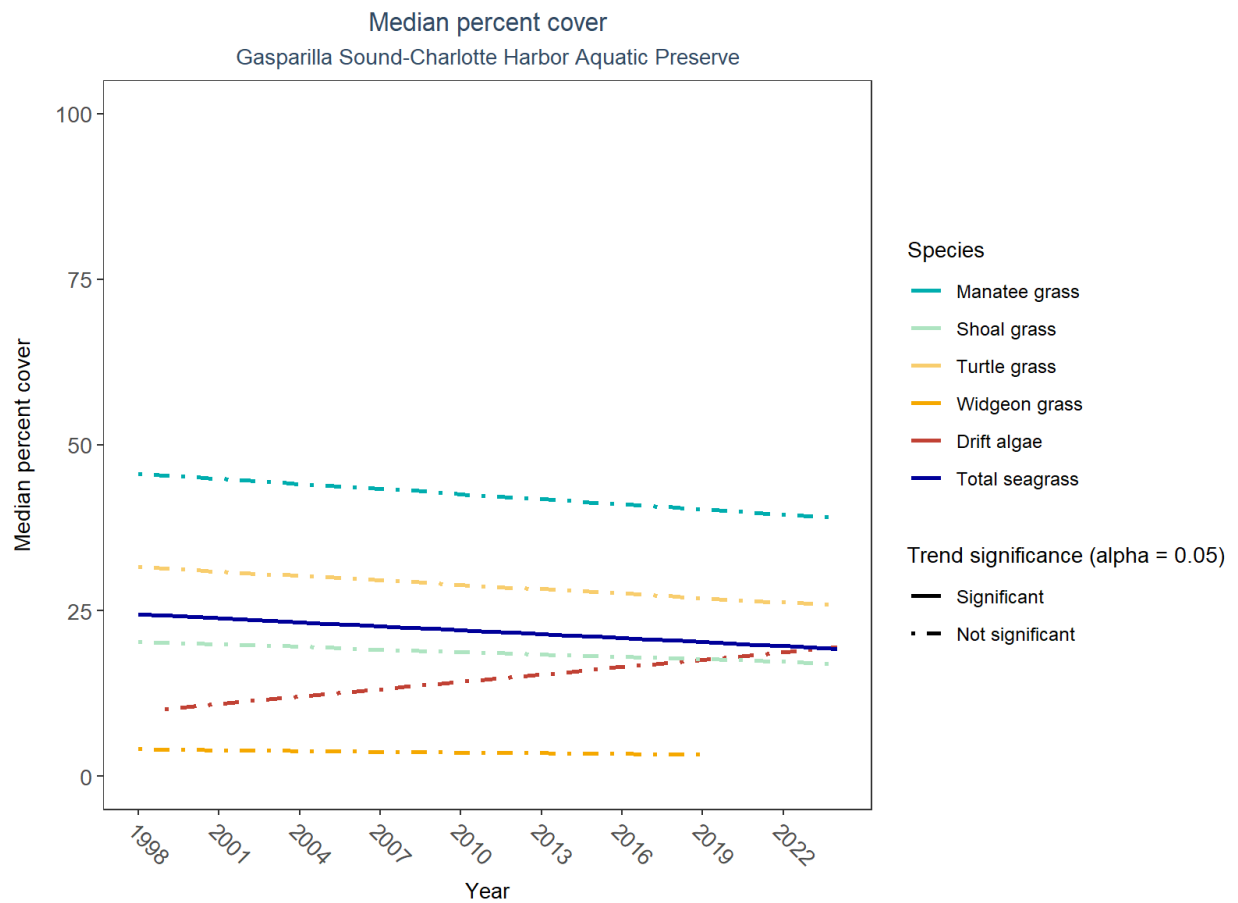


Table 13: SAV LME Results for Gasparilla Sound-Charlotte Harbor Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2007 - 2024	-11.38	0.91	0.26
Drift algae	No detectable trend	1999 - 2024	8.35	0.37	0.18
Shoal grass	No detectable trend	1998 - 2024	20.79	-0.12	0.08
Star grass	Insufficient data	-	-	-	-
Widgeon grass	No detectable trend	1998 - 2019	4.26	-0.04	0.80
Manatee grass	No detectable trend	1998 - 2024	46.72	-0.26	0.62
Turtle grass	No detectable trend	1998 - 2024	32.49	-0.22	0.14
Total seagrass	Decreasing trend	1998 - 2024	25.25	-0.20	0.01

An annual decrease in percent cover was observed for total seagrass (-0.20%). Manatee grass, shoal grass, turtle grass, widgeon grass, attached algae, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for star grass due to insufficient data.



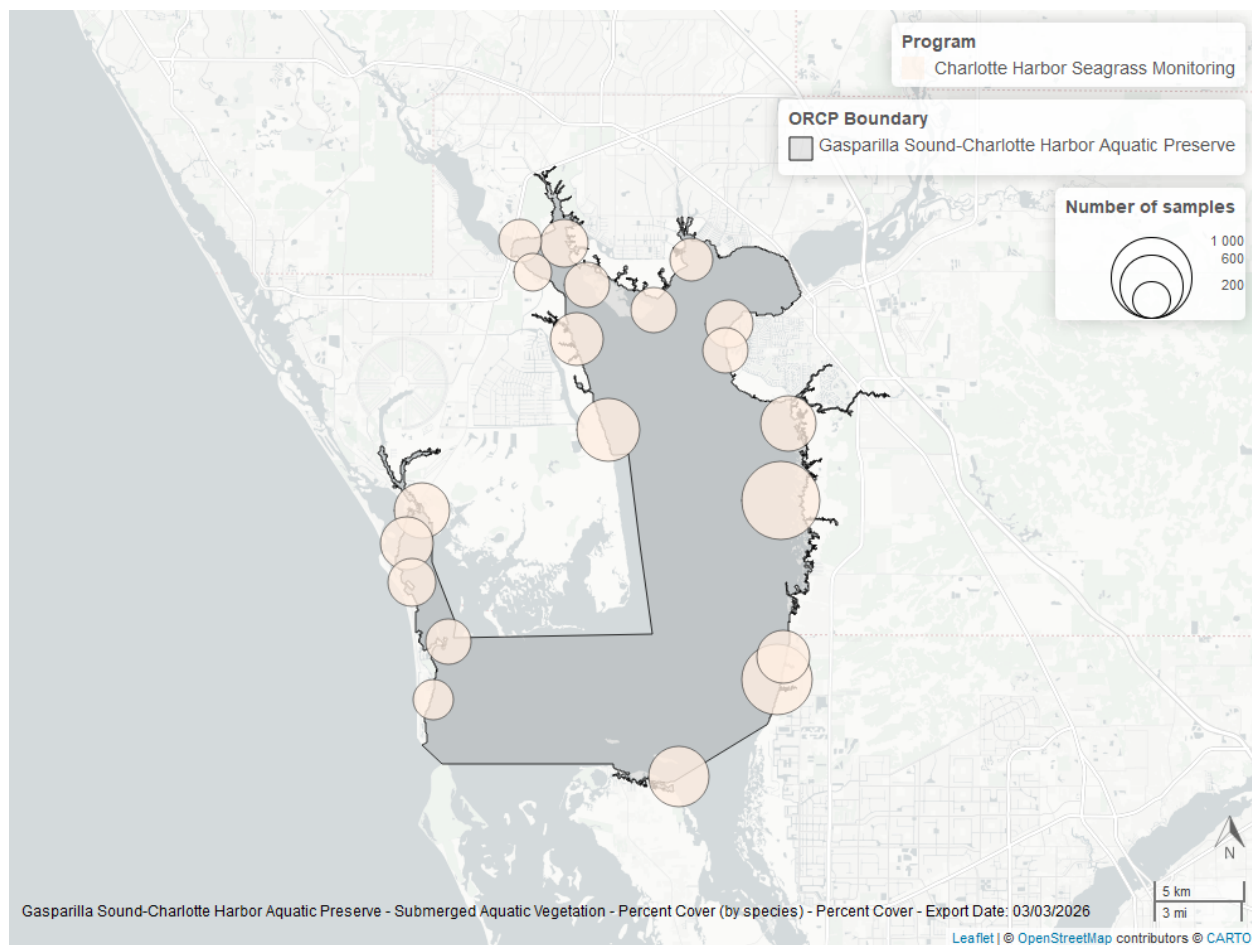
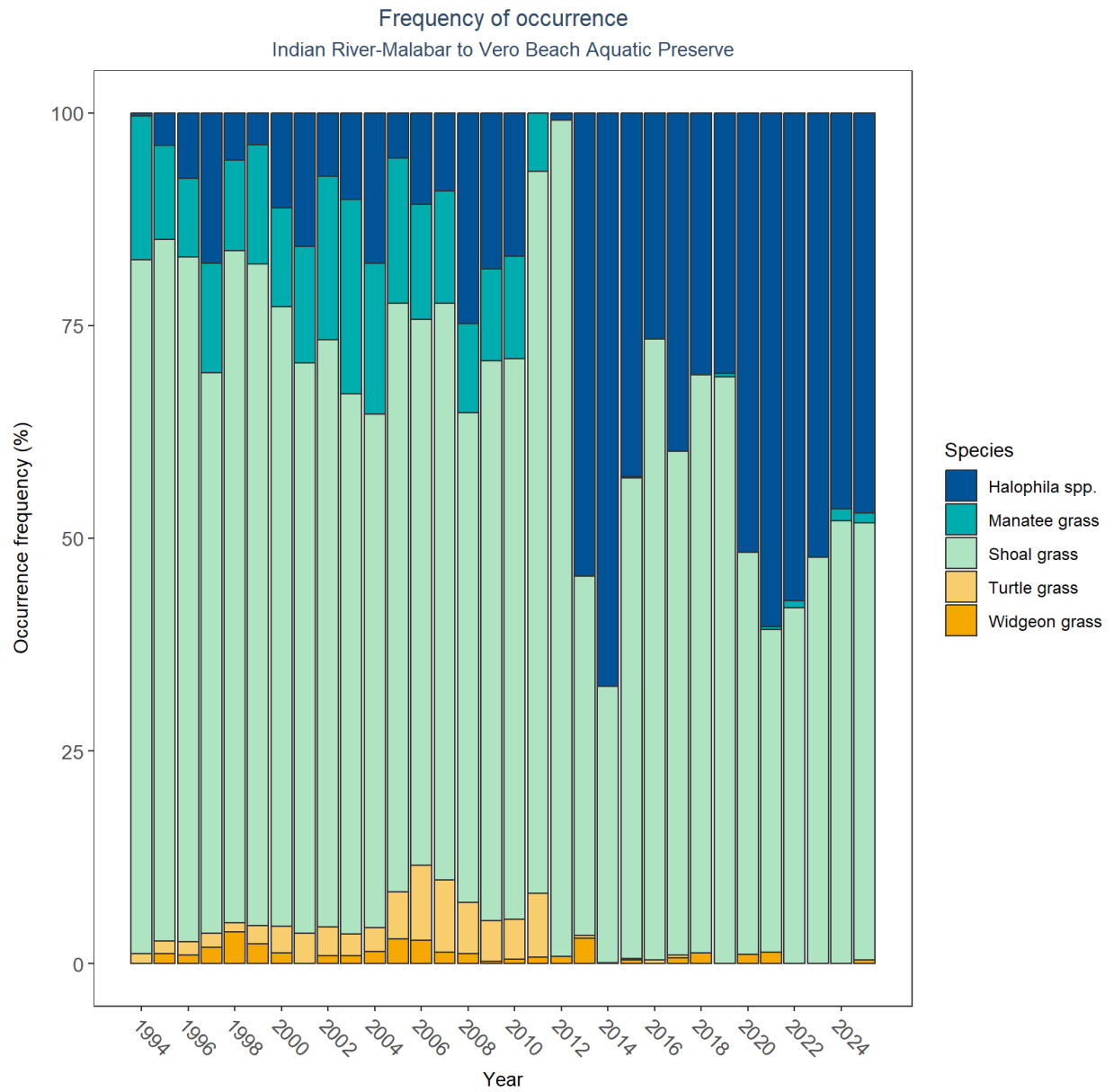


Figure 13: Map showing SAV sampling sites within the boundaries of *Gasparilla Sound-Charlotte Harbor Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Indian River-Malabar to Vero Beach Aquatic Preserve



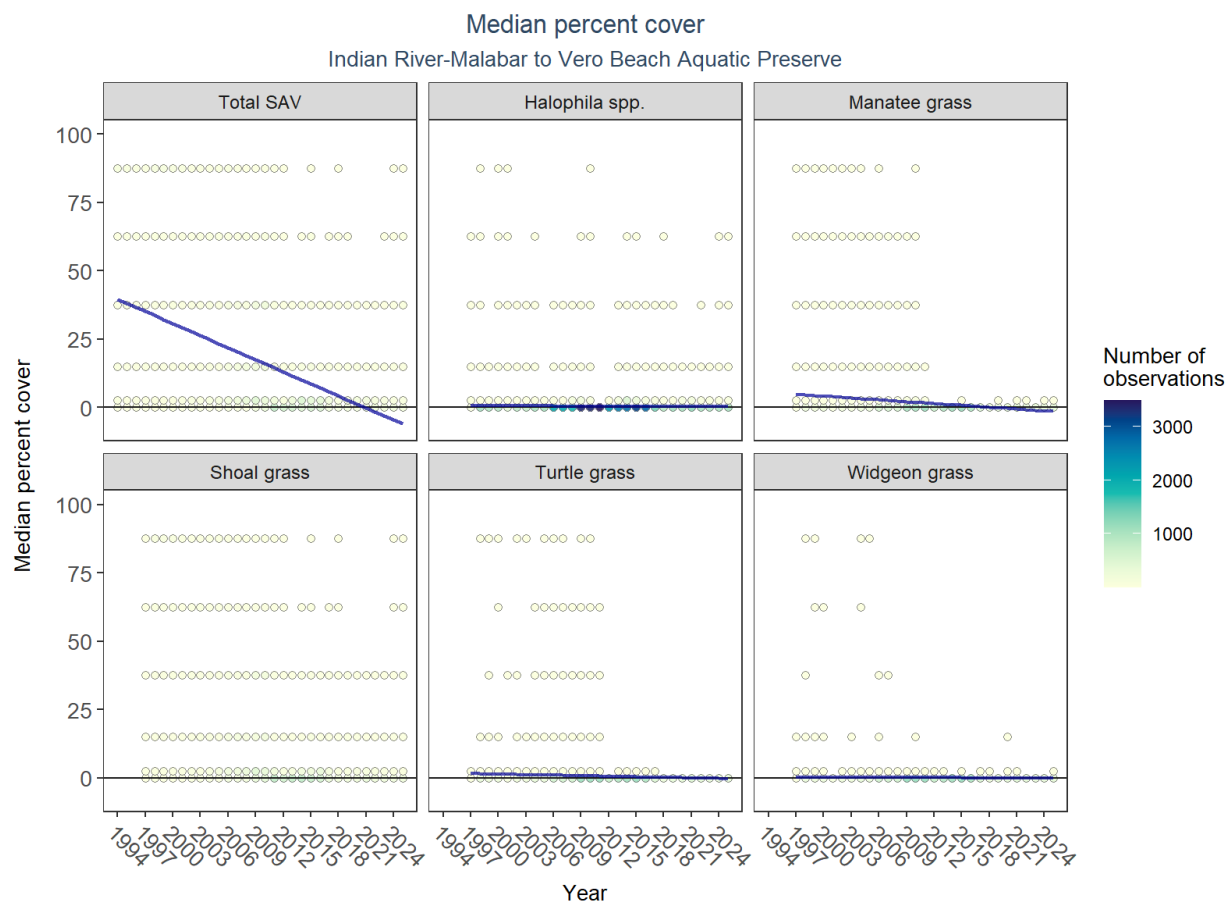
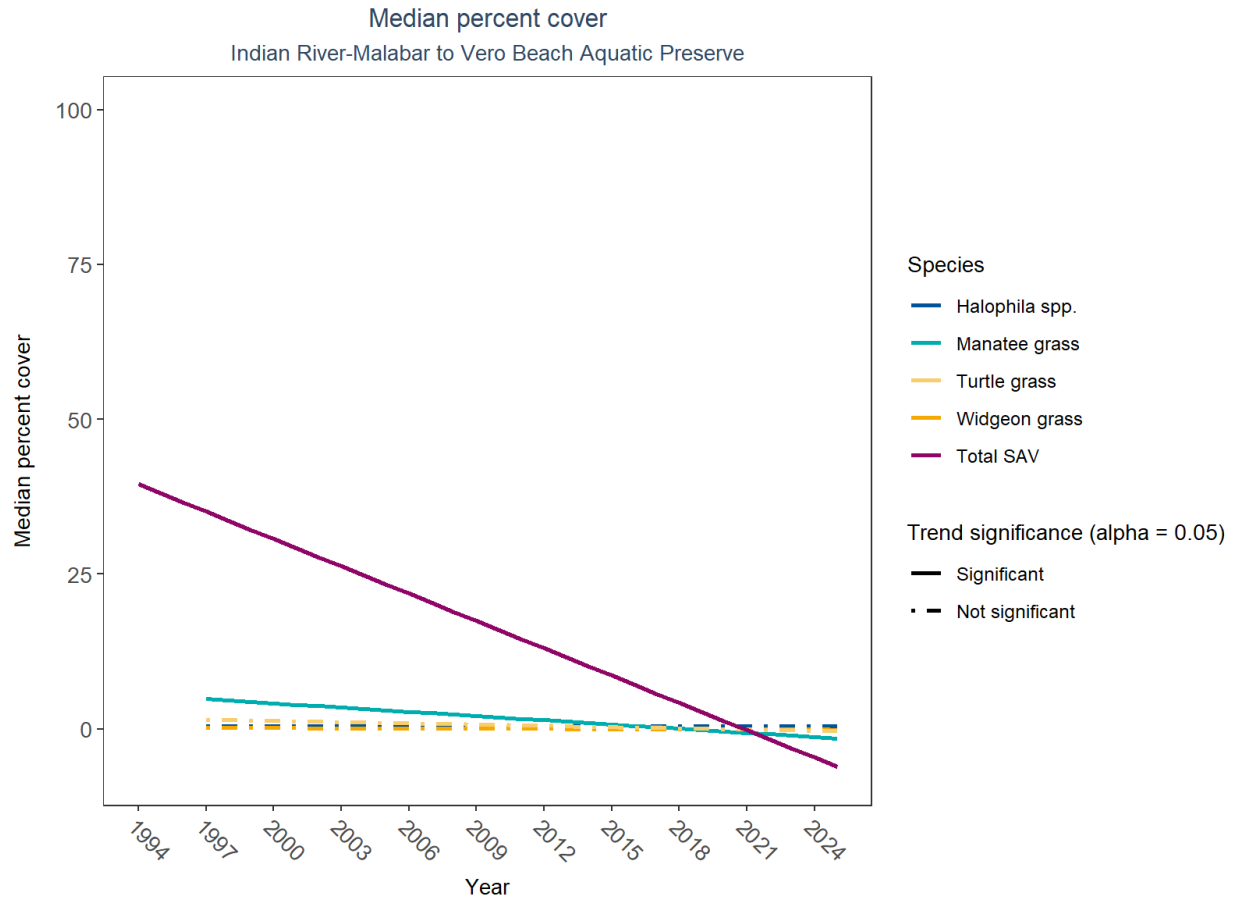


Table 14: SAV LME Results for Indian River-Malabar to Vero Beach Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Model did not fit the available data	-	-	-	-
Halophila spp.	No detectable trend	1997 - 2025	0.49	0.00	0.79
Widgeon grass	No detectable trend	1997 - 2025	0.24	-0.01	0.10
Manatee grass	Decreasing trend	1997 - 2025	5.52	-0.23	0.00
Turtle grass	No detectable trend	1997 - 2025	1.73	-0.07	0.10
Total SAV	Decreasing trend	1994 - 2025	39.51	-1.47	0.00
Total seagrass	Insufficient data	-	-	-	-

Annual decreases in percent cover were observed for total SAV (-1.47%) and manatee grass (-0.23%). *Halophila* spp., turtle grass, and widgeon grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for total seagrass and drift algae due to insufficient data, and the model could not be fitted for shoal grass.



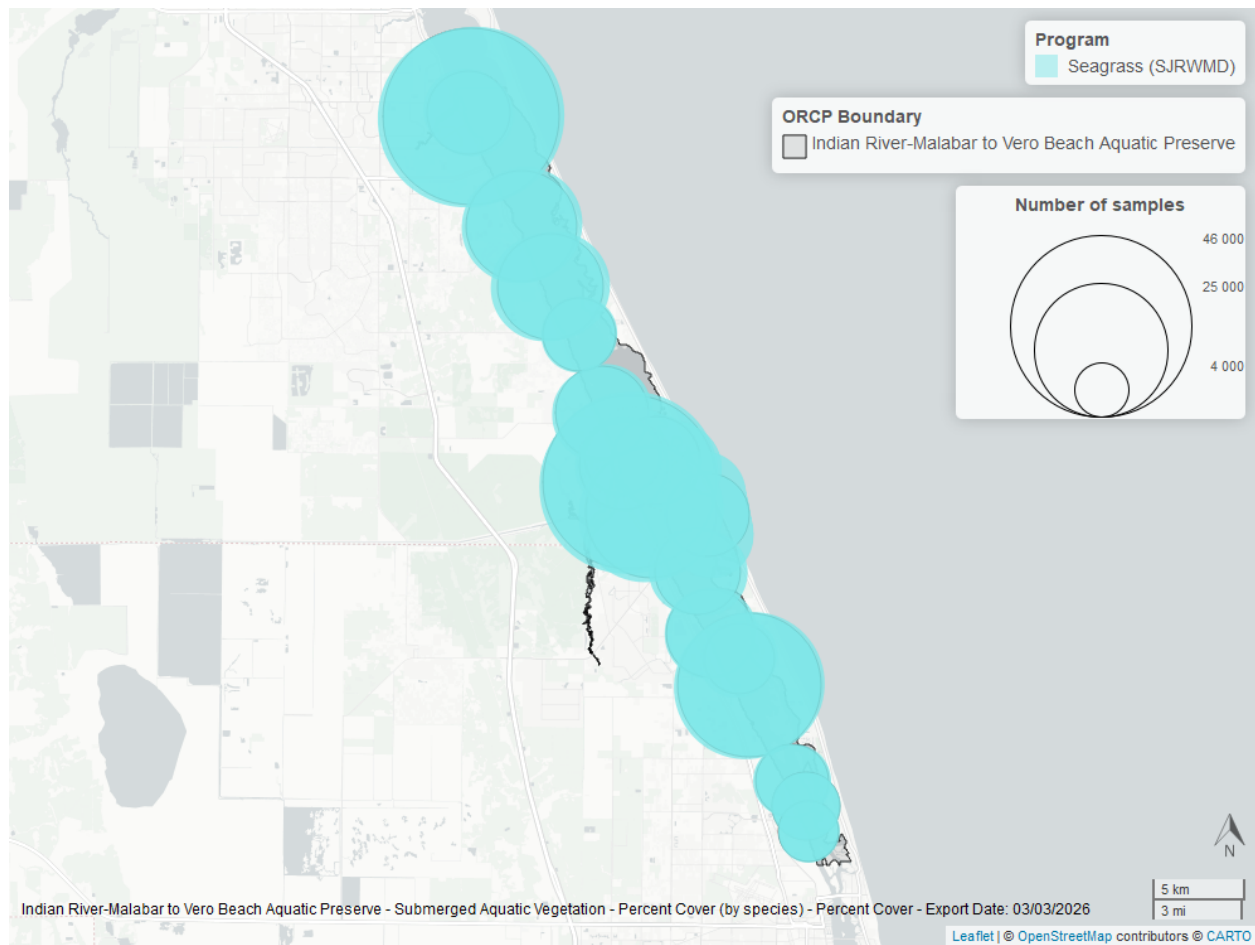
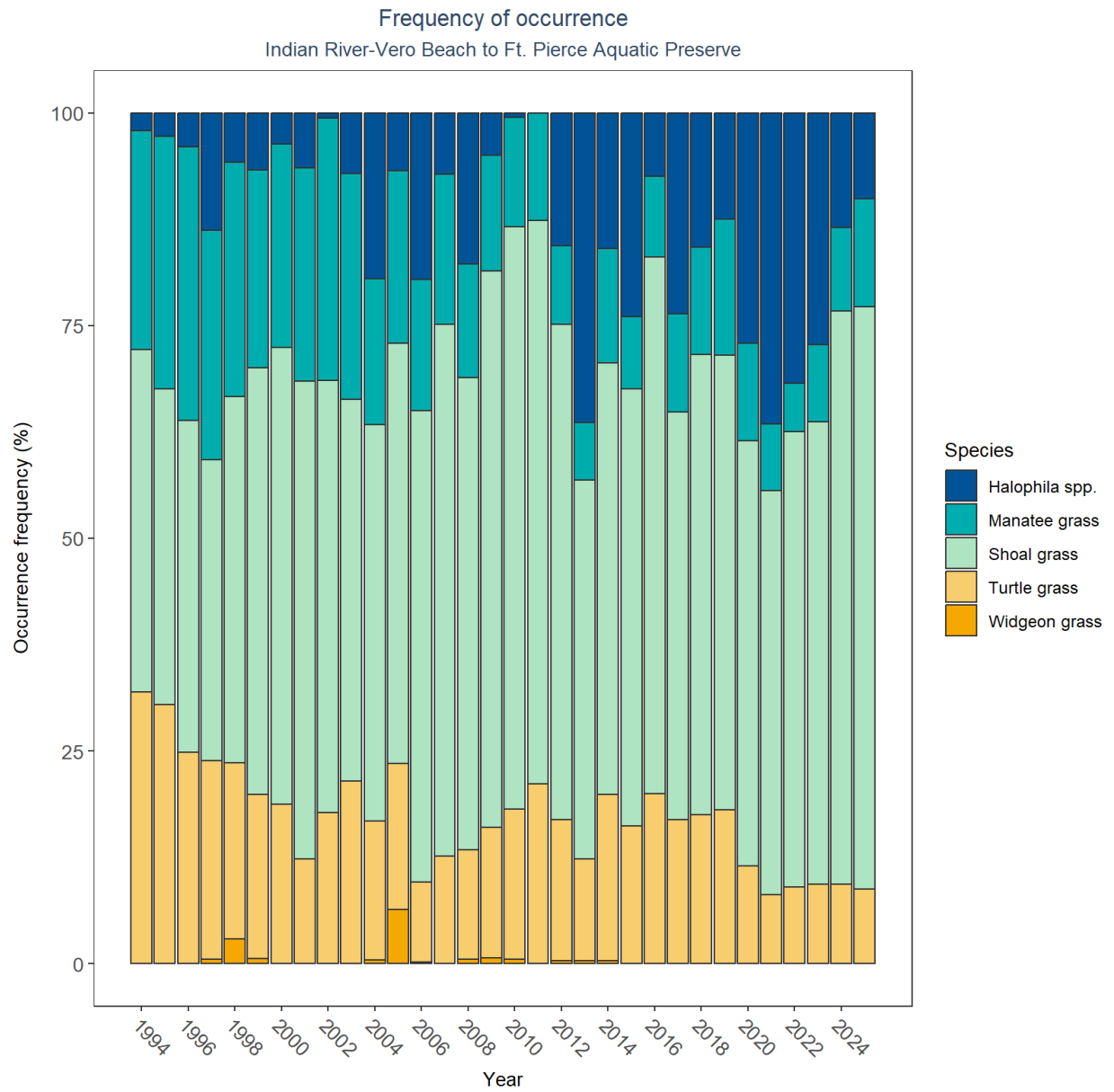


Figure 14: Map showing SAV sampling sites within the boundaries of *Indian River-Malabar to Vero Beach Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Indian River-Vero Beach to Ft. Pierce Aquatic Preserve



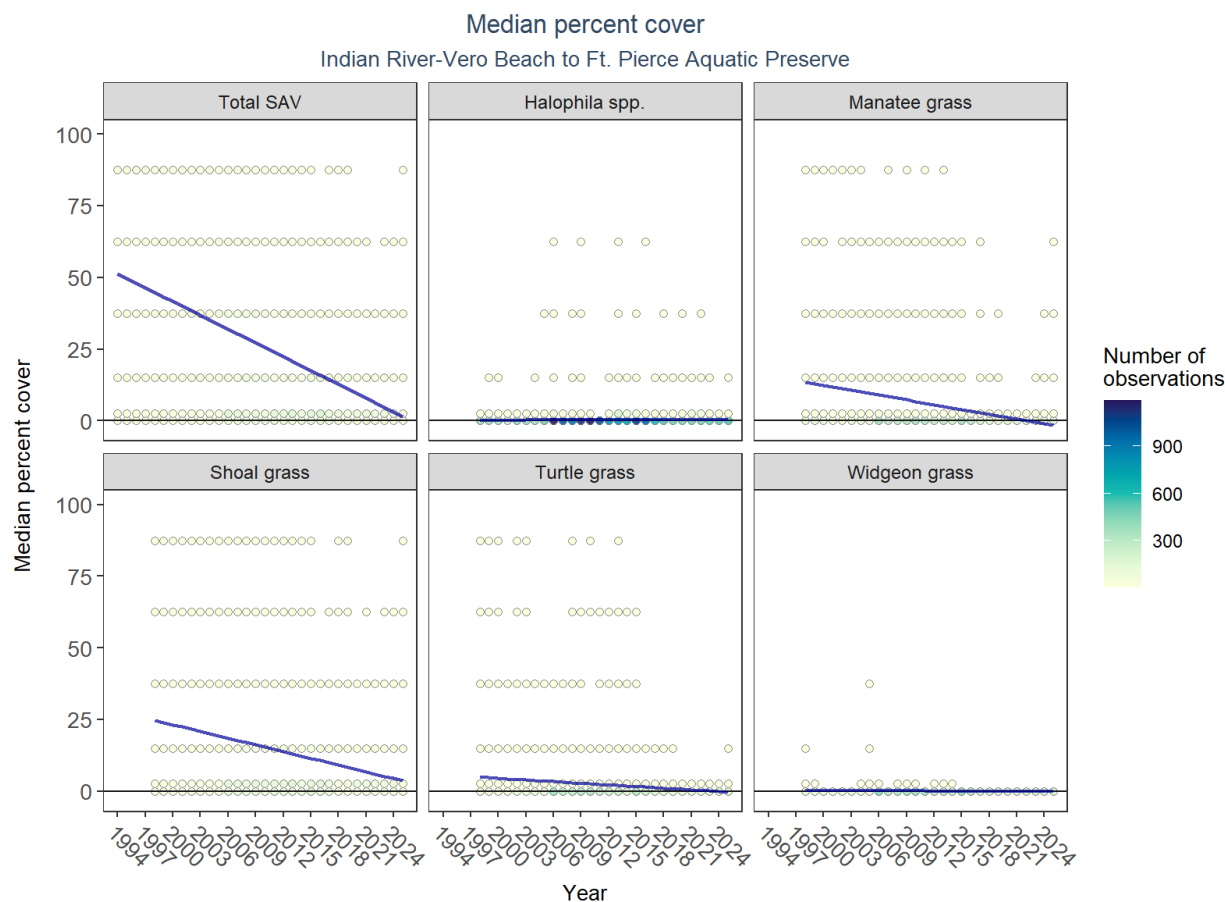
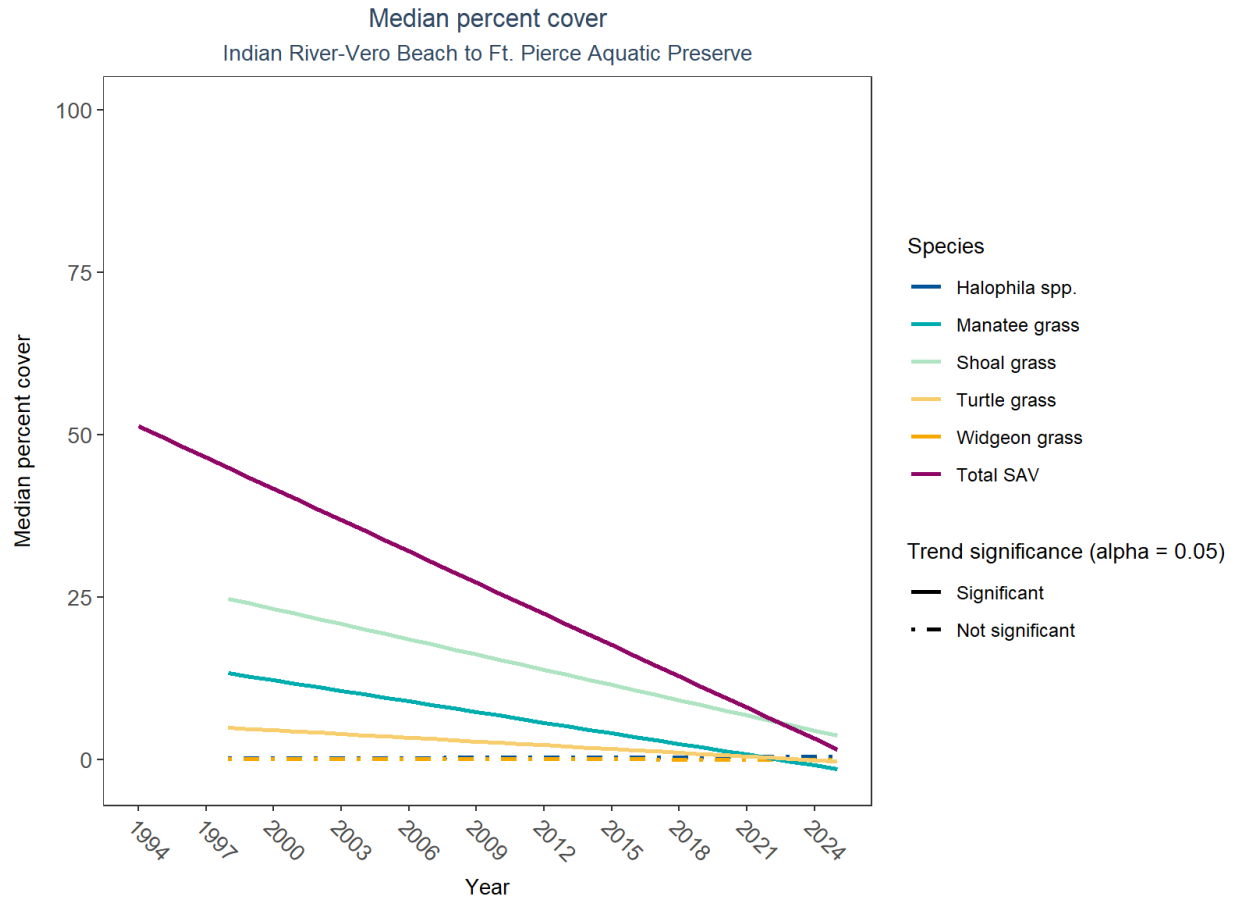


Table 15: SAV LME Results for Indian River-Vero Beach to Ft. Pierce Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Decreasing trend	1998 - 2025	27.88	-0.78	0.00
Halophila spp.	No detectable trend	1998 - 2025	0.12	0.01	0.09
Widgeon grass	No detectable trend	1998 - 2025	0.13	0.00	0.12
Manatee grass	Decreasing trend	1998 - 2025	15.48	-0.55	0.04
Turtle grass	Decreasing trend	1998 - 2025	5.65	-0.19	0.01
Total SAV	Decreasing trend	1994 - 2025	51.26	-1.60	0.00
Total seagrass	Insufficient data	-	-	-	-

Annual decreases in percent cover were observed for total SAV (-1.60%), manatee grass (-0.55%), shoal grass (-0.78%), and turtle grass (-0.19%). No detectable change in percent cover was observed for *Halophila* spp. and widgeon grass. Trends in percent cover could not be evaluated for total seagrass and drift algae due to insufficient data.



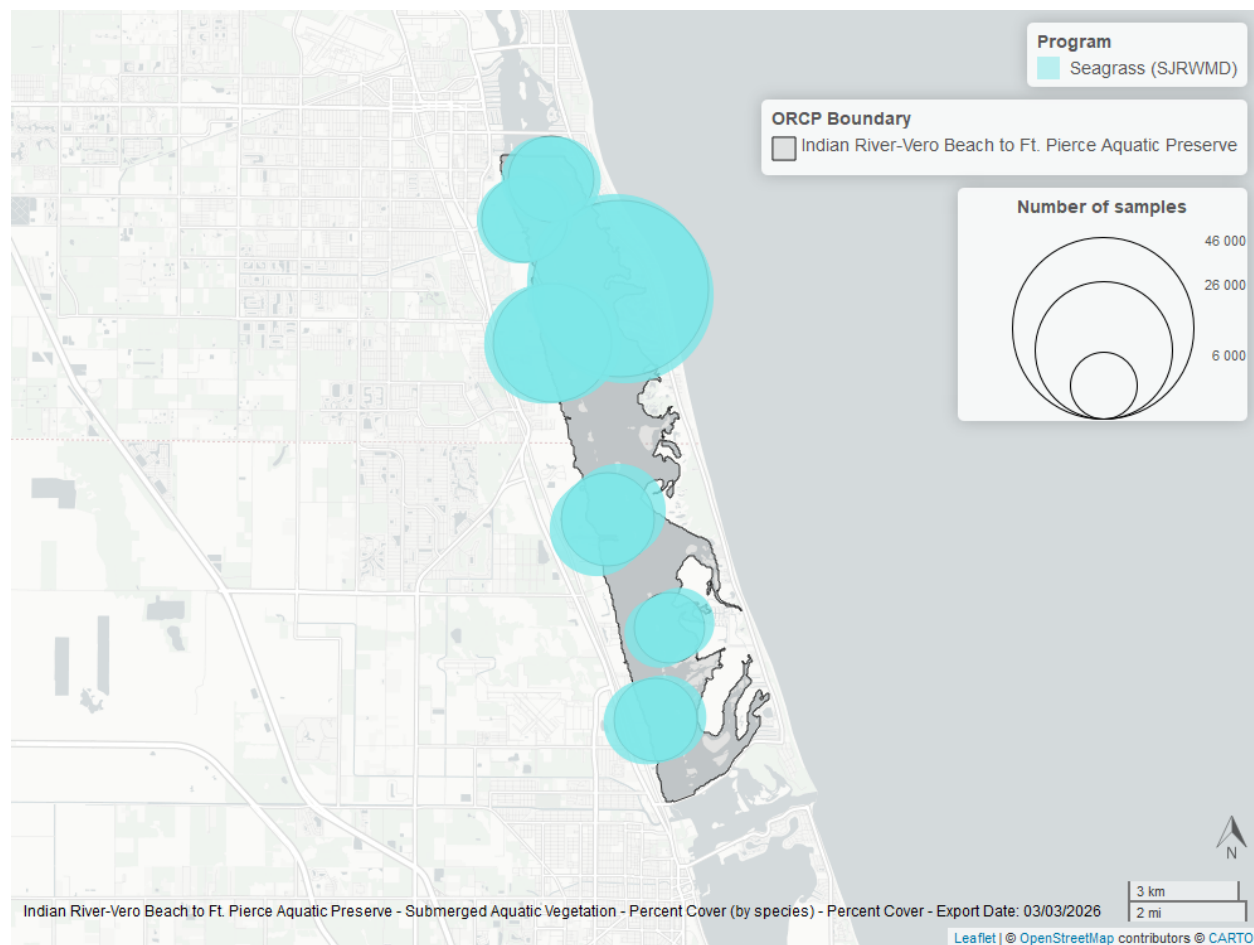
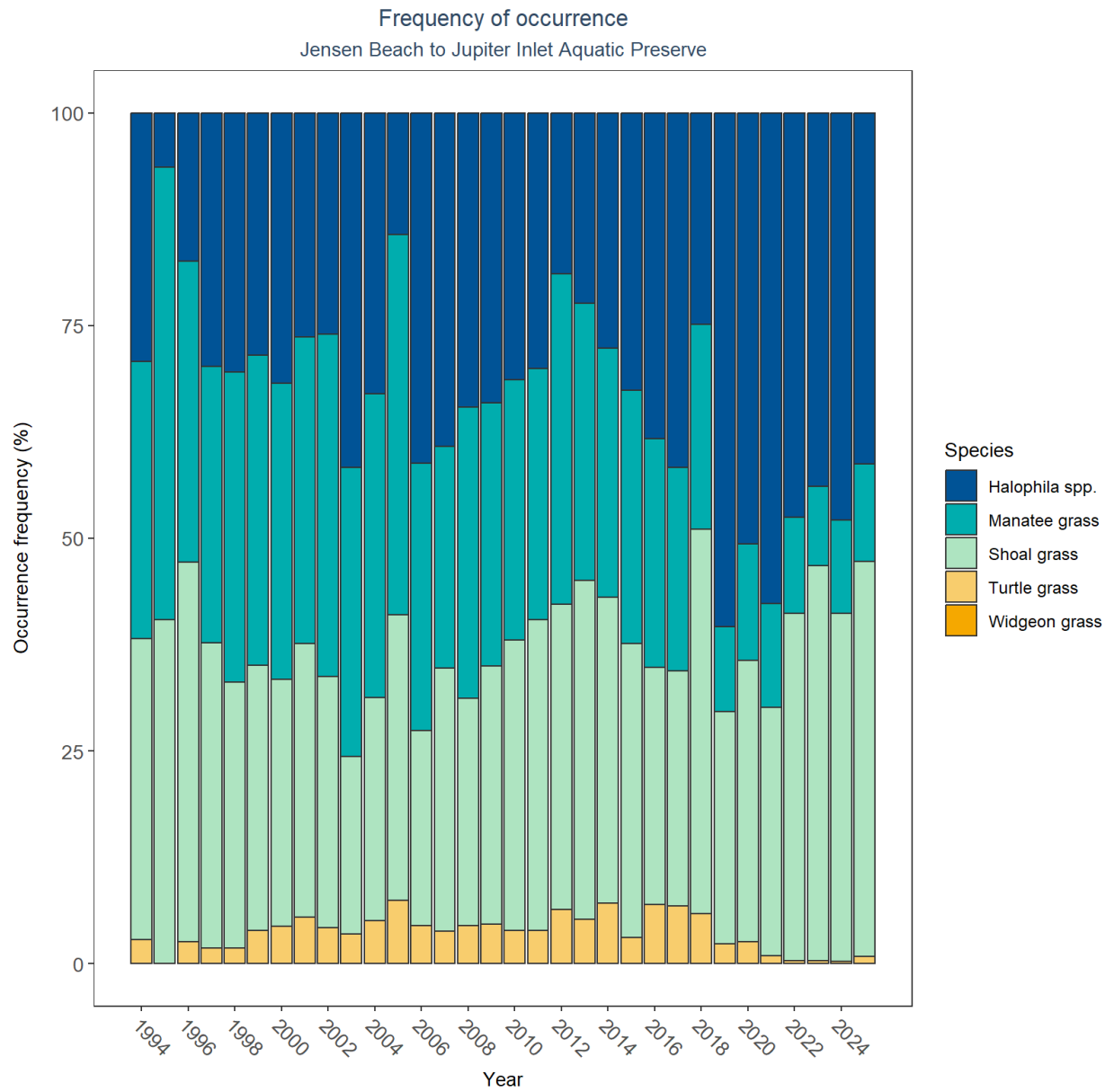


Figure 15: Map showing SAV sampling sites within the boundaries of *Indian River-Vero Beach to Ft. Pierce Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Jensen Beach to Jupiter Inlet Aquatic Preserve



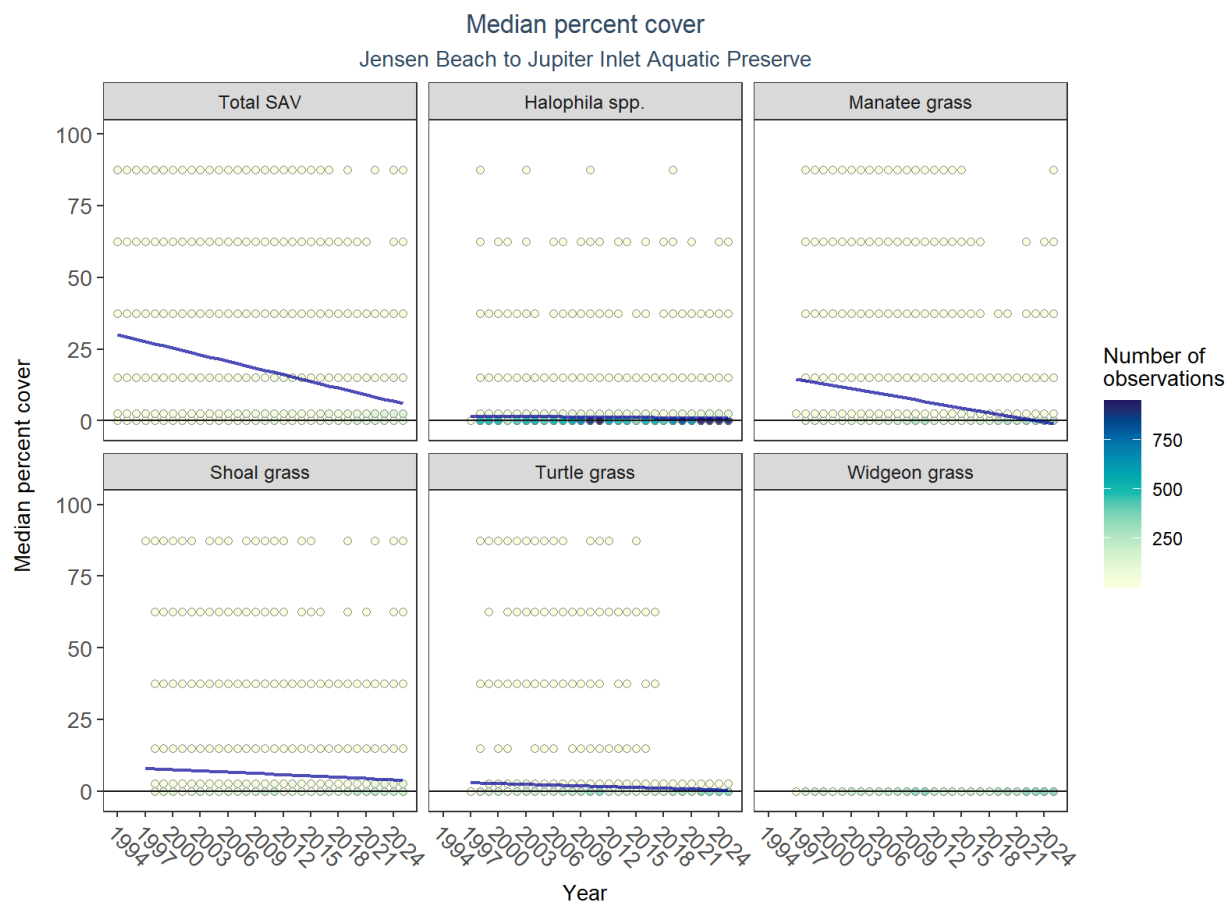
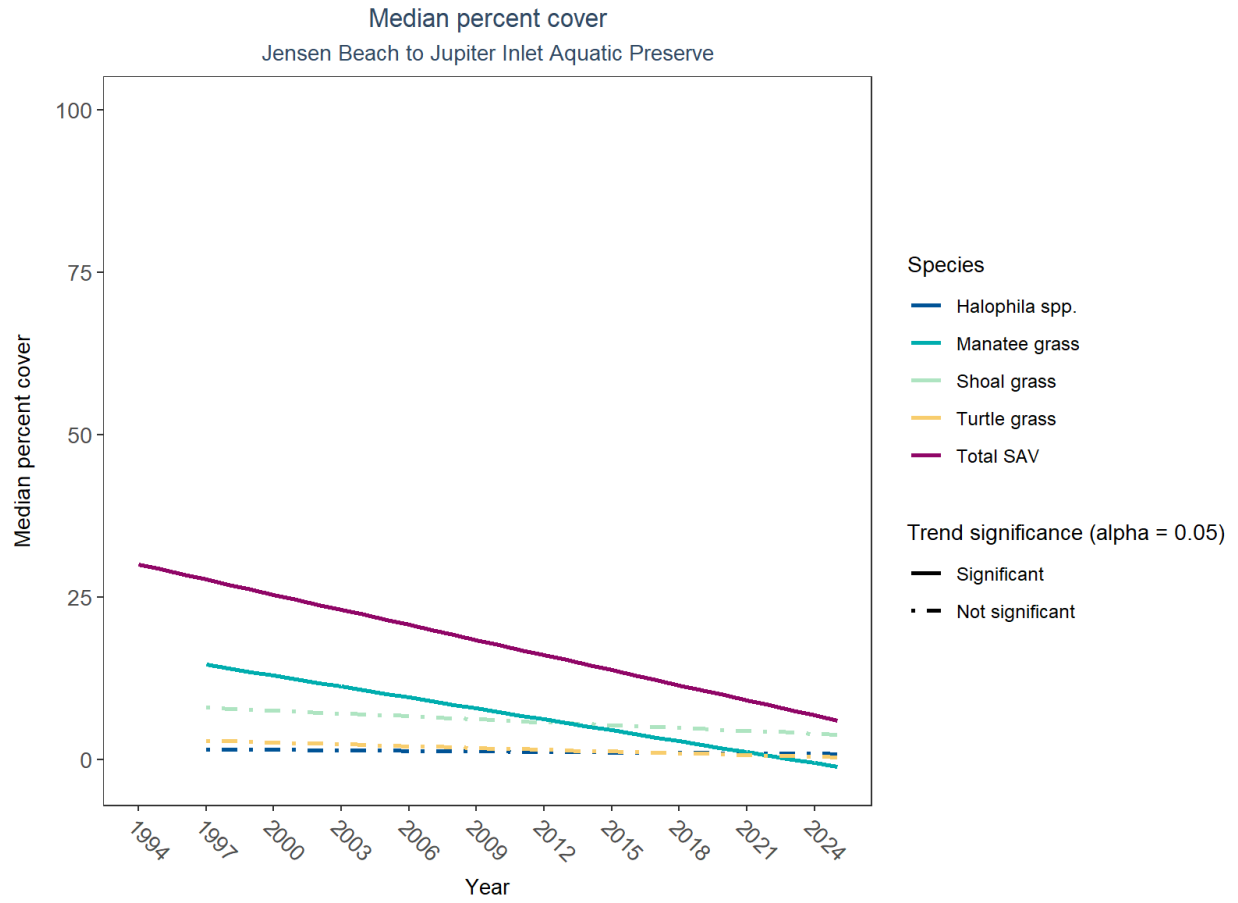


Table 16: SAV LME Results for Jensen Beach to Jupiter Inlet Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	No detectable trend	1997 - 2025	8.41	-0.15	0.10
Halophila spp.	No detectable trend	1997 - 2025	1.66	-0.03	0.06
Widgeon grass	Model did not fit the available data	-	-	-	-
Manatee grass	Decreasing trend	1997 - 2025	16.27	-0.56	0.00
Turtle grass	No detectable trend	1997 - 2025	3.16	-0.09	0.07
Total SAV	Decreasing trend	1994 - 2025	30.00	-0.77	0.00
Total seagrass	Insufficient data	-	-	-	-

Annual decreases in percent cover were observed for total SAV (-0.77%) and manatee grass (-0.56%). *Halophila* spp., shoal grass, and turtle grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for total seagrass and drift algae due to insufficient data, and the model could not be fitted for widgeon grass.



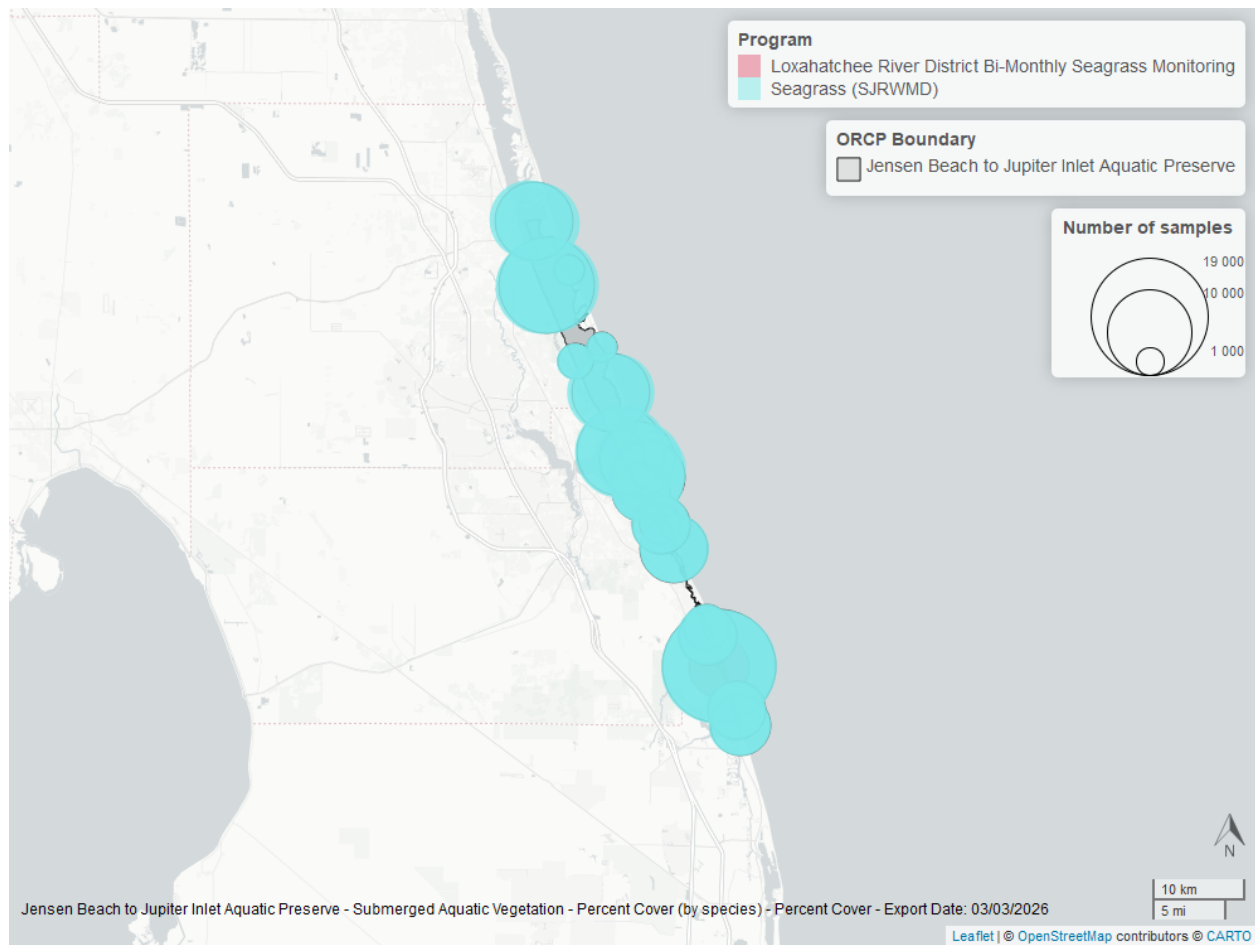
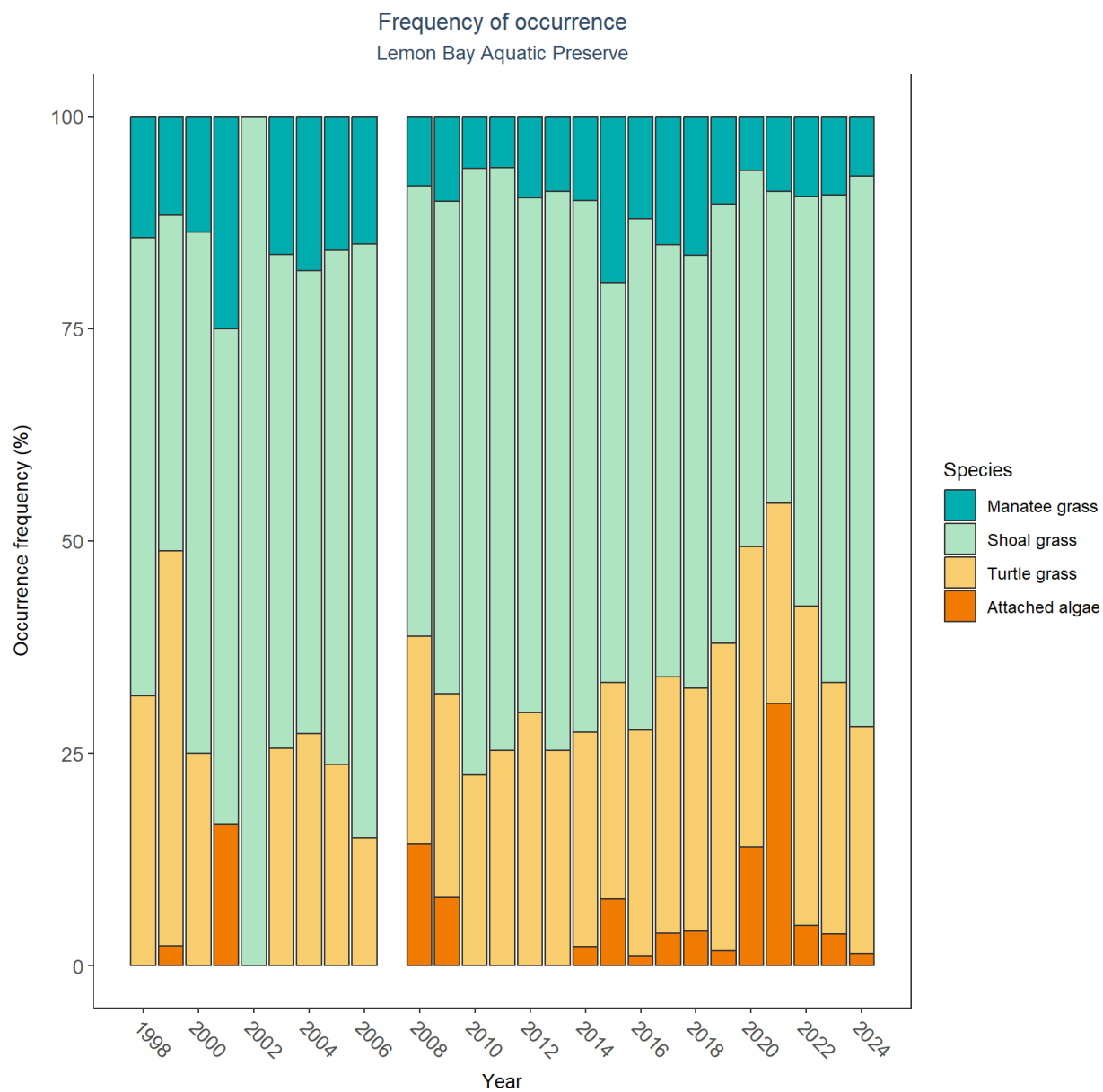


Figure 16: Map showing SAV sampling sites within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Lemon Bay Aquatic Preserve



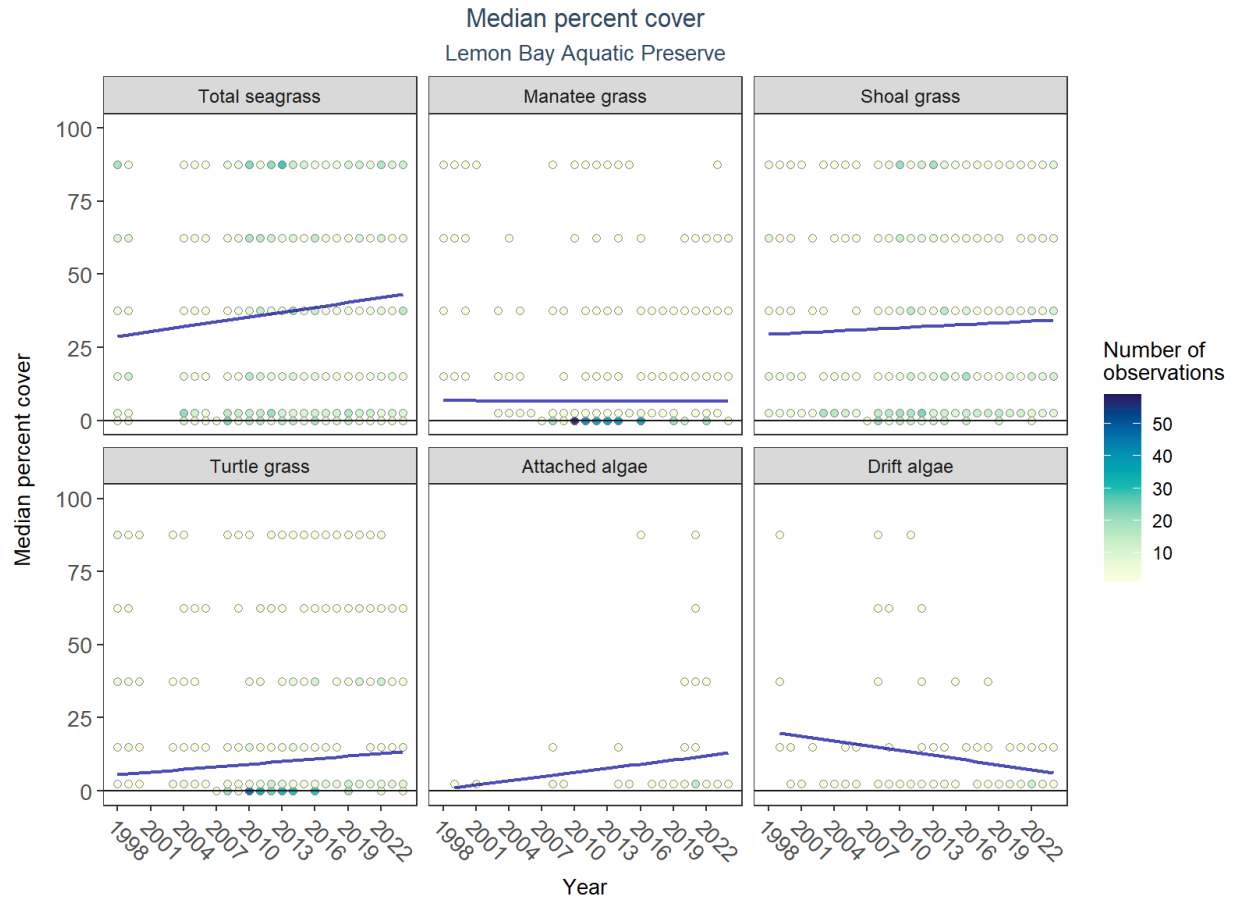
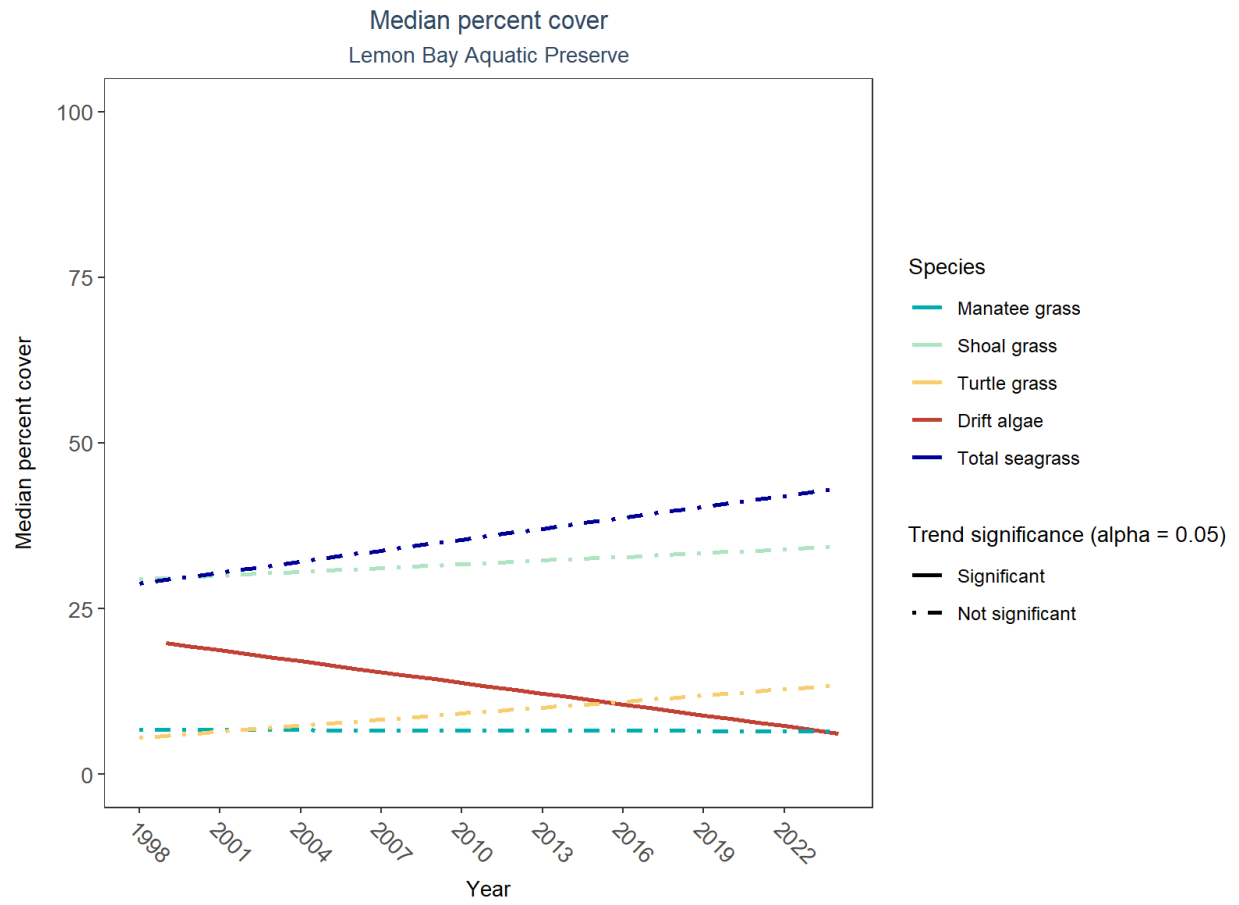


Table 17: SAV LME Results for Lemon Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	1999 - 2024	-1.17	0.47	0.26
Drift algae	Decreasing trend	1999 - 2024	22.53	-0.54	0.02
Shoal grass	No detectable trend	1998 - 2024	28.69	0.19	0.60
Manatee grass	No detectable trend	1998 - 2024	6.78	-0.01	0.96
Turtle grass	No detectable trend	1998 - 2024	4.32	0.30	0.45
Total seagrass	No detectable trend	1998 - 2024	26.64	0.55	0.25

An annual decrease in percent cover was observed for drift algae (-0.54%). Total seagrass, manatee grass, shoal grass, turtle grass, and attached algae showed no detectable change in percent cover.



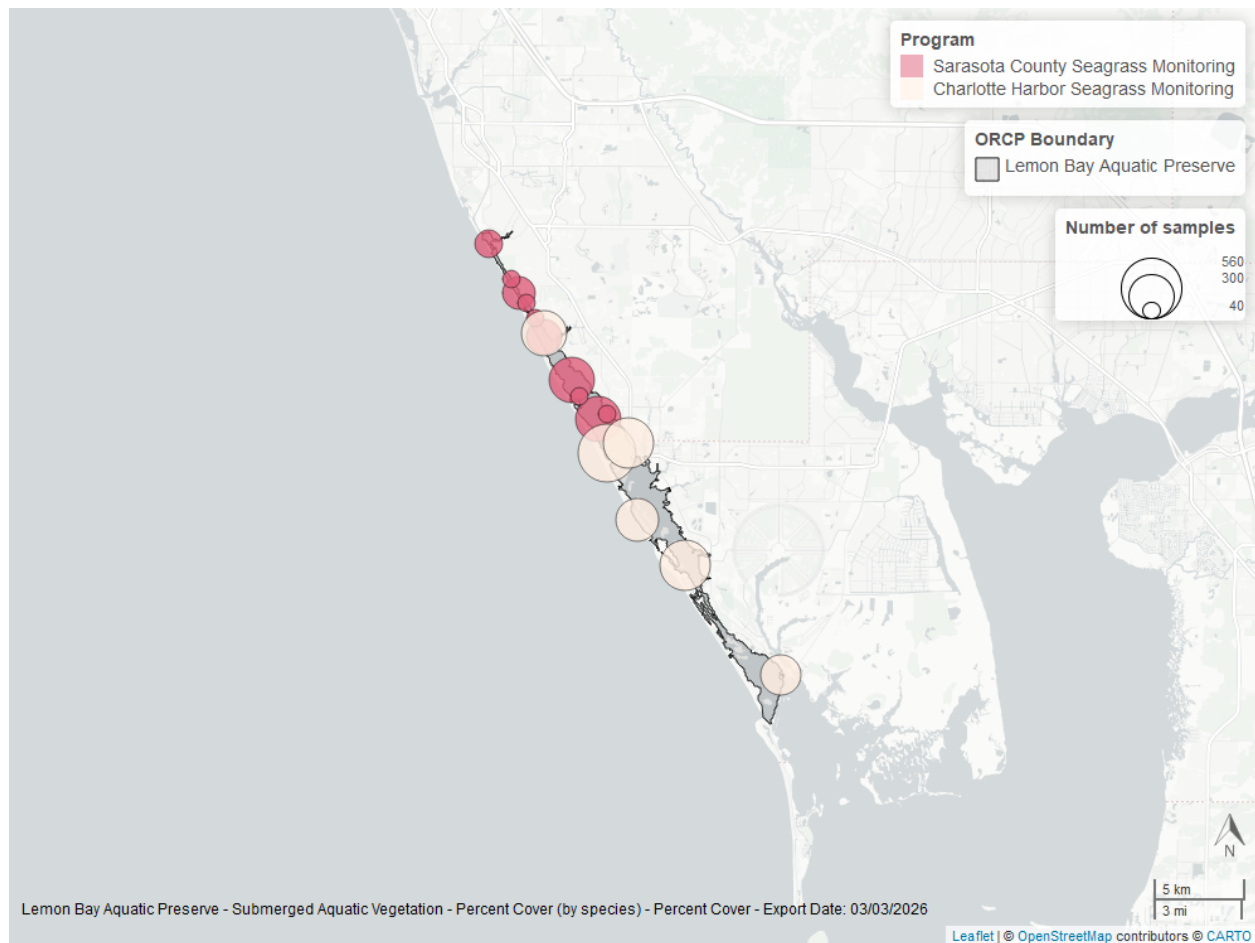
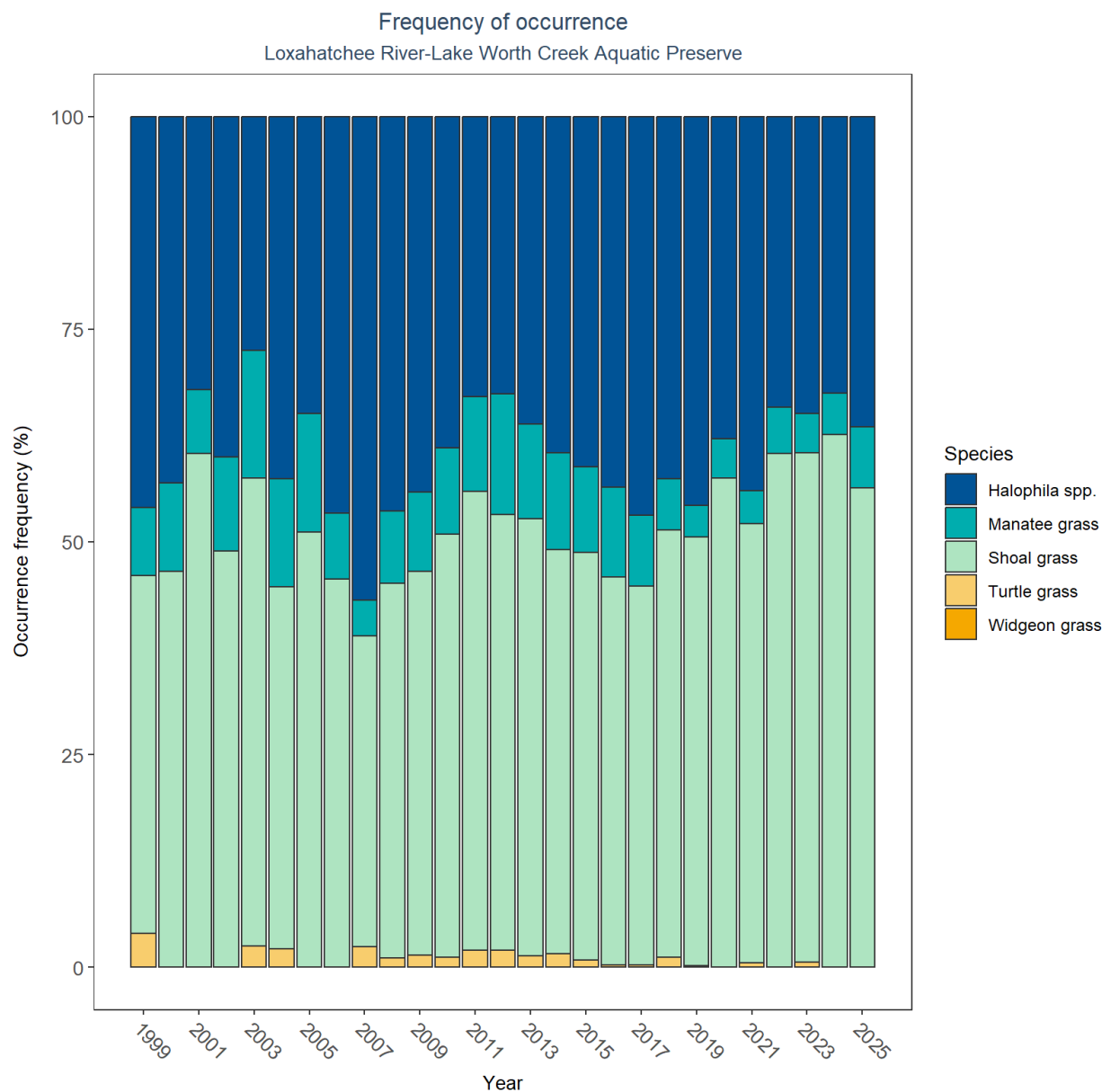


Figure 17: Map showing SAV sampling sites within the boundaries of *Lemon Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Lignumvitae Key Aquatic Preserve

Not enough data to perform analyses in Lignumvitae Key Aquatic Preserve

Loxahatchee River-Lake Worth Creek Aquatic Preserve



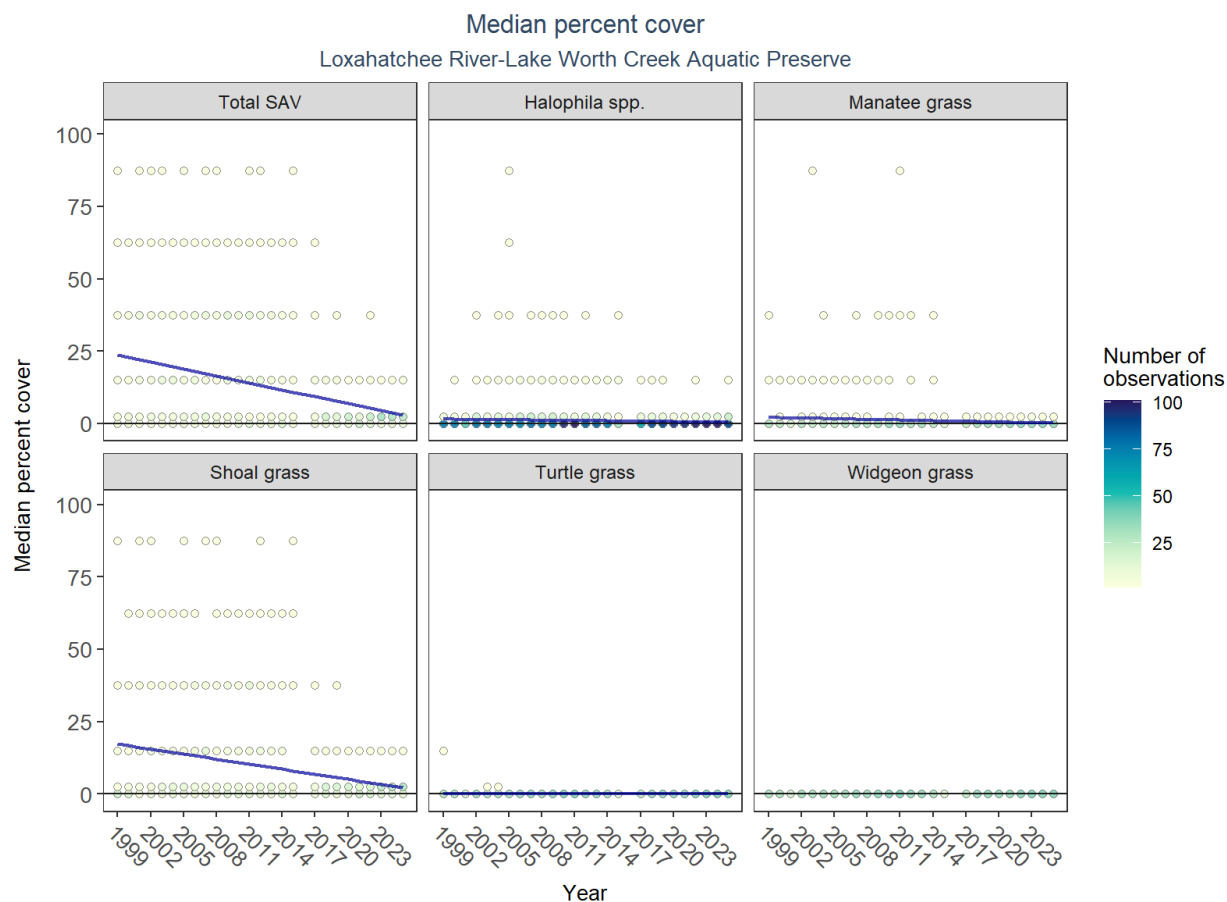
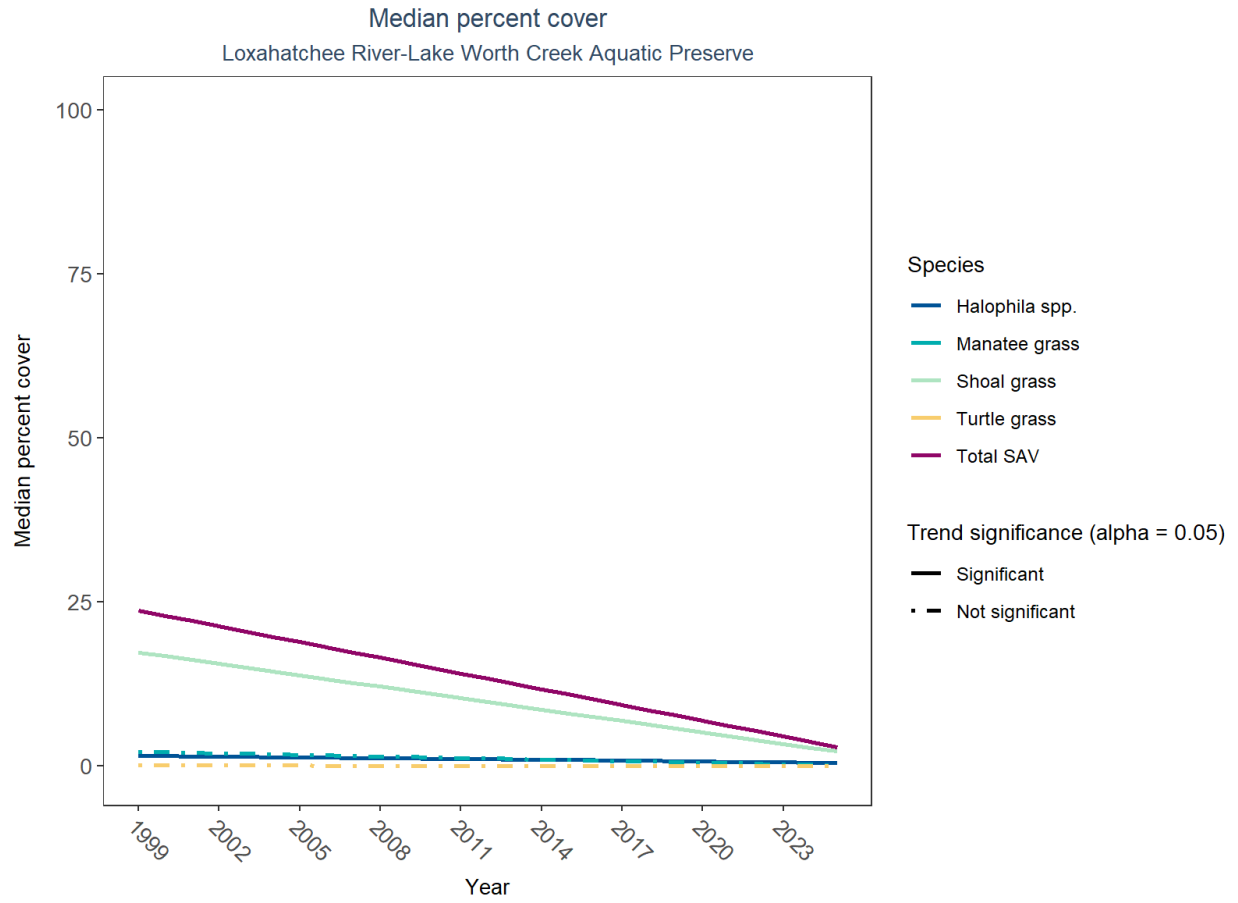


Table 18: SAV LME Results for Loxahatchee River-Lake Worth Creek Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Decreasing trend	1999 - 2025	20.20	-0.58	0.00
Halophila spp.	Decreasing trend	1999 - 2025	1.71	-0.04	0.00
Widgeon grass	Model did not fit the available data	-	-	-	-
Manatee grass	No detectable trend	1999 - 2025	2.54	-0.08	0.34
Turtle grass	No detectable trend	1999 - 2025	0.09	0.00	0.30
Total SAV	Decreasing trend	1999 - 2025	27.66	-0.80	0.00
Total seagrass	Insufficient data	-	-	-	-

Annual decreases in percent cover were observed for total SAV (-0.80%), *Halophila* spp. (-0.04%), and shoal grass (-0.58%). No detectable change in percent cover was observed for manatee grass and turtle grass. Trends in percent cover could not be evaluated for total seagrass and drift algae due to insufficient data, and the model could not be fitted for widgeon grass.



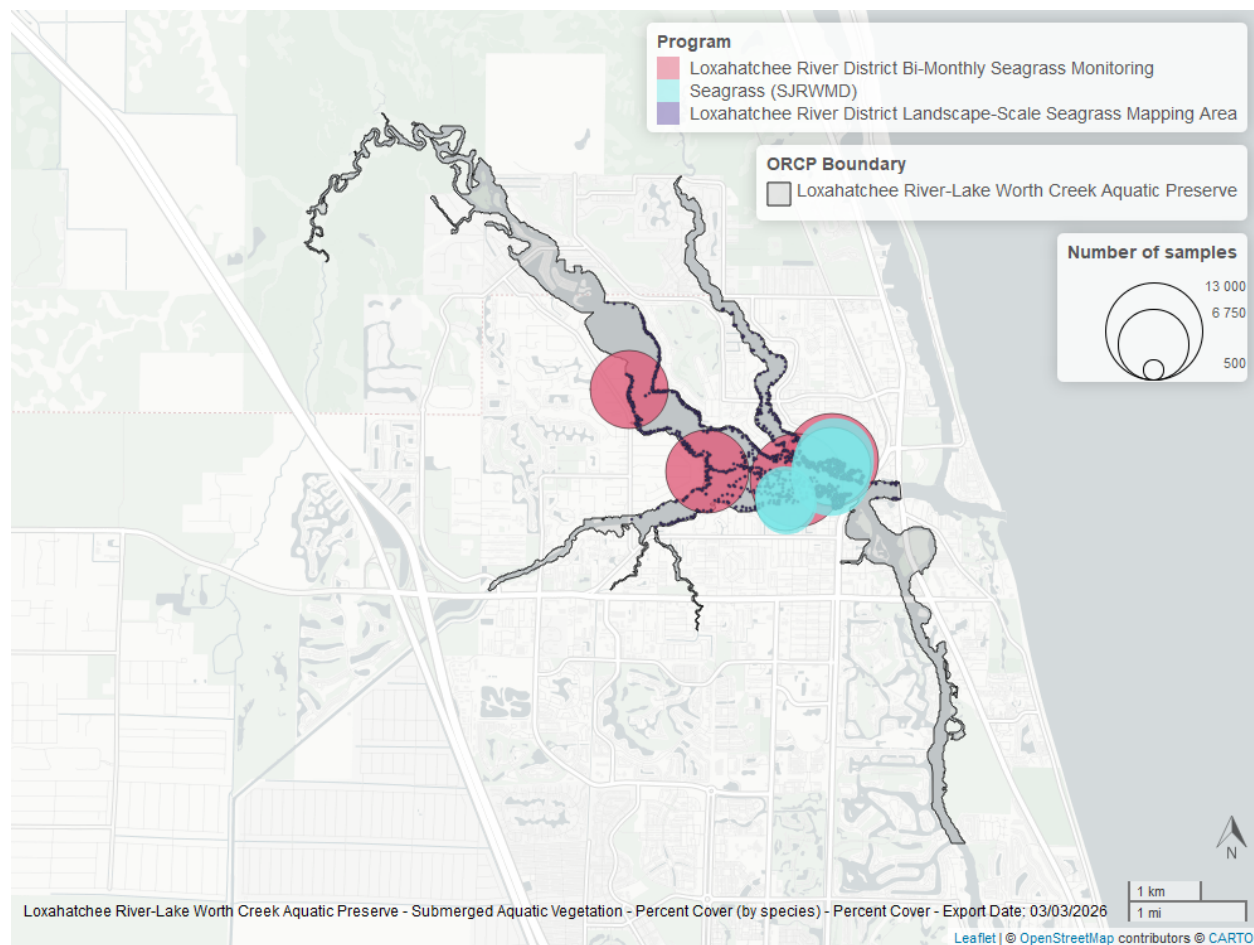
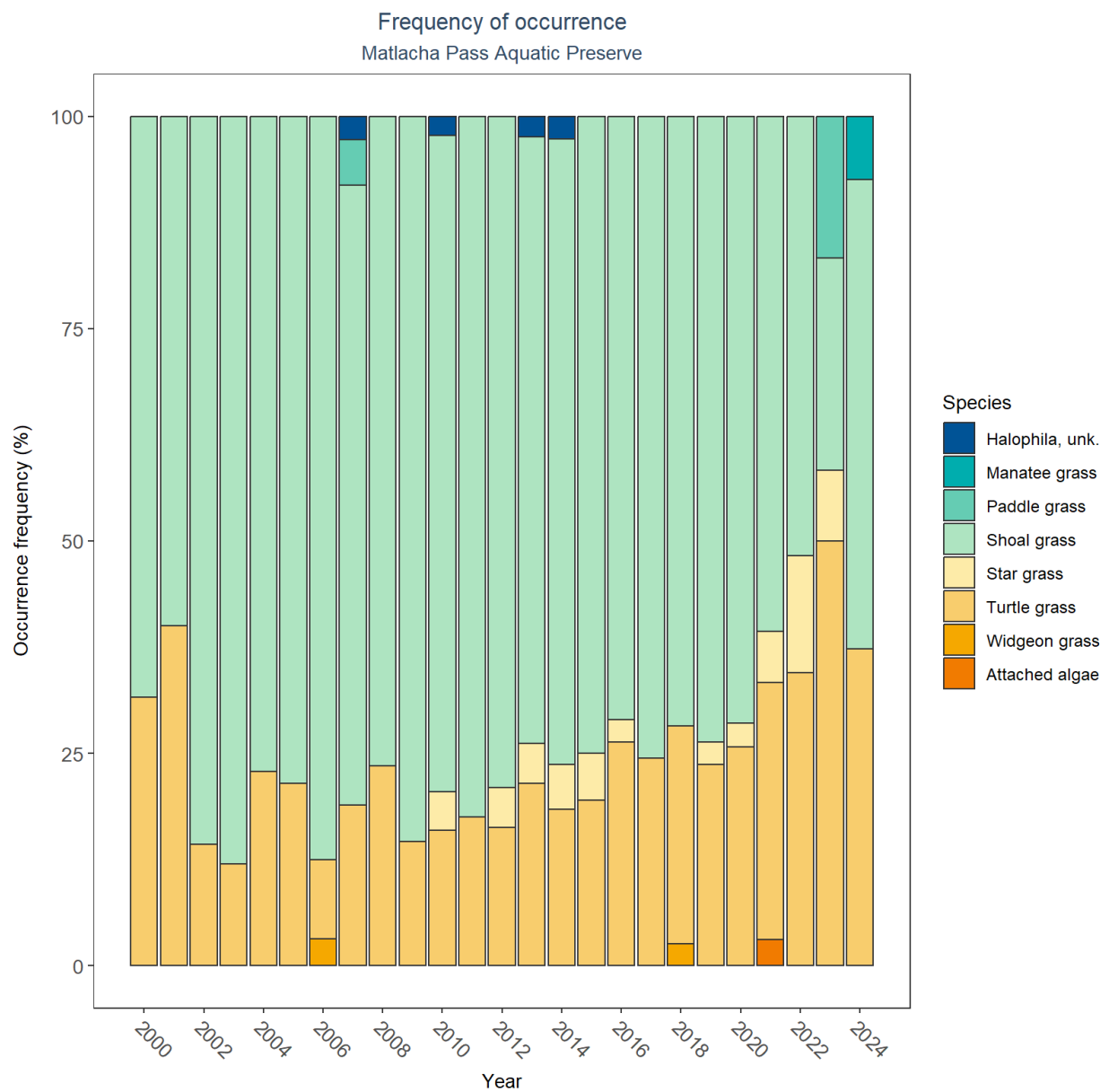


Figure 18: Map showing SAV sampling sites within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Matlacha Pass Aquatic Preserve



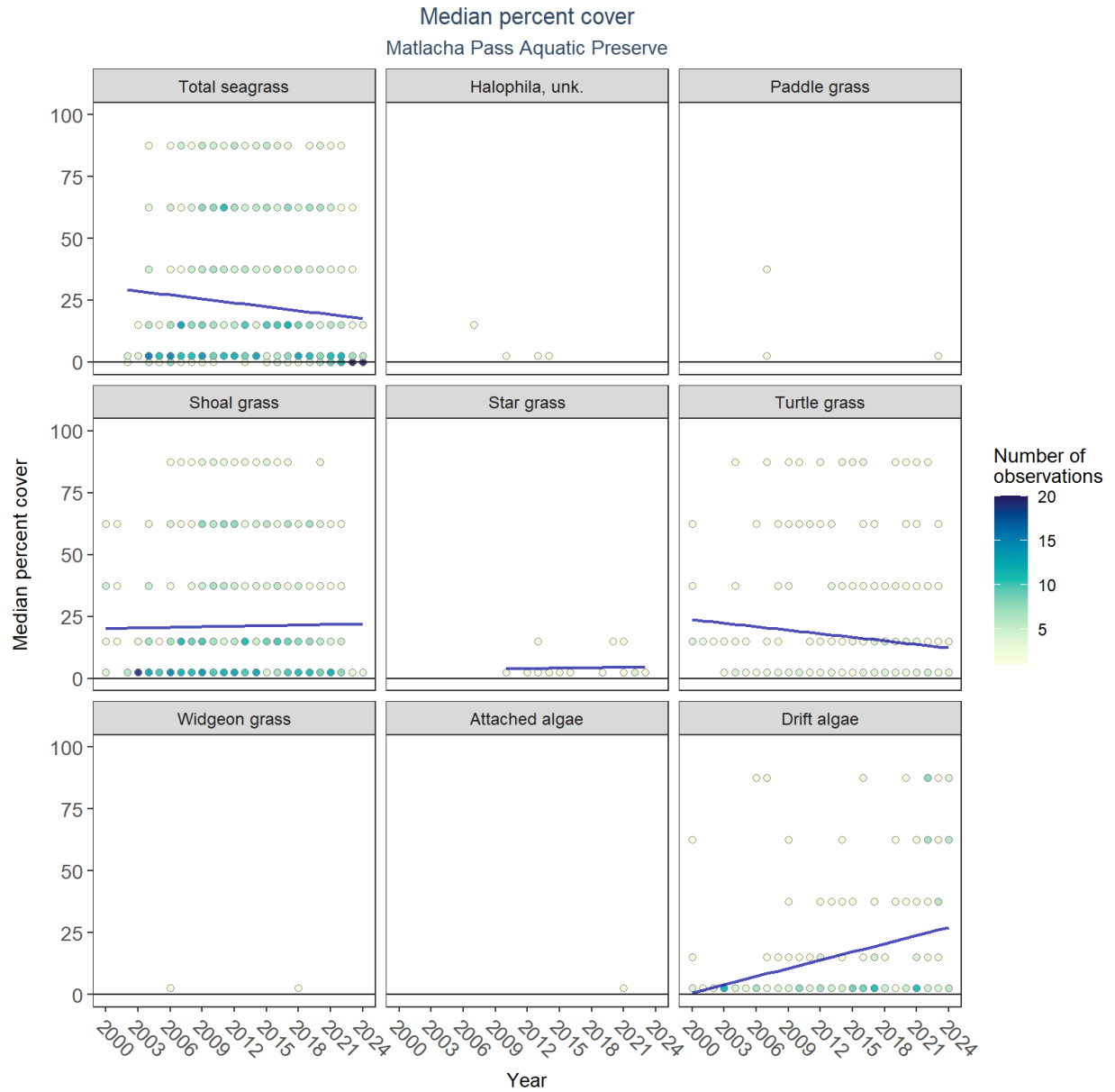
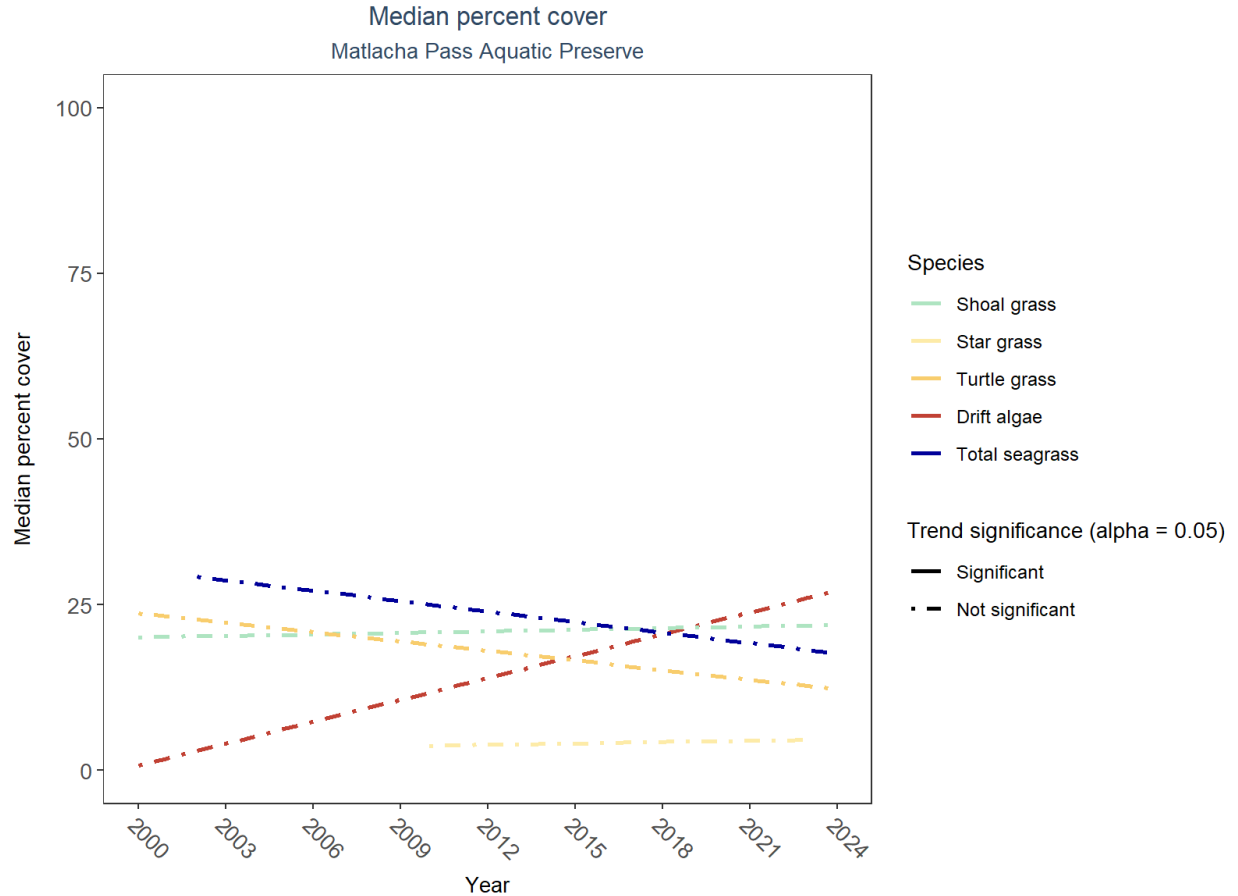


Table 19: SAV LME Results for Matlacha Pass Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	Insufficient data	-	-	-	-
Drift algae	No detectable trend	2000 - 2024	-5.87	1.10	0.10
Shoal grass	No detectable trend	2000 - 2024	19.63	0.08	0.88
Paddle grass	Insufficient data	-	-	-	-
Star grass	No detectable trend	2010 - 2023	2.64	0.07	0.78
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	Insufficient data	-	-	-	-
Turtle grass	No detectable trend	2000 - 2024	26.60	-0.48	0.16
Total seagrass	No detectable trend	2002 - 2024	33.45	-0.53	0.13
Halophila, unk.	Insufficient data	-	-	-	-

Total seagrass, shoal grass, star grass, turtle grass, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for unknown *Halophila*, manatee grass, paddle grass, widgeon grass, and attached algae due to insufficient data.



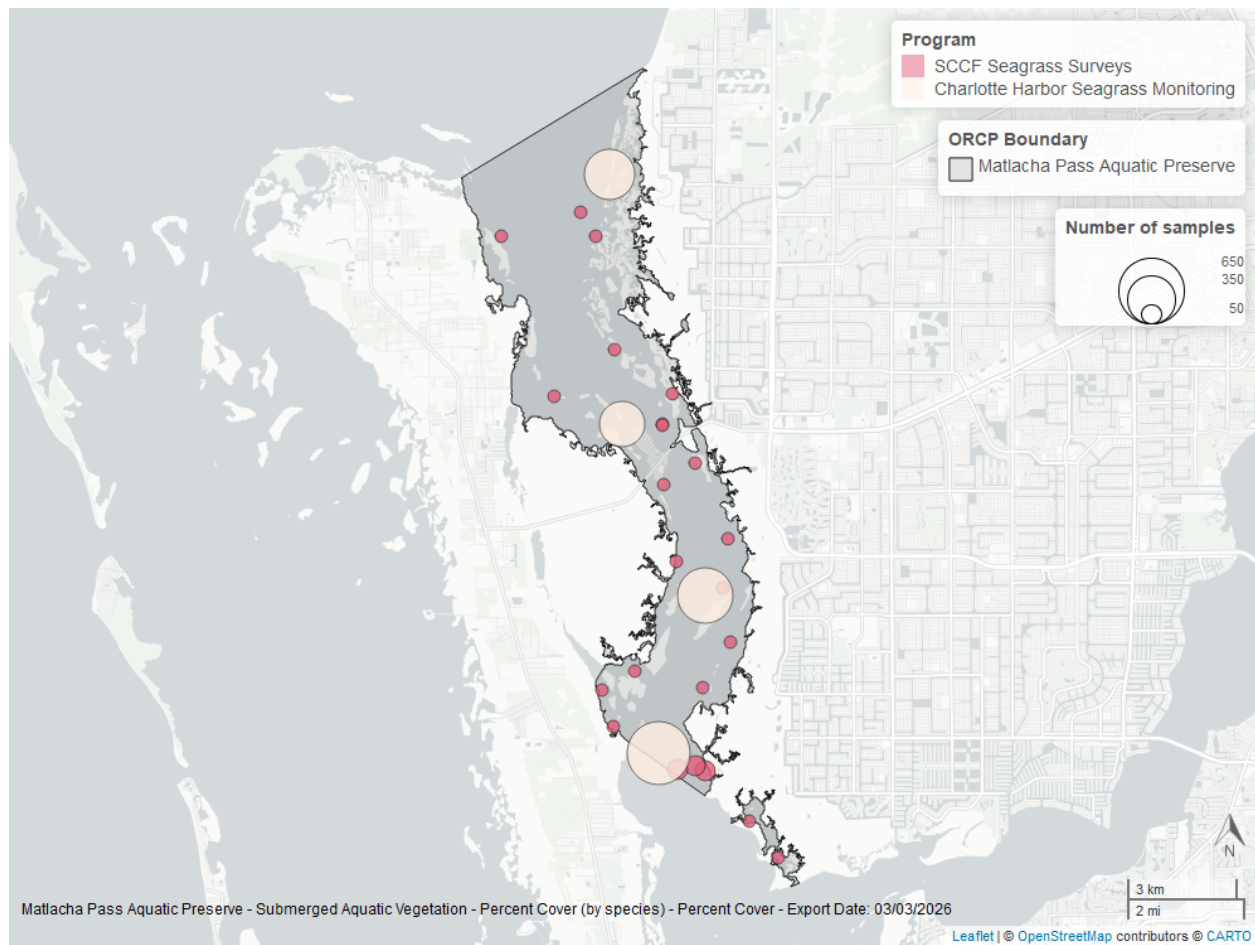
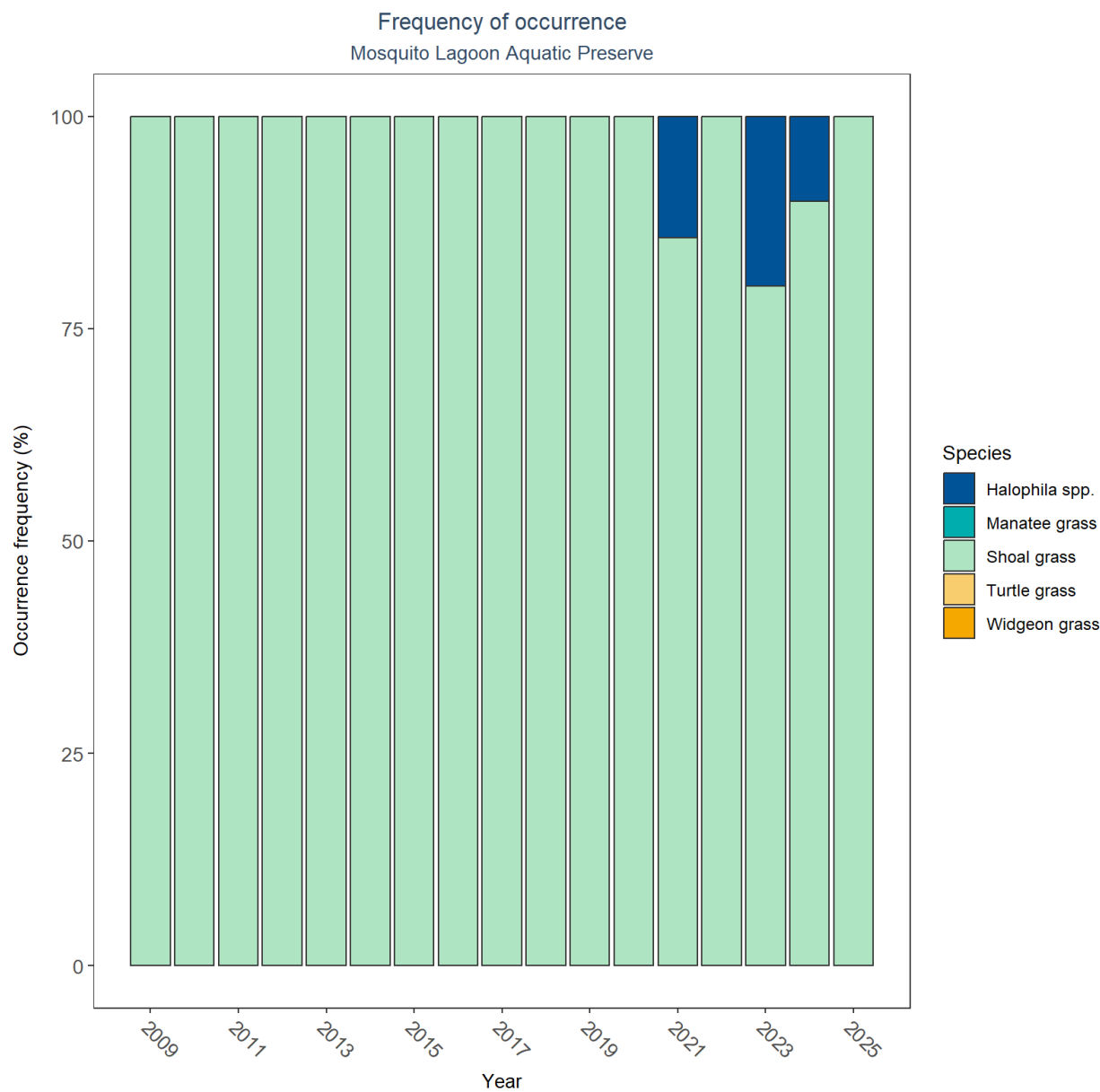


Figure 19: Map showing SAV sampling sites within the boundaries of *Matlacha Pass Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Mosquito Lagoon Aquatic Preserve



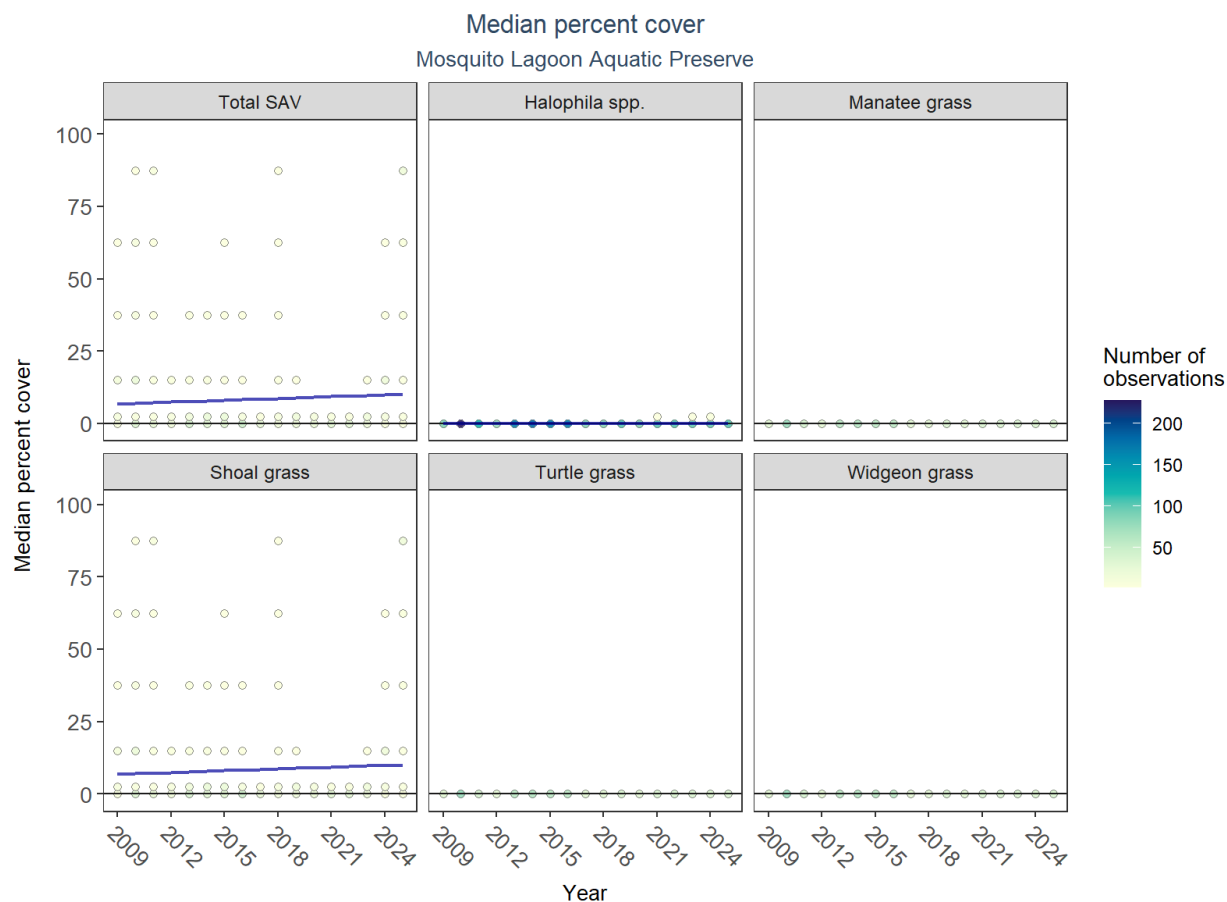
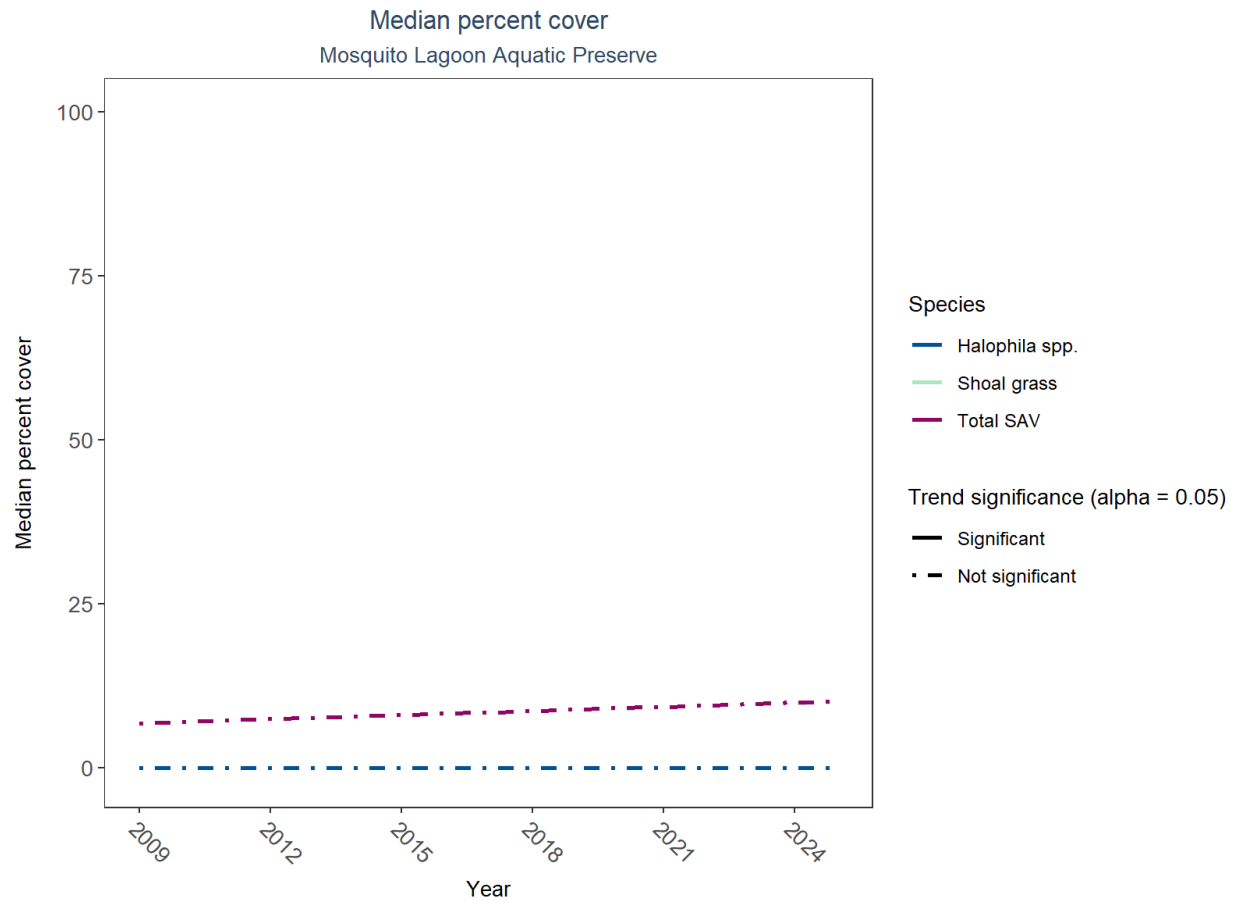


Table 20: SAV LME Results for Mosquito Lagoon Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	No detectable trend	2009 - 2025	3.71	0.21	0.24
Halophila spp.	No detectable trend	2009 - 2025	-0.05	0.00	0.16
Widgeon grass	Model did not fit the available data	-	-	-	-
Manatee grass	Model did not fit the available data	-	-	-	-
Turtle grass	Model did not fit the available data	-	-	-	-
Total SAV	No detectable trend	2009 - 2025	3.67	0.21	0.24
Total seagrass	Insufficient data	-	-	-	-

Total SAV, *Halophila* spp., and shoal grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for total seagrass and drift algae due to insufficient data, and a model could not be fitted for manatee grass, turtle grass, and widgeon grass.



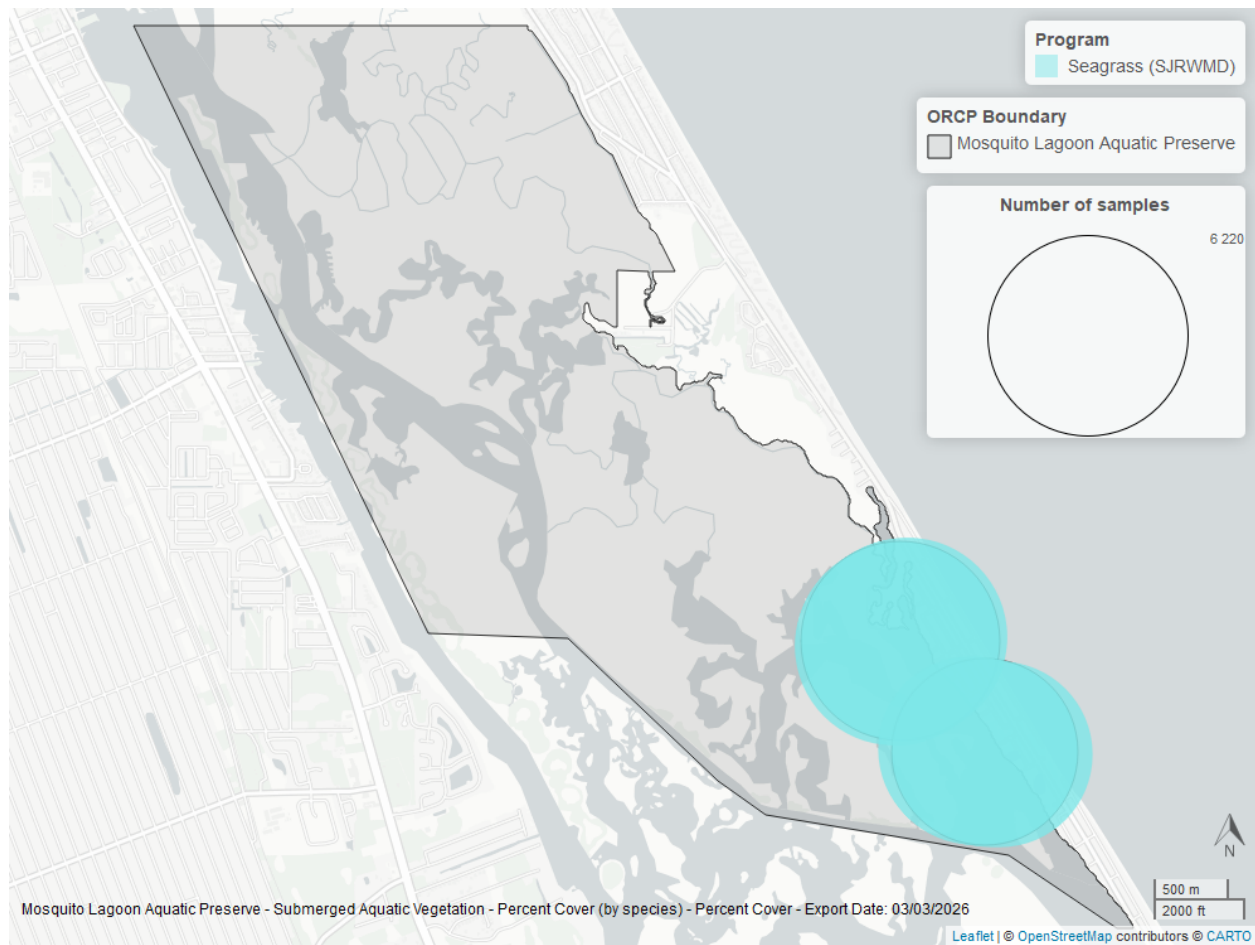
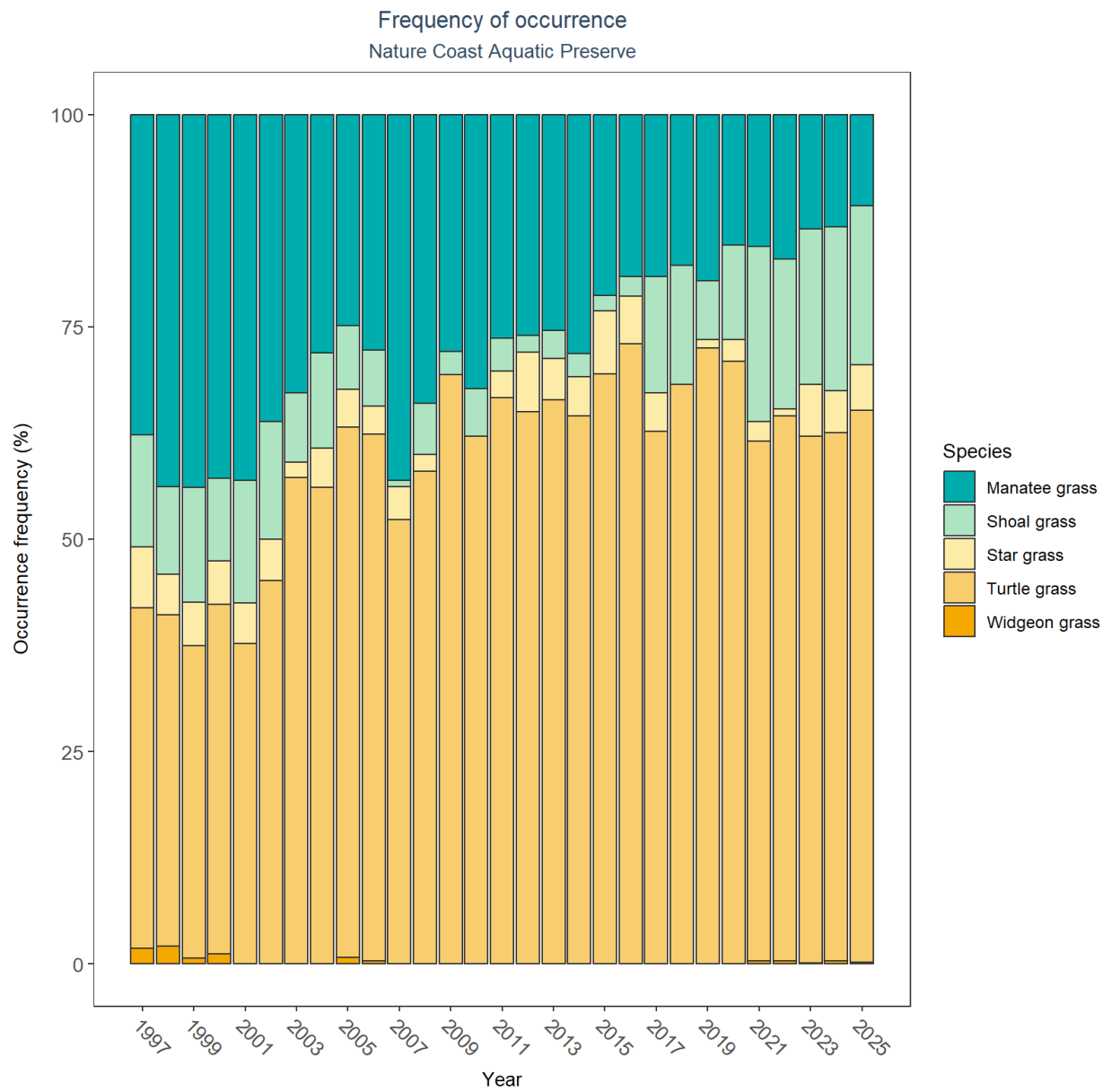


Figure 20: Map showing SAV sampling sites within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Nature Coast Aquatic Preserve



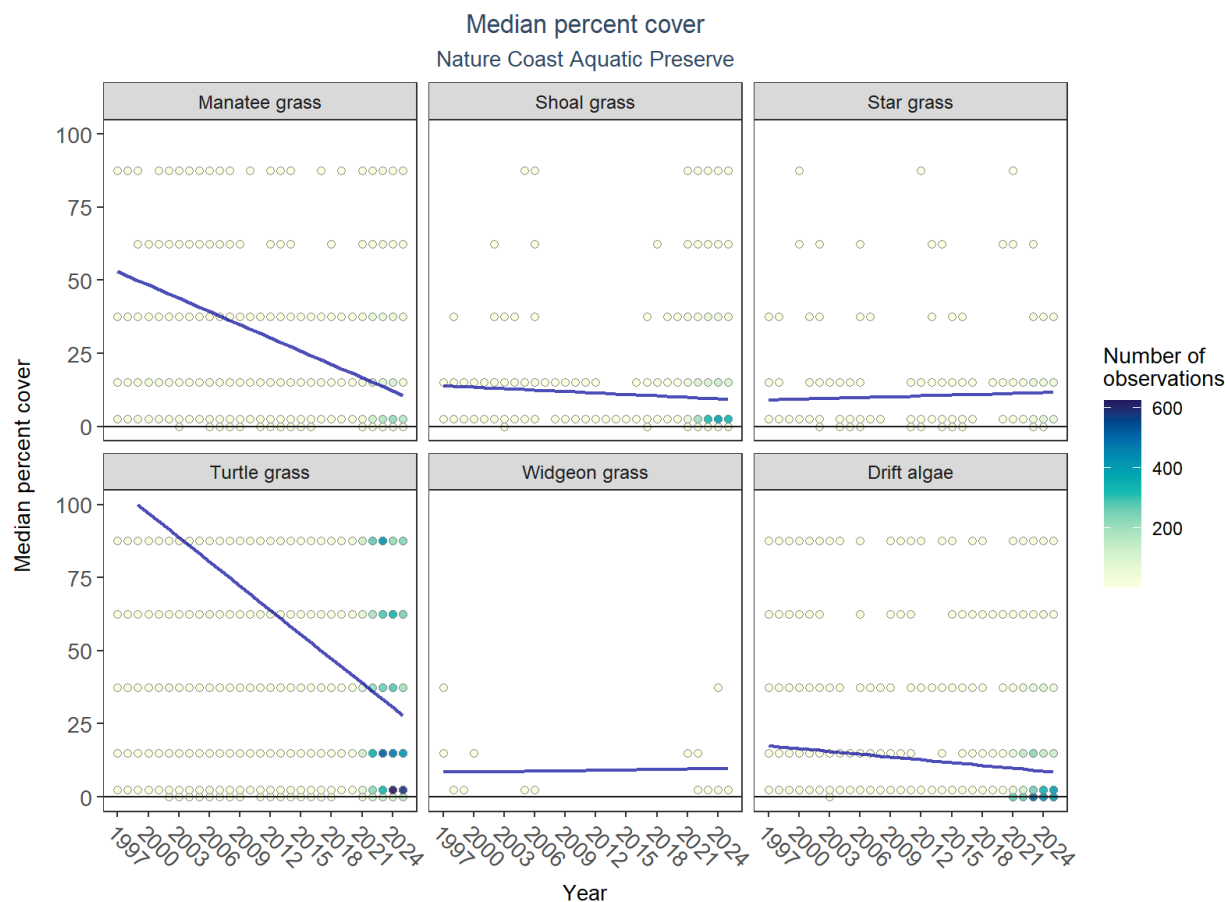
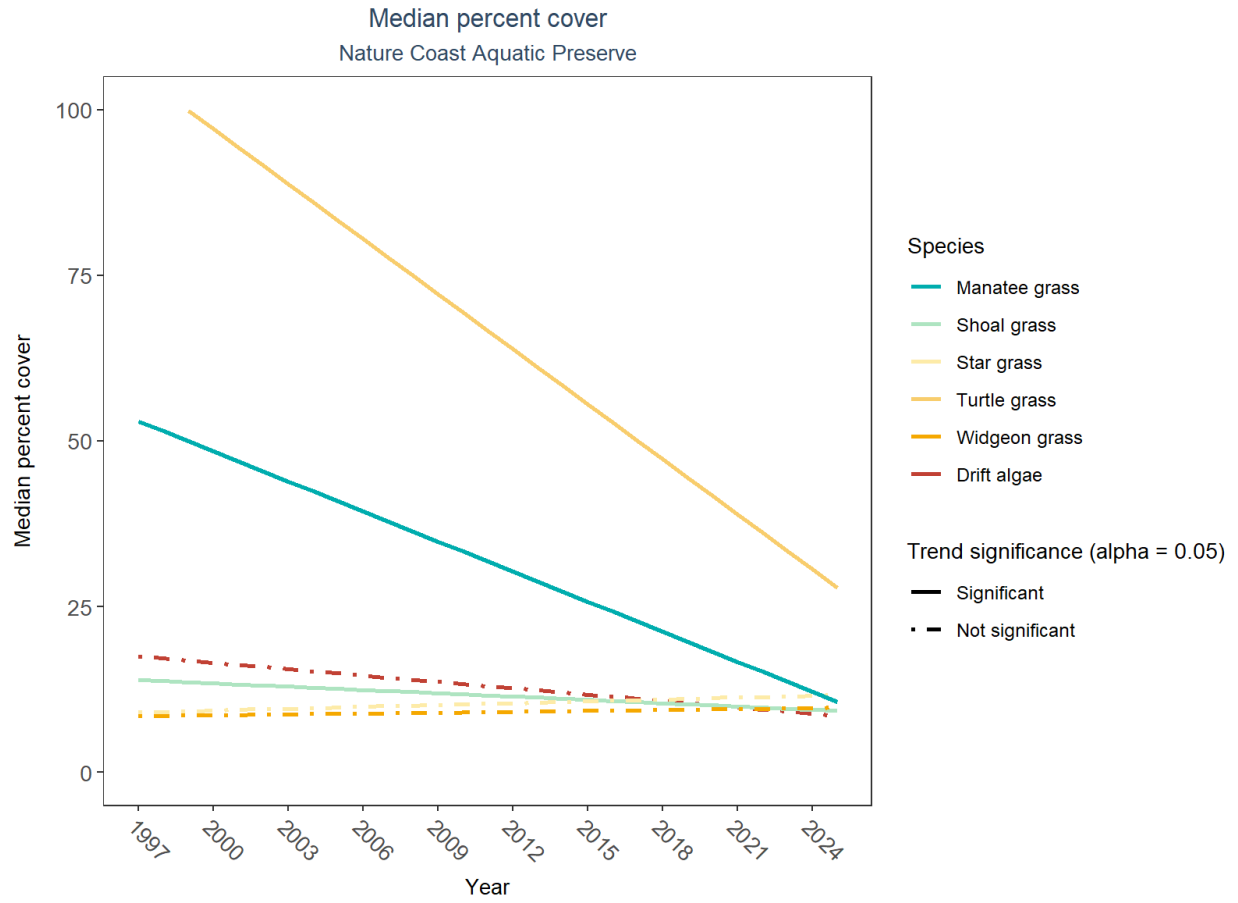


Table 21: SAV LME Results for Nature Coast Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	1997 - 2025	18.46	-0.32	0.43
Shoal grass	Decreasing trend	1997 - 2025	14.44	-0.17	0.04
Star grass	No detectable trend	1997 - 2025	8.79	0.09	0.39
Widgeon grass	No detectable trend	1997 - 2025	8.39	0.04	0.97
Manatee grass	Decreasing trend	1997 - 2025	57.52	-1.51	0.00
Turtle grass	Decreasing trend	1997 - 2025	113.70	-2.77	0.00

Annual decreases in percent cover were observed for manatee grass (-1.51%), shoal grass (-0.17%), and turtle grass (-2.77%). Star grass, widgeon grass, and drift algae showed no detectable change in percent cover.



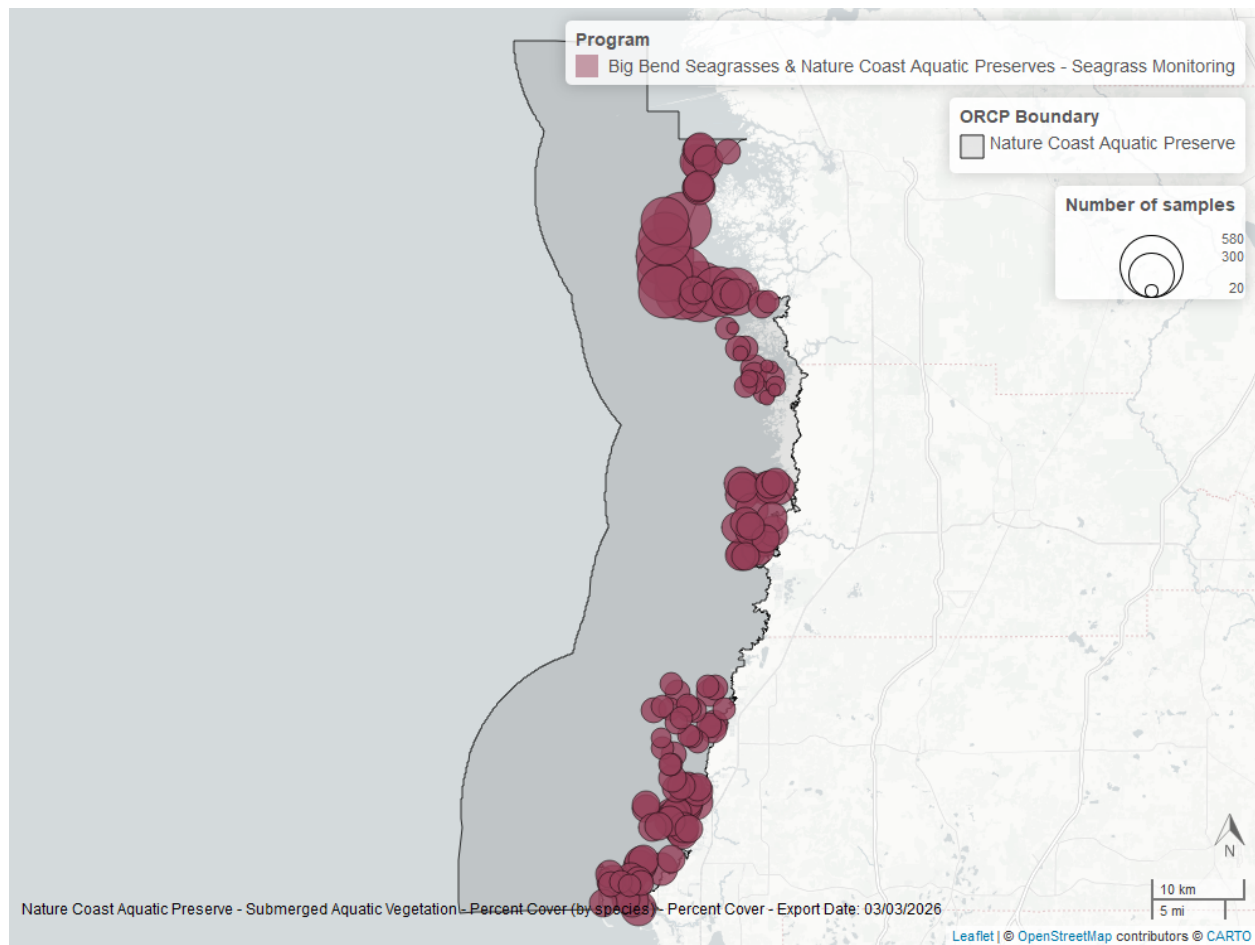
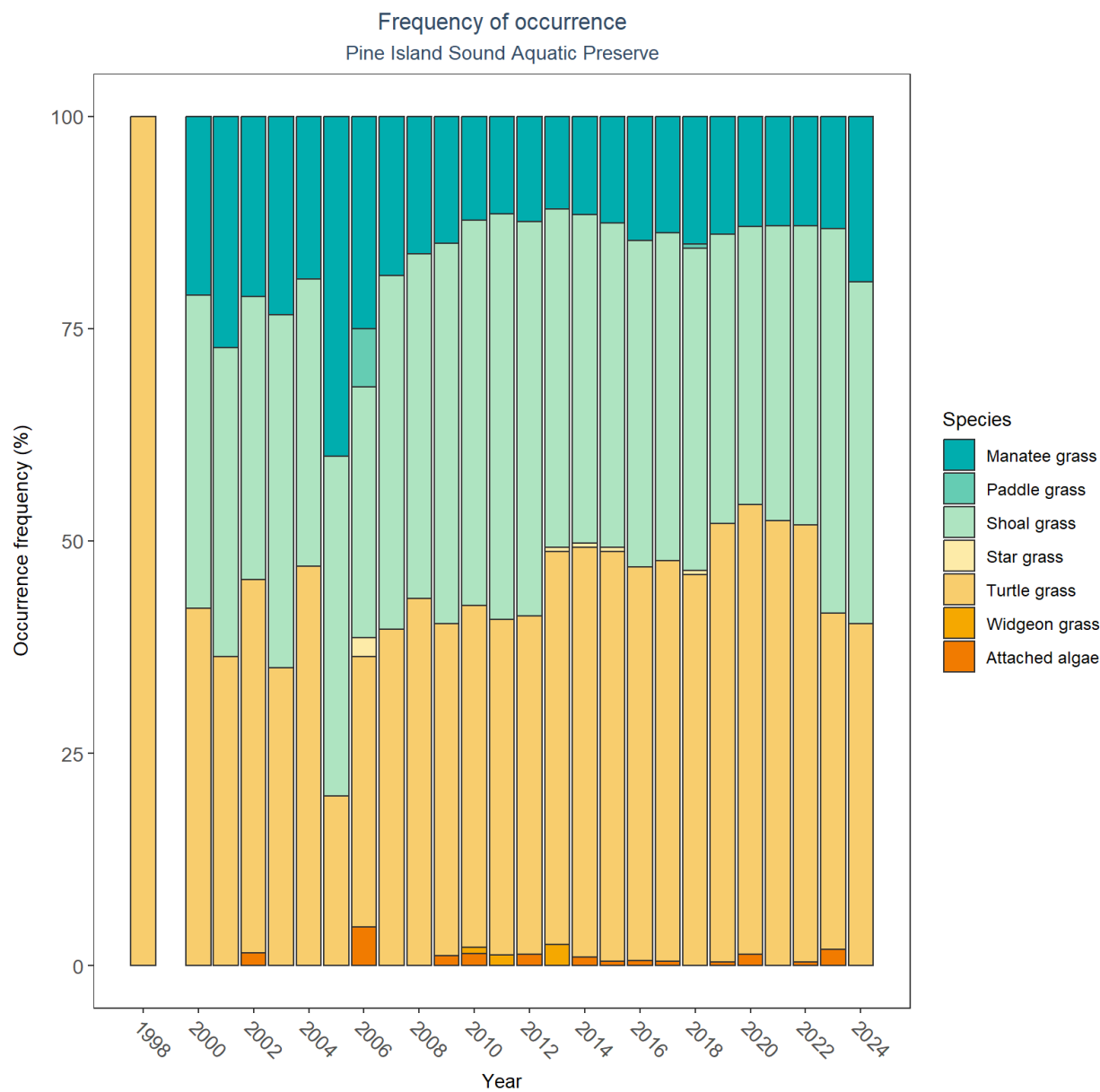


Figure 21: Map showing SAV sampling sites within the boundaries of *Nature Coast Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Pine Island Sound Aquatic Preserve



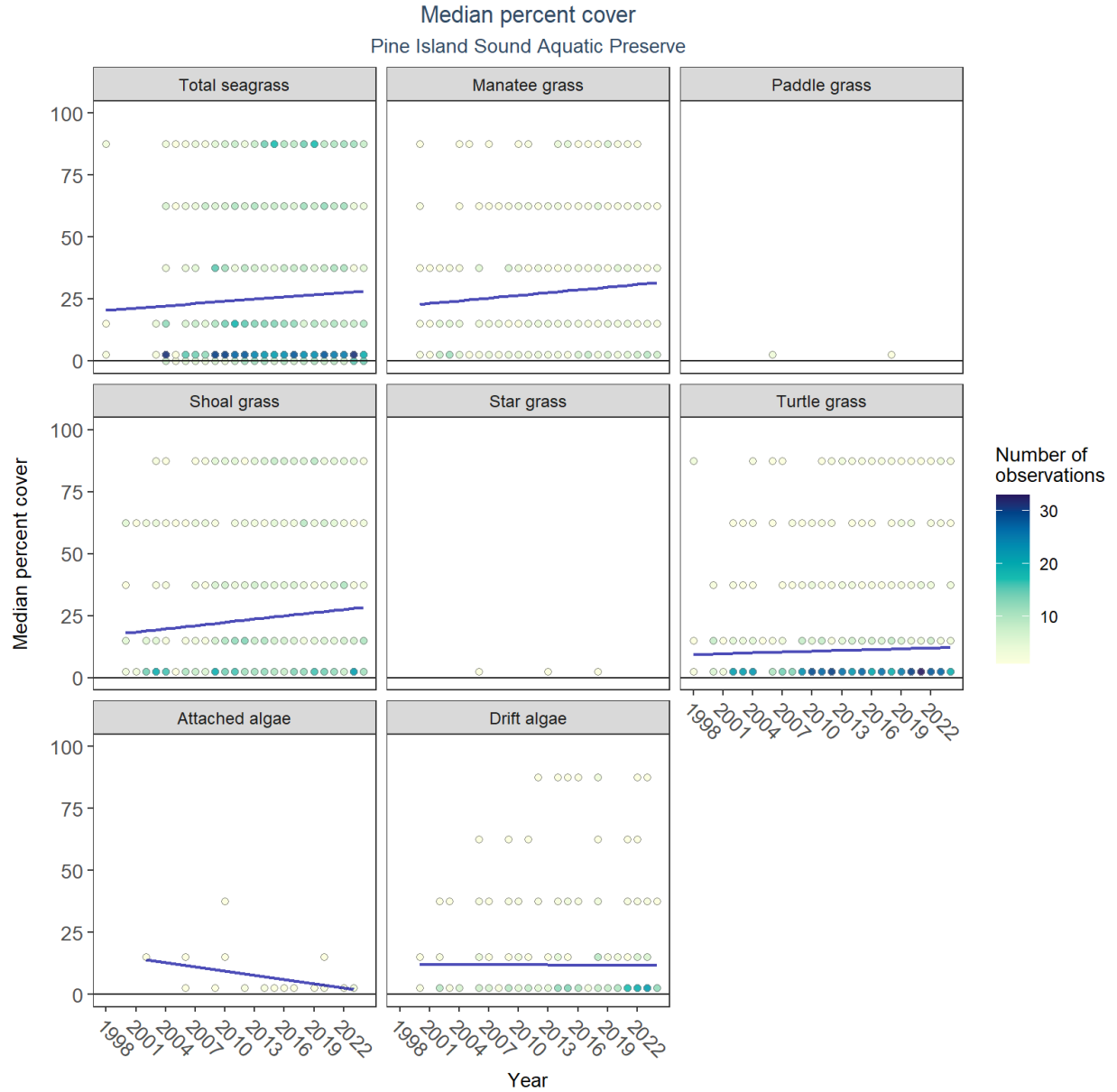
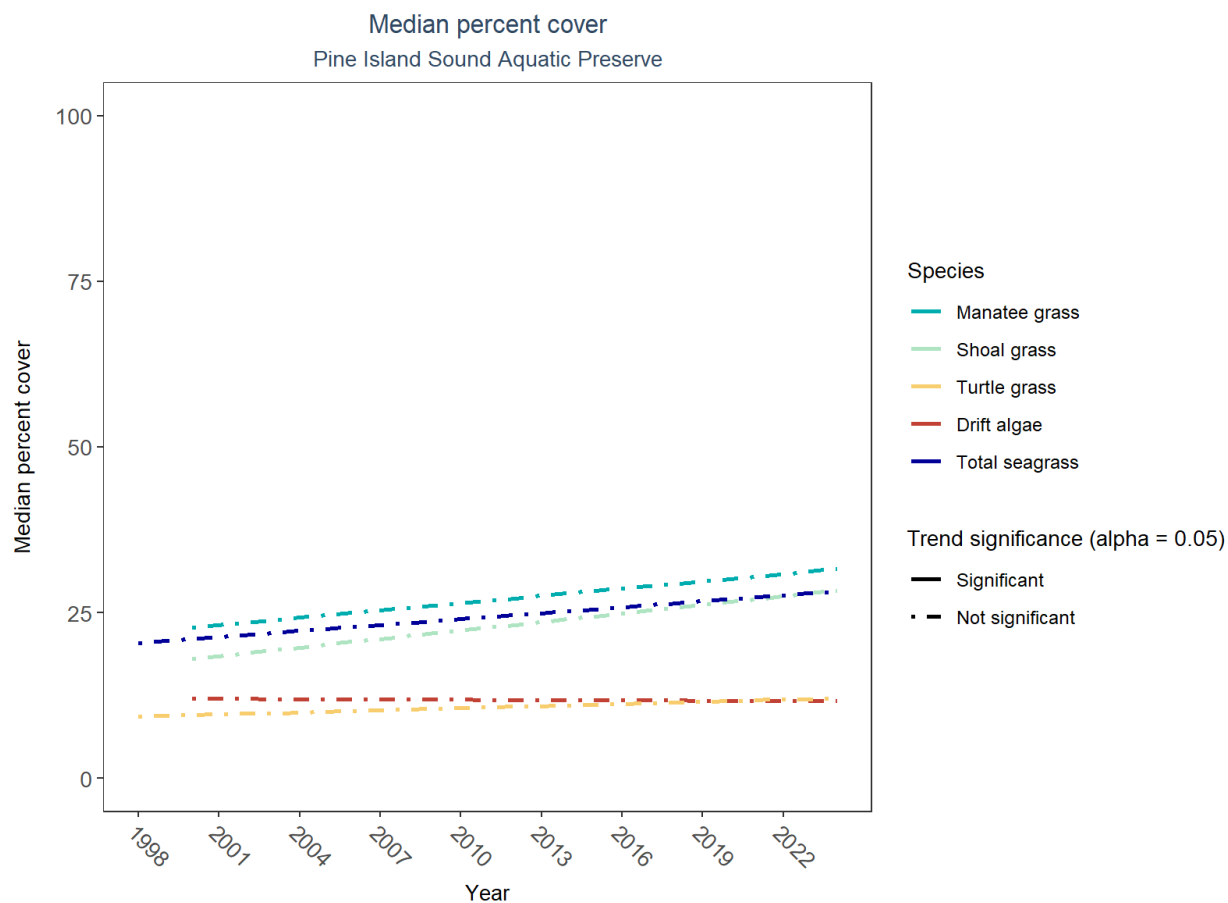


Table 22: SAV LME Results for Pine Island Sound Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2002 - 2023	18.49	-0.57	0.07
Drift algae	No detectable trend	2000 - 2024	12.13	-0.01	0.93
Shoal grass	No detectable trend	2000 - 2024	15.44	0.43	0.22
Paddle grass	Insufficient data	-	-	-	-
Star grass	Insufficient data	-	-	-	-
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	No detectable trend	2000 - 2024	20.59	0.37	0.28
Turtle grass	No detectable trend	1998 - 2024	8.91	0.11	0.46
Total seagrass	No detectable trend	1998 - 2024	19.27	0.30	0.33

Total seagrass, manatee grass, shoal grass, turtle grass, attached algae, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for paddle grass, star grass, and widgeon grass due to insufficient data.



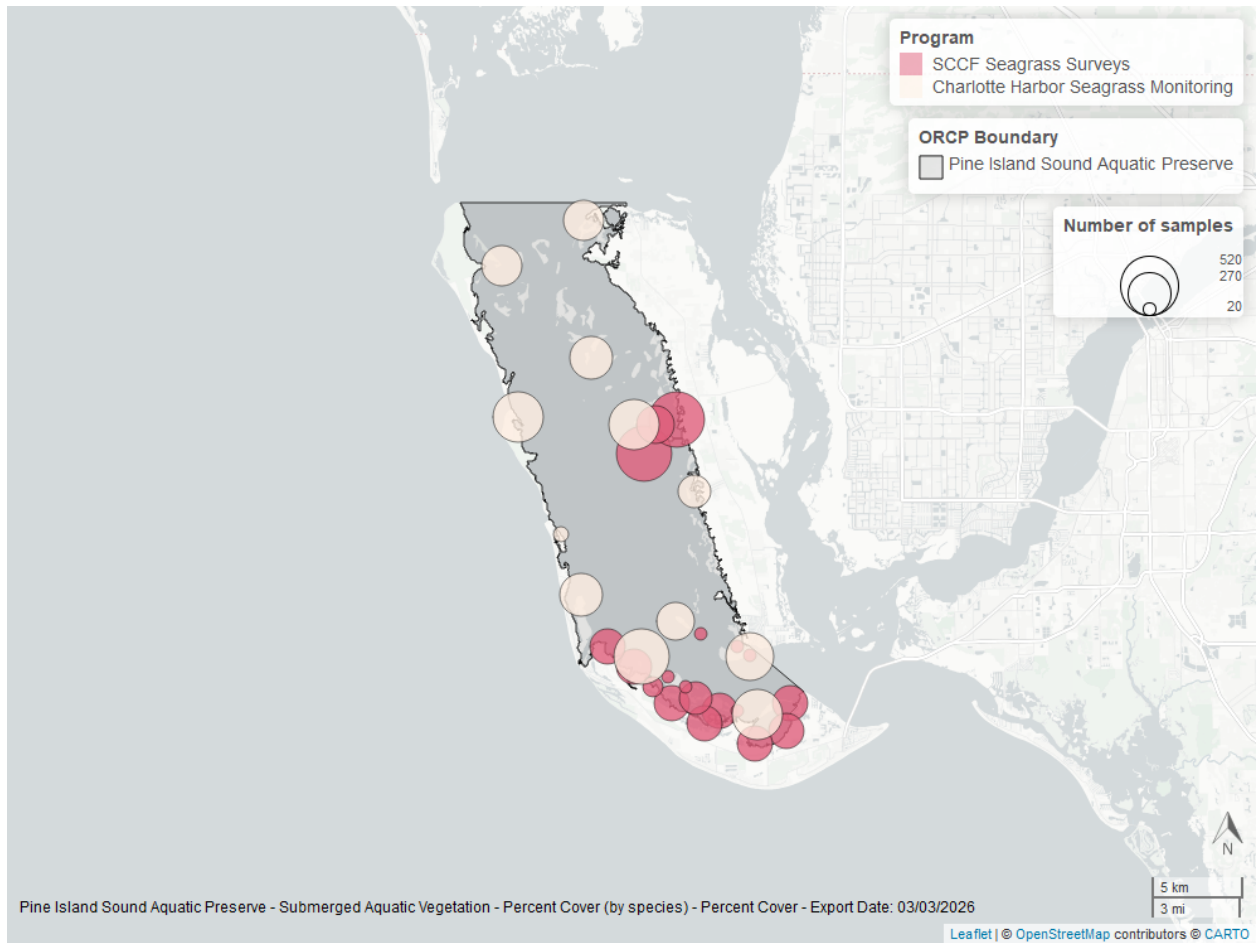
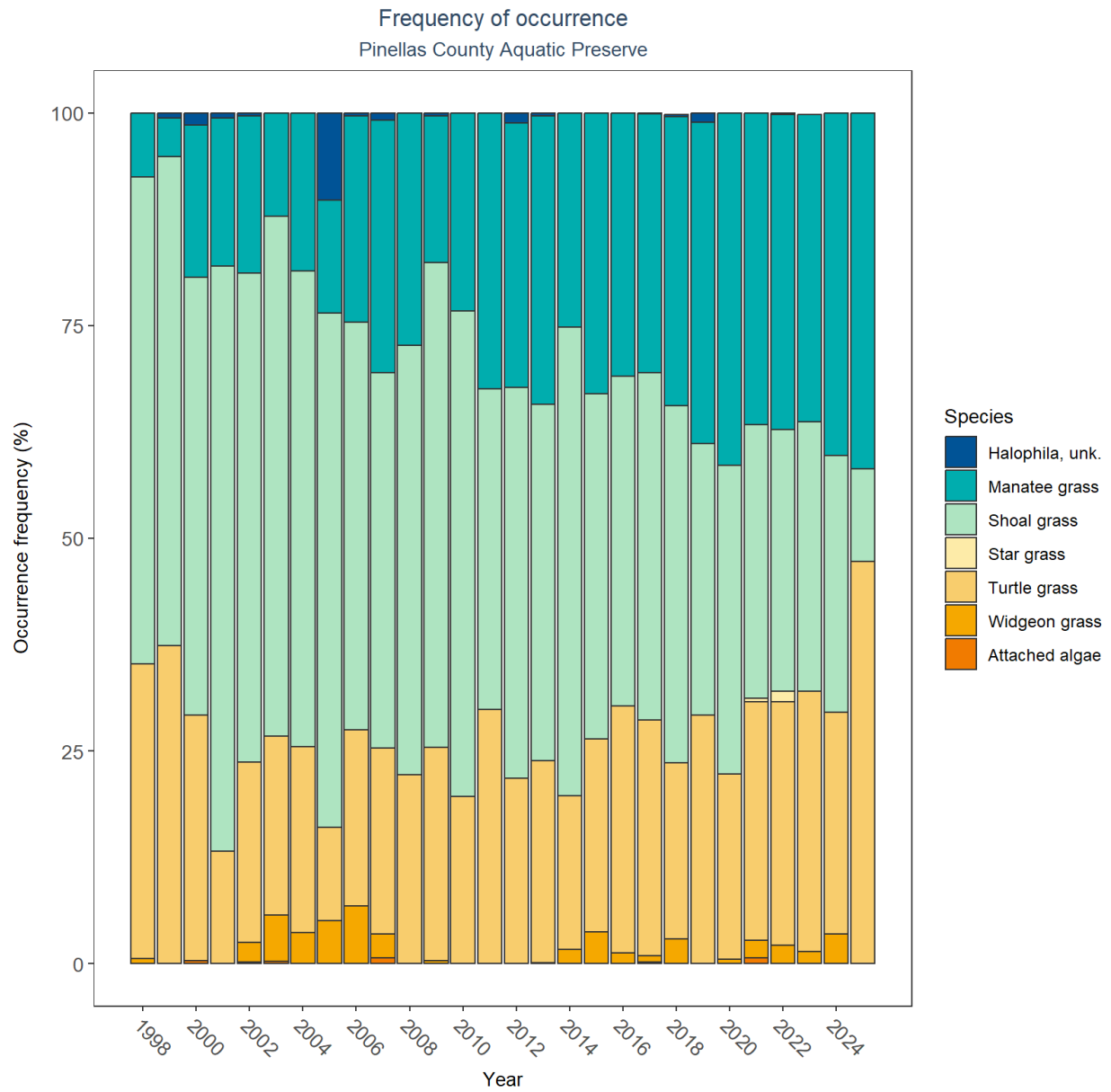


Figure 22: Map showing SAV sampling sites within the boundaries of *Pine Island Sound Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Pinellas County Aquatic Preserve



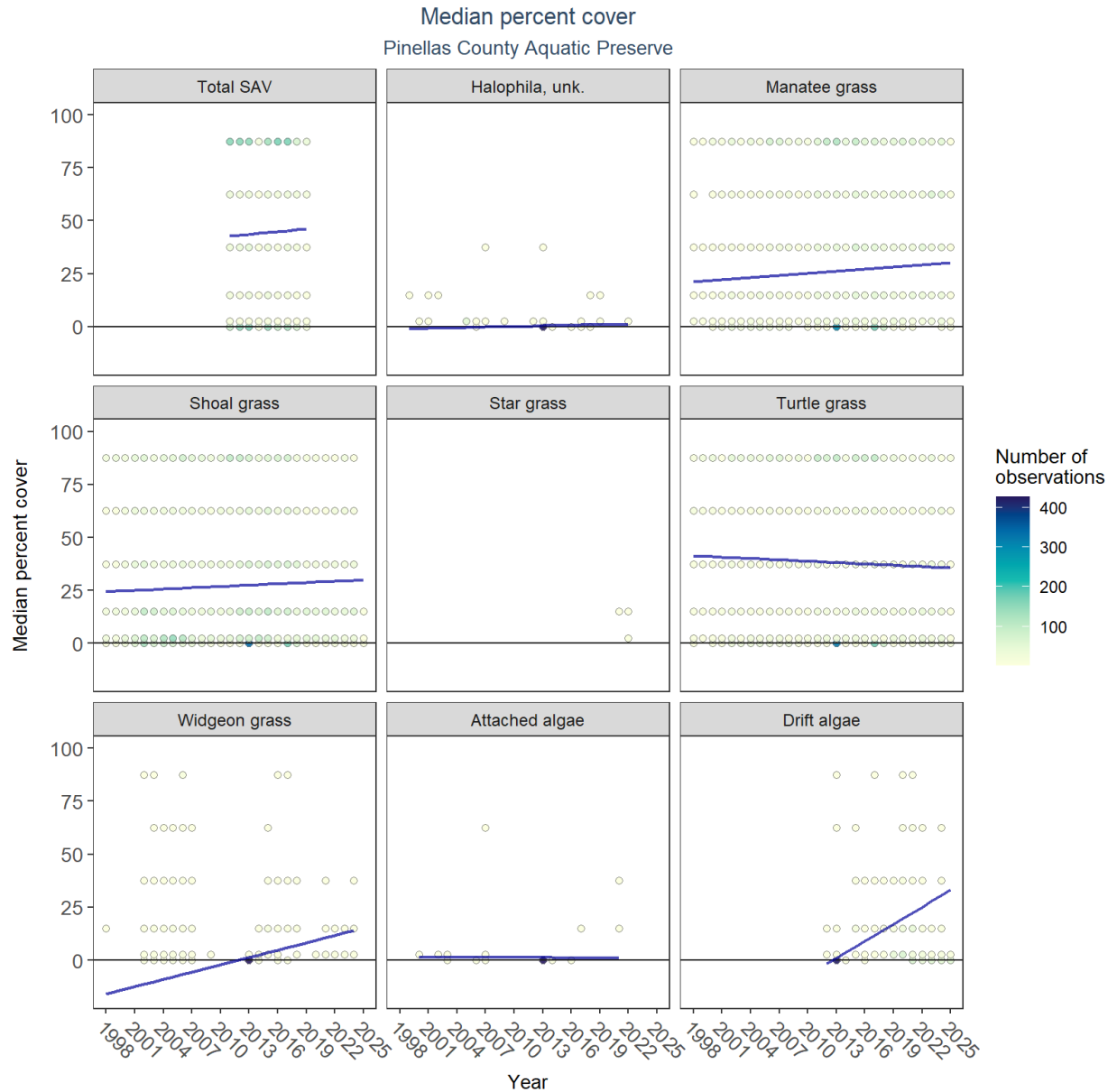
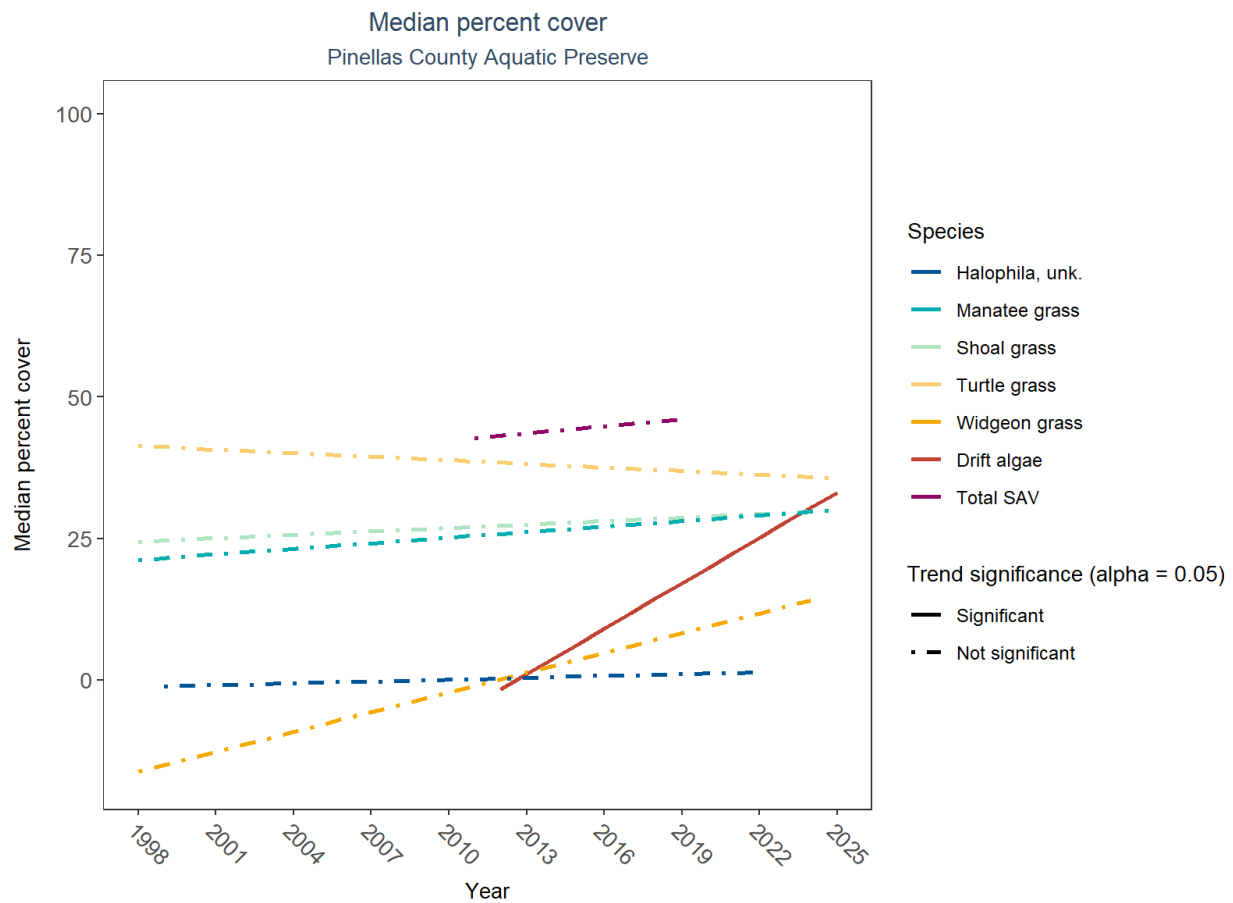


Table 23: SAV LME Results for Pinellas County Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2000 - 2021	1.35	-0.01	0.96
Drift algae	Increasing trend	2012 - 2025	-49.76	2.67	0.00
Shoal grass	No detectable trend	1998 - 2025	23.58	0.21	0.15
Star grass	Insufficient data	-	-	-	-
Widgeon grass	No detectable trend	1998 - 2024	-20.72	1.16	0.37
Manatee grass	No detectable trend	1998 - 2025	19.93	0.33	0.11
Turtle grass	No detectable trend	1998 - 2025	42.22	-0.21	0.20
Total SAV	No detectable trend	2011 - 2019	35.61	0.42	0.57
Halophila, unk.	No detectable trend	1999 - 2022	-1.64	0.11	0.66

An annual increase in percent cover was observed for drift algae (2.67%). Total SAV, unknown *Halophila*, manatee grass, shoal grass, turtle grass, widgeon grass, and attached algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for star grass due to insufficient data.



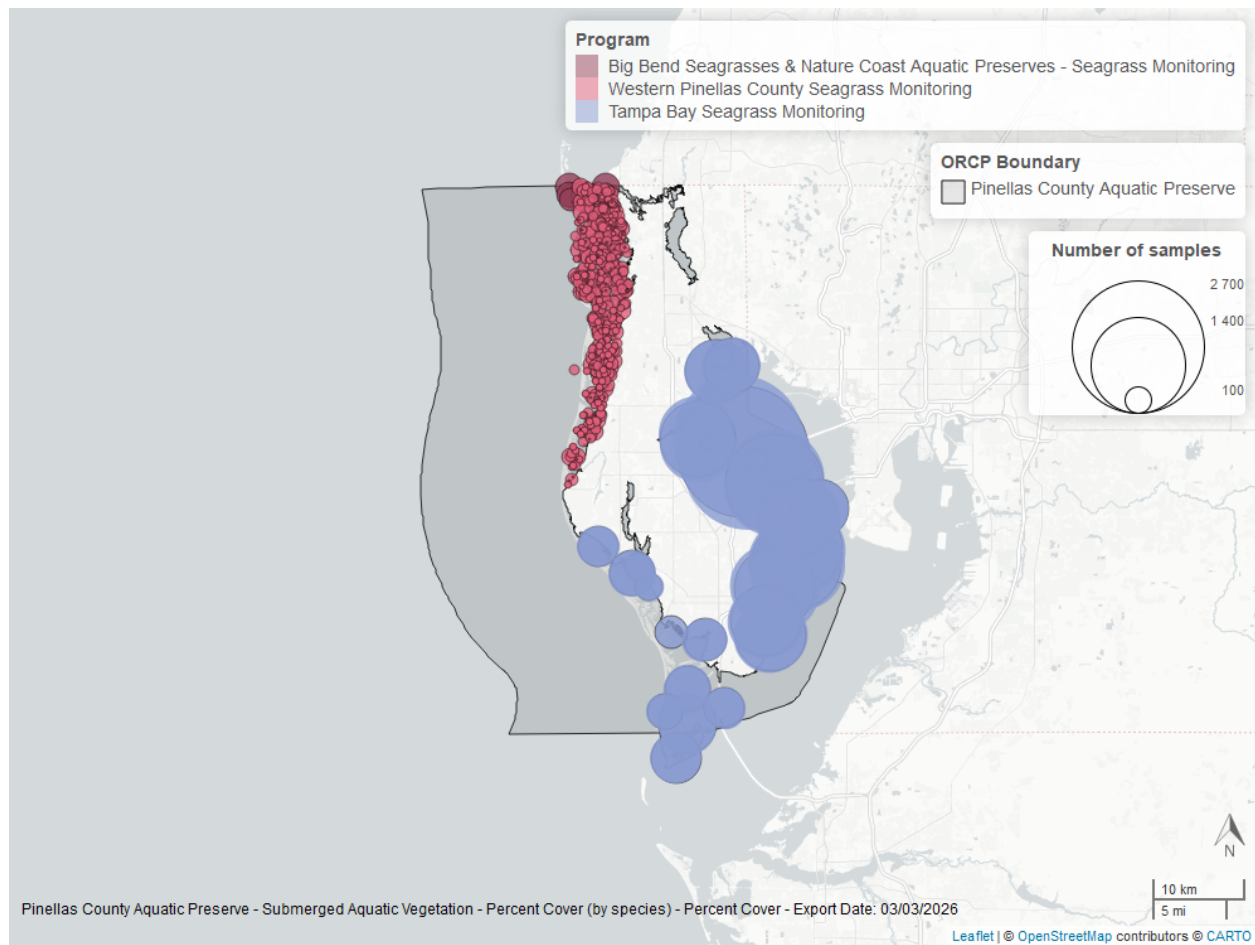
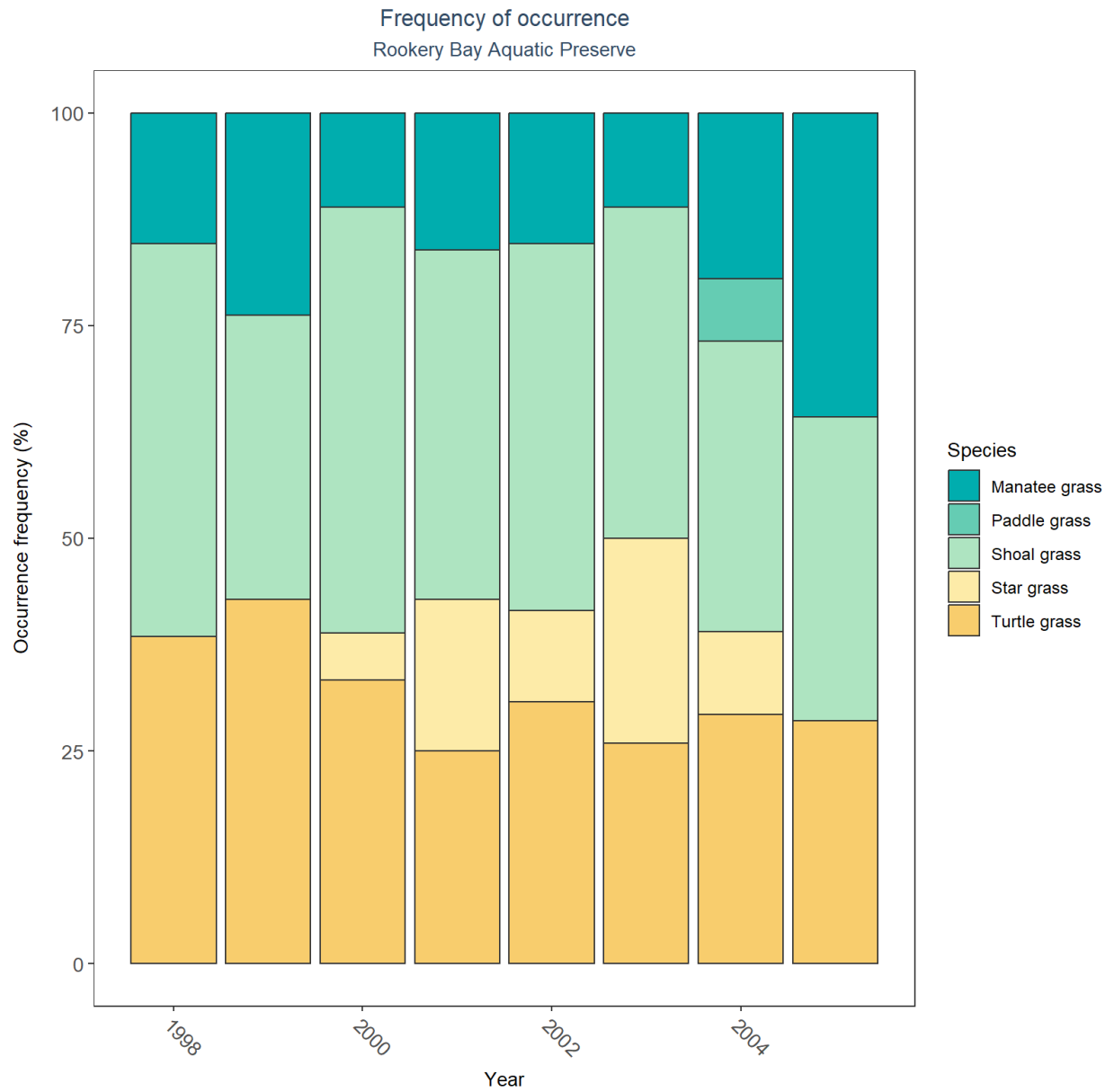


Figure 23: Map showing SAV sampling sites within the boundaries of *Pinellas County Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Rookery Bay Aquatic Preserve



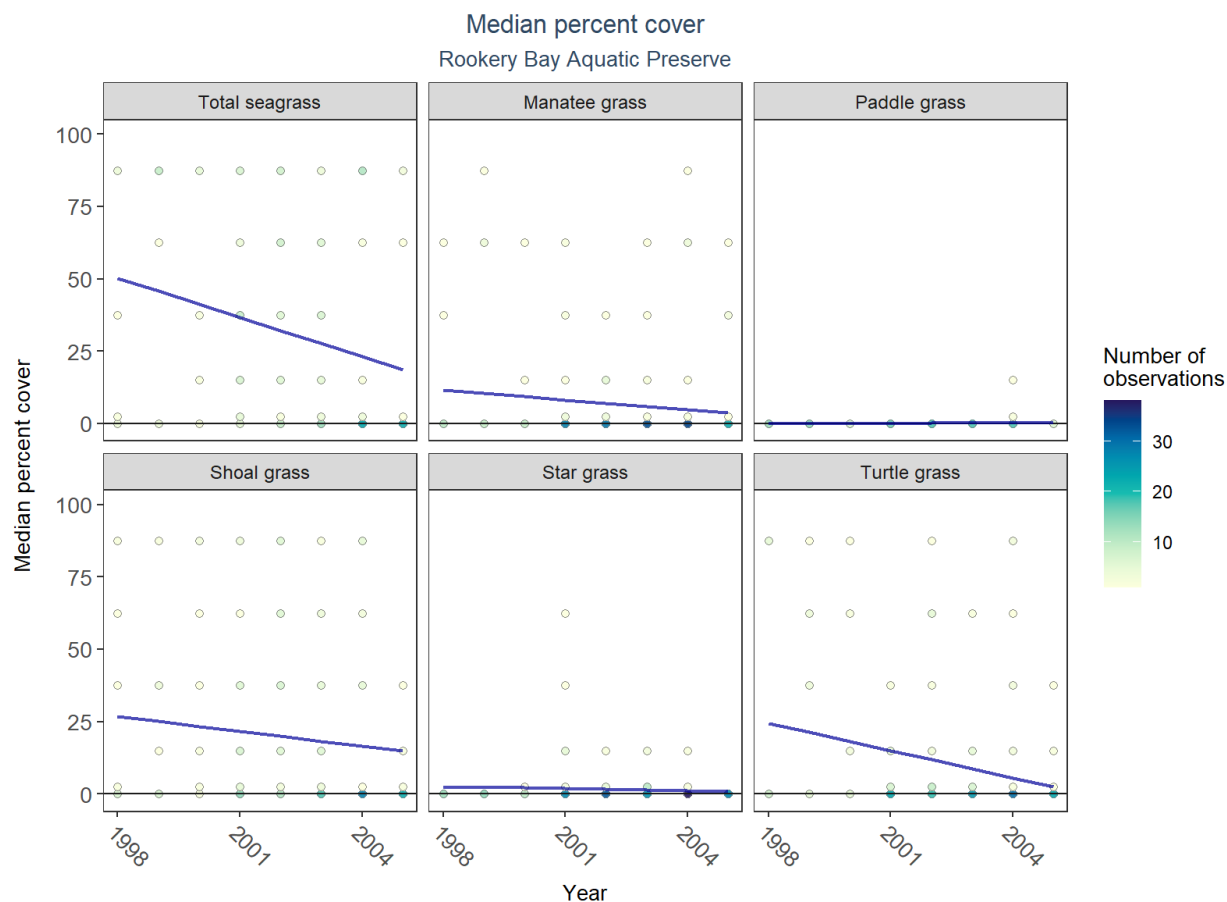
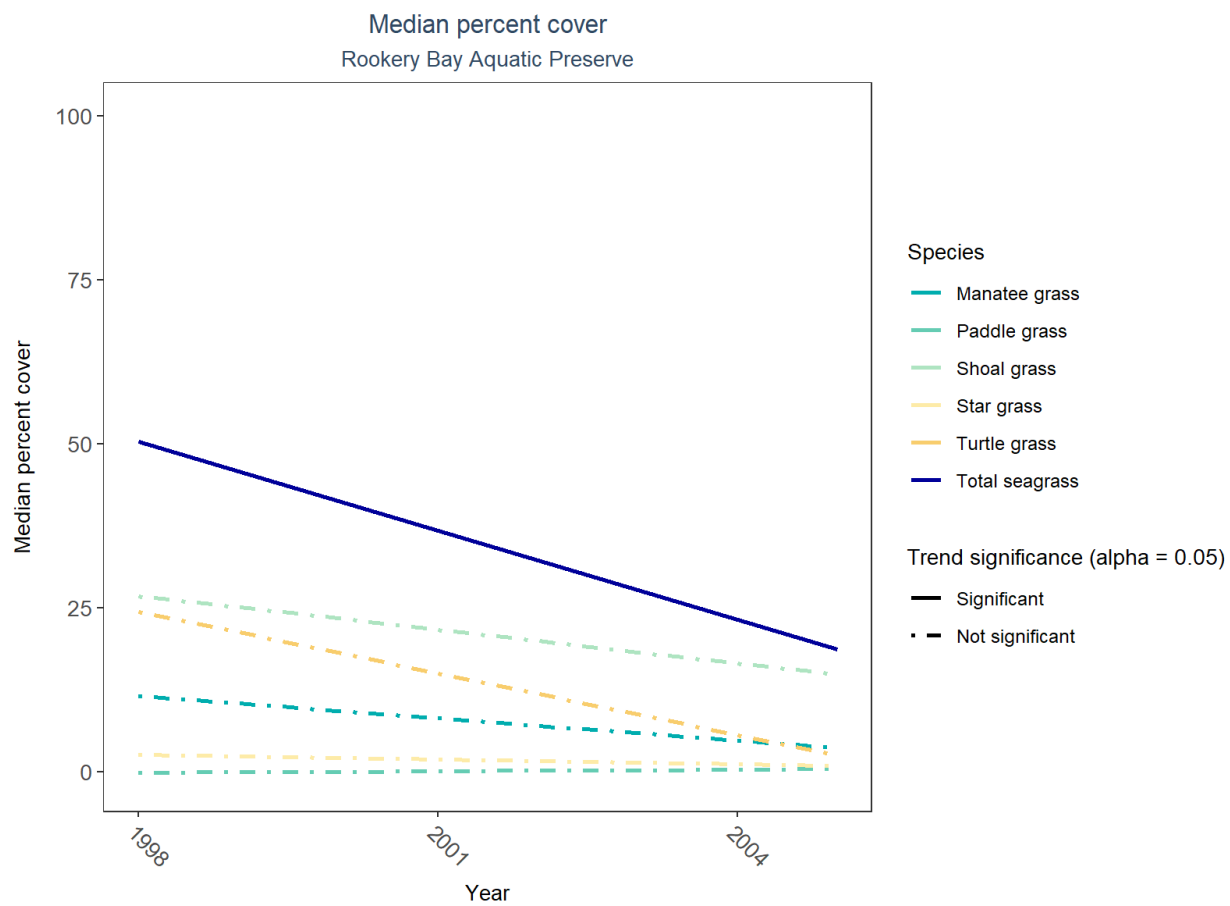


Table 24: SAV LME Results for Rookery Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	1998 - 2005	33.63	-1.71	0.25
Paddle grass	No detectable trend	1998 - 2005	-0.44	0.08	0.36
Star grass	No detectable trend	1998 - 2005	3.53	-0.23	0.44
Manatee grass	No detectable trend	1998 - 2005	16.10	-1.13	0.40
Turtle grass	No detectable trend	1998 - 2005	36.99	-3.14	0.07
Total seagrass	Decreasing trend	1998 - 2005	68.37	-4.52	0.02

An annual decrease in percent cover was observed for total seagrass (-4.52%). Manatee grass, paddle grass, shoal grass, star grass, and turtle grass showed no detectable change in percent cover.



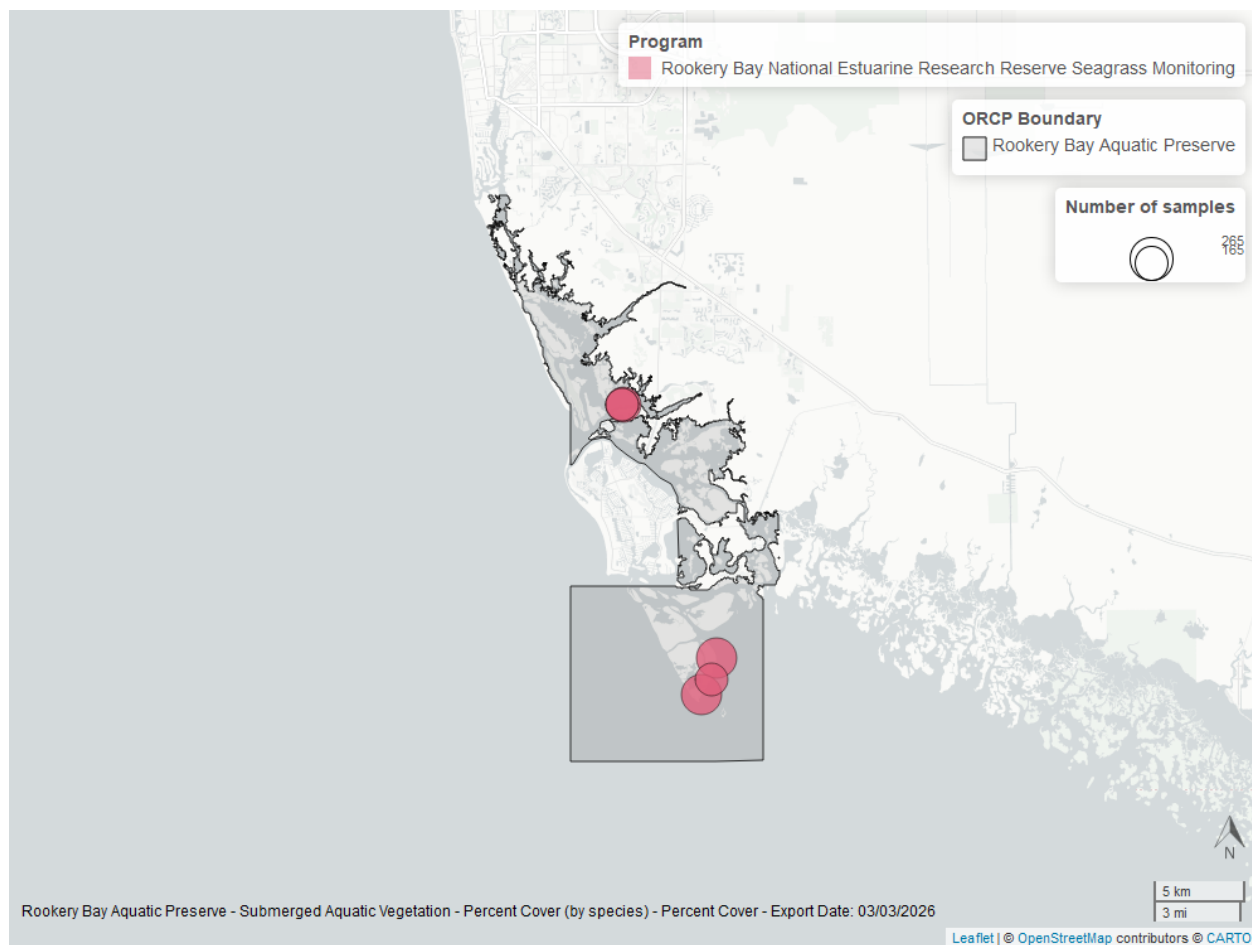
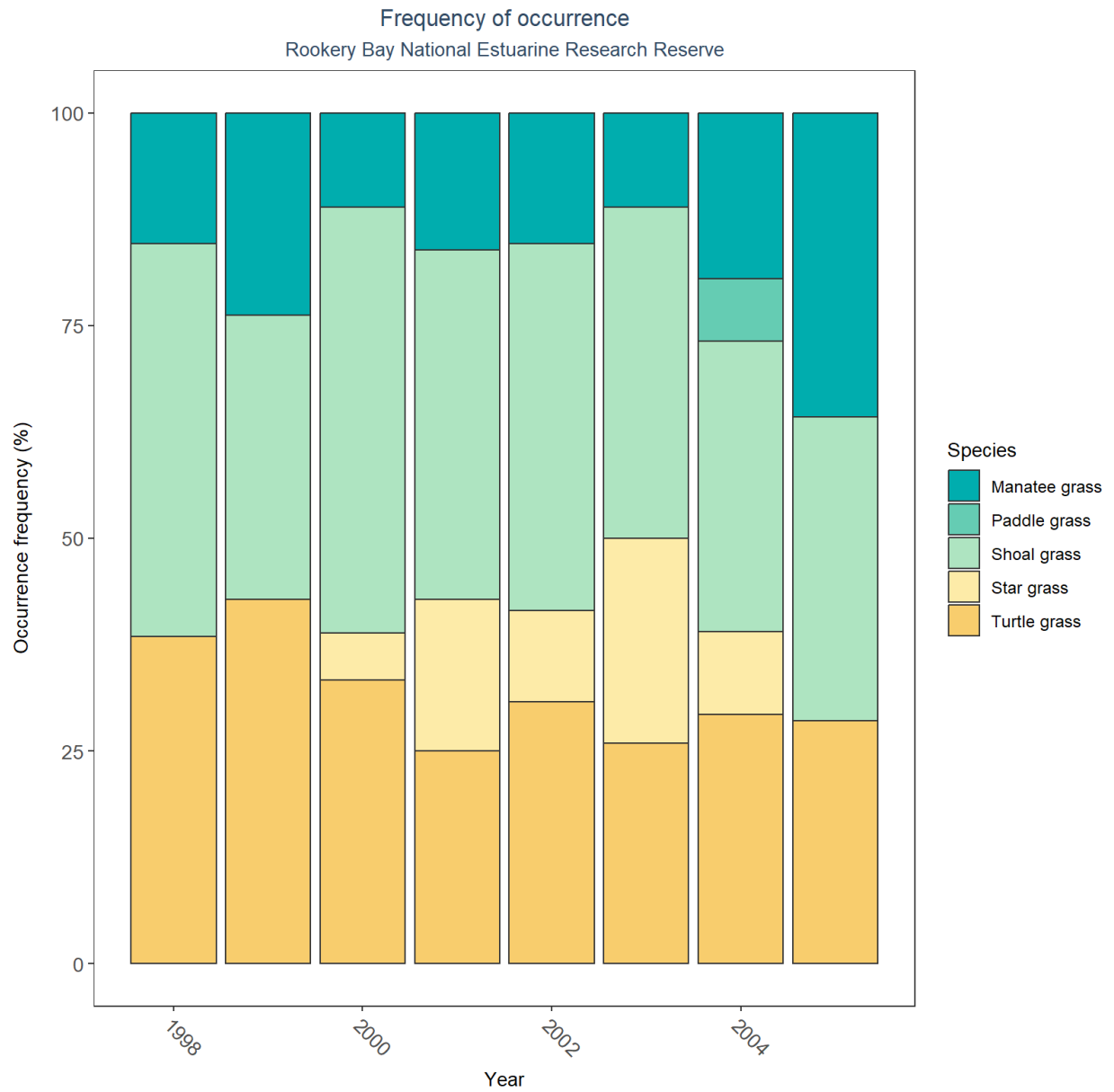


Figure 24: Map showing SAV sampling sites within the boundaries of *Rookery Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Rookery Bay National Estuarine Research Reserve



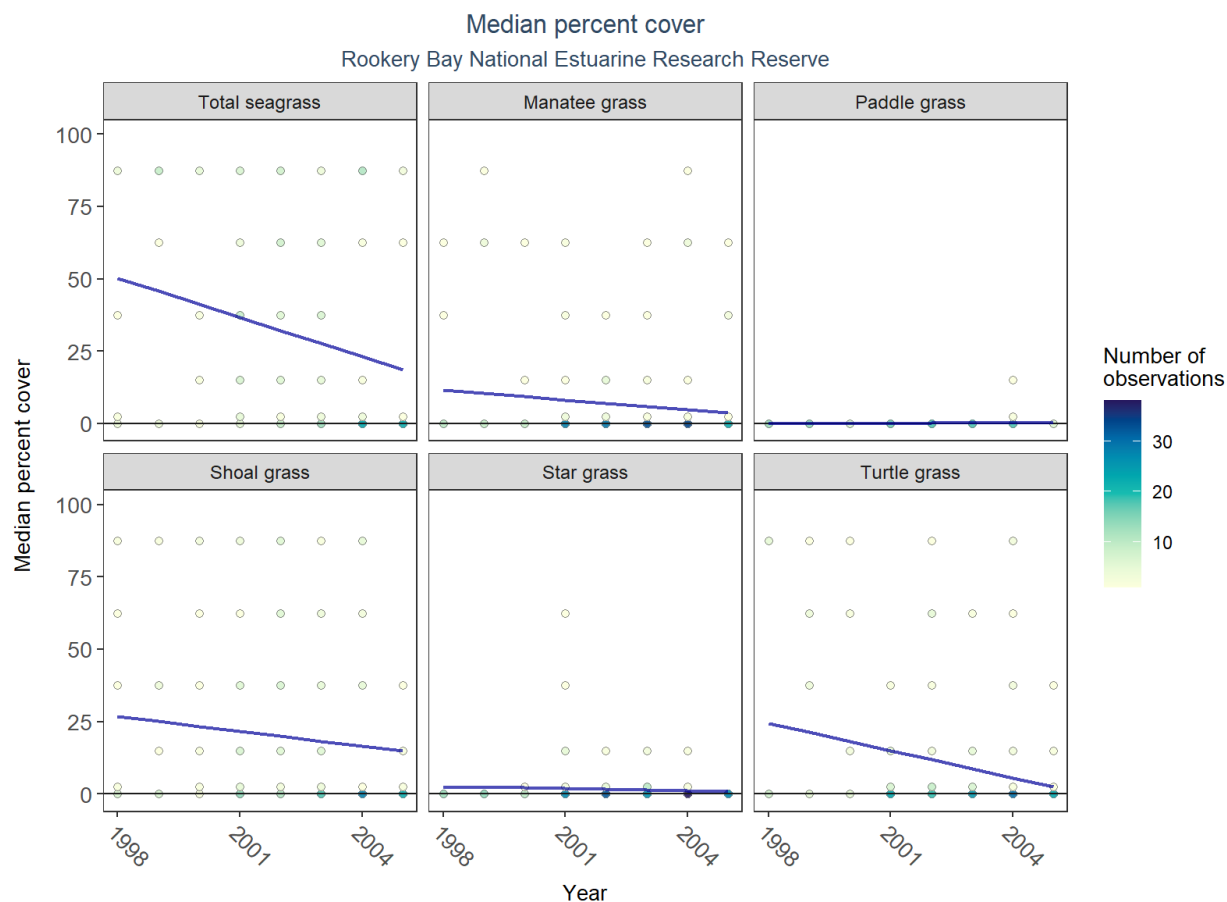
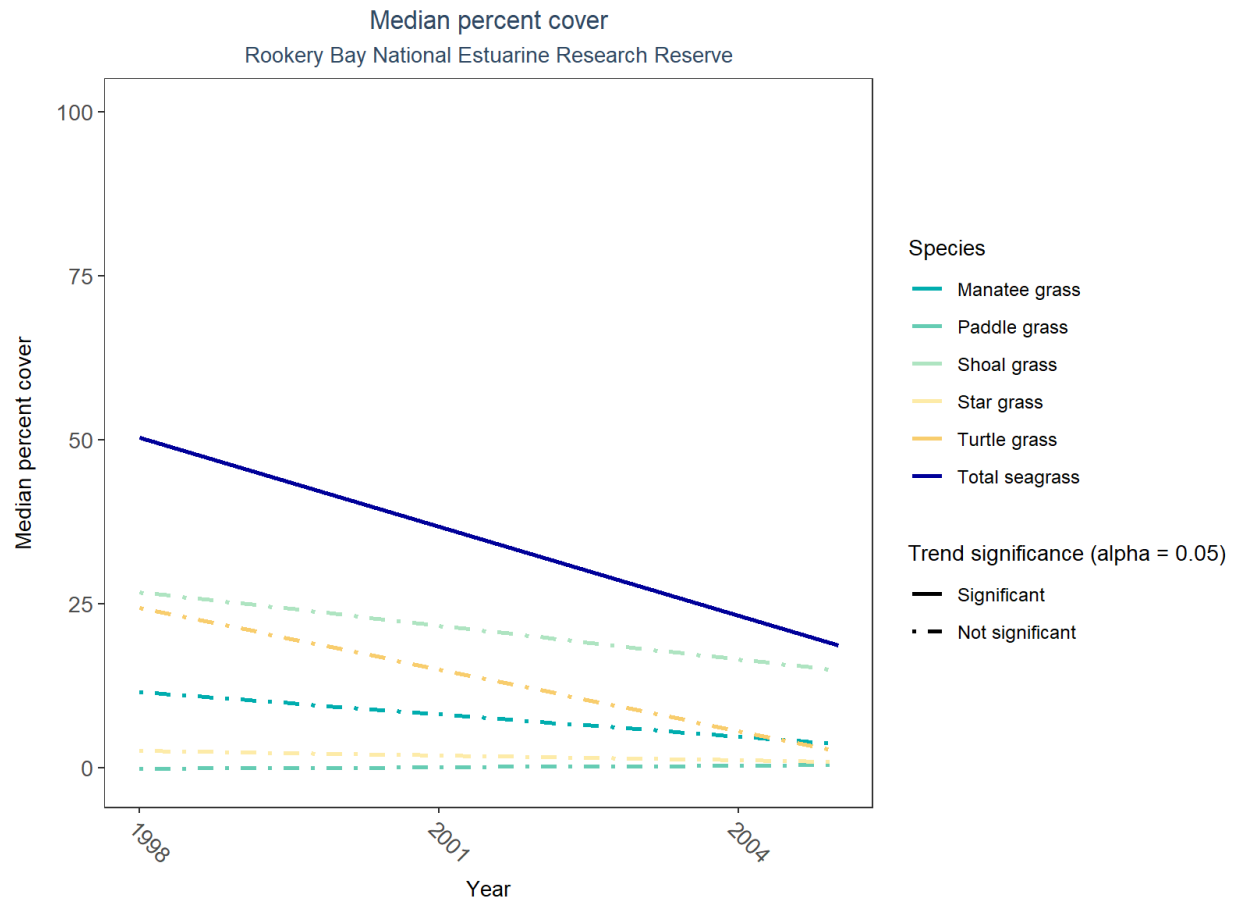


Table 25: SAV LME Results for Rookery Bay National Estuarine Research Reserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	1998 - 2005	33.63	-1.71	0.25
Paddle grass	No detectable trend	1998 - 2005	-0.44	0.08	0.36
Star grass	No detectable trend	1998 - 2005	3.53	-0.23	0.44
Manatee grass	No detectable trend	1998 - 2005	16.10	-1.13	0.40
Turtle grass	No detectable trend	1998 - 2005	36.99	-3.14	0.07
Total seagrass	Decreasing trend	1998 - 2005	68.37	-4.52	0.02

An annual decrease in percent cover was observed for total seagrass (-4.52%). Manatee grass, paddle grass, shoal grass, star grass, and turtle grass showed no detectable change in percent cover.



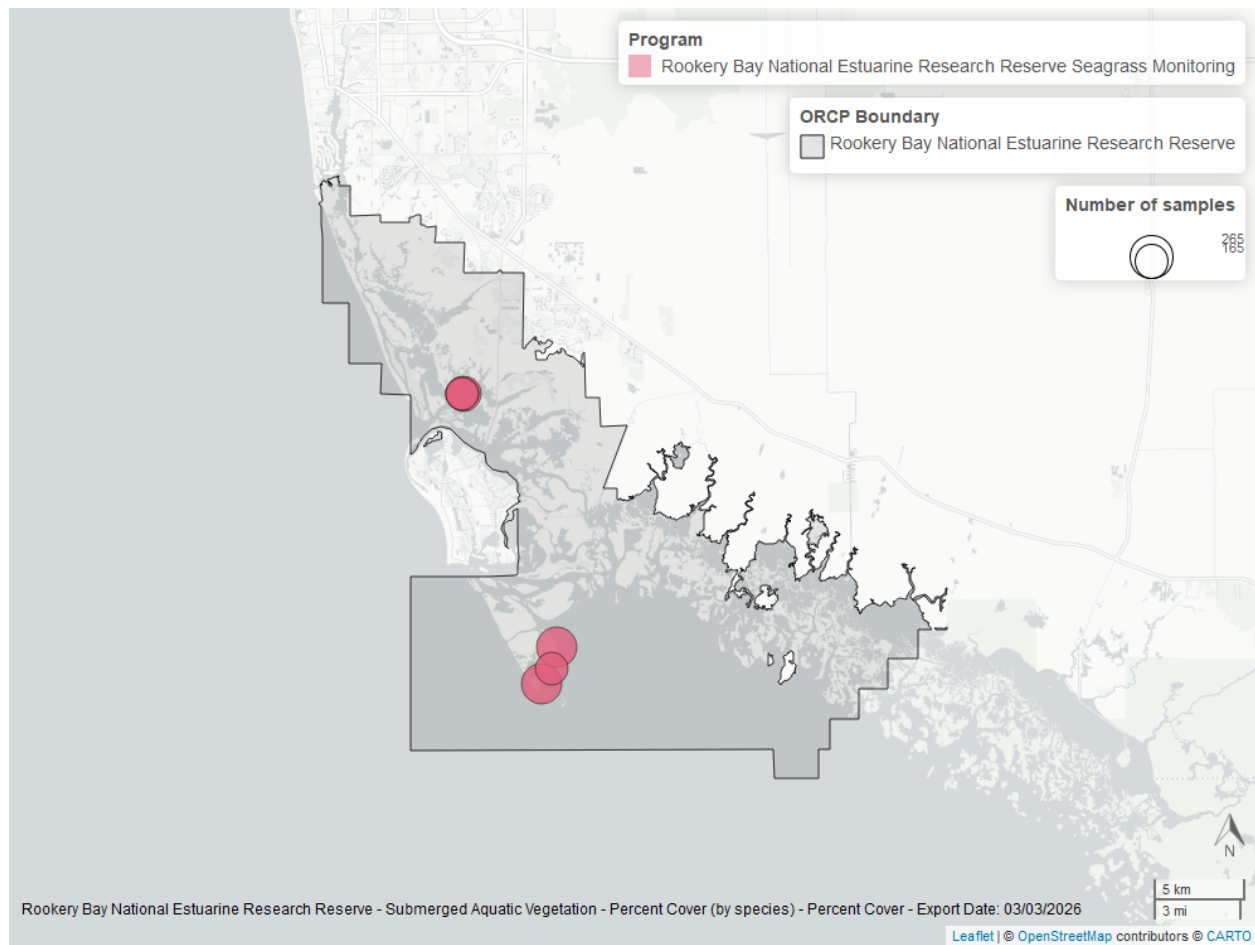
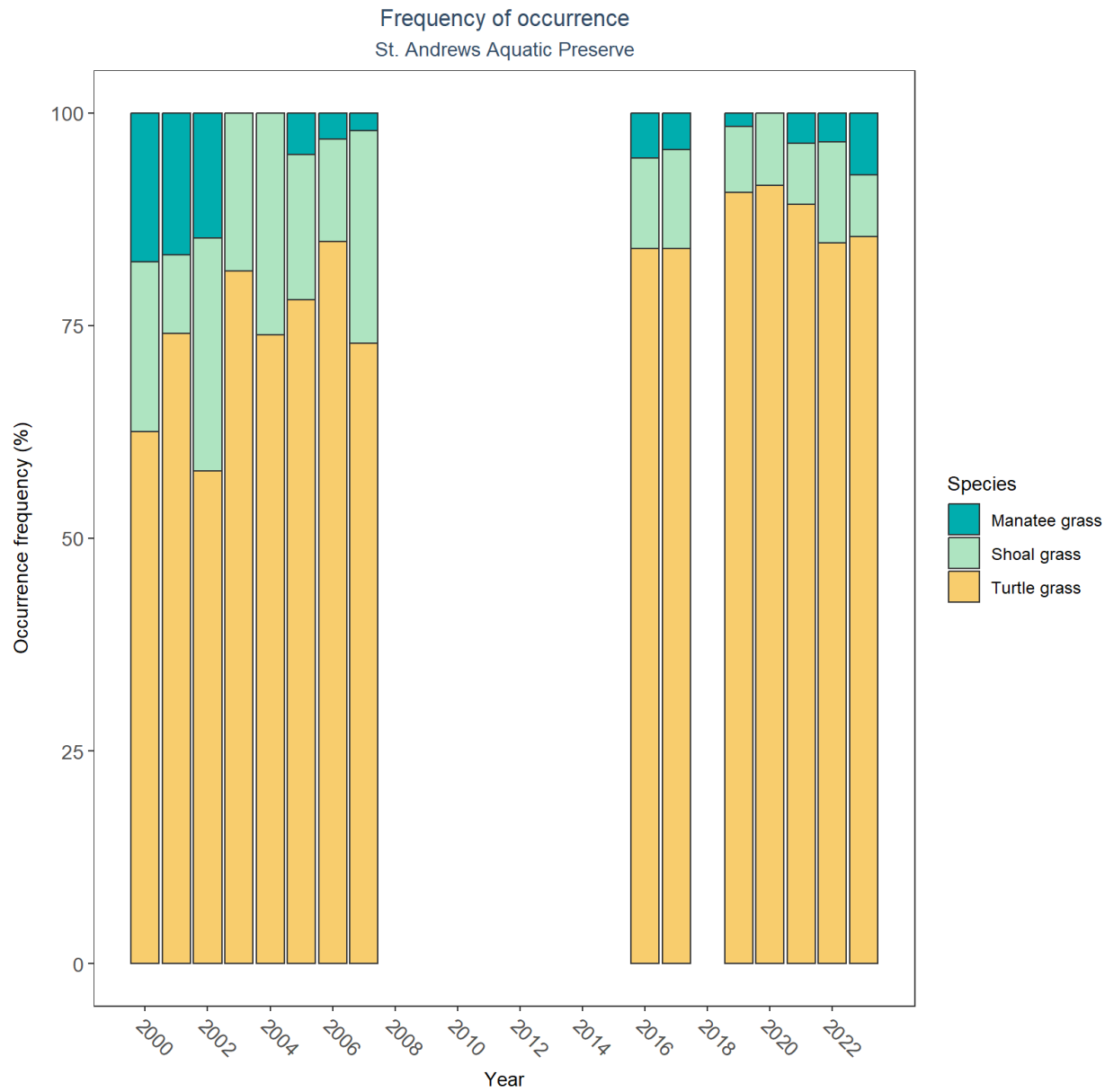


Figure 25: Map showing SAV sampling sites within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The point size reflects the number of samples at a given sampling site.

St. Andrews Aquatic Preserve



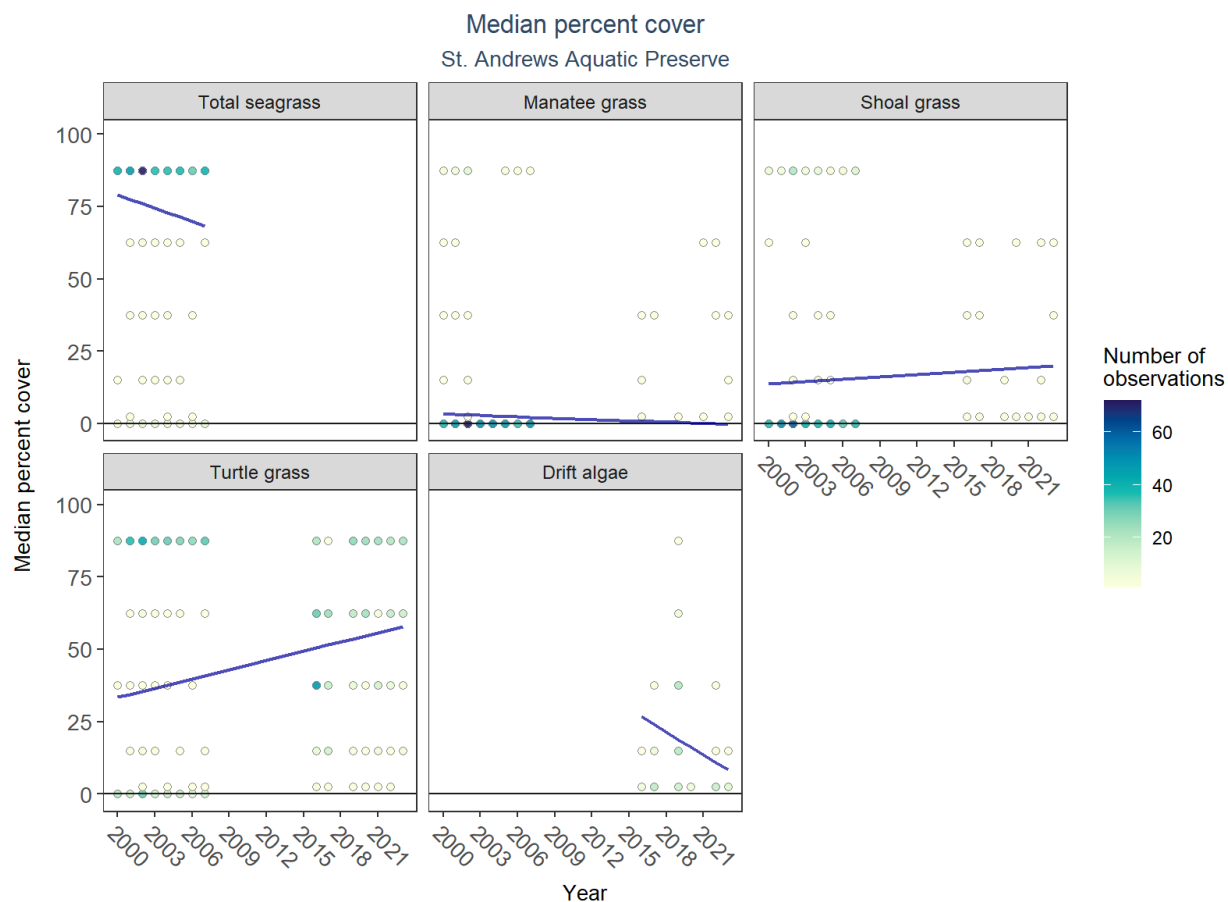
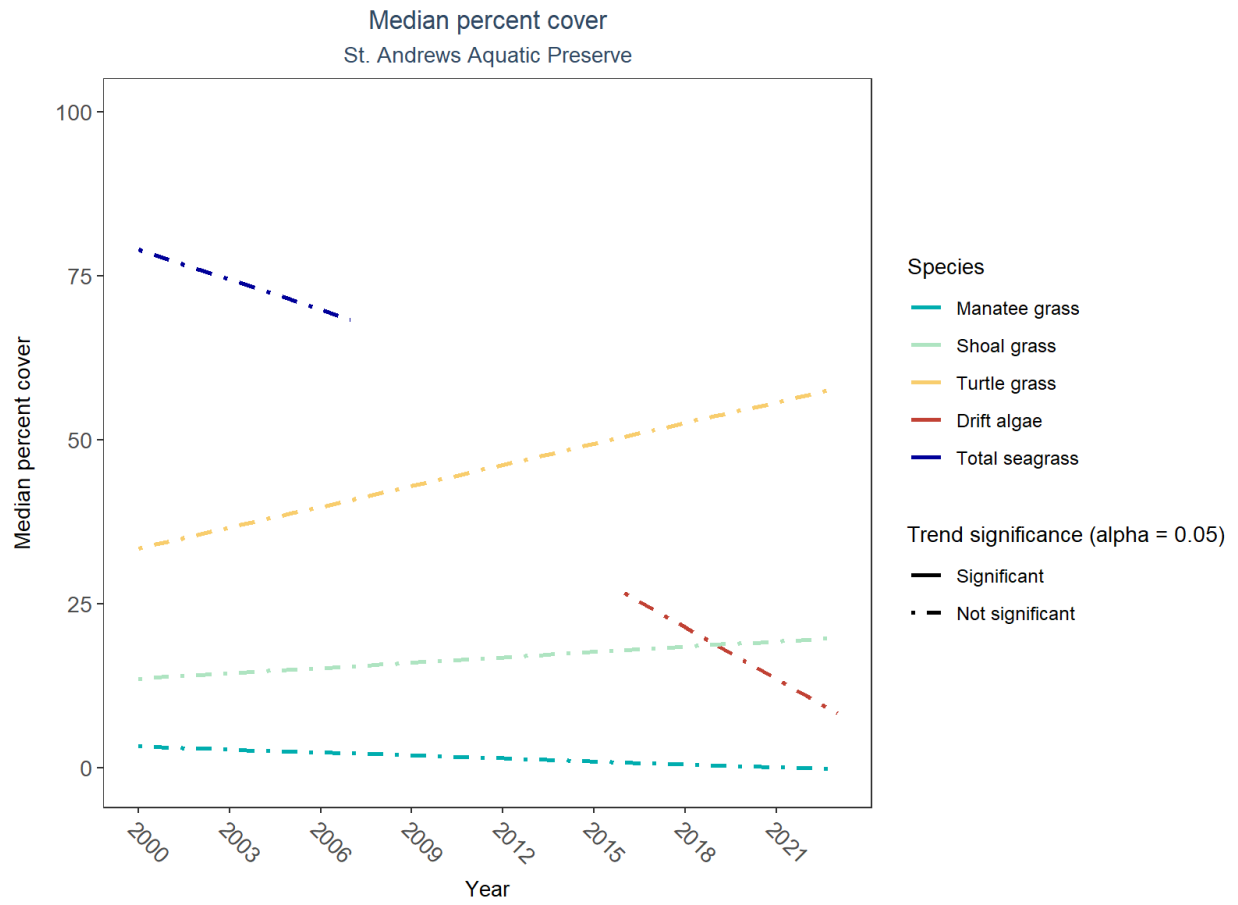


Table 26: SAV LME Results for St. Andrews Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	2016 - 2023	84.42	-2.62	0.07
Shoal grass	No detectable trend	2000 - 2023	11.99	0.27	0.49
Manatee grass	No detectable trend	2000 - 2023	4.17	-0.15	0.82
Turtle grass	No detectable trend	2000 - 2023	27.04	1.06	0.13
Total seagrass	No detectable trend	2000 - 2007	88.14	-1.52	0.15

Total seagrass, manatee grass, shoal grass, turtle grass, and drift algae showed no detectable change in percent cover.



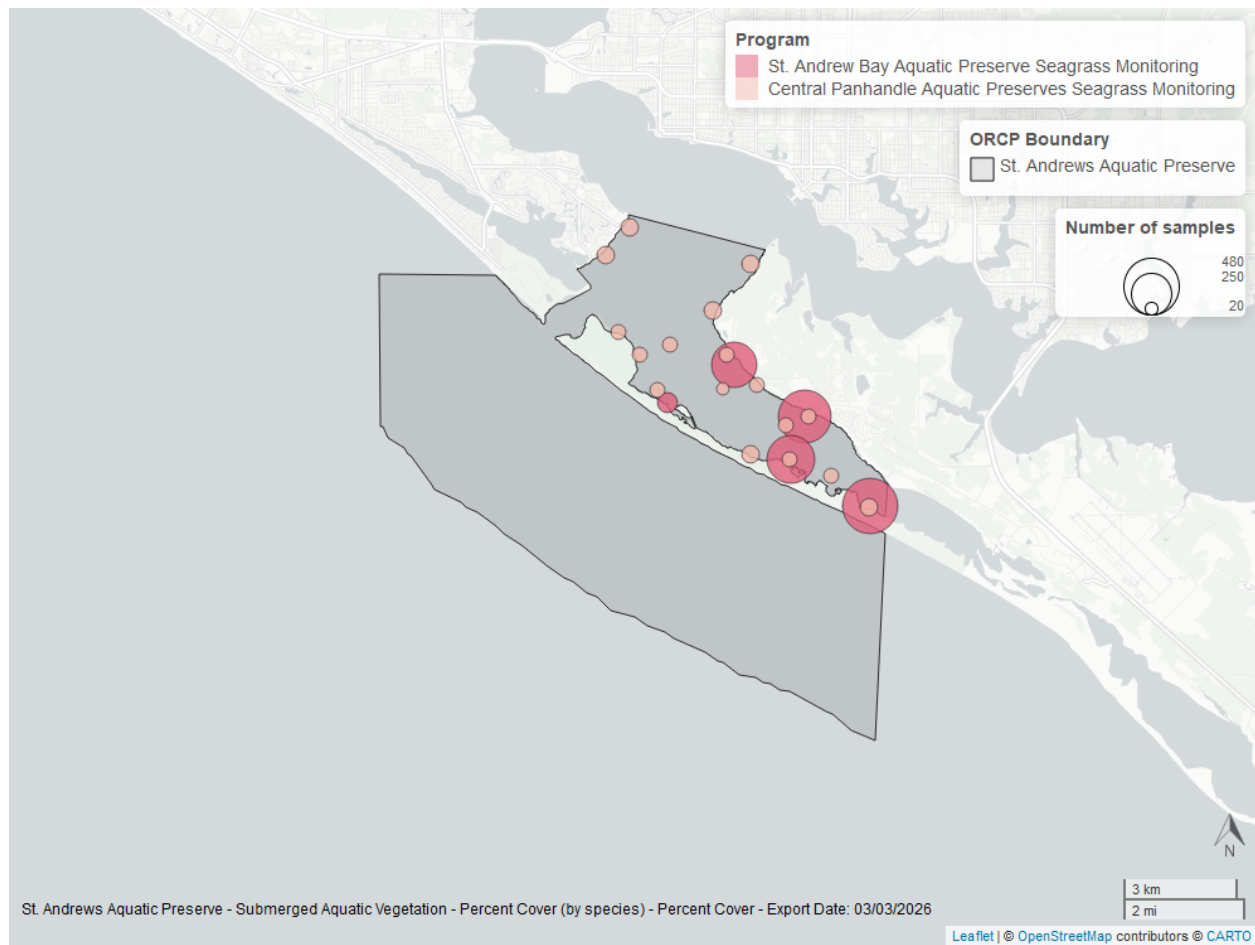
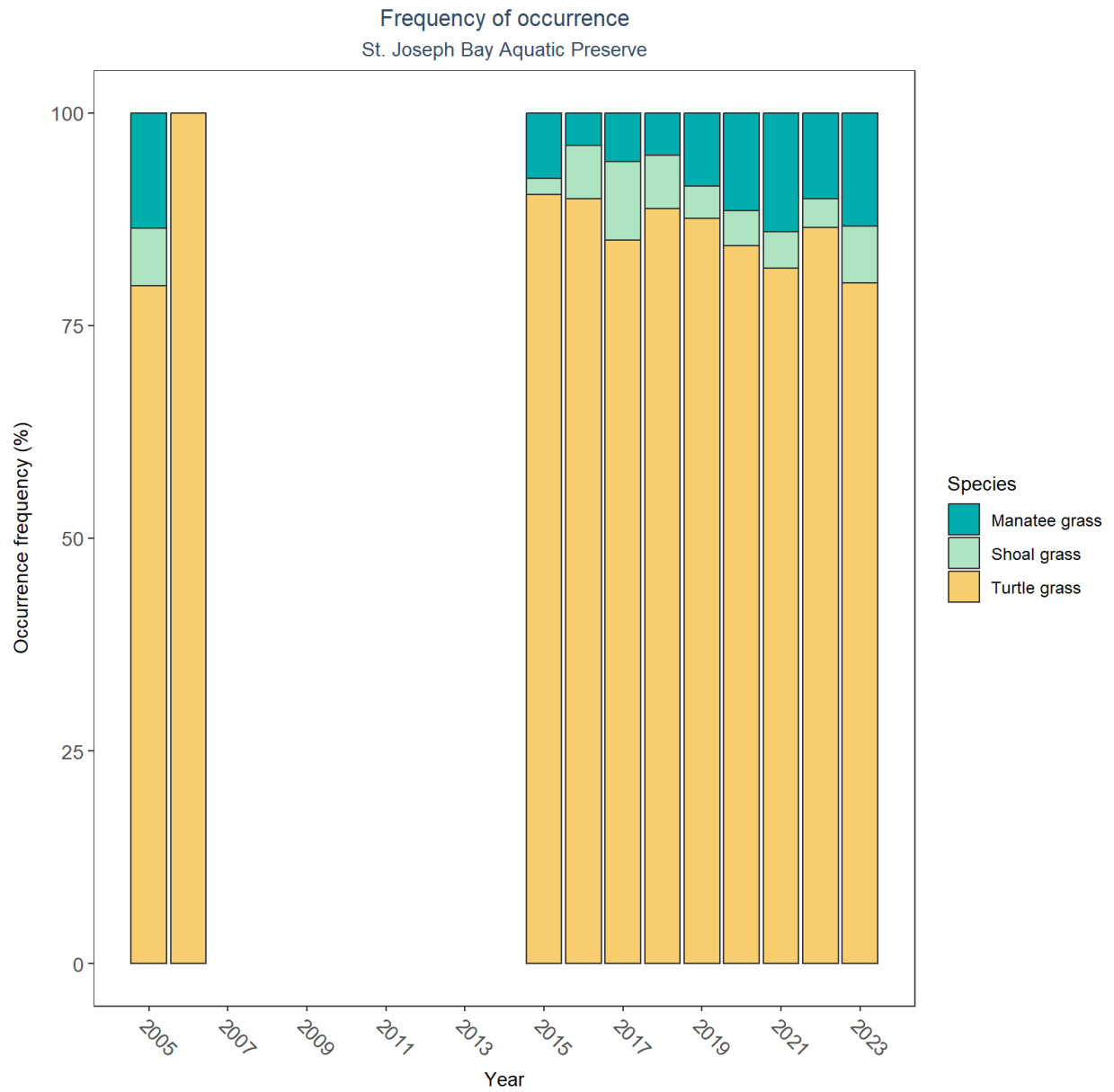


Figure 26: Map showing SAV sampling sites within the boundaries of *St. Andrews Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

St. Joseph Bay Aquatic Preserve



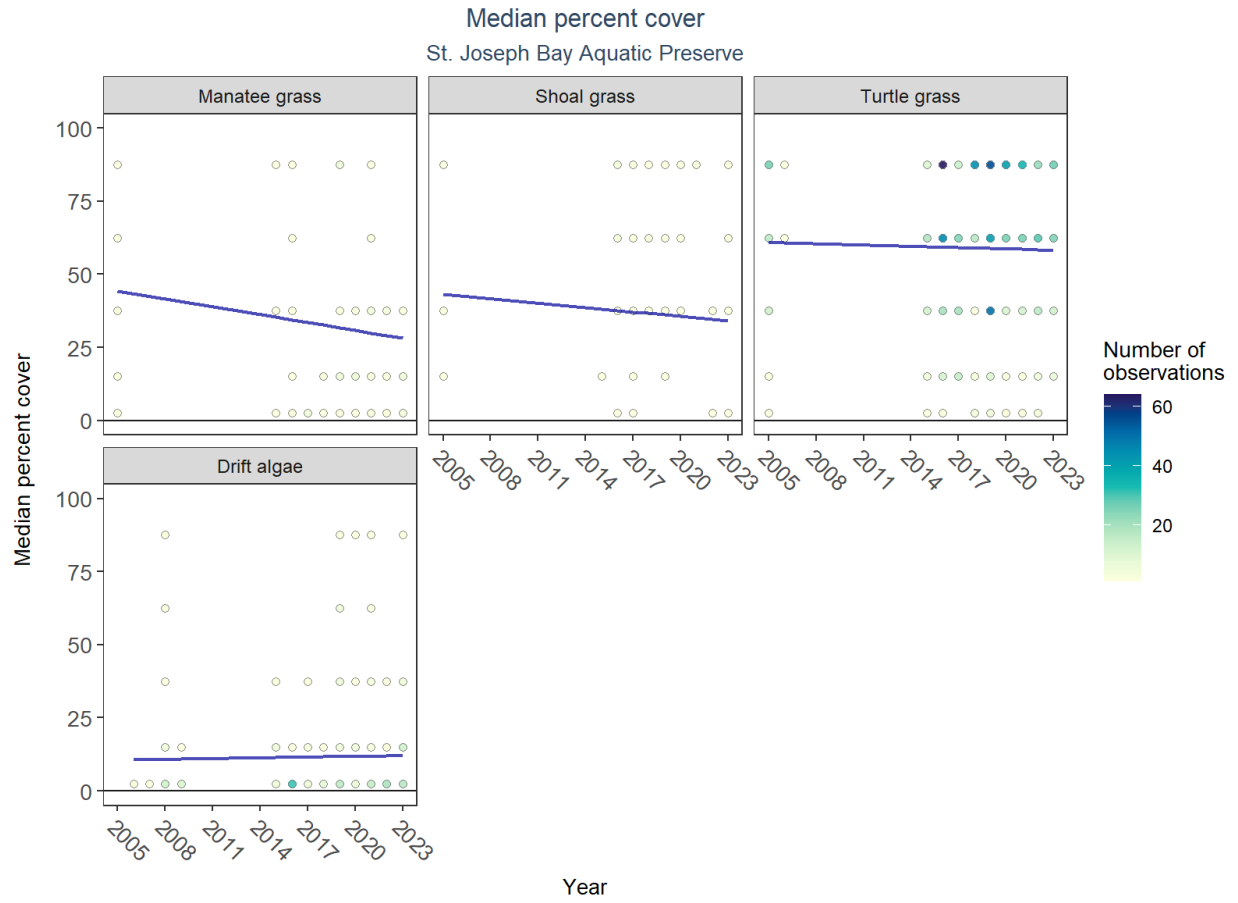
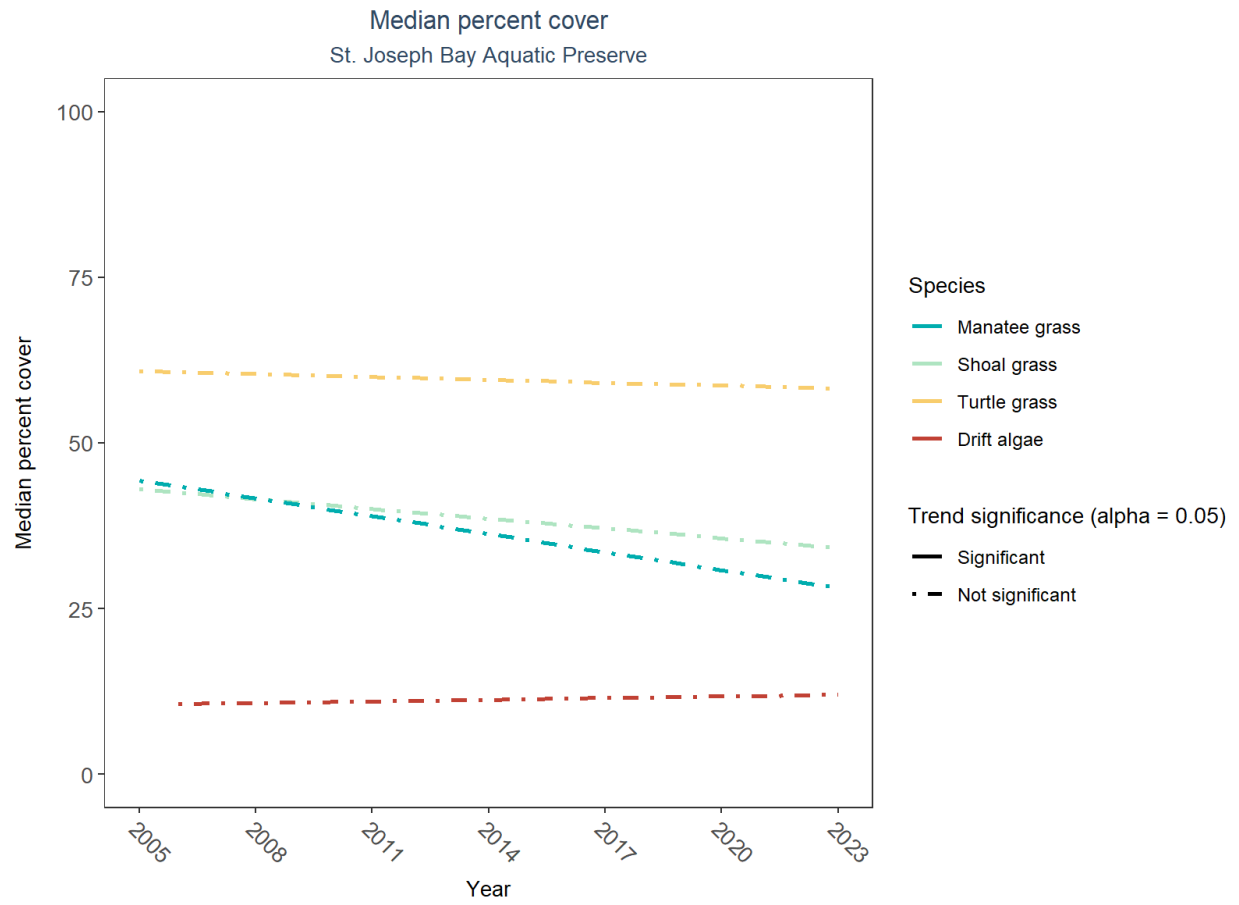


Table 27: SAV LME Results for St. Joseph Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	2006 - 2023	9.61	0.08	0.83
Shoal grass	No detectable trend	2005 - 2023	48.45	-0.49	0.50
Manatee grass	No detectable trend	2005 - 2023	54.20	-0.90	0.26
Turtle grass	No detectable trend	2005 - 2023	62.38	-0.14	0.69

Manatee grass, shoal grass, turtle grass, and drift algae showed no detectable change in percent cover.



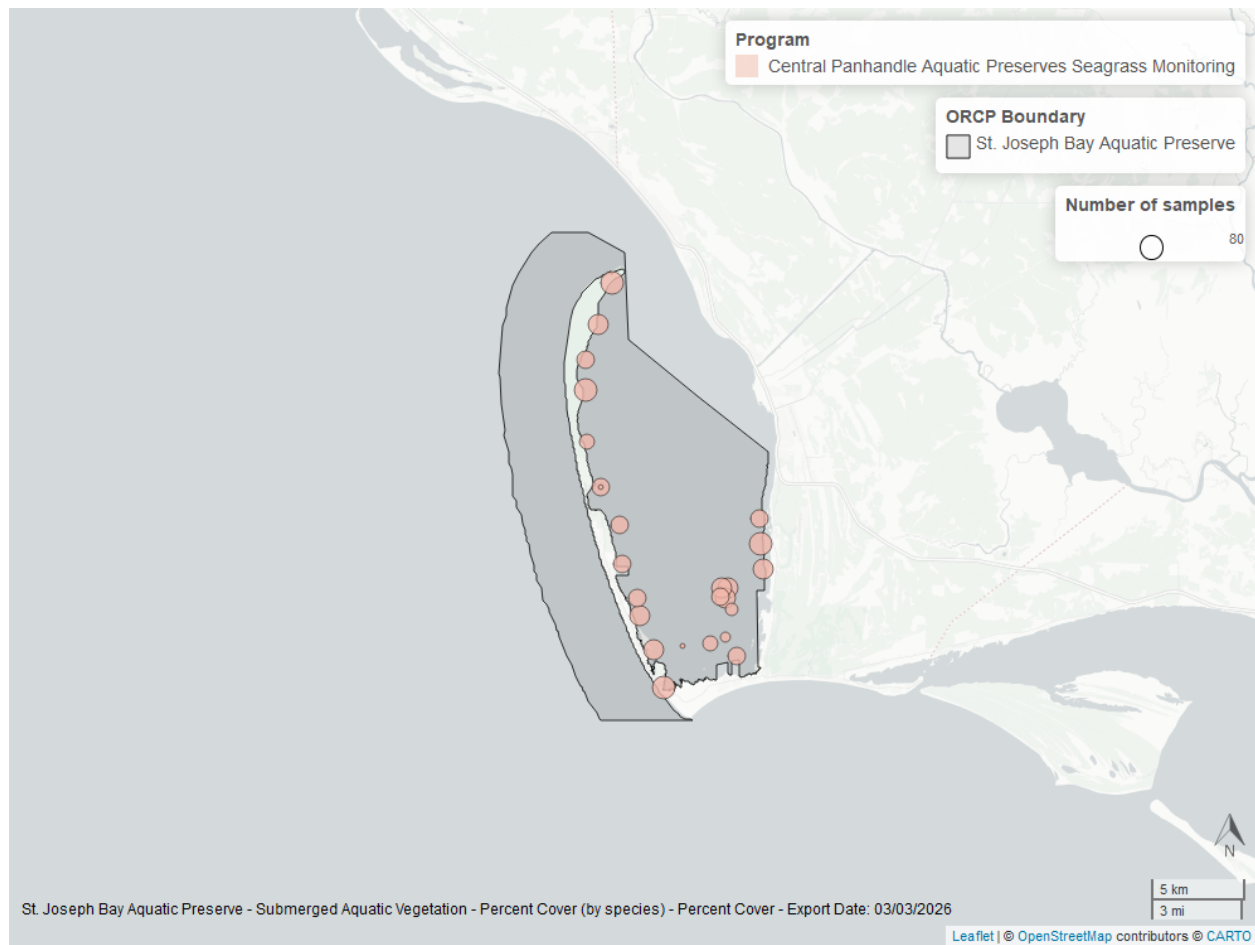
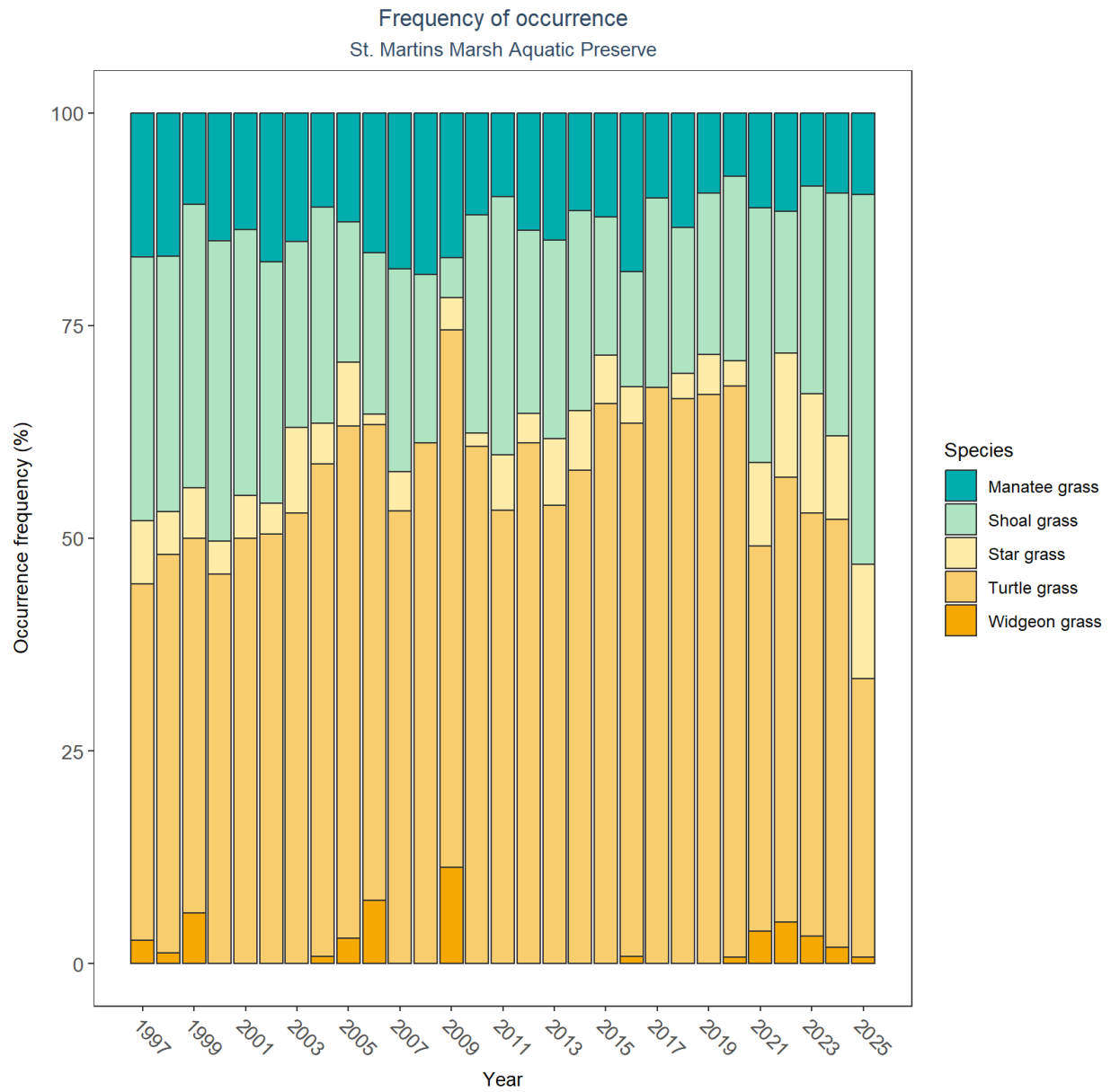


Figure 27: Map showing SAV sampling sites within the boundaries of *St. Joseph Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

St. Martins Marsh Aquatic Preserve



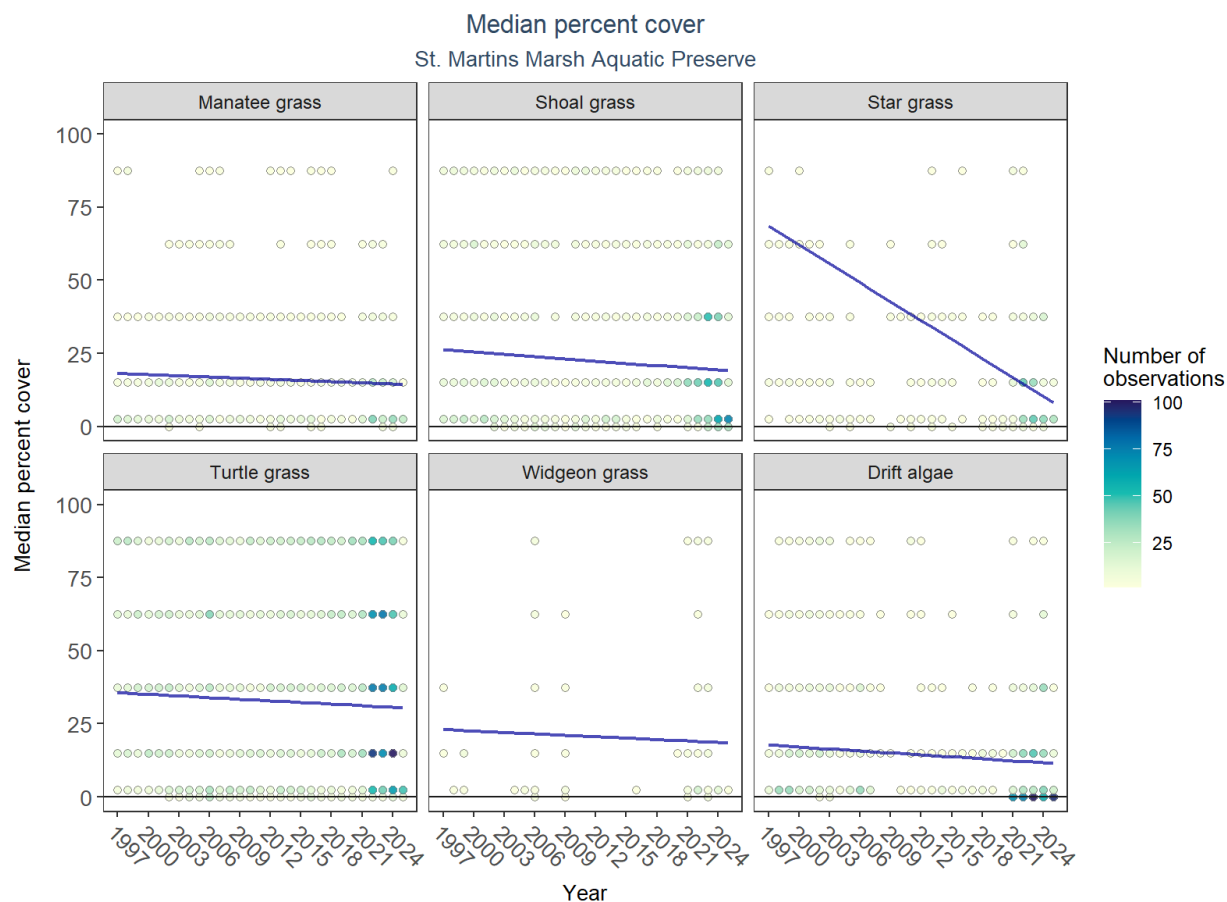
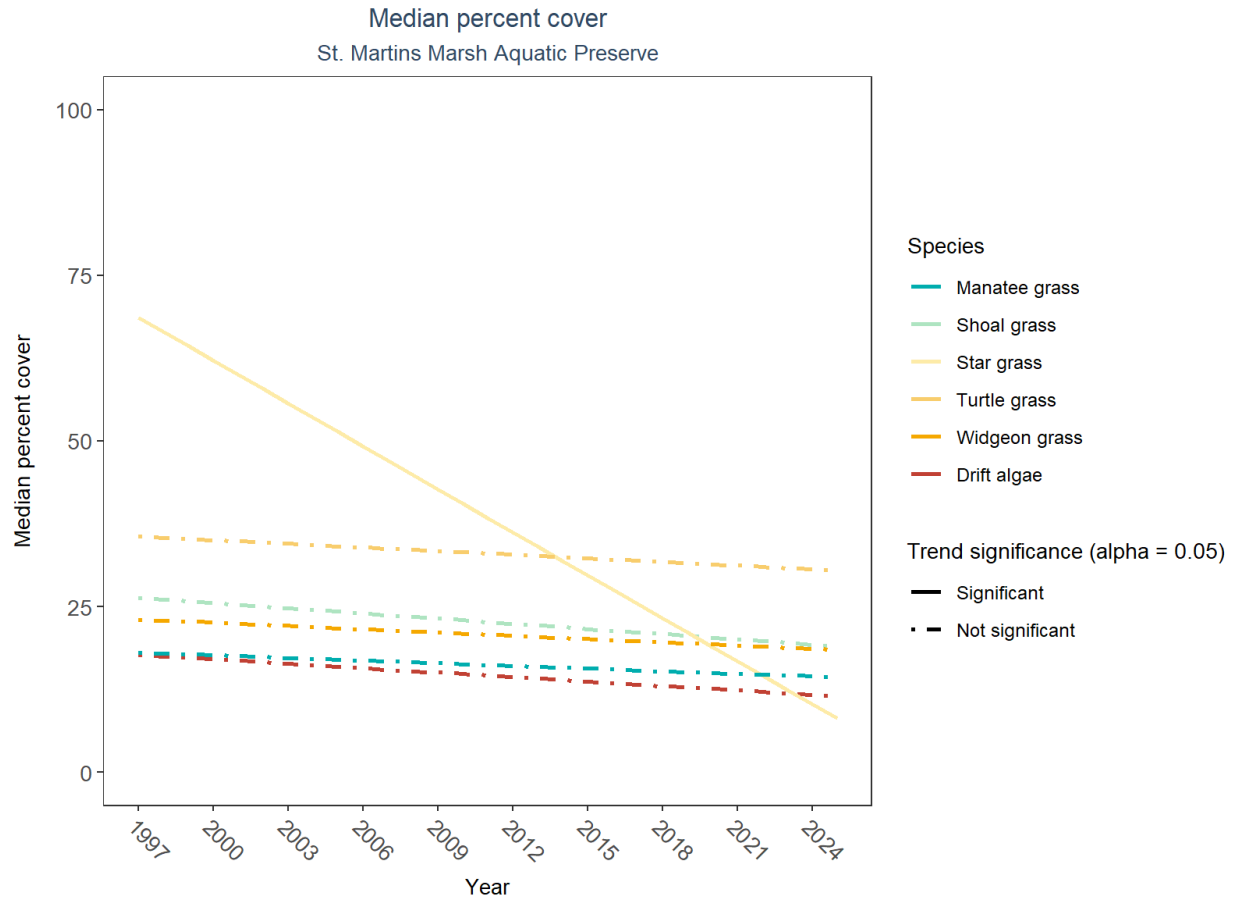


Table 28: SAV LME Results for St. Martins Marsh Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	1997 - 2025	18.42	-0.22	0.10
Shoal grass	No detectable trend	1997 - 2025	27.14	-0.26	0.14
Star grass	Decreasing trend	1997 - 2025	75.14	-2.16	0.04
Widgeon grass	No detectable trend	1997 - 2025	23.54	-0.16	0.66
Manatee grass	No detectable trend	1997 - 2025	18.49	-0.13	0.51
Turtle grass	No detectable trend	1997 - 2025	36.16	-0.18	0.25

An annual decrease in percent cover was observed for star grass (-2.16%). Manatee grass, shoal grass, turtle grass, widgeon grass, and drift algae showed no detectable change in percent cover.



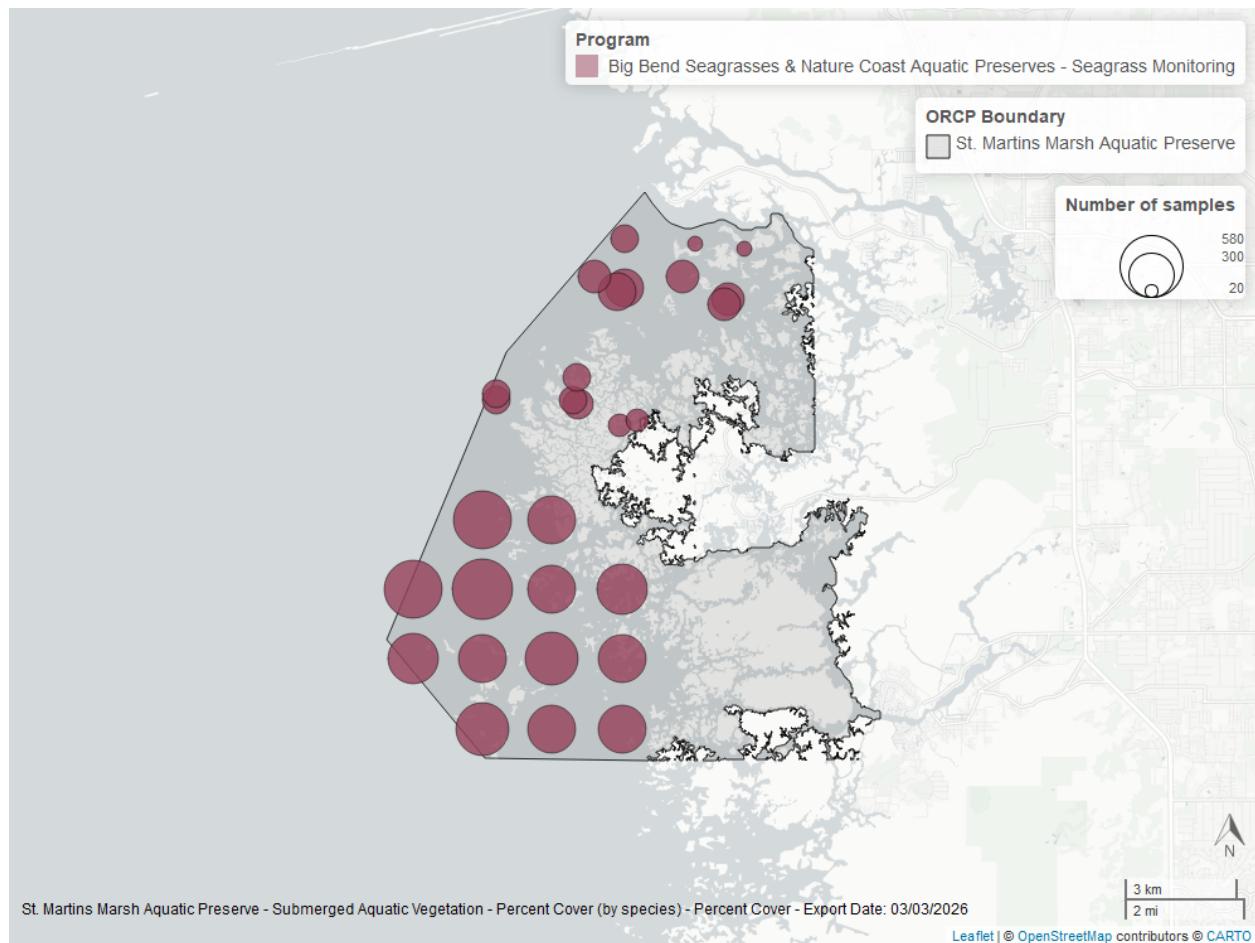
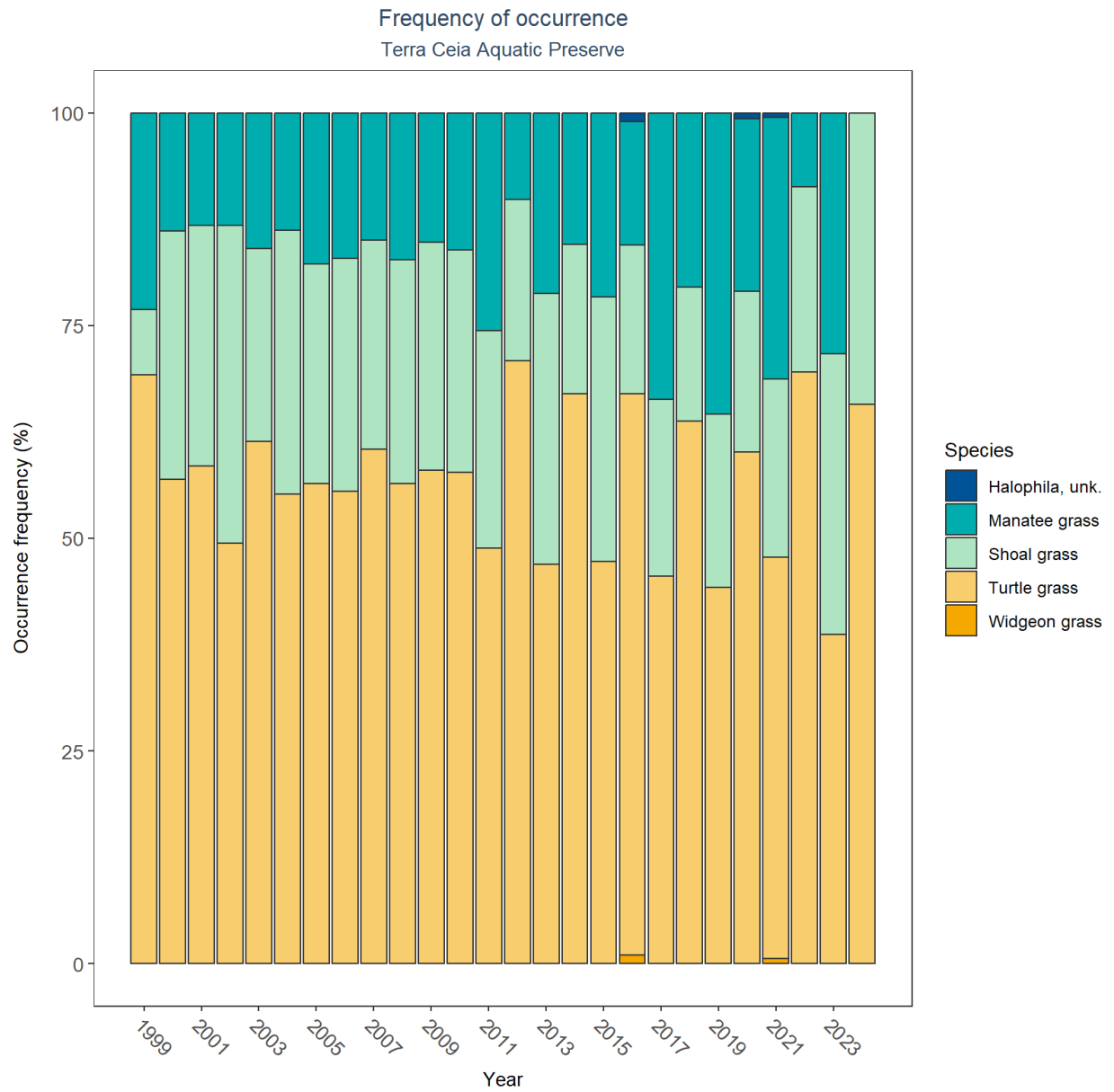


Figure 28: Map showing SAV sampling sites within the boundaries of *St. Martins Marsh Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Terra Ceia Aquatic Preserve



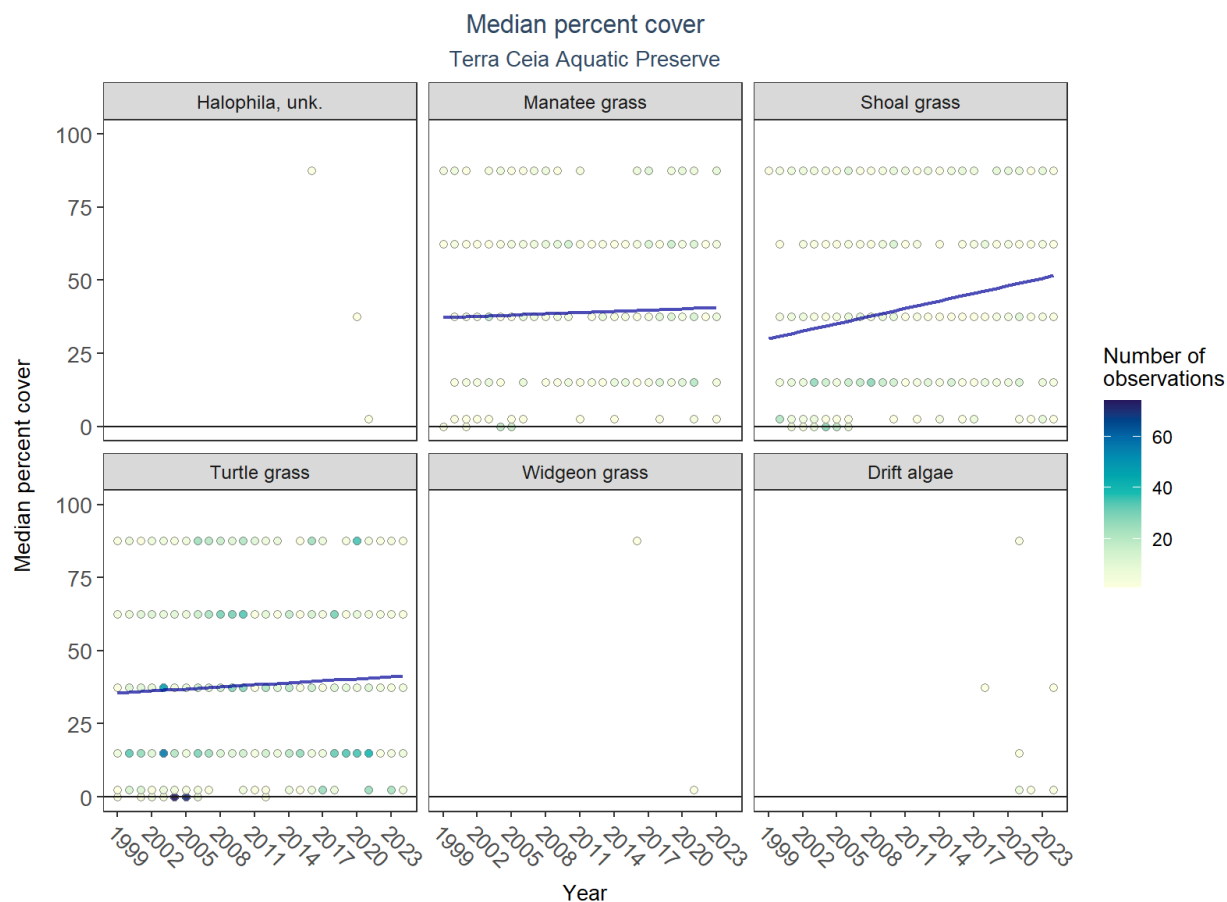
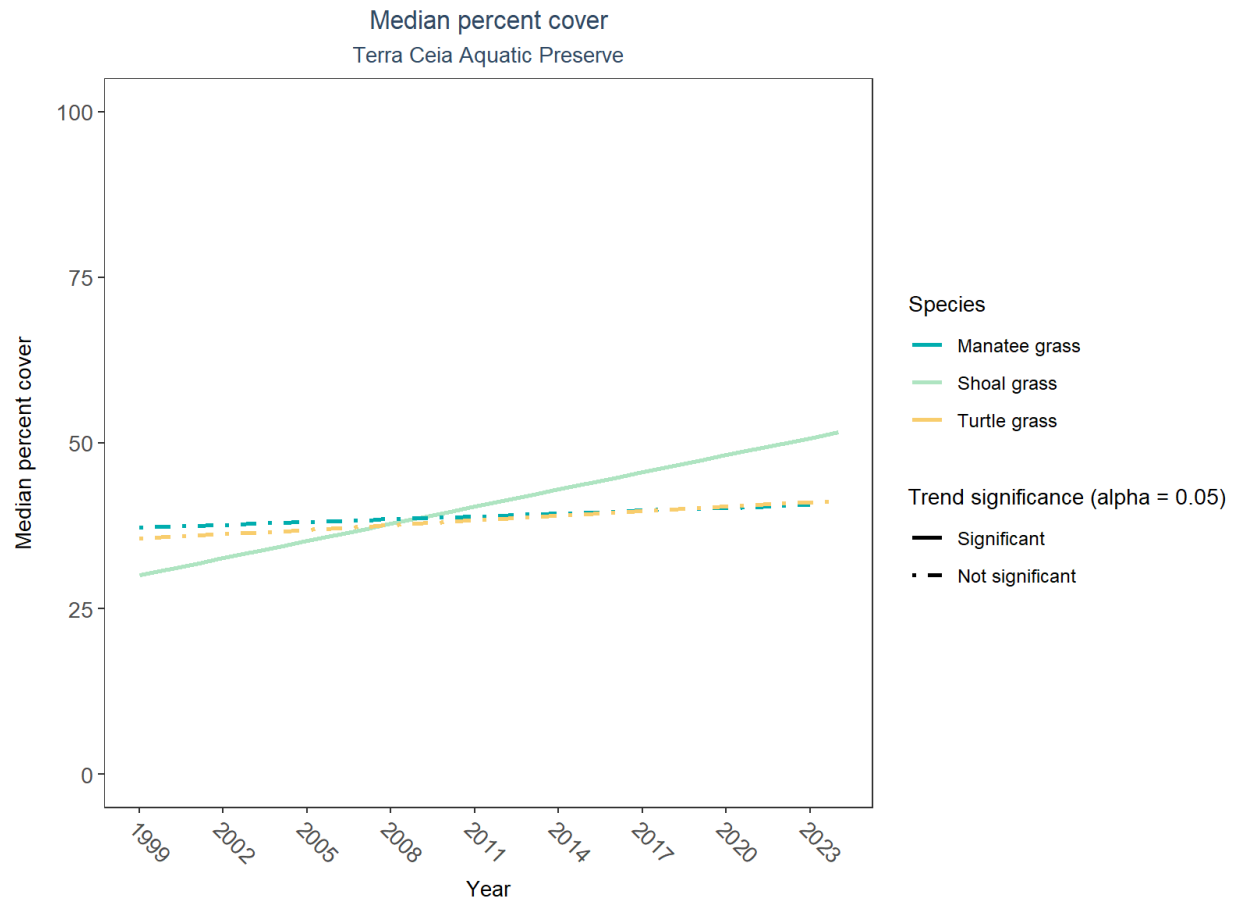


Table 29: SAV LME Results for Terra Ceia Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	-	-	-	-
Shoal grass	Increasing trend	1999 - 2024	25.74	0.86	0.00
Widgeon grass	Insufficient data	-	-	-	-
Manatee grass	No detectable trend	1999 - 2023	36.54	0.14	0.71
Turtle grass	No detectable trend	1999 - 2024	34.43	0.23	0.45
Halophila, unk.	Insufficient data	-	-	-	-

An annual increase in percent cover was observed for shoal grass (0.86%). No detectable change in percent cover was observed for manatee grass and turtle grass. Trends in percent cover could not be evaluated for unknown *Halophila*, widgeon grass, and drift algae due to insufficient data.



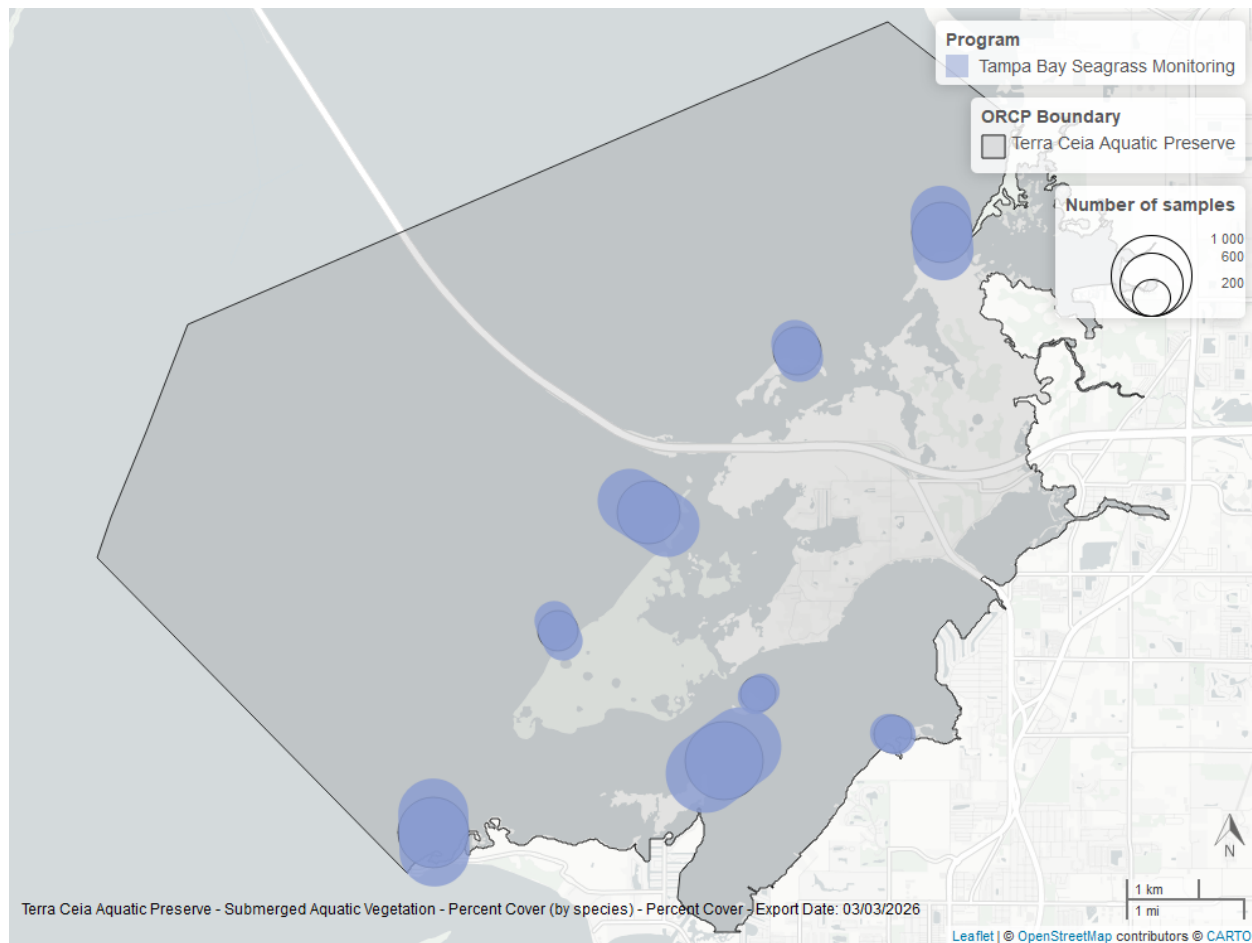


Figure 29: Map showing SAV sampling sites within the boundaries of *Terra Ceia Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Summary of SEACAR_SAV_BB_script_website.R

- Objective: Import and format SAV data, create summary plots and maps of changes over time, model changes over time using Bayesian and mixed-effects models. Results are separate for each managed area, species, and parameter (e.g., Braun Blanquet, percent cover, etc.).
- Packages: bayesplot, brms, broom.mixed, data.table, grid, gridExtra, gtable, nlme, scales, sf, tictoc, tidybayes, tidyverse
- File inputs: All_SAV_Parameters-2024-Sep-19.txt, seacar_dbo_SampleLocation_Point.shp, seacar_dbo_SampleLocation_Line.shp, ORCP_Managed_Areas.shp, Counties_-_Detailed_Shoreline.shp, MAPolygons_corners.csv
- Steps by line number:
 - 1 - 40: load libraries and import SAV file, create necessary sub-folders.
 - 43 - 132: format SAV data including renaming columns, removing NA values, and reformatting abundance/cover values. For the latter, this included removal of NA values and those out of range, and ensuring appropriate values for Braun Blanquet, modified Braun Blaunqet, percent occurrence, and percent cover.
 - 136: Import ManagedArea.csv to obtain abbreviated ManagedAreaNames.
 - 139 - 193: function for plotting model predictions (multiplots).
 - 196 - 288: function for plotting model predictions with simplified trendlines on a single plot

- (trendplots).
- 292 - 307: function used throughout script to allow for swift modification of species labels between “common” and “scientific” names. The script defaults to using scientific throughout the backend (i.e. `.rds` objects with species names default to scientific), with the `usenames` variable (line 468) converting the display of species names to the desired display type.
 - 310: EDA variable to specify what the script should produce. Setting to “plots” creates data exploration plots.
 - 313: `scope_plots` variable can be set to `TRUE` or `FALSE`. Set to `TRUE` to render spatio-temporal scope plots for each `ManagedArea`. Resulting plots are placed in the ‘output/Figures/SAV_temporal_scope_plots/’ folder.
 - 323 - 324: create `failedmods` data table to store failed model results.
 - 327 - 499: setup parameter list and objects for looping through parameters to create models and summary output, parameters include Braun Blanquet, median percent cover, visual percent cover, percent occurrence, frequency of occurrence. Write results to output `.xlsx` files.
 - * 351 - 395: setting up palettes for both species (`spcols`) and ProgramNames (`prcols`).
 - 517 - 1192: loop through parameters to create models and summary output, the following is an outline of steps in this loop.
 - * 530 - 1518: loop through managed areas using parameter from outside loop
 - * 539 - 553: create and save plot of parameter score for managed area over time by species
 - * 555 - 568: create and save plot of parameter score for managed area over time by program ID
 - * 570 - 583: create and save plot of parameter score for managed area over time grouped by species, by program ID
 - * 585 - 598: create and save plot of quadrat sizes for managed area over time by species
 - * 600 - 613: create and save plot of quadrat sizes for managed area over time by program ID
 - * 615 - 628: create and save plot of method for managed area over time by species
 - * 630 - 643: create and save plot of method for managed area over time by program ID
 - * 645 - 659: create and save plot of method for managed area by quadrat size and species
 - * 661 - 675: create and save plot of method for managed area by quadrat size and program ID
 - * 677 - 707: create and save plots of grid values over time by species and program ID if data available
 - * 709 - 739: create and save plots of depth values over time by species and program ID if data available
 - * 743 - 759: create and save a plot legend of species
 - * 762 - 785: loop through species to create and save a plot of parameter score over time
 - * 787 - 808: create and save a plot of totals for the species of parameter score over time
 - * 811 - 827: create and save a plot legend of species
 - * 830 - 853: loop through species to create and save a plot of parameter score as boxplots over time
 - * 855 - 875: create and save a plot of boxplots for the species of parameter score over time
 - * 898 - 908: setup empty objects for model results
 - * 912 - 1029: Loop through species to fit models, with separate exception statements for different parameters. The modeling workflow is similar for each parameter, with minor exceptions. The general goal of each is to assess trends in a parameter over time for a particular species and managed area. Each workflow includes error handling if models did not converge, produces summary tables of model fit, and summary plots showing model result. The models vary in the Gaussian distribution family for the response variable depending on parameter. Random effects (e.g., for `LocationID`) are used for all models.
 - * 1033 - 1124: create base plot of seagrass percent cover data over time for managed area. Add model fits if applicable, and saves object
 - 1036 - 1046: reads and sets modeled data for plots
 - * 1127 - 1185: create and save barplots of parameter results for managed area over time by species, only for Braun Blanquet and percent cover, save model results from prior loops
 - * 1197 - 1265: export `.png` of all plot types, setting height dynamically for `multiplots`.
 - * 1273 - 1277: zip all images into subfolders and create `SAV_Figures_[usenames].zip` where

`usenames` is either “scientific” or “common” based on the `usenames` variable setting.